On the Collatz Conjecture

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Martingales, Momentum
and Continued Fractions

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On the Collatz Conjecture: Martingales, Momentum and Continued Fractions.

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Independently published.

For the students who keep computing the orbit of 27.

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Preface

Take a positive integer. If it is even, halve it; if it is odd, triple it and add one. Repeat. The Collatz conjecture says that every starting value eventually reaches 1. It can be explained to a child in a minute, checked by a laptop for every number a person could write down in a lifetime, and it has resisted every serious attempt at proof since the 1930s. This book is about why.

We came to the problem from outside number theory. One of us works in Bayesian statistics and probability, the other in optimization and machine learning, and what drew us in was a simple observation that anyone trained in either field makes within an hour of looking at the map. The logarithm of the iterate behaves like a random walk with negative drift. Each odd step multiplies by about $3/2$, each even step divides by 2, and if the parities were fair coin flips the expected change in $\log x$ per step would be $\frac{1}{2}\log\frac{3}{2} - \frac{1}{2}\log 2 = \log(\sqrt{3}/2) \approx -0.144$. That is a Lyapunov function decreasing in expectation, the most familiar object in optimization and control, and it predicts the growth of the stopping times of real orbits to within about two percent. None of it is a proof, and the reason it is not a proof is the subject of the book.

What the book does

We read the Collatz dynamics as an optimization problem and push the reading as far as it will go. Three things come out.

The first is an accounting of what a proof would have to contain. The conjecture splits into two halves that require completely different tools. The divergence half asks whether $\log x_n$ can drift upward forever along a single orbit. It is equivalent to a statement about Cesàro averages of the dyadic valuations $v_2(3y+1)$ along that orbit, namely that the lower limit of their running mean exceeds $\log_2 3 \approx 1.585$ (Conjecture 11.6), and it is a one sided law of large numbers for a deterministic sequence that nobody knows how to prove. The cycle half asks whether an orbit can close up. It is not an averaging question at all: a cycle forces a product of numbers slightly larger than 3 to equal a power of 2, so the rational approximations of $\log_2 3$ enter, although, as we show, they cannot decide it on their own.

The second is a set of structural theorems, most of them short and several of them classical. We prove that the Syracuse exponent is exactly Geometric$(\frac{1}{2})$ on residue classes, that under the Bernoulli model the iterates form a multiplicative martingale while their logarithms form a strict supermartingale (the gap between

the two is Jensen's inequality), that the parity autocorrelation function of the standard map is $\rho(h) = (-\frac{1}{2})^h$ in the model, and that no monotone Lyapunov function certifies descent over any fixed number of strides. On cycles we give a short mediant lemma for the one sided distances $\lceil r \log_2 3 \rceil - r \log_2 3$ which, with a self contained proof of the classical description of the one sided best approximations, shows that only the upper rungs of the semiconvergent ladder of $\log_2 3$ ever need to be checked. Combined with verification up to 2^{68} it gives, unconditionally, that every nontrivial cycle has at least 72 057 431 991 odd elements. The method goes back to Crandall and Eliahou, the bound is that of Hercher and Puchert, and we claim no priority for any numerical bound; the current record, $r \geq 1.375 \times 10^{11}$, is Hercher's, made unconditional by Bařina's verification to 2^{71}. What we offer is a short rigorous route to the bound and an exact integer certificate for it. We then prove that Diophantine information about $\log_2 3$ alone cannot finish the cycle half, since infinitely many cycle lengths survive every verification range, and we show with a window lemma that an arithmetic improvement of the basic cycle inequality by a factor slightly larger than $r/3$, combined with a finite verification, would. On the computational side, nested, split and vertical likelihood sampling reach regions of exponentially small prior mass in a number of steps polynomial in the depth, provided each level set can be sampled exactly. We prove that a perfect version of split sampling samples the exceptional residue classes exactly, at any depth and in time polynomial in the depth. Beyond the bit length of the integers the level set samplers eventually stop, and for split sampling the census shows why: in the boxes we computed, the slow integers are a small number of long orbits together with their tributaries, and a local sampler leaves the tree it starts in only about as often as rejection sampling finds a new one.

The third is an algorithm. The Collatz map alternates an ascent, the odd step, with a corrective descent, the run of halvings, and the ratio of the two logarithmic step sizes is $\beta = \log(3/2)/\log 2 = \log_2 3 - 1 \approx 0.585$. We turn this into Switching MM, a first order method with fixed momentum β that extrapolates only on odd iterations. On quadratics we prove that two of its steps are exactly a gradient step of size α followed by one of size $(1+\beta)\alpha$, so that it is a cyclic step size method in disguise, and we compute its stationary variance under gradient noise in closed form. The noise amplification of Switching MM relative to gradient descent at the same step is bounded by $5/3$ for every $\beta < 1$, whereas in the slow modes every step momentum amplifies noise by $1/(1-\beta)$, which is 10 at the machine learning default $\beta = 0.9$. That is the precise sense in which the parity schedule is a low tuning default. For any smooth strongly convex function we prove a linear rate whose factor is the exact worst case for the distance to the minimizer, and a gain weight theorem shows that no periodic momentum schedule of any period can beat gradient descent's noise floor in the slow modes at matched speed: momentum trades speed for noise in exact proportion, and unequal weighting of gradients pays a Cauchy and Schwarz penalty.

What the book does not do

It does not prove the Collatz conjecture. We say this at the outset and repeat it where it matters, because the literature on the problem contains many claimed proofs and the optimization language makes it easy to mistake a heuristic for an argument. Several of the most instructive passages in the book are pitfalls: arguments that look like progress and are not. The Cesàro criterion $\bar{k}_N > \log_2 3$ holds automatically at the stride N at which an orbit reaches 1, so checking it numerically on terminating orbits carries no information. Convergent denominators alone do not locate the records of the cycle bound, because the relevant distance is one sided. In the stochastic model, the linear majorizer of the logarithm has positive mean increment while the logarithm itself has negative drift. Two constants near one half, a momentum rate that pairs an ascent and a descent measured at different levels and the modulus of an eigenvalue of a chain conditioned on a set that it leaves, agree by accident (Remark 5.5).

How to read it

Part I sets up the problem: the history and the three standard maps (Chapter 1), the arithmetic of parity vectors and the Syracuse map including Terras' theorem and the density result (Chapter 2), and the stochastic models with their martingales (Chapter 3). Part II is the optimization reading: Lyapunov functions and their limits (Chapter 4); majorization and momentum (Chapter 5); majorization pressed to its limit on the Collatz map, where it reproduces the cycle inequality and shows that no correction of $\log x$ by a function of the last d binary digits of x, for any fixed d, certifies descent (Chapter 6); Switching MM as an optimizer (Chapter 7) and its behaviour under noise, with experiments (Chapter 8); and nonlinear difference equations and momentum MM, with difference inequalities for the inverse tree and a linear rate for Switching MM on smooth strongly convex functions that is tight for the distance to the minimizer (Chapter 9). Part III treats cycles through continued fractions, with a section on the sibling problem of Erdős 125 settled by machine checked proof search and on our own Lean statements (Chapter 10); the divergence half through Cesàro averages, mixing and Tao's theorem (Chapter 11); a synthesis of what a proof would require (Chapter 12); a computational path, with what the MM view implies for automated certificates (Chapter 13); what nested, split and vertical likelihood sampling can and cannot do, with the tree structure of the slow integers (Chapter 14); and a list of open problems and computational projects (Chapter 15). A gallery of computed pictures follows Part III, and two appendices tabulate the constants and describe the code.

A reader interested only in the optimizer can read Chapters 5, 7 and 8 and then Chapter 9, which uses their results, turning back to Part I and Chapter 4 for the stride arithmetic behind the momentum constant. A reader interested

only in the number theory can go from Part I directly to Chapters 10 and 11, but before Chapters 12 and 13 needs the impossibility results of Chapters 4 and 6 and the first two sections of Chapter 9 (with Section 9.4 for two comparisons in Chapter 13). Every chapter ends with notes, and every chapter except Chapter 15 with exercises.

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We thank participants at the Chicago Booth Statistics Workshop and colleagues who listened to early versions of these arguments and pointed out the ones that were wrong. The computations were done in Python with mpmath, NumPy, SciPy, SymPy, Numba, CVXPY and scikit-learn, and the figures were drawn with Matplotlib; the formal statements in Chapter 10 were checked in Lean 4.

Chicago and Fairfax, October 2026 N. G. P. V. S.

Notation

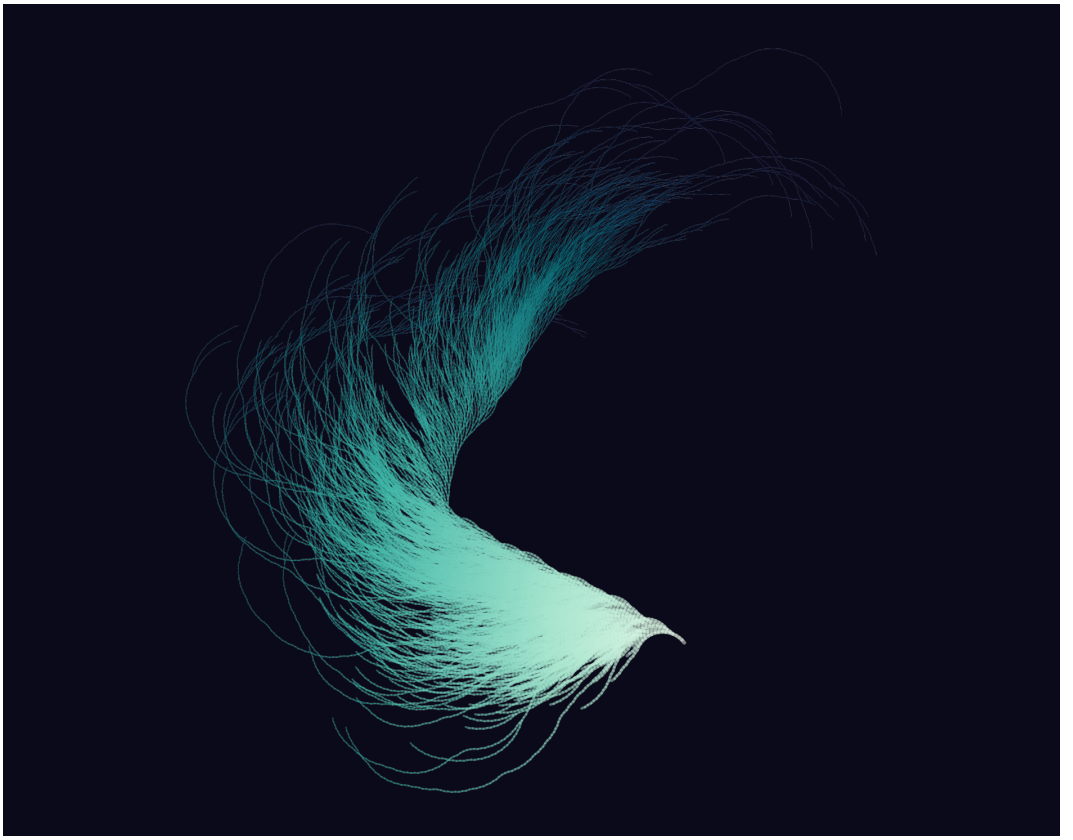
The table lists the symbols that a reader meets in more than one chapter, or that a later chapter uses without redefining. When a letter carries different meanings in different parts of the book, its row says so. Symbols local to one proof or one section are defined where they occur.

Symbol	Meaning	Defined in
T_0, $T_0^{(k)}$	standard map, $n/2$ for even n and $3n+1$ for odd n, and its kth iterate	Definition 2.1
T, $T^{(k)}$	accelerated map, $n/2$ for even n and $(3n+1)/2$ for odd n, and its kth iterate	Definition 2.1
S, $S^{(k)}$	Syracuse map on odd n, $S(n) = (3n+1)/2^{v_2(3n+1)}$, and its kth iterate; one step of S is a stride	Definition 2.1
$v_2(m)$	the 2 adic valuation: the exponent of the largest power of 2 dividing $m \neq 0$	Definition 2.1
k, k_j, $\bar{k}_N$	halving exponent $v_2(3x+1)$ of the stride from an odd x; $\bar{k}_N$ is the mean of $k_1, \ldots, k_N$ along an orbit of S	Section 2.1, Chapter 11
$v_k(n)$, $m_k(n)$, Φ_k	parity vector of n under T (first k steps), its number of ones (the odd steps), and the Terras bijection $n \bmod 2^k \mapsto v_k(n)$ of residue classes onto parity vectors	Definition 2.6, Theorem 2.8
$c_k(n)$	the integer in $2^k T^{(k)}(n) = 3^{m_k(n)} n + c_k(n)$, determined by $v_k(n)$	Proposition 2.11
$\mathbb{Z}_2$, μ, E	the 2 adic integers, their Haar probability measure μ, and the closed null set E of points with infinite coefficient stopping time	Section 2.5, Section 13.3
$\sigma(n)$	stopping time: least $k \geq 1$ with $T^{(k)}(n) < n$; unrelated to the standard deviation σ below	Definition 2.22
$\sigma_\infty(n)$	total stopping time: least k with $T_0^{(k)}(n) = 1$ in Chapter 1, but with $T^{(k)}(n) = 1$ in Chapters 3, 14 and 15	Section 1.1
$\tau(n)$	coefficient stopping time: least $k \geq 1$ with $3^{m_k(n)} < 2^k$; in Chapters 3 and 4, τ is instead a first passage time of a model walk	Definition 2.22
Bernoulli, stride and renewal models	coin models of T (factors $\xi_i \in \{\frac{1}{2}, \frac{3}{2}\}$), of S (factors $3 \cdot 2^{-K_j}$) and of T_0 (parity chain B_n)	Definition 3.1
δ^*, $\log \frac{3}{4}$	model drift of $\log x$: $\delta^* = \log(\sqrt{3}/2)$ per step of T, $\log \frac{3}{4}$ per stride of S, $\frac{1}{3} \log \frac{3}{4}$ per step of T_0; δ^* is not $\delta(r)$	Propositions 3.4 and 3.7 and Corollary 2.16
σ^2, C, γ_0	variance $\frac{1}{4}(\log 3)^2$ of a log step of T, index of dispersion $C = \sigma^2/(\delta^*)^2 \approx 14.58$, and slope $\gamma_0 = 1/\lvert\delta^*\rvert \approx 6.95$ of the model total stopping time of T in $\log n$; in Chapter 8, σ^2 is the gradient noise variance	Proposition 3.8, Section 8.1

Symbol	Meaning	Defined in
$\log_2 3$, θ	$\log_2 3 = 1.58496\ldots$ is the Cesàro threshold for $\bar{k}_N$; θ is $\log_2 3$ in Chapter 10, $1/\log_2 3$ in Theorem 2.26, and an exponent in Proposition 3.4	Section 1.1, Chapter 10
r, s, M, Λ	cycle of S: r odd elements, s halvings (one turn is r steps of S, s of T, $r+s$ of T_0), least element M, $\Lambda = s\log 2 - r\log 3$; written m, k in Chapters 2 and 15	Lemma 10.1, Remark 10.2
$\delta(r)$, Λ_r, $F(r)$	$\delta(r) = \lceil r\theta \rceil - r\theta$, $\Lambda_r = \delta(r)\log 2 \le \Lambda$, $F(r) = r/(3\Lambda_r) > M$	Equation (10.5), Equation (10.6)
a_k, p_k/q_k, $u_{k,j}$	partial quotients, convergents $1, 2, \frac{3}{2}, \frac{8}{5}, \frac{19}{12}, \ldots$ and ladder rungs $q_{k-1} + jq_k$ of $\log_2 3 = [1; 1, 1, 2, 2, 3, \ldots]$	Section 10.3, Table 10.2
ν_k	irrationality exponent $-\log\lvert\theta - p_k/q_k\rvert / \log q_k$ of the kth convergent; unrelated to ν_1, ν_2 in Section 8.8	Section 10.7
X	verified range: every nontrivial cycle has all its elements at least X; $X = 2^{71}$ is established	Definition 10.3
V	Lyapunov function (certificate), the standard potential being $V(x) = \log x$; in Chapter 8, V_{GD}, V_{SMM} and $V(\beta)$ are variances	Definition 4.1
φ, G_d	residue correction in the potential $V(x) = \log x + \varphi(x \bmod 2^d)$, and the residue graph with edge weights $\log\frac{3}{2}$ (odd) and $-\log 2$ (even)	Section 6.2, Theorem 6.3
μ, L, κ, λ, e_t	f is μ strongly convex with L Lipschitz gradient and $\kappa = L/\mu$ is its condition number; λ is a Hessian eigenvalue and $e_t = x_t - x^*$ the error; μ is also the Haar measure, and L an energy in Chapter 14	Section 5.1
L, μ, κ (stride)	stride constants of Chapter 5: $L = \log\frac{5}{3}$ the largest ascent, $\mu = \lvert\log\frac{3}{4}\rvert$ the mean descent, $\kappa = L/\mu \approx 1.776$ a ratio of increments and not a condition number	Definition 5.3
α, a, β	step size, usually $1/L$; the normalized step $a = \alpha\lambda$; momentum coefficient β, equal to $\log(3/2)/\log 2 = \log_2 3 - 1$ in SMM; in Chapter 14, β is an inverse temperature	Section 7.1, Section 5.6
$\mathrm{SMM}_{\mathrm{det}}$, $\mathrm{SMM}_{\mathrm{state}}$, G_h	Switching MM: $\mathrm{SMM}_{\mathrm{det}}$ extrapolates on odd iterations only, two steps from an even index being $G_{(1+\beta)\alpha} \circ G_\alpha$ with $G_h(x) = x - h\nabla f(x)$; $\mathrm{SMM}_{\mathrm{state}}$ injects only when progress is slow	Algorithm 7.1, Theorem 7.6
m_{SMM}, $V(\beta)$	slow mode step multiplier $(2+\beta)/2$ and noise amplification $(2+2\beta+\beta^2)/(2+\beta)$ of $\mathrm{SMM}_{\mathrm{det}}$	Corollary 7.9, Equation (8.14)

Part I

The Problem



The Collatz coral: the orbits of the first 20 000 *integers, drawn from* 1 *outward with a small turn at every step, one way at even values and the other at odd values; colour is total stopping time.*

Chapter 1

The $3x+1$ Problem

Take a positive integer. If it is even, halve it; if it is odd, triple it and add one. Repeat. Starting from 7 the sequence runs

$$7,\ 22,\ 11,\ 34,\ 17,\ 52,\ 26,\ 13,\ 40,\ 20,\ 10,\ 5,\ 16,\ 8,\ 4,\ 2,\ 1,$$

and once it reaches 1 it falls into the loop $1 \to 4 \to 2 \to 1$ forever. The $3x+1$ problem asks whether this always happens. Every starting value that anyone has ever tried, and the number of starting values tried is now 2^{71}, eventually reaches 1. Nobody knows why.

The problem can be explained to a child and has defeated some of the strongest mathematicians of the last ninety years. It sits at an unusual junction. The rule is arithmetic, but the behaviour it produces looks random; the obvious probabilistic heuristic predicts the answer with great accuracy, but the heuristic is not a proof and no one knows how to turn it into one; and generalizations of the problem that look only slightly more complicated are provably undecidable. This book is about that junction, read through the lens of martingales, optimization and Diophantine approximation. This first chapter states the problem, recounts its history, works through examples, surveys what is known, and lays out the plan of the book.

1.1 The conjecture

We write $\mathbb{N} = \{1, 2, 3, \dots\}$ for the positive integers. The *standard Collatz map* $T_0 : \mathbb{N} \to \mathbb{N}$ is

$$T_0(n) = \begin{cases} n/2, & n \text{ even}, \\ 3n+1, & n \text{ odd}. \end{cases} \tag{1.1}$$

We write $T_0^{(k)}$ for the kth iterate, with $T_0^{(0)}(n) = n$, and call the sequence $n, T_0(n), T_0^{(2)}(n), \dots$ the *orbit* or *trajectory* of n.

Conjecture 1.1 *(The Collatz conjecture). For every $n \in \mathbb{N}$ there is $k \geq 0$ with $T_0^{(k)}(n) = 1$.*

The least such k is the *total stopping time* of n, written $\sigma_\infty(n)$, with the convention $\sigma_\infty(n) = \infty$ if no such k exists. The conjecture says $\sigma_\infty(n) < \infty$ for every n.

It is useful to see at once what could go wrong. The orbit of n is a sequence of positive integers, so exactly one of three things happens.

(i) The orbit reaches 1, and from then on cycles through $1, 4, 2$.

(ii) The orbit is unbounded: $\limsup_k T_0^{(k)}(n) = \infty$. Since the orbit cannot revisit a value without becoming periodic, in this case all its values are distinct and in fact $T_0^{(k)}(n) \to \infty$.

(iii) The orbit is bounded but never reaches 1. A bounded orbit takes only finitely many values, so it must eventually repeat a value, and then it is periodic from that point on. Hence the orbit enters a cycle that does not contain 1.

The conjecture is the assertion that alternatives (ii) and (iii) never occur. We shall call these the *divergence half* and the *cycle half* of the problem. They look like two aspects of one question, but one of the main messages of this book is that they are different in kind and call for different tools.

That the problem is genuinely arithmetic, and not a consequence of some soft principle about tripling and halving, is shown by running the same rule on the negative integers. Formula (1.1) makes sense on all of $\mathbb{Z}$, and there it has at least three cycles besides the fixed point 0: the cycle $-1 \to -2 \to -1$, the cycle

$$-5 \to -14 \to -7 \to -20 \to -10 \to -5,$$

and a cycle of length 18 through -17, containing 7 odd terms, whose most negative value is -272. On the negative integers the conjecture is therefore false, in the sense that there is more than one cycle. Notice the counts: the cycle through -5 has 2 odd steps and 3 halvings, and $2^3 < 3^2$; the cycle through -17 has 7 odd steps and 11 halvings, and $2^{11} < 3^7$. For the trivial positive cycle, 1 odd step and 2 halvings, the inequality is reversed: $2^2 > 3$. We shall see in Chapter 10 that a positive cycle with r odd steps and s halvings must satisfy $2^s > 3^r$ with $s \log 2 - r \log 3$ extremely small, which is a statement about rational approximations s/r to $\log_2 3$. The fractions $3/2$ and $11/7$ that appear in the negative cycles approximate $\log_2 3 = 1.58496\ldots$ from below.

Variants of the map. Since $3n + 1$ is even whenever n is odd, one loses nothing by combining each tripling with the halving that must follow it. This gives the *accelerated map* $T(n) = (3n + 1)/2$ for odd n and $T(n) = n/2$ for even n. Going further, one may divide out all factors of 2 at once, which gives the *Syracuse map* $S(n) = (3n + 1)/2^{v_2(3n+1)}$ on odd integers, where $v_2(m)$ is the exponent of the largest power of 2 dividing m. The three maps are equivalent for the conjecture, and each is the natural object for some purpose. Chapter 2 sets them side by side; for now we work with T_0, which is the map in the statement.

1.2 A brief history

The problem is usually attributed to Lothar Collatz, who studied iterations of simple arithmetic functions as a student in Germany in the 1930s and is said to

have recorded problems of this type in his notebooks then. He did not publish it. The problem was in circulation by the early 1950s [53, pp. 3 to 4] and spread by word of mouth for decades, and as it travelled it collected names. Helmut Hasse lectured on it, and it became known as *Hasse's algorithm*; it reached Syracuse University, and became the *Syracuse problem*; Shizuo Kakutani circulated it at Yale and Chicago, and it became *Kakutani's problem*; Stanislaw Ulam spread it at Los Alamos and elsewhere, and it became *Ulam's problem*. It is also called the $3x + 1$ problem, the $3n + 1$ problem, and the Collatz problem, the name we use. The multiplicity of names is itself a historical record: the problem was rediscovered or relayed by many people before anyone wrote about it in print.

The standard account of the history is the survey of Lagarias [53] in the *American Mathematical Monthly*, which is still the best introduction to the mathematics. Lagarias recounts Kakutani's remark that for about a month everybody at Yale worked on the problem without result, and he records Paul Erdős's verdict: "Mathematics is not yet ready for such problems." Erdős offered a prize for a solution, and the remark has been quoted ever since, sometimes as a warning and sometimes as a challenge. We take it as a description of the situation. The tools that have produced partial results, namely density arguments, stochastic models, Diophantine approximation and 2 adic dynamics, are each strong enough to explain why the conjecture should be true and each falls short of proving it for a reason that can be stated precisely, and a large part of this book states those reasons.

The literature has grown large. Lagarias has maintained annotated bibliographies [57, 58] covering the work from 1963 onward, and he edited the volume *The Ultimate Challenge* [55], published by the American Mathematical Society in 2010, which collects survey articles on the history, the stochastic models, the computational verification, the generalizations, and the connections with ergodic theory and logic, together with reprints of key papers. The reader who wants a single reference besides this book should start there. The research monograph of Wirsching [107] treats the problem as a dynamical system and is a second good entry point.

1.3 Worked orbits

The best way to feel the problem is to compute. Every number quoted in this section was computed by the authors with the scripts listed in Appendix B.

Example 1.2 *(The orbit of 27).* Small starting values usually reach 1 quickly: 3 takes 7 steps, 7 takes 16 steps and climbs to 52, 9 takes 19 steps. The value 27 is the first surprise. Its orbit begins

$$27,\ 82,\ 41,\ 124,\ 62,\ 31,\ 94,\ 47,\ 142,\ 71,$$
$$214,\ 107,\ 322,\ 161,\ 484,\ 242,\ 121,\ 364,\ \ldots$$

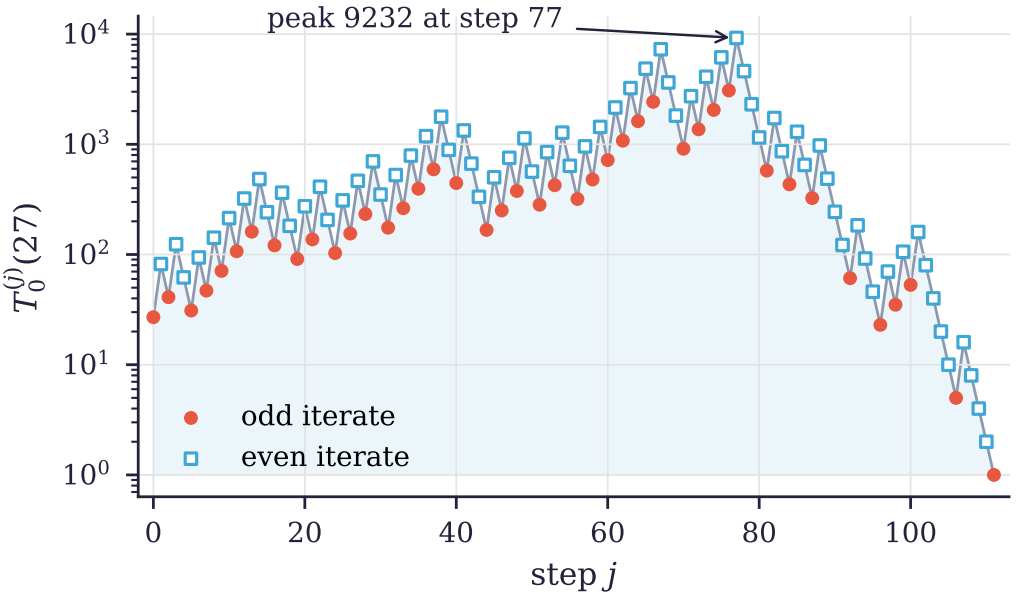


FIGURE 1.1. The orbit of 27 under the standard map T_0, on a logarithmic scale. Iterates are marked by parity: every odd iterate x is followed by a rise to $3x+1$ and every even iterate by a halving, so the odd iterates are the points where the path turns upward. The orbit takes 111 steps to reach 1 and peaks at 9232 after 77 steps.

and wanders upward for a long time. It reaches its maximum 9232 at step 77 and finally arrives at 1 after $\sigma_\infty(27) = 111$ steps under T_0. Of these 111 steps, 41 are triplings and 70 are halvings. Figure 1.1 shows the whole orbit on a logarithmic scale.

The counts in this example already contain most of the arithmetic we shall need. Because each tripling is followed by at least one halving, the same orbit takes $111 - 41 = 70$ steps under the accelerated map T and 41 steps under the Syracuse map S. Each tripling multiplies by slightly more than 3 and each halving divides by exactly 2, so an orbit with 41 triplings and 70 halvings that starts at 27 and ends at 1 must have $27 \cdot 3^{41}$ a little below 2^{70}. Indeed $27 \cdot 3^{41}/2^{70} \approx 0.834$, and the gap from 1 is made up by the additive ones. We make this exact in Chapter 2, where we show that $2^k T^{(k)}(n) = 3^m n + c_k(n)$ under the accelerated map, with m the number of odd steps among the first k and $c_k(n)$ an explicit integer, positive as soon as $m > 0$, built from the positions of the odd steps.

The orbit of 27 is also a reminder that orbits merge. The orbit of 97 reaches 94 after 13 steps; the orbit of 27 reaches 94 after 6 steps; from 94 onward the two coincide. Hence $\sigma_\infty(97) = 13 + 105 = 118$ and the orbit of 97 peaks at 9232 as well. The orbits that reach 1 form a tree rooted at 1, once the edge from 1 to 4 is removed; it is a component of the *Collatz graph*, which joins each n to $T_0(n)$, and

TABLE 1.1. Selected delay records for the standard map T_0: total stopping time $\sigma_\infty(n)$ and the largest value in the orbit.

n	$\sigma_\infty(n)$	$\max_k T_0^{(k)}(n)$
7	16	52
9	19	52
27	111	9 232
97	118	9 232
871	178	190 996
6 171	261	975 400
77 031	350	21 933 016
837 799	524	2 974 984 576

the conjecture says that this tree contains every positive integer.

Records. A starting value n is a *delay record* if $\sigma_\infty(n) > \sigma_\infty(n')$ for every $n' < n$. The delay records up to 10^5 are

1, 2, 3, 6, 7, 9, 18, 25, 27, 54, 73, 97, 129, 171, 231,
313, 327, 649, 703, 871, 1161, 2223, 2463, 2919, 3711,
6171, 10971, 13255, 17647, 23529, 26623,
34239, 35655, 52527, 77031.

Table 1.1 lists a selection with their total stopping times and peaks. The longest orbit below 10^4 starts at 6171 and takes 261 steps; below 10^5 it starts at 77031 and takes 350 steps; below 10^6 it starts at 837799 and takes 524 steps. Peaks can be enormous relative to the starting value: below 10^4 the highest point reached by any orbit is 27 114 424, from $n = 9663$, and below 10^5 it is 1 570 824 736, from $n = 77671$.

How long do orbits take? Figure 1.2 plots $\sigma_\infty(n)$ for all $n \le 10^5$ against $\log n$. Two features stand out. The points fill a band whose centre grows linearly in $\log n$, and the band is wide, with a visible fine structure of streaks caused by merging orbits. The linear growth is predicted by the simplest probabilistic model. Under T_0 an odd value is always followed by an even one, and on average about one step in three is odd. If we pretend that the parities are random with $\mathbb{P}(\text{odd}) = 1/3$, each step changes $\log x$ by $\log 3$ (ignoring the $+1$) with probability $1/3$ and by $-\log 2$ with probability $2/3$, so the mean change per step is

$$\tfrac{1}{3}\log 3 - \tfrac{2}{3}\log 2 = \tfrac{1}{3}\log\tfrac{3}{4} \approx -0.0959.$$

An orbit starting at n should therefore need about

$$\frac{3\log n}{\log(4/3)} \approx 10.43 \log n$$

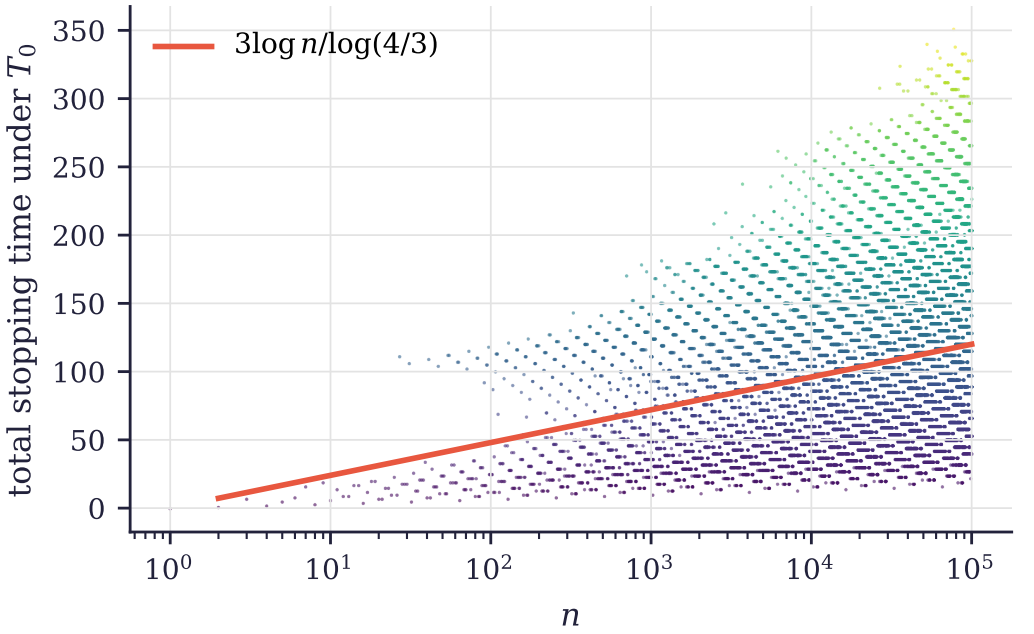


FIGURE 1.2. Total stopping times $\sigma_\infty(n)$ under T_0 for all $n \le 10^5$, with n on a logarithmic scale. The solid line is the prediction $3\log n/\log(4/3)$ of the random parity model. The streaks are families of starting values whose orbits merge after a few steps.

steps, which is the line drawn in the figure. The least squares line through the origin fitted to $\sigma_\infty(n)$ against $\log n$ over $2 \le n \le 10^4$ has slope 10.36, and over the range $2 \le n \le 10^5$ of Figure 1.2 it has slope 10.23. For the accelerated map, whose heuristic slope is $2/\log(4/3) = 1/|\delta^*| \approx 6.952$ with $\delta^* = \log(\sqrt{3}/2)$, the fitted slopes over the same two ranges are 6.920 and 6.837. The constant δ^* will be a central character in Chapter 3.

The agreement between heuristic and computation is good, and it is worth being clear about what it does and does not mean. It is consistent with the parities of the accelerated map along a typical orbit having the frequencies of fair coin flips, so that one step of T_0 in three is odd. It does not mean that any particular orbit must descend, and it gives no handle at all on the question of whether some exceptional orbit could avoid 1 forever. The heuristic averages over starting values, while the conjecture is about every starting value. That gap recurs in every chapter of this book.

1.4 Computational verification

Because the conjecture is a universal statement about integers, it invites computation, and the problem has been checked by computer for almost as

long as there have been computers. Each verification establishes that every n up to some bound X reaches 1. The techniques are clever: one need only check that each n eventually falls below itself, since smaller values have already been handled; residue classes that provably fall below their starting value in a few steps can be skipped wholesale, which is the idea behind the stopping time analysis of Chapter 2; and the accelerated and Syracuse forms let one take several steps at once using precomputed tables indexed by residues modulo a power of 2.

The bounds have grown by many orders of magnitude. In the 2000s Oliveira e Silva verified the conjecture up to $5 \cdot 2^{60} \approx 5.76 \times 10^{18}$, an effort described in his chapter of the Lagarias volume [76]. In 2021 Bařina [6] reported verification up to $2^{68} \approx 2.95 \times 10^{20}$, using a carefully optimized algorithm on graphics processors, and in 2025 he extended it to $2^{71} \approx 2.36 \times 10^{21}$ [7]. Most of the computations in this book use the range 2^{68}: whenever we say *the verified range* we mean that every $n \leq 2^{68}$ is known to reach 1. We shall see that the sharp form of the cycle bound below is the same at 2^{71}.

Verification has two consequences beyond the obvious one. First, any orbit that fails to reach 1 must stay above 2^{68} forever, because if it ever dropped to 2^{68} or below it would join a verified orbit. In particular a nontrivial cycle would have least element above 2^{68}. Second, and less obviously, this lower bound on the least element can be converted into a lower bound on the length of a cycle, by an argument going back to Crandall [18] and Eliahou [21] that uses the continued fraction of $\log_2 3$. In Chapter 10 we carry out this argument in detail. A first version, the block form of the bound, which uses only the convergents of $\log_2 3$, shows that with the verified range 2^{68} every nontrivial cycle contains at least 6 586 818 670 odd terms. A sharper version, which uses a short mediant lemma together with the classical description of the best one sided approximations [31, Theorem 4.5(ii)], both proved in Chapter 10, to show that the extremal cases always occur at intermediate fractions of the continued fraction lying above $\log_2 3$, gives unconditionally

$$r \;\geq\; 72\,057\,431\,991$$

for the number r of odd terms in any nontrivial cycle. It coincides with the bound $r > 7.2 \times 10^{10}$ that Hercher [38] credits to Hercher and Puchert [37, 84]. Applied to Eliahou's verification range 2^{40}, the same method returns exactly his lower bound 17 087 915 on the length of a cycle of the accelerated map. The current record is larger still: Hercher [38, Corollary 29] showed, by a sharper accounting of the structure of a cycle, that verification up to $1536 \cdot 2^{60} = 3 \cdot 2^{69}$ gives $r \geq 1.375 \times 10^{11}$, and Bařina's verification to 2^{71} [7] has made this unconditional. Cycles, if they exist, are very long and live very high up.

Verification can never prove the conjecture, and it is not even clear what evidential weight it carries. Orbits that misbehave, if they exist, might start at numbers so large that no computation will ever reach them. The heuristic of

Section 1.3 predicts that the longest orbits below N take time of order $\log N$, so nothing in the data suggests a change of regime, but one should remember that the problem has a large supply of quantities that grow like $\log N$ or $\log\log N$, and phenomena governed by such quantities can hide far beyond the reach of any computer.

1.5 Undecidability: Conway's theorem

Why should such a simple problem be hard? One answer is that the class of problems it belongs to contains problems that are not merely hard but impossible. In 1972 Conway [16] considered *generalized Collatz maps*: fix a modulus $p \geq 2$ and rational numbers $a_0, \ldots, a_{p-1}$ and $b_0, \ldots, b_{p-1}$, chosen so that

$$g(n) = a_i n + b_i \qquad \text{whenever } n \equiv i \pmod{p}$$

maps integers to integers. The Collatz map is the case $p = 2$, $a_0 = 1/2$, $b_0 = 0$, $a_1 = 3$, $b_1 = 1$. Conway showed that this class is rich enough to simulate any computation: given a Turing machine one can construct such a g whose iterates encode the steps of the machine. It follows that there is no algorithm which, given g in this class and a starting value n, decides whether the orbit of n under g eventually reaches 1. Conway's construction in fact needs only the maps with all $b_i = 0$, which multiply by a rational depending on the residue class.

Conway later packaged this observation as a programming language, FRACTRAN [17]. A FRACTRAN program is a finite list of fractions $q_1, \ldots, q_t$; the state is a positive integer n; one step replaces n by $q_i n$ for the first i such that $q_i n$ is an integer, and the program halts if there is none. Each such step is a generalized Collatz map with all $b_i = 0$ and modulus the least common multiple of the denominators. Conway exhibited a FRACTRAN program of fourteen fractions whose orbit starting at 2 passes through exactly the powers of 2 with prime exponent, in increasing order, and showed that FRACTRAN is a universal model of computation. Kurtz and Simon [52] sharpened the logical classification, showing that the generalized problem of deciding whether every orbit of a given g reaches 1 is complete for the class Π^0_2 of the arithmetic hierarchy.

None of this says that the original problem is undecidable, and indeed a single statement such as Conjecture 1.1 is either true or false; the undecidability concerns algorithms that work uniformly over the whole class. What Conway's theorem does say is that any proof of the Collatz conjecture must use something specific about the numbers 2 and 3. Soft arguments that would apply equally to every map of the form $a_i n + b_i$ cannot work, because for some maps in the class the corresponding statement is false, and for every sound, recursively axiomatized theory there are maps for which it is true but not provable in that theory. This is the theoretical counterpart of the observation about negative integers above. In this book the specific arithmetic of 2 and 3 enters in three places: the Geometric

law of $v_2(3n+1)$ in Chapter 2, through the comparison of its mean 2 with $\log_2 3$ (the same law holds for $v_2(5n+1)$, but $\log_2 5 > 2$), the identity $\frac{1}{2}\cdot\frac{1}{2}+\frac{1}{2}\cdot\frac{3}{2}=1$ behind the martingale of Chapter 3, and the continued fraction of $\log_2 3$ in Chapter 10.

1.6 What is known: density theorems

Since proving the conjecture for every n is out of reach, a natural weakening is to prove it for almost every n, in some sense of density. Recall that a set $A \subseteq \mathbb{N}$ has *natural density* δ if $\#\{n \le x : n \in A\}/x \to \delta$ as $x \to \infty$, and *logarithmic density* δ if

$$\frac{1}{\log x} \sum_{n \le x,\ n \in A} \frac{1}{n} \longrightarrow \delta.$$

Natural density 1 implies logarithmic density 1, but not conversely, so logarithmic density statements are weaker.

Terras and Everett. The first theorem of this type concerns the *stopping time* $\sigma(n)$, the least $k \ge 1$ with $T^{(k)}(n) < n$ under the accelerated map. Terras [99, 100] and, independently, Everett [22] proved that the set of n with $\sigma(n) < \infty$ has natural density 1. Terras proved more: for each k the set $\{n : \sigma(n) \le k\}$ has a natural density $F(k)$, and $F(k) \to 1$. The key idea, common to both, is that the first k parities of the orbit of n depend only on n modulo 2^k, and that every pattern of k parities occurs for exactly one residue class [53, Theorem B]. We prove all of this in Chapter 2, where it is Theorem 2.8 and Theorem 2.26. A stopping time result is weaker than it looks: it says that almost every orbit eventually goes below its starting value, which, applied repeatedly, does not show that almost every orbit reaches 1, because the exceptional set at each stage could catch the orbit.

Smaller and smaller. Later work showed that almost all orbits go much lower. Korec [49] proved that for every $\theta > \log 3/\log 4 \approx 0.7925$ the set of n whose orbit contains a value below n^θ has natural density 1. The exponent $\log 3/\log 4$ has the same source as the drift $\log(3/4)$ of the Syracuse map that we compute in Chapter 2.

Krasikov and Lagarias. A different kind of result counts the integers that are known to reach 1. Writing $\pi_1(x)$ for the number of $n \le x$ whose orbit contains 1, Krasikov and Lagarias [51] proved, by a system of difference inequalities derived from the structure of the Collatz graph modulo powers of 3, that $\pi_1(x) \ge x^{0.84}$ for all sufficiently large x. This is far from density 1, but it is a statement about orbits that actually reach 1, not merely orbits that descend.

Tao. The strongest result to date is due to Tao. For the smallest value attained by the orbit of n under the standard map we write

$$\mathrm{Col}_{\min}(n) = \min_{k \ge 0} T_0^{(k)}(n).$$

Theorem 1.3 *(Tao [96, Theorem 1.3]). Let $f : \mathbb{N} \to \mathbb{R}$ be any function with $f(n) \to \infty$ as $n \to \infty$. Then*

$$\mathrm{Col}_{\min}(n) < f(n)$$

for all n in a set of logarithmic density 1.

Taking $f(n) = \log\log\log\log n$, for example, the theorem says that almost every orbit, in logarithmic density, eventually dips below $\log\log\log\log n$. This is the sense of the title of Tao's paper: almost all orbits attain almost bounded values. It is a remarkable strengthening of the Terras and Korec results, which give the weaker bounds $\mathrm{Col}_{\min}(n) < n$ and $\mathrm{Col}_{\min}(n) < n^{\theta}$. The proof works with the Syracuse map. Its deep input is a fine scale mixing estimate [96, Proposition 1.14]: the residues modulo powers of 3 of the Syracuse iterates of a suitably random starting value are governed by a certain *Syracuse random variable* that encodes the first few Syracuse steps, and this variable is approximately equidistributed at fine 3 adic scales, although far from uniform at coarse ones [96, Remark 1.15]. The key estimate is a decay bound for its characteristic function. The fine scale mixing is then fed into an iterative argument that pushes the orbit down from scale n to scale n^{α} for some $\alpha < 1$, many times over.

Three limitations of Theorem 1.3 illustrate the general difficulty. It concerns logarithmic density, not natural density. It shows that orbits become small, not that they reach 1; to conclude that they reach 1 one would need to know that every orbit below some fixed bound reaches 1, which is a finite computation, but the bound $f(n)$ depends on n and the exceptional set depends on f. Most importantly, it is a statement about almost all starting values, and like every density theorem it is silent about any individual orbit. The conjecture asserts something about every integer, and no density statement, however strong, reaches every integer.

1.7 The two halves and the heuristic

We can now state the heuristic that organizes the subject, and the reason it is not a proof.

Under the accelerated map, an odd step multiplies by roughly $3/2$ and an even step divides by 2. If the parities along an orbit behaved like independent fair coin flips, then $\log T^{(k)}(n)$ would be a random walk with increments $\log(3/2)$ and $-\log 2$, each with probability $\frac{1}{2}$, and drift

$$\delta^{*} = \tfrac{1}{2}\log\tfrac{3}{2} - \tfrac{1}{2}\log 2 = \log\frac{\sqrt{3}}{2} \approx -0.1438.$$

A random walk with negative drift goes to $-\infty$, so almost every orbit would descend, and since the orbit is bounded below by 1 it would end in the trivial cycle. This is the heuristic of Crandall [18] and of Lagarias and Weiss [59], and Chapter 3 develops it carefully. Its predictions agree closely with computation

where they can be tested. It predicts the slope of Figure 1.2, $3/\log(4/3)$ steps of T_0 per unit of $\log n$, which is $1/|\delta^*|$ steps of T, to within about two percent; it predicts the Geometric distribution of $v_2(3n+1)$, which in Chapter 2 we prove exactly for residue classes; and it predicts the autocorrelations $(-\frac{1}{2})^h$ of the parity sequence of T_0 (Section 3.8), which orbits of large integers reproduce over the range where the model is exact, while full orbits of small integers reproduce them only at lags 1 and 2, where the agreement is forced or reflects a single stride.

It is not a proof because the parity sequence of an orbit is not random. It is a deterministic function of the starting value. The model is exactly right for a uniformly random starting residue modulo 2^k and the first k steps, which is the content of the Terras bijection, and it is no longer justified after that because the orbit of a fixed integer is a single sample path, not an ensemble. No known theorem converts the statistics of residue classes into statements about a single path. Every serious attack on the divergence half is, in one form or another, an attempt to close that gap.

The two halves of the conjecture interact with the heuristic differently. The divergence half asks whether $\log T^{(k)}(n)$ can drift upward forever. This is a law of large numbers question: in Syracuse form, an orbit can be unbounded only if the running average of the halving exponents $v_2(3y+1)$ along it has lower limit at most $\log_2 3 \approx 1.585$ (Proposition 11.3 and Corollary 11.4), so the divergence half is equivalent to the Cesàro liminf conjecture (Conjecture 11.6) that this lower limit always exceeds $\log_2 3$; under the Geometric law the average is 2. The cycle half asks whether an orbit can close up. That is not an averaging question at all. A cycle imposes an exact multiplicative identity, which after taking logarithms says that $s\log 2 - r\log 3$ is positive and very small for the number r of odd steps and s of halvings in the cycle. The first tools are Diophantine, through the rational approximations of $\log_2 3$, but we shall see that they can never suffice on their own: arithmetic information about the parity word is also needed.

1.8 Machine checked mathematics

Machine checking is changing how results about problems like this one are verified, and we use it in this book. Proof assistants such as Lean allow a theorem and its proof to be written in a formal language and checked mechanically by a small trusted kernel. In 2026 Google DeepMind described *AlphaProof Nexus* [101], a system that couples a large language model to the Lean compiler: the model edits a Lean file containing an unproved statement, receives the compiler's messages, and continues until a complete proof is found or a budget is exhausted. Run over 353 formalized problems from the collection of problems posed by Erdős, it produced complete formal solutions to nine, among them Erdős problem 125 on the lower density of the sumset of the integers whose base 3 digits lie in $\{0,1\}$ and those whose base 4 digits lie in $\{0,1\}$; the answer is that the lower density is zero. That proof turns on good rational approximations to $\log 4/\log 3 = 2/\log_2 3$,

the same kind of Diophantine object that appears in the cycle problem, where it plays the opposite role. For Erdős 125 the existence of good approximations drives the density down. For Collatz cycles good approximations are what a cycle needs, they exist for infinitely many cycle lengths at every verification range (Proposition 10.20), and excluding cycles needs arithmetic information about the parity word that no bound on the quality of approximation supplies.

We claim nothing more for this than it delivers. AlphaProof Nexus does not address the Collatz conjecture, and we know of no automated result that bears on it directly. What machine checking does offer is a reliable record of exactly what has and has not been proved. In this spirit we have written Lean statements of the Collatz conjecture, of its descent form and of the cycle problem, together with machine checked proofs of some elementary ingredients, among them the affine identity for iterates proved in Chapter 2 and the inequality $3^r < 2^s$ satisfied by any cycle of positive integers; these are described in Section 10.10, following our note [78].

1.9 What this book does

We should be direct about the status of what follows. This book does not prove the Collatz conjecture. It does four other things.

It *organizes* the known heuristics into a single framework in which the dynamics are read as an optimization method: a potential function $\log x$ that the iteration tries to decrease, a move that increases it (the odd step), and moves that decrease it (the halvings). In that language the two halves of the conjecture become the two ways an optimizer can fail: the iterates run away, or they enter a limit cycle.

It *makes the constants explicit*: the drift $\delta^* = \log(\sqrt{3}/2)$, the step variance, the Syracuse drift $\log(3/4)$, the stride ratio $\kappa = \log(5/3)/|\log(3/4)| \approx 1.776$ of the largest ascent of a stride to its mean descent, the parity autocorrelation $(-\frac{1}{2})^h$, and the exact mixing time of the model chain on the units modulo 3^a, in which the halving exponents are independent Geometric$(\frac{1}{2})$ draws. These matter because any heuristic for the conjecture must be consistent with them; a heuristic predicting a different drift can be discarded at once.

It *proves structural results*, most of them short and several of them classical. Some are positive, such as the Terras bijection, the exact Geometric law, the martingale duality and the closed form of the parity autocorrelations in the model. Some are negative: no monotone Lyapunov function certifies descent over any fixed number of Syracuse strides, the natural majorization of $\log x$ drifts the wrong way in the model, and the martingale of the model fails to be uniformly integrable. Each negative result shows where a natural line of attack stops.

And it *sharpens* the questions. For the cycle half it proves that Diophantine approximation of $\log_2 3$, however strong, combined with any finite verification can exclude only finitely many cycle lengths, a point that Halbeisen and Hungerbühler [30, §5] made in another form; it then shows, through a window

lemma, that an arithmetic improvement of the basic cycle inequality by a factor slightly larger than $r/3$, combined with a finite verification, would exclude all cycles. The divergence half is restated exactly as a conjecture about the lower limit of the Cesàro averages of halving exponents (Conjecture 11.6), which we can state but cannot prove.

Plan of the book. The book has three parts.

Part I, *The Problem*, lays the foundations. Chapter 2 sets out the three maps, parity vectors, the Terras bijection, the exact Geometric law of $v_2(3n+1)$, the 2 adic extension and its conjugacy to the shift, and the density theorem of Terras and Everett with a complete proof. Chapter 3 develops the stochastic model and the martingale duality: under the Bernoulli model the iterates form a multiplicative martingale, because $\frac{1}{2}\cdot\frac{1}{2}+\frac{1}{2}\cdot\frac{3}{2}=1$ exactly, while their logarithms form a strict supermartingale with drift δ^*. The gap is Jensen's inequality. We also compute the parity autocorrelations in closed form and show that the martingale is not uniformly integrable.

Part II, *The Optimization Reading*, takes the analogy with optimization seriously and pushes it until it breaks. Chapter 4 treats $\log x$ as a Lyapunov function, proves the exact stride descent inequality, and proves that no monotone Lyapunov function certifies descent over any fixed number of strides; it also shows that under the stochastic model the linear majorizer of the MM framework [40, 60] has positive mean increment while the potential has negative drift. Chapter 5 makes the analogy with Nesterov momentum [72, 95] precise at the level of strides and is explicit about its limits: the stride ratio is a ratio of increments, not a condition number. Chapter 6 presses the majorization idea on the Collatz map itself: the multiplicative majorizer reproduces the cycle inequality, and three impossibility results show that no correction of $\log x$ by a function of x modulo a fixed power of 2 certifies descent, that any correction that certifies it must be unbounded, and that a penalty proportional to the number of trailing ones in the binary expansion of x fails at a positive fraction of odd integers for every weight. Chapters 7 and 8 then run the analogy in the other direction. The Collatz arithmetic suggests an optimizer, *Switching MM*, which alternates a plain gradient step with a momentum step according to the parity of the iteration counter, with momentum constant

$$\beta=\frac{\log(3/2)}{\log 2}=\log_2 3-1\approx 0.585,$$

the ratio of the ascent of an odd step to the descent of an even step. We show that on quadratics two steps of Switching MM are exactly two gradient steps with step sizes α and $(1+\beta)\alpha$, so that the method is a cyclic step size schedule, and we compute its stationary variance under gradient noise exactly. Its noise amplification relative to gradient descent at the same step is bounded by $5/3$ for every $\beta<1$, whereas in the slow modes every step momentum amplifies noise by $1/(1-\beta)$. Nothing in these two chapters bears on the conjecture; they

are an instance of number theory suggesting a design. Chapter 9 closes Part II by treating both the Collatz recursion and the momentum methods as nonlinear difference equations: it derives certified growth exponents for the inverse tree from difference inequalities in the style of Krasikov and Lagarias, computes the joint spectral radius that governs the stability of Switching MM under arbitrary switching, which is below one at every parameter value computed, and proves a linear rate for Switching MM on smooth strongly convex functions that is the exact worst case for the distance to the minimizer.

Part III, *Cycles and Divergence*, returns to the conjecture. Chapter 10 treats the cycle half through the continued fraction of $\log_2 3$, derives the cycle length bounds quoted in Section 1.4 from the verified range, and places them in the literature from Crandall [18] and Steiner [94] through Eliahou, Simons and de Weger [92], and Hercher and Puchert. The key ingredient is a mediant lemma which, with the one sided theorem of that chapter, gives a short proof of the classical fact that the first cycle length not excluded by a given verification range is always a rung of the semiconvergent ladder that approximates $\log_2 3$ from above; it makes the computation rigorous and yields the unconditional bound $r \geq 72\,057\,431\,991$. The method is that of Crandall and Eliahou, the bound coincides with that of Hercher and Puchert, and it remains below Hercher's current record. An exact integer certificate for the bound is given. The chapter then proves that Diophantine information about $\log_2 3$ alone cannot finish the cycle half, and locates the least element of any cycle in a window whose width, a factor of about $r/3$, measures exactly what is missing. Chapter 11 treats the divergence half through the Cesàro averages of halving exponents, proves that the natural Cesàro criterion holds automatically at the stride at which an orbit terminates (so that verifying it numerically on terminating orbits carries no information), formulates the conjecture that would close this half, and shows that the model chain on the units modulo 3^a, in which the halving exponents are independent Geometric$(\frac{1}{2})$ draws, is exactly stationary after a strides, a restatement in Markov chain form of an identity of Tao [96, eq. (1.21)]. Chapter 12 assembles what a proof would have to contain, Chapter 13 sets out what computation can and cannot contribute, with the guidance the optimization reading gives it, Chapter 14 tests level set samplers against the map and finds the horizon at which they fail, and Chapter 15 collects open problems and computational projects. A gallery of computed pictures, *A Gallery of Orbits*, follows Part III, and the appendices tabulate the constants and describe the code.

How to read this book. A reader interested mainly in the optimizer can read Chapters 5, 7 and 8 and then Chapter 9, which uses their results, turning back to Part I and Chapter 4 for the stride arithmetic from which the momentum constant is derived. A reader interested mainly in the number theory can go from Chapter 2 directly to Chapter 10, and needs Chapter 3 before Chapter 11; before Chapters 12 and 13 this reader also needs the impossibility results of Chapters 4 and 6 and

the first two sections of Chapter 9, on which Chapter 13 builds; its comparisons with Switching MM also cite Section 9.4. Every chapter ends with notes on the literature, and every chapter except Chapter 15 with exercises. The numbers we compute can be reproduced with the code described in Appendix B.

Notes

The history in Section 1.2 follows Lagarias [53] and the introductory chapters of [55], which also discuss the many names of the problem and their origins in more detail than we have. The annotated bibliographies [57, 58] are the quickest way into the specialized literature.

The stochastic models that underlie the heuristic of Section 1.7 are developed by Lagarias and Weiss [59] and surveyed by Kontorovich and Lagarias [47]; the survey also treats the $5x+1$ problem, where the analogous model predicts divergence for almost all orbits, a useful sanity check on any argument for the $3x+1$ case.

The original paper of Conway [16] is short and readable, and the FRACTRAN paper [17] contains the prime generating program and the universality argument. The undecidability results show that the arithmetic of 2 and 3 must enter any proof, and the reader can locate exactly where it enters each argument in this book.

Tao's paper [96] is long but carefully written; its introduction gives an excellent account of why density methods stop short of the full conjecture. The equidistribution it proves is 3 adic and holds only at fine scales [96, Proposition 1.14 and Remark 1.15]; the equidistribution in Chapter 2 is 2 adic and much easier. Both are statements about ensembles of starting values.

Exercises

Exercise 1.4. Find all cycles of T_0 on the integers n with $|n| \le 1000$. For each cycle record the number r of odd terms and the number s of even terms, and check that $2^s > 3^r$ for positive cycles and $2^s < 3^r$ for negative ones. Explain this sign rule by multiplying the relations $T_0(x)/x$ around the cycle.

Exercise 1.5. Show that if $n \equiv 4 \pmod 6$ then n has two preimages under T_0, namely $2n$ and $(n-1)/3$, and otherwise only one. Deduce that the Collatz graph branches only at vertices congruent to 4 modulo 6, and use this to explain why the orbits of 27 and 97 in Section 1.3 merge.

Exercise 1.6. Under the random parity model for T_0 in which an odd step is followed by an even step and an even step is followed by an odd or even step with equal probability, show that the long run fraction of odd steps is $1/3$. Deduce the heuristic slope $3/\log(4/3)$ for $\sigma_\infty(n)$ against $\log n$, and check numerically that the fraction of odd steps along the orbit of 27 is $41/111 \approx 0.369$.

Exercise 1.7. Consider the generalized Collatz map $g(n) = n/2$ for n even and $g(n) = 5n + 1$ for n odd. Follow the orbits of $n \leq 100$ until they repeat or exceed 10^{15}, find the two cycles not containing 1, and count the orbits that are still growing when you stop. Repeat the heuristic computation of Section 1.7 for this map and show that the predicted drift of $\log x$ is positive.

Exercise 1.8. Write a program that verifies Conjecture 1.1 for all $n \leq 10^7$ by checking only that each $n \geq 2$ eventually falls below n. Show that it suffices to test odd $n \equiv 3 \pmod 4$, and estimate the speedup obtained by also skipping the residue classes modulo 2^k that fall below their starting value within k accelerated steps (after reading Chapter 2; see Table 2.2).

Chapter 2

Parity Vectors and the Syracuse Map

The rule "halve if even, triple and add one if odd" has two ingredients, a multiplicative one and a parity test, and everything interesting about the problem comes from the way they interact. This chapter isolates the parity test. We record the sequence of parities along an orbit as a binary word, the *parity vector*, and show that over a block of k steps the parity vector and the residue of the starting value modulo 2^k determine each other completely. This theorem of Terras and Everett is the single most useful fact about the problem. It explains why the iteration looks random: a uniformly random residue modulo 2^k produces a parity vector of k independent fair coin flips, exactly and not approximately. It gives the exact Geometric law of the halving exponent $v_2(3n+1)$, the exact drift $\log(3/4)$ of the Syracuse map, and, after passing to the 2 adic integers, a conjugacy of the Collatz map with the binary shift. And it yields the first genuine theorem about the conjecture, that almost every positive integer eventually falls below itself.

All of these results are statements about ensembles of starting values. None of them says anything about a particular orbit, and the reader should keep in mind throughout the chapter where exactly that limitation enters.

2.1 Three maps

Chapter 1 introduced three versions of the iteration. We now fix them formally.

Definition 2.1 *(The three Collatz maps).*

(i) The *standard map* $T_0 : \mathbb{Z} \to \mathbb{Z}$ is $T_0(n) = n/2$ for even n and $T_0(n) = 3n+1$ for odd n.

(ii) The *accelerated map* $T : \mathbb{Z} \to \mathbb{Z}$ is $T(n) = n/2$ for even n and $T(n) = (3n+1)/2$ for odd n.

(iii) The *Syracuse map* S is defined on odd integers by $S(n) = (3n+1)/2^{v_2(3n+1)}$, where $v_2(m)$ denotes the exponent of the largest power of 2 dividing $m \neq 0$. It maps odd integers to odd integers.

We write $T_0^{(k)}$, $T^{(k)}$, $S^{(k)}$ for the kth iterates, with the zeroth iterate the identity.

The maps T_0 and T are defined on all of $\mathbb{Z}$, the map S on all odd integers (since $3n+1 \neq 0$ for integer n), and all three send positive integers to positive integers.

The relations between them are immediate from the definitions. For every integer n,

$$T(n) = T_0^{(1+b)}(n), \qquad b = n \bmod 2, \tag{2.1}$$

because for odd n the value $3n+1$ is even and T simply performs the forced halving at once. For odd n, with $k = v_2(3n+1)$,

$$S(n) = T^{(k)}(n) = T_0^{(k+1)}(n), \tag{2.2}$$

since $T(n) = (3n+1)/2$ and the next $k-1$ values $(3n+1)/2^i$, $1 \le i < k$, are even. Thus the orbit of n under T is obtained from its orbit under T_0 by deleting the even value that follows each odd value, and the orbit of an odd n under S is the subsequence of odd values in either.

Proposition 2.2 *(Equivalence of the three maps). Let $n \ge 1$. The orbit of n under T_0 contains* 1 *if and only if its orbit under T contains* 1*. If n is odd, this happens if and only if its orbit under S contains* 1*. If $n = 2^a n'$ with n' odd, the orbit of n contains* 1 *if and only if the orbit of n' does.*

Proof. The values deleted in passing from the T_0 orbit to the T orbit are of the form $3y+1$ with $y \ge 1$ odd, hence at least 4; so 1 is never deleted, and the two orbits contain 1 together. For odd n, the S orbit is the set of odd values in the T orbit, and 1 is odd. Finally $T^{(a)}(2^a n') = n'$. □

Each map has its uses. The standard map is the one in the statement of the conjecture, and it has one structural property the others lack.

Lemma 2.3 *(No consecutive odd steps). If n is odd then $T_0(n)$ is even. Consequently the parity sequence $b_j = T_0^{(j)}(n) \bmod 2$ of any orbit of T_0 never contains two consecutive ones.*

Proof. If n is odd then $3n$ is odd, so $3n+1$ is even. □

Lemma 2.3 is trivial, but it is easy to transfer it by mistake to the accelerated map.

Remark 2.4 *(A pitfall: odd runs under the accelerated map).* Under T consecutive odd steps do occur. For odd n, $T(n) = (3n+1)/2$ is odd exactly when $3n+1 \equiv 2 \pmod 4$, that is, when $n \equiv 3 \pmod 4$. Thus $T(3) = 5$, $T(7) = 11$, $T(11) = 17$, and $T^{(2)}(7) = 17 > 7$, an ascent over two steps. A tempting but false argument runs: "an odd step multiplies by about $3/2$ and is followed by an even step dividing by 2, so any two consecutive steps of T that begin with an odd one multiply by at most about $3/4$." The first clause is true and the second is false, because under T the halving that follows the tripling has already been absorbed into the odd step.

In fact runs of odd steps under T can be arbitrarily long: for $j \geq 1$ and any odd a, induction on i gives

$$T^{(i)}(2^j a - 1) = 3^i 2^{j-i} a - 1, \qquad 0 \leq i \leq j,$$

so the orbit of $2^j a - 1$ begins with j consecutive odd steps and climbs to $3^j a - 1$. Any descent inequality over a fixed number of steps of T must accommodate these runs. We return to this point in Chapter 4: a Syracuse stride from an odd $x \geq 3$, defined below, descends exactly when it contains at least two halvings (Theorem 4.7), and no fixed number of strides gives descent for every starting value (Corollary 4.15).

The Syracuse map compresses each passage from one odd value to the next, which we call a *stride*, into a single step. For odd x write $k = v_2(3x+1) \geq 1$. Then

$$\log S(x) - \log x = \log\Big(3 + \frac{1}{x}\Big) - k \log 2. \tag{2.3}$$

Since $2 < 3 + 1/x \leq 4$ for $x \geq 1$, the increment is positive when $k = 1$ and negative when $k \geq 2$ (for $x = 1$, $k = 2$ and the increment is 0). The whole question of descent is therefore a question about how often the halving exponent k equals 1, and about the sizes of the larger values.

Example 2.5 *(The orbit of 27 under the three maps).* The orbit of 27 reaches 1 after 111 steps of T_0, 70 steps of T, and 41 steps of S. The 41 halving exponents along the Syracuse orbit are

$$1,2,1,1,1,1,2,2,1,2,1,1,2,1,1,1,2,3,1,1,2,$$
$$1,2,1,1,1,1,1,3,1,1,1,4,2,2,4,3,1,1,5,4,$$

with sum 70 and mean $70/41 \approx 1.707$. The mean is below 2, the value predicted by the Geometric law of Section 2.4; the climb itself happens in the first 32 strides, whose exponents have mean $44/32 = 1.375 < \log_2 3$ and which take the orbit to its largest odd value 3077. The overall mean is above $\log_2 3 \approx 1.585$, which is forced, as we shall see in Chapter 11, for any orbit that reaches 1.

2.2 Parity vectors and the Terras bijection

From now on, unless stated otherwise, we work with the accelerated map T on $\mathbb{Z}$.

Definition 2.6 *(Parity vector).* For $n \in \mathbb{Z}$ and $k \geq 1$, the *parity vector* of length k of n is

$$v_k(n) = \big(n \bmod 2,\ T(n) \bmod 2,\ \ldots,\ T^{(k-1)}(n) \bmod 2\big) \in \{0,1\}^k.$$

We write $m_k(n)$ for the number of ones in $v_k(n)$, the number of odd steps among the first k, and $m_0(n) = 0$.

For example $v_3(3) = (1,1,0)$, since $3 \mapsto 5 \mapsto 8$, and $v_4(5) = (1,0,0,0)$, since $5 \mapsto 8 \mapsto 4 \mapsto 2$. The basic computation is the following lemma, which says how the orbit of x responds when x is perturbed by a multiple of a power of 2.

LEMMA 2.7 *(Perturbation by a power of two). Let $x, a \in \mathbb{Z}$ and $j \geq 0$. For $0 \leq i \leq j$,*

$$T^{(i)}(x + 2^j a) = T^{(i)}(x) + 3^{m_i(x)}\, 2^{j-i}\, a.$$

In particular, for $i < j$ the values $T^{(i)}(x + 2^j a)$ and $T^{(i)}(x)$ have the same parity, so $v_j(x + 2^j a) = v_j(x)$.

Proof. Induction on i. For $i = 0$ there is nothing to prove. Suppose the identity holds for some $i < j$, and put $y = T^{(i)}(x)$ and $y' = T^{(i)}(x + 2^j a) = y + 3^{m_i(x)} 2^{j-i} a$. Since $j - i \geq 1$, $y' \equiv y \pmod 2$. If y is even then $T(y') = y/2 + 3^{m_i(x)} 2^{j-i-1} a$ and $m_{i+1}(x) = m_i(x)$. If y is odd then

$$T(y') = \frac{3y+1}{2} + 3^{m_i(x)+1} 2^{j-i-1} a,$$

and $m_{i+1}(x) = m_i(x) + 1$. In both cases the identity holds at $i+1$. □

THEOREM 2.8 *(Terras [99], Everett [22]). For every $k \geq 1$, the parity vector $v_k(n)$ depends only on $n \bmod 2^k$, and the induced map*

$$\Phi_k : \mathbb{Z}/2^k\mathbb{Z} \to \{0,1\}^k, \qquad n \bmod 2^k \mapsto v_k(n),$$

is a bijection.

Proof. That $v_k(n)$ depends only on $n \bmod 2^k$ is the case $j = k$ of Lemma 2.7. For injectivity, suppose $0 \leq x < x' < 2^k$ and $v_k(x) = v_k(x')$. Let $j = v_2(x' - x)$, so $0 \leq j < k$ and $x' = x + 2^j a$ with a odd. By Lemma 2.7 with $i = j$,

$$T^{(j)}(x') - T^{(j)}(x) = 3^{m_j(x)} a,$$

which is odd. Hence $T^{(j)}(x)$ and $T^{(j)}(x')$ have different parities, and since $j \leq k-1$ the vectors $v_k(x)$ and $v_k(x')$ differ in position j, a contradiction. An injective map between two sets with 2^k elements is a bijection. □

The proof shows a little more, which we record because it is the form used in Section 2.5: if $v_2(x' - x) = j$ then the parity vectors of x and x' agree in positions $0, \ldots, j-1$ and differ in position j.

EXAMPLE 2.9 *(Small moduli).* For $k = 3$ the bijection Φ_3 is

$n \bmod 8$	0	1	2	3	4	5	6	7
$v_3(n)$	000	101	010	110	001	100	011	111

Every word of length 3 appears exactly once. Note that 7 has parity vector 111, three odd steps in a row, in line with Remark 2.4.

Corollary 2.10 *(Exact randomness of the first k parities). Let n be uniformly distributed on $\{1,2,\dots,2^kN\}$ for some $N \geq 1$, or on $\mathbb{Z}/2^k\mathbb{Z}$. Then the entries of $v_k(n)$ are independent Bernoulli random variables with parameter $\frac{1}{2}$. Consequently $m_k(n)$ has the binomial law with parameters k and $\frac{1}{2}$.*

Proof. In both cases $n \bmod 2^k$ is uniform on $\mathbb{Z}/2^k\mathbb{Z}$, so by Theorem 2.8 $v_k(n)$ is uniform on $\{0,1\}^k$, which is the law of k independent fair bits. □

Corollary 2.10 is the precise sense in which the Collatz iteration is random. It is exact, but it has a horizon: the first k parities of n are random when n is random modulo 2^k, and no statement is made about step $k+1$ and beyond unless n is random modulo a higher power of 2. For n random modulo 2^k, each step uses up one bit of the residue, and after k steps the randomness is exhausted.

2.3 The affine form of an iterate

The second basic fact is that k steps of T act on n as an affine map whose coefficients depend only on the parity vector.

Proposition 2.11 *(Affine form). For $n \in \mathbb{Z}$ and $k \geq 0$,*

$$2^k\, T^{(k)}(n) = 3^{m_k(n)}\, n + c_k(n), \tag{2.4}$$

where $c_0(n) = 0$ and

$$c_{k+1}(n) = \begin{cases} 3\,c_k(n) + 2^k, & T^{(k)}(n) \text{ odd}, \\ c_k(n), & T^{(k)}(n) \text{ even}. \end{cases} \tag{2.5}$$

Explicitly, if the odd steps occur at positions $0 \leq i_1 < i_2 < \cdots < i_m \leq k-1$ of the parity vector, with $m = m_k(n)$, then

$$c_k(n) = \sum_{j=1}^{m} 3^{\,m-j}\, 2^{\,i_j}. \tag{2.6}$$

In particular $c_k(n)$ depends only on $v_k(n)$, it is a nonnegative integer, and it is positive as soon as $m_k(n) > 0$.

Proof. Induction on k; the case $k = 0$ is trivial. Write $x = T^{(k)}(n)$ and $m = m_k(n)$, and assume $2^k x = 3^m n + c_k(n)$. If x is even then $2^{k+1}T(x) = 2^k x$, so the identity holds at $k+1$ with $m_{k+1} = m$ and $c_{k+1} = c_k$. If x is odd then

$$2^{k+1}T(x) = 2^k(3x+1) = 3 \cdot 2^k x + 2^k = 3^{m+1}n + 3c_k(n) + 2^k,$$

and $m_{k+1} = m+1$. Unwinding the recursion, the odd step at position i_j contributes 2^{i_j}, which is then multiplied by 3 once for each of the $m-j$ later odd steps; this is (2.6). □

The identity (2.4), with the positivity of c_k, is simple enough to be checked by machine, and we have done so: it is one of the statements proved in the Lean proof assistant, by exactly the induction above, in the formalization described in Chapter 10. There is a second, multiplicative way to see the same quantity. Multiplying the ratios $T(y)/y$ along the orbit, each even value contributes $1/2$ and each odd value y contributes $(3+1/y)/2$, so for $n \geq 1$

$$\frac{2^k T^{(k)}(n)}{3^m n} = \prod_{\substack{0 \leq i < k \\ T^{(i)}(n) \text{ odd}}} \Big(1 + \frac{1}{3T^{(i)}(n)}\Big), \tag{2.7}$$

and hence $c_k(n) = 3^m n\big(\prod(1 + 1/(3T^{(i)}(n))) - 1\big)$, the product running over the same odd values. The additive ones of the iteration are thus a small multiplicative correction when the odd values are large, and (2.7) is the form we use in Chapter 10.

Example 2.12. The orbit of 27 reaches 1 after 70 steps of T, with $m_{70}(27) = 41$. Then (2.4) reads $2^{70} = 3^{41} \cdot 27 + c_{70}(27)$, and indeed the sum (2.6) evaluates to $c_{70}(27) = 195\,820\,718\,533\,800\,070\,543 = 2^{70} - 27 \cdot 3^{41}$, about $0.166 \cdot 2^{70}$.

Two consequences of Proposition 2.11 will be used repeatedly.

Corollary 2.13 *(Explicit inverse of the Terras map). For $w \in \{0,1\}^k$ with m ones at positions $i_1 < \cdots < i_m$, the unique residue $n \bmod 2^k$ with $v_k(n) = w$ is*

$$n \equiv -3^{-m}\, c(w) \equiv -\sum_{j=1}^{m} 3^{-j} 2^{i_j} \pmod{2^k},$$

where $c(w)$ is the sum (2.6) *and 3^{-1} denotes the inverse of 3 modulo 2^k.*

Proof. Reduce (2.4) modulo 2^k: $0 \equiv 3^m n + c(w)$, and 3 is invertible modulo 2^k. □

For example, $w = (1,1,0)$ has $c(w) = 3 \cdot 2^0 + 2^1 = 5$ and $m = 2$; since $9 \equiv 1 \pmod 8$ we get $n \equiv -5 \equiv 3 \pmod 8$, as in Example 2.9.

Corollary 2.14 *(Cycles). Suppose $n \geq 1$ and $T^{(k)}(n) = n$ for some $k \geq 1$, and let $m = m_k(n)$. Then $m \geq 1$, $3^m < 2^k$, and*

$$n = \frac{c_k(n)}{2^k - 3^m}. \tag{2.8}$$

Proof. A cycle of positive integers cannot consist of halvings alone, so $m \geq 1$ and $c_k(n) > 0$. By (2.4), $(2^k - 3^m)n = c_k(n) > 0$. □

For the trivial cycle, $n = 1$, $k = 2$, $w = (1,0)$ gives $c = 1$ and $2^2 - 3 = 1$. The same formula holds on $\mathbb{Z}$ without the sign conclusion: the negative cycle $-5 \mapsto -7 \mapsto -10 \mapsto -5$ of T has $w = (1,1,0)$, $c = 5$ and $2^3 - 3^2 = -1$, so $n = -5$. Formula (2.8) turns the search for cycles with a given parity word into a divisibility test, and the inequality $3^m < 2^k$ is the starting point of Chapter 10.

TABLE 2.1. Residue classes for the halving exponent $k = v_2(3n+1)$ of odd n. The second column gives the class modulo 2^j on which $k \geq j$, the third the class modulo 2^{j+1} on which $k = j$ exactly, and the last its proportion among the odd residues.

j	$k \geq j$	$k = j$	$\mathbb{P}(k = j)$
1	$n \equiv 1 \pmod 2$	$n \equiv 3 \pmod 4$	1/2
2	$n \equiv 1 \pmod 4$	$n \equiv 1 \pmod 8$	1/4
3	$n \equiv 5 \pmod 8$	$n \equiv 13 \pmod{16}$	1/8
4	$n \equiv 5 \pmod{16}$	$n \equiv 5 \pmod{32}$	1/16
5	$n \equiv 21 \pmod{32}$	$n \equiv 53 \pmod{64}$	1/32
6	$n \equiv 21 \pmod{64}$	$n \equiv 21 \pmod{128}$	1/64
7	$n \equiv 85 \pmod{128}$	$n \equiv 213 \pmod{256}$	1/128
8	$n \equiv 85 \pmod{256}$	$n \equiv 85 \pmod{512}$	1/256

2.4 The Geometric law of the halving exponent

We turn to the Syracuse map and the halving exponent $k = v_2(3n+1)$. Define

$$a_j = \frac{4^{\lceil j/2 \rceil} - 1}{3}, \qquad j \geq 1,$$

so that $a_1 = a_2 = 1$, $a_3 = a_4 = 5$, $a_5 = a_6 = 21$, $a_7 = a_8 = 85$. Since $3a_j + 1 = 4^{\lceil j/2 \rceil}$ is divisible by 2^j, a_j is the inverse of -3 modulo 2^j.

PROPOSITION 2.15 *(Exact Geometric law). Let n be odd and $j \geq 1$. Then $v_2(3n+1) \geq j$ if and only if $n \equiv a_j \pmod{2^j}$. Consequently, if n is uniformly distributed on the odd residues modulo 2^M, then*

$$\mathbb{P}\big(v_2(3n+1) \geq j\big) = 2^{-(j-1)}, \qquad 1 \leq j \leq M,$$
$$\mathbb{P}\big(v_2(3n+1) = j\big) = 2^{-j}, \qquad 1 \leq j \leq M-1.$$

Thus $v_2(3n+1)$ has the Geometric$(\frac{1}{2})$ law on $\{1, 2, \ldots\}$ up to the truncation imposed by the modulus, with mean 2 and variance 2 in the limit.

Proof. $v_2(3n+1) \geq j$ means $3n \equiv -1 \pmod{2^j}$, and since 3 is invertible modulo 2^j this congruence has the unique solution $n \equiv a_j$; it is odd because a_j is odd. Among the 2^{M-1} odd residues modulo 2^M, exactly 2^{M-j} reduce to a_j modulo 2^j, giving probability $2^{M-j}/2^{M-1} = 2^{-(j-1)}$. Subtracting consecutive values gives the law of $\{v_2(3n+1) = j\}$ for $j \leq M-1$. The Geometric$(\frac{1}{2})$ law on $\{1, 2, \ldots\}$ has mean $\sum_{j \geq 1} 2^{-(j-1)} = 2$ and variance $(1 - \frac{1}{2})/(\frac{1}{2})^2 = 2$. □

Table 2.1 lists the classes for small j. The law is classical: it is the one stride case of Corollary 2.10. For odd n, $T(n) = (3n+1)/2$ and the halving exponent equals one plus the number of zeros that follow the leading one of the parity

vector before the next one. Under the uniform law on parity vectors that number of zeros is Geometric on $\{0,1,2,\dots\}$ with parameter $\frac{1}{2}$.

Over the odd integers below 2^{20}, the exponent $k = j$ occurs exactly 2^{19-j} times for each $j \le 19$, and the mean of k is 1.9999981; the left panel of Figure 2.1 shows the agreement with 2^{-j}. This is not a statistical fit. It is Proposition 2.15 with $M = 20$; a single odd $n < 2^{20}$ has exponent larger than 19.

COROLLARY 2.16 *(Exact Syracuse drift). Let K have the Geometric($\frac{1}{2}$) law on $\{1,2,\dots\}$. Then the multiplicative part $\log 3 - K \log 2$ of the Syracuse increment* (2.3) *has*

$$\mathbb{E}[\log 3 - K \log 2] = \log 3 - 2\log 2 = \log \tfrac{3}{4} \approx -0.2877,$$
$$\operatorname{Var}(\log 3 - K \log 2) = 2(\log 2)^2 \approx 0.9609.$$

Proof. Immediate from $\mathbb{E}K = 2$ and $\operatorname{Var} K = 2$. □

The correction $\log(1 + 1/(3x))$ in (2.3) is at most $1/(3x)$ and disappears for large x. Over the odd $n < 2^{20}$ the empirical mean of $\log S(n) - \log n$ is -0.287676, against $\log(3/4) = -0.287682$, and exactly half of the increments are positive, namely those with $k = 1$. So half of all Syracuse steps go up, and yet on the logarithmic scale the average step goes down by a factor of $3/4$ (the average of $S(n)/n$ itself is 1.0000047); the asymmetry comes from the long tail of the Geometric law.

A Syracuse step from x consumes $k = v_2(3x+1)$ steps of T, one odd and $k-1$ even. Dividing the Syracuse drift by the mean number 2 of accelerated steps per Syracuse step gives the drift per step of T,

$$\tfrac{1}{2}\log \tfrac{3}{4} = \log \frac{\sqrt{3}}{2} = \delta^* \approx -0.1438,$$

the constant of the heuristic in Chapter 1. The step variance of T under the Bernoulli model is $\frac{1}{4}(\log 3)^2 \approx 0.3017$, because the two increments $\log(3/2)$ and $-\log 2$ differ by $\log 3$. These two constants drive the stochastic analysis of Chapter 3.

REMARK 2.17 *(What the Geometric law does not say).* Proposition 2.15 describes one Syracuse step from a random odd residue. Along a single orbit the successive exponents $k_1, k_2, \dots$ are deterministic. Along an orbit that reaches 1 they are eventually all equal to 2, since $S(1) = 1$; along a cycle with r odd values and s halvings in a period their running averages tend to s/r, which lies strictly between $\log_2 3$ and 2 unless the cycle is that of 1 (summing (2.3) around the cycle gives $s\log 2 = \sum \log(3 + 1/y)$, a sum of r terms in $(\log 3, \log 4]$, with $\log 4$ only at $y = 1$); and along a divergent orbit their upper limit is at most $\log_2 3$ (Corollary 11.4). So whether the running averages tend to 2 along every orbit is the whole conjecture, and whether their lower limit exceeds $\log_2 3$ along every

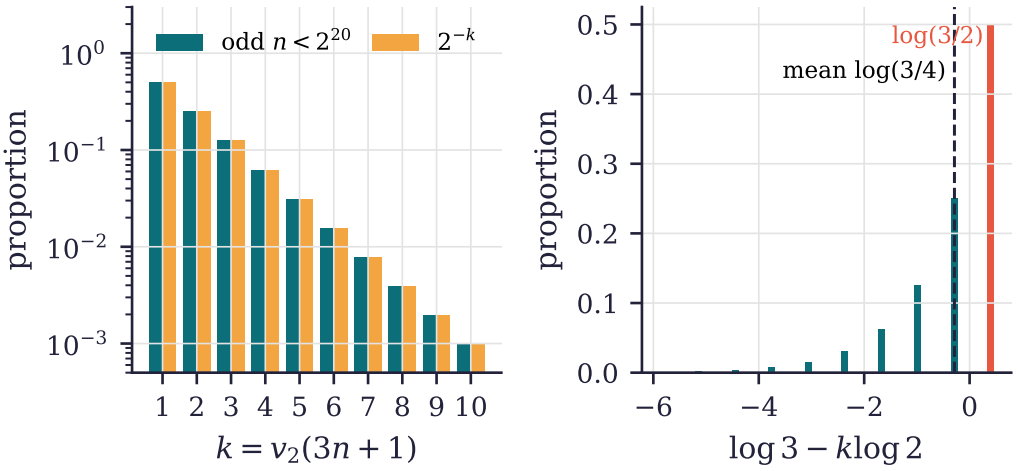


FIGURE 2.1. Left: the proportion of odd $n < 2^{20}$ with $v_2(3n+1) = k$ (dark bars) against the Geometric probabilities 2^{-k} (light bars); they agree exactly. Right: the induced law of $\log 3 - k \log 2$, the multiplicative part of the Syracuse increment, with the single ascending value $\log(3/2)$, the only bar right of zero, and the mean $\log(3/4)$ dashed.

orbit is exactly the divergence half (Conjecture 11.6). Taking logarithms in the Syracuse orbit $x_t = S^{(t)}(n)$,

$$\log x_t = \log n + t \log 3 - (k_1 + \cdots + k_t) \log 2 + \sum_{s<t} \log\Big(1 + \frac{1}{3x_s}\Big),$$

so an orbit along which $(k_1 + \cdots + k_t)/t$ stays above $\log_2 3 + \varepsilon$ for all large t is bounded. Chapter 11 studies this criterion and its limitations.

2.5 The 2 adic extension and the shift

Theorem 2.8 is a statement about every power of 2 at once, and the natural setting for such statements is the ring $\mathbb{Z}_2$ of 2 adic integers. We recall what we need. Every $x \in \mathbb{Z}_2$ has a unique expansion $x = \sum_{i \geq 0} x_i 2^i$ with $x_i \in \{0, 1\}$; the integers sit inside $\mathbb{Z}_2$ (negative integers having expansions with infinitely many ones, for instance $-1 = \sum_{i \geq 0} 2^i$), and so does every rational number with odd denominator. The 2 adic absolute value is $|x|_2 = 2^{-v_2(x)}$, and the balls $a + 2^k \mathbb{Z}_2$ are both open and closed. Odd integers, and in particular 3, are units of $\mathbb{Z}_2$. The Haar probability measure μ on $\mathbb{Z}_2$ is the unique Borel probability measure invariant under translation; it gives every ball $a + 2^k \mathbb{Z}_2$ mass 2^{-k}, and under μ the digits $x_0, x_1, \ldots$ are independent fair bits.

The accelerated map extends to $\mathbb{Z}_2$ by the same formula,

$$T(x) = \begin{cases} x/2, & x \in 2\mathbb{Z}_2, \\ (3x+1)/2, & x \notin 2\mathbb{Z}_2, \end{cases}$$

and Lemma 2.7 holds verbatim for $x, a \in \mathbb{Z}_2$, with the same proof. Let $\Sigma = \{0,1\}^{\{0,1,2,\dots\}}$ be the space of infinite binary sequences with the shift $\mathrm{sh}(w_0, w_1, \dots) = (w_1, w_2, \dots)$, the metric $d(w, w') = 2^{-\min\{i : w_i \neq w'_i\}}$, and the Bernoulli product measure β under which the coordinates are independent fair bits. Define the *parity vector map*

$$Q : \mathbb{Z}_2 \to \Sigma, \qquad Q(x) = \big(T^{(i)}(x) \bmod 2\big)_{i \geq 0}.$$

Theorem 2.18 *(Conjugacy to the shift). The map Q has the following properties.*

(i) *Q is a bijective isometry from $(\mathbb{Z}_2, |\cdot|_2)$ onto (Σ, d).*

(ii) *$Q \circ T = \mathrm{sh} \circ Q$.*

(iii) *Q carries the Haar measure μ to the Bernoulli measure β.*

(iv) *If $w \in \Sigma$ has ones at positions $d_1 < d_2 < \cdots$ (finitely or infinitely many), then*

$$Q^{-1}(w) = -\sum_{j \geq 1} 3^{-j} 2^{d_j},$$

the series converging in $\mathbb{Z}_2$.

The conjugacy (i) to (iii) was first stated explicitly by Lagarias [53, Theorem L], who records that it had been observed by Terras and Pomerance; it rests on the bijection of Terras [99] and Everett [22]. The explicit inverse (iv) is due to Bernstein [10]. Since each part follows in a few lines from what we have proved, we include the proofs.

Proof. (i) Let $x \neq y$ in $\mathbb{Z}_2$ with $|x - y|_2 = 2^{-j}$, so $y = x + 2^j a$ with a a unit. By Lemma 2.7, $T^{(i)}(x)$ and $T^{(i)}(y)$ have the same parity for $i < j$, while $T^{(j)}(y) - T^{(j)}(x) = 3^{m_j(x)} a$ is a unit, so the parities differ at $i = j$. Hence $d(Q(x), Q(y)) = 2^{-j} = |x - y|_2$. An isometry is injective and continuous. For surjectivity, let $w \in \Sigma$. By Theorem 2.8 there is for each k a unique residue $r_k \bmod 2^k$ with $v_k(r_k) = (w_0, \dots, w_{k-1})$; since v_k of a residue modulo 2^{k+1} is the truncation of its v_{k+1}, we have $r_{k+1} \equiv r_k \pmod{2^k}$. The compatible residues define $x \in \mathbb{Z}_2$ with $x \equiv r_k \pmod{2^k}$ for all k, and then the first k entries of $Q(x)$ are $v_k(r_k)$, so $Q(x) = w$.

(ii) The ith entry of $Q(T(x))$ is the parity of $T^{(i+1)}(x)$, which is the $(i+1)$st entry of $Q(x)$.

(iii) By (i), Q maps each ball $x + 2^k\mathbb{Z}_2$ onto the cylinder of sequences beginning with $v_k(x)$. Both have mass 2^{-k}, under μ and β respectively. Cylinders form a

π system generating the Borel σ algebra of Σ, so the probability measures β and $\mu \circ Q^{-1}$, which agree on cylinders, are equal.

(iv) Let $x = Q^{-1}(w)$. Reducing (2.4) modulo 2^k in $\mathbb{Z}_2$ gives, as in Corollary 2.13, $x \equiv -\sum_{j \le m_k(x)} 3^{-j} 2^{d_j} \pmod{2^k}$. If w has infinitely many ones the terms satisfy $|3^{-j}2^{d_j}|_2 = 2^{-d_j} \to 0$, so the series converges; if finitely many, it is a finite sum. Letting $k \to \infty$ gives the formula. □

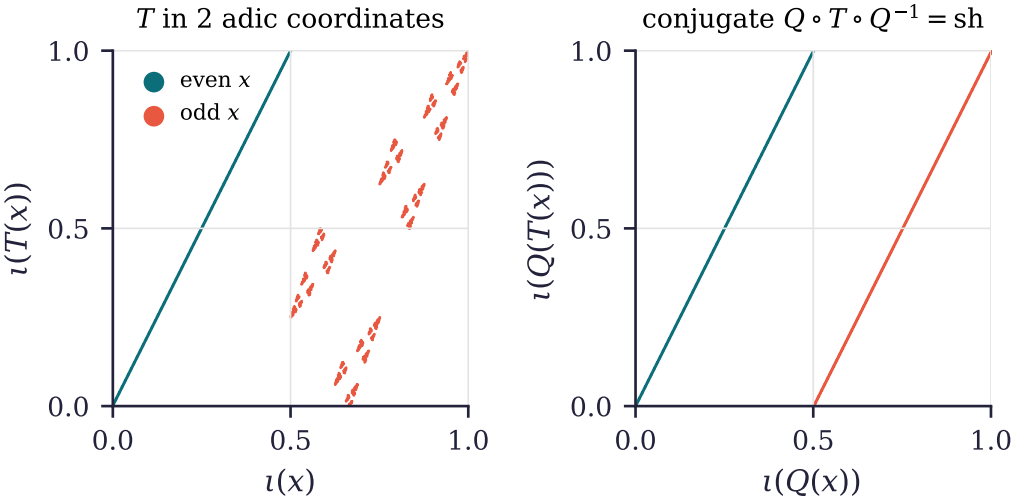


FIGURE 2.2. The conjugacy of Theorem 2.18, drawn on the residues modulo 2^{16}. Left: the graph of T in 2 adic coordinates, a point $x = \sum x_i 2^i$ being placed at $\iota(x) = \sum x_i 2^{-i-1} \in [0,1)$; even points lie on one line of slope 2 and odd points scatter. Right: in the coordinates given by the parity vector Q, the same map becomes the doubling map, the shift, with even and odd points on its two branches.

For example, $Q(0) = 000\cdots$, and $Q(-1) = 111\cdots$ since $T(-1) = -1$; indeed $-\sum_{j\ge1} 3^{-j}2^{j-1} = -1$ in $\mathbb{Z}_2$. Similarly $Q(1) = 1010\cdots$, since $1 \mapsto 2 \mapsto 1$, and $-\sum_{j\ge1} 3^{-j}4^{j-1} = -\frac{1}{3}(1-\frac{4}{3})^{-1} = 1$, the geometric series converging 2 adically because $|4/3|_2 = 1/4$.

Part (iii) contains a measure preservation statement that we isolate, with a direct proof that does not use the conjugacy.

PROPOSITION 2.19 *(Haar measure is invariant). For every Borel set $A \subseteq \mathbb{Z}_2$, $\mu(T^{-1}(A)) = \mu(A)$.*

Proof. It suffices to check balls $A = a + 2^k\mathbb{Z}_2$, since they form a π system generating the Borel sets. A point x lies in $T^{-1}(A)$ either because x is even and $x/2 \in A$, that is $x \in 2a + 2^{k+1}\mathbb{Z}_2$, or because x is odd and $(3x+1)/2 \in A$, that is

$$x \in 3^{-1}(2a-1) + 2^{k+1}\mathbb{Z}_2,$$

where we used that 3 is a unit; the centre $3^{-1}(2a-1)$ is odd, so this ball consists of odd points. The two balls are disjoint and each has measure 2^{-k-1}, so $\mu(T^{-1}(A)) = 2^{-k} = \mu(A)$. □

Since the Bernoulli shift is mixing, Theorem 2.18 shows that T is a mixing, in particular ergodic, transformation of $(\mathbb{Z}_2, \mu)$ (Lagarias [53, Theorem K], as a special case of a result of Matthews and Watts [67]). This is the cleanest statement of the randomness of the Collatz map, and it is simultaneously the cleanest statement of why randomness does not suffice: the positive integers form a set of Haar measure zero in $\mathbb{Z}_2$, and ergodic theory is silent about null sets. Inside $\mathbb{Z}_2$ the conjecture takes the following form.

Corollary 2.20. *An integer $n \geq 1$ satisfies $T^{(k)}(n) = 1$ if and only if $\mathrm{sh}^k Q(n) = 1010\cdots$. Hence the Collatz conjecture is equivalent to the statement that the parity vector of every positive integer ends in the periodic tail $1010\cdots$.*

Proof. By (ii), $\mathrm{sh}^k Q(n) = Q(T^{(k)}(n))$, and by (i) this equals $Q(1)$ if and only if $T^{(k)}(n) = 1$. □

Remark 2.21 *(Rational points and periodic sequences).* If w is eventually periodic, then the series in (iv) is eventually a geometric series and $Q^{-1}(w)$ is a rational number with odd denominator. The converse, that Q maps every rational number with odd denominator to an eventually periodic sequence, is the *periodicity conjecture* of Lagarias [53, 54]. It would imply that no orbit of a positive integer is unbounded: if $Q(n)$ is eventually periodic, say $\mathrm{sh}^{k+p} Q(n) = \mathrm{sh}^k Q(n)$, then $T^{(k+p)}(n) = T^{(k)}(n)$ by injectivity of Q, and the orbit is eventually periodic.

2.6 Stopping times and the density theorem

Definition 2.22 *(Stopping times).* For $n \geq 1$ the *stopping time* is $\sigma(n) = \min\{k \geq 1 : T^{(k)}(n) < n\}$, with $\sigma(n) = \infty$ if there is no such k. The *coefficient stopping time* is

$$\tau(n) = \min\{k \geq 1 : 3^{m_k(n)} < 2^k\},$$

the first time the multiplicative coefficient in (2.4) drops below 1.

If every $n \geq 2$ has finite stopping time then the conjecture follows by strong induction, since the orbit of n reaches a smaller value whose orbit reaches 1. Conversely, under the conjecture every $n \geq 2$ has finite stopping time. Note that $\sigma(1) = \infty$ trivially, because the orbit of 1 never goes below 1. The coefficient stopping time is a function of the parity vector alone, and therefore of a residue class: if $\tau(n) = k$ then $\tau(n') = k$ for every $n' \equiv n \pmod{2^k}$.

The following facts, and the exact counts of Section 2.6.1, are due to Terras [99] (his Proposition 1.8 and Theorems 1.9, 1.14 and 2.1); see also [53, Theorem C].

Proposition 2.23 *(Stopping times and coefficient stopping times).*

(i) $\tau(n) \le \sigma(n)$ *for every* $n \ge 1$.

(ii) If $\tau(n) = k$ *and* $m = m_k(n)$, *then* $\sigma(n') = k$ *for every* $n' \equiv n \pmod{2^k}$ *with* $n' > c_k(n)/(2^k - 3^m)$.

(iii) If $\tau(n) < \infty$, *then* $\tau(n) = \lfloor m \log_2 3 \rfloor + 1$ *with* $m = m_{\tau(n)}(n)$, *and the last entry of* $v_{\tau(n)}(n)$ *is* 0.

Proof. (i) If $T^{(k)}(n) < n$ then by (2.4) $3^{m_k} n + c_k = 2^k T^{(k)}(n) < 2^k n$, and since $c_k \ge 0$ this forces $3^{m_k} < 2^k$. So $\tau(n) \le k$ for every such k.

(ii) For n' in the class, $\tau(n') = k$ and $c_k(n') = c_k(n)$, so $T^{(k)}(n') < n'$ if and only if $3^m n' + c_k(n) < 2^k n'$, which is the stated condition. Then $\sigma(n') \le k$, and $\sigma(n') \ge \tau(n') = k$ by (i).

(iii) Let $k = \tau(n) < \infty$ and $m = m_k(n)$. If $k = 1$ then n is even, $m = 0$ and $k = \lfloor 0 \rfloor + 1$. If $k \ge 2$ then $3^{m_{k-1}} \ge 2^{k-1}$, and in fact $3^{m_{k-1}} > 2^{k-1}$ since a power of 3 equals a power of 2 only when both are 1. If the last step were odd we would have $m = m_{k-1} + 1$ and $3^m > 3 \cdot 2^{k-1} > 2^k$, contradicting $3^m < 2^k$. So the last step is even, $m = m_{k-1}$, and $2^{k-1} < 3^m < 2^k$, which says $k - 1 < m \log_2 3 < k$, that is $k = \lfloor m \log_2 3 \rfloor + 1$. □

By (iii), finite coefficient stopping times can take only the values 1, 2, 4, 5, 7, 8, 10, 12, 13, 15, 16, 18, 20, and so on; the gaps at 3, 6, 9, 11, 14, 17, 19, and so on, are the integers not of the form $\lfloor m \log_2 3 \rfloor + 1$. By (ii) the same restriction applies to $\sigma(n)$ for all sufficiently large n in each class with finite τ, and Proposition 2.25 below shows that every value in the list is actually taken by some residue class.

Example 2.24. Even n have $\sigma(n) = 1$. If $n \equiv 1 \pmod 4$ then the parity vector of length 2 is $(1,0)$, so $\tau(n) = 2$, $c_2 = 1$, and $T^{(2)}(n) = (3n+1)/4 < n$ for all $n > 1$. If $n \equiv 3 \pmod{16}$ then $v_4(n) = (1,1,0,0)$, so $m = 2$, $c_4 = 3 + 2 = 5$, $2^4 - 3^2 = 7$, and $T^{(4)}(n) = (9n+5)/16 < n$ for every $n \ge 1$ in the class. In the other classes modulo 16, namely $7, 11, 15 \pmod{16}$, the first four steps do not yet decide.

Figure 2.3 shows the same phenomenon on actual integers. The stopping times $\sigma(n)$ of $2 \le n < 2^{13}$ are arranged in rows of 64, so that each column is a residue class modulo 64. In 56 of the 64 columns the stopping time is constant: it is the coefficient stopping time of the class, reached within six steps. The remaining eight columns are the classes modulo 64 whose first six parities do not yet decide, and only there does the stopping time vary from row to row.

2.6.1 Exact counts

The number of residue classes that stop at each step can be computed exactly, without enumerating residues, by a dynamic programme over the number of odd steps. Call a word $w \in \{0,1\}^k$ *alive* if $3^{m_j(w)} > 2^j$ for $1 \le j \le k$, where $m_j(w)$ is the number of ones among its first j letters; by Theorem 2.8 the alive words of length k are the parity vectors of the residue classes modulo 2^k with $\tau > k$. (Since a power

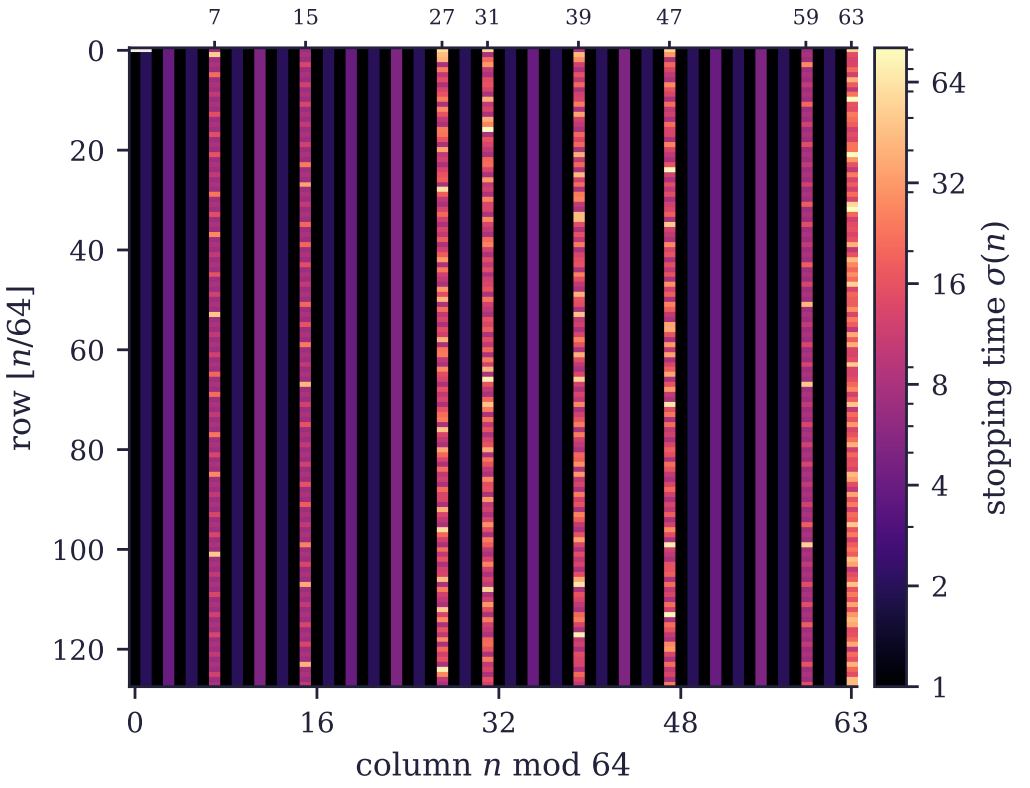


FIGURE 2.3. Stopping times $\sigma(n)$ for $2 \le n < 2^{13}$, arranged with n in row $\lfloor n/64 \rfloor$ and column n mod 64, on a logarithmic colour scale ($n = 0, 1$ are left blank). Fifty six columns are constant, because $\tau \le 6$ on them and $\sigma = \tau$ there; the eight columns 7, 15, 27, 31, 39, 47, 59, 63 marked at the top are the classes with $\tau > 6$, a proportion $1/8$ of all classes.

of 3 never equals a positive power of 2, the alternative to alive is $3^{m_j} < 2^j$ for some j.) Write

$$N_k(m) = \#\{w \in \{0,1\}^k \text{ alive} : m_k(w) = m\},$$
$$R_k = \#\{r \bmod 2^k : \tau(r) = k\}.$$

PROPOSITION 2.25 *(Exact stopping counts).*

(i) $N_0(0) = 1$, *and for* $k \ge 1$ *and* $m \ge 0$,

$$N_k(m) = \big(N_{k-1}(m) + N_{k-1}(m-1)\big)\,[\,3^m > 2^k\,],$$

with $N_{k-1}(-1) = 0$, *where* $[P]$ *is* 1 *if* P *holds and* 0 *otherwise. The proportion of residues modulo* 2^k *with* $\tau > k$ *is* $1 - F(k) = 2^{-k}\sum_m N_k(m)$, *and among them the number of odd steps in the first* k *has the law* $m \mapsto N_k(m)/\sum_{m'} N_k(m')$.

(ii) There is at most one integer $m_k^* \geq 0$ with $2^{k-1} \leq 3^{m_k^*} < 2^k$; equality on the left occurs only for $k = 1$ and $m_1^* = 0$. If it exists then $R_k = N_{k-1}(m_k^*) \geq 1$, and otherwise $R_k = 0$. Consequently $R_k > 0$ if and only if $k = \lfloor m \log_2 3 \rfloor + 1$ for some integer $m \geq 0$.

(iii) The counts satisfy $2^k\big(1 - F(k)\big) = 2^k\big(1 - F(k-1)\big) - R_k$ and $F(k) = \sum_{j \leq k} R_j 2^{-j}$.

Proof. (i) A word of length k is alive exactly when its prefix of length $k-1$ is alive and $3^{m_k} > 2^k$. Splitting according to the last letter, which is 0 or 1, gives the recursion. The statements about residues follow from Theorem 2.8, which makes the parity vector of a uniform residue modulo 2^k uniform on $\{0,1\}^k$.

(ii) The condition $2^{k-1} \leq 3^m < 2^k$ says that m lies in the half open interval $\big[(k-1)/\log_2 3,\, k/\log_2 3\big)$, of length $1/\log_2 3 < 1$, so it has at most one solution. A residue has $\tau = k$ exactly when its parity word of length k has an alive prefix of length $k-1$ and $3^{m_k} < 2^k$. By Proposition 2.23(iii) its last letter is then 0 and $m = m_{k-1} = m_k$ satisfies $2^{k-1} \leq 3^m < 2^k$ (for $k = 1$ the word is 0 and $m = 0$), so $m = m_k^*$. Conversely, every alive word of length $k-1$ with m_k^* ones, followed by a 0, has $\tau = k$. Hence $R_k = N_{k-1}(m_k^*)$. The word $1^m 0^{k-1-m}$ with $m = m_k^*$ is alive, since $3^j > 2^j$ for $1 \leq j \leq m$ and $3^m > 2^j$ for $m < j \leq k-1$, so $R_k \geq 1$. Finally m_k^* exists if and only if $k - 1 \leq m \log_2 3 < k$ for some integer $m \geq 0$, that is, $k = \lfloor m \log_2 3 \rfloor + 1$.

(iii) Each alive word of length $k-1$ has two extensions, and exactly the R_k extensions counted in (ii) fail to be alive. The second identity follows by summing. $\square$

Table 2.2 lists R_k and $1 - F(k)$ for $k \leq 40$, computed from the recursion with exact integer arithmetic, and Figure 2.4 draws the first seven levels as a tree. For small k the classes are easily listed: $\tau = 1$ on 0 mod 2, $\tau = 2$ on 1 mod 4, $\tau = 4$ on 3 mod 16, $\tau = 5$ on 11 and 23 mod 32, $\tau = 7$ on 7, 15 and 59 mod 128, and $\tau = 8$ on 39, 79, 95, 123, 175, 199 and 219 mod 256. The recursion has been checked against direct enumeration of all residues for $k \leq 20$, and the equivalence in (ii) against the recursion for $k \leq 400$ (script `ch02_exact.py`). Figure 2.5 displays the whole alive tree to depth nine, coloured by the height $m_k \log 3 - k \log 2$, the logarithm of the coefficient $3^{m_k}/2^k$ in (2.4); a class stops at the first ring where its height becomes negative.

We can now prove the theorem of Terras and Everett, in the quantitative form given by Lagarias [53, Theorem D]. Write

$$F(k) = 2^{-k}\,\#\{r \bmod 2^k : \tau(r) \leq k\}, \qquad \theta = \frac{\log 2}{\log 3} \approx 0.6309,$$

and let

$$D = \theta \log(2\theta) + (1-\theta)\log\big(2(1-\theta)\big) \approx 0.03469$$

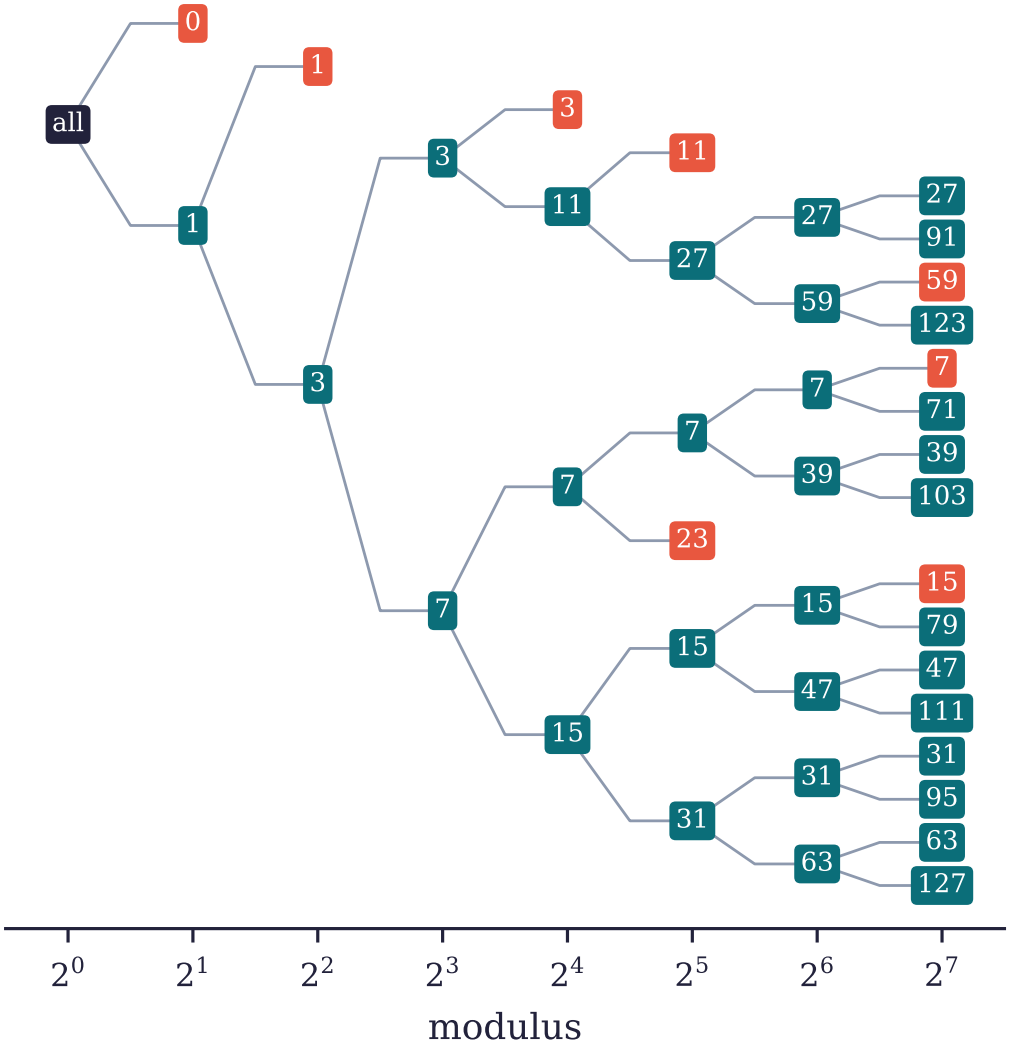


FIGURE 2.4. The stopping tree. A node at modulus 2^j is a residue class modulo 2^j whose orbit has not stopped before step j. Lighter nodes stop at step j ($\tau = j$) and are leaves; each darker node r splits into the classes r and $r + 2^j$ modulo 2^{j+1}. At modulus 2^7, 13 of the 128 classes remain.

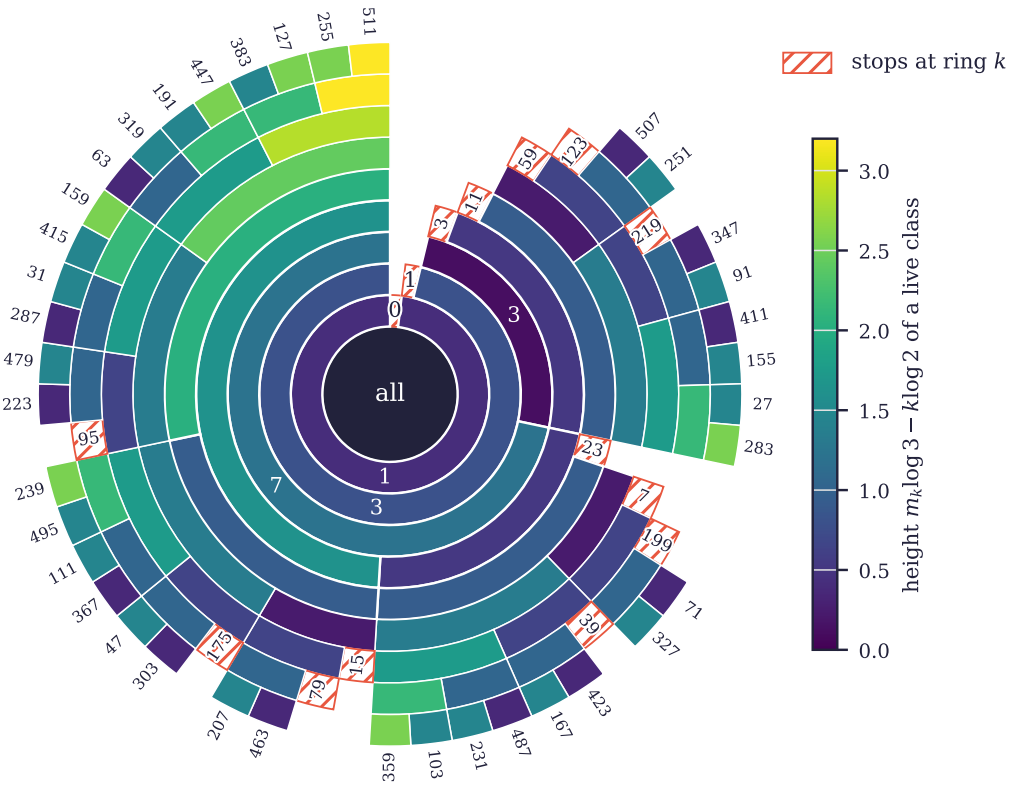


FIGURE 2.5. The Terras parity tree as a sunburst. Ring k consists of residue classes modulo 2^k, equivalently parity words of length k, and the two children of a class in ring $k+1$ are its two lifts. A class that stops at ring k ($\tau = k$) is a hatched wedge labelled by its residue with nothing outside it; live classes are shaded by their height $m_k \log 3 - k \log 2$. Angular width is proportional to the number of leaves below a node, not to Haar measure. The outer labels are the 38 classes modulo 2^9 with $\tau > 9$.

TABLE 2.2. Number R_k of residue classes modulo 2^k with coefficient stopping time $\tau = k$, and the proportion $1 - F(k)$ of residues modulo 2^k with $\tau > k$, from Proposition 2.25. The rows with $R_k = 0$ are the k not of the form $\lfloor m \log_2 3 \rfloor + 1$. Terras [99] tabulates the same distribution to $k = 900$ (his Table A).

k	R_k	$1 - F(k)$	k	R_k	$1 - F(k)$
1	1	0.500000	21	8045	0.022226
2	1	0.250000	22	0	0.022226
3	0	0.250000	23	17 637	0.020123
4	1	0.187500	24	51 033	0.017082
5	2	0.125000	25	0	0.017082
6	0	0.125000	26	108 950	0.015458
7	3	0.101562	27	312 455	0.013130
8	7	0.074219	28	0	0.013130
9	0	0.074219	29	663 535	0.011894
10	12	0.062500	30	0	0.011894
11	0	0.062500	31	1 900 470	0.011009
12	30	0.055176	32	5 936 673	0.009627
13	85	0.044800	33	0	0.009627
14	0	0.044800	34	13 472 296	0.008843
15	173	0.039520	35	39 993 895	0.007679
16	476	0.032257	36	0	0.007679
17	0	0.032257	37	87 986 917	0.007039
18	961	0.028591	38	0	0.007039
19	0	0.028591	39	257 978 502	0.006569
20	2652	0.026062	40	820 236 724	0.005823

be the Kullback and Leibler divergence of the Bernoulli(θ) law from the Bernoulli($\frac{1}{2}$) law. (In this section only, θ denotes $1/\log_2 3$, the threshold frequency of odd steps; in Chapter 10 it denotes $\log_2 3$ itself.)

THEOREM 2.26 *(Terras [99, 100], Everett [22], Lagarias [53, Theorem D]). For each $k \geq 1$ the set $\{n \geq 1 : \sigma(n) \leq k\}$ has natural density $F(k)$, and*

$$1 - F(k) \leq e^{-Dk}.$$

Consequently the set of $n \geq 1$ with finite stopping time has natural density 1.

Proof. Step 1: the coefficient stopping time. The set $A_k = \{n \geq 1 : \tau(n) \leq k\}$ is a union of residue classes modulo 2^k: whether $\tau(n) \leq k$ is decided by $v_k(n)$, hence by n mod 2^k. A union of R classes modulo 2^k has natural density $R/2^k$, so A_k has density $F(k)$.

Step 2: comparison with the stopping time. Let $B_k = \{n \geq 1 : \sigma(n) \leq k\}$. By Proposition 2.23(i), $B_k \subseteq A_k$. Conversely, each class r mod 2^j with $\tau = j \leq k$ is contained in B_k apart from the finitely many $n \leq c_j(r)/(2^j - 3^{m_j(r)})$, by

Proposition 2.23(ii). There are finitely many such classes, so $A_k \setminus B_k$ is finite, and B_k has the same density $F(k)$ as A_k.

Step 3: reduction to a binomial tail. By Corollary 2.10, $1 - F(k)$ is the probability that a uniformly random word $w \in \{0,1\}^k$ satisfies $3^{m_j(w)} \geq 2^j$ for every $j \leq k$, where $m_j(w)$ is the number of ones among the first j letters. In particular such a word has $3^{m_k(w)} \geq 2^k$, that is $m_k(w) \geq \theta k$. Hence

$$1 - F(k) \leq \mathbb{P}(X \geq \theta k), \qquad X \sim \text{Binomial}(k, \tfrac{1}{2}).$$

Step 4: the Chernoff bound. For every $\lambda > 0$, Markov's inequality applied to $e^{\lambda X}$ gives

$$\mathbb{P}(X \geq \theta k) \leq e^{-\lambda\theta k}\, \mathbb{E}e^{\lambda X} = \Big(e^{-\lambda\theta}\, \frac{1 + e^{\lambda}}{2}\Big)^k.$$

Since $\theta > \frac{1}{2}$ we may take $e^{\lambda} = \theta/(1-\theta) > 1$. Then $(1+e^{\lambda})/2 = 1/(2(1-\theta))$ and $e^{-\lambda\theta} = ((1-\theta)/\theta)^{\theta}$, and the logarithm of the bracket is

$$\begin{aligned} -\theta \log \frac{\theta}{1-\theta} &- \log\big(2(1-\theta)\big) \\ &= -\theta \log(2\theta) - (1-\theta)\log\big(2(1-\theta)\big) = -D. \end{aligned}$$

So $1 - F(k) \leq e^{-Dk}$.

Step 5: conclusion. The set $B = \{n : \sigma(n) < \infty\}$ contains B_k for every k, so its lower density is at least $F(k) \geq 1 - e^{-Dk}$ for every k. Since $D > 0$, the lower density of B is 1, and hence B has natural density 1. □

Figure 2.6 compares the exact proportion $1 - F(k)$, computed by dynamic programming over the number of ones in the parity word, with the bounds used in the proof. The bound is valid but loose: at $k = 100$ the exact value is 2.39×10^{-4}, the binomial tail is 3.32×10^{-3}, and the Chernoff bound is 3.12×10^{-2}. The ratio $(1 - F(k))/e^{-Dk}$ falls from 0.052 at $k = 20$ to 0.0077 at $k = 100$ and 0.0032 at $k = 200$. Step 3 throws away the requirement that the walk $j \mapsto m_j \log 3 - j \log 2$ stay nonnegative at every intermediate time and keeps only the endpoint. This loses at most a factor k (Exercise 2.33), so the exponent D cannot be improved, as Lagarias [53] observes. The endpoint probability is of order $k^{-1/2}e^{-Dk}$, and for a random walk with negative drift the probability of staying above a barrier for k steps is, heuristically, smaller by a further factor of order k^{-1}, so of order $k^{-3/2}e^{-Dk}$. The local exponent of the observed ratio to e^{-Dk}, -1.19 between $k = 20$ and $k = 100$ and -1.41 between $k = 400$ and $k = 800$, approaches $-3/2$ slowly; we do not need or prove this refinement.

Two further remarks place the theorem in context.

Remark 2.27 *(The coefficient stopping time conjecture).* We always have $\tau \leq \sigma$ by Proposition 2.23, with equality for all sufficiently large members of each class. Terras [99] conjectured that every $n \geq 2$ has $\sigma(n) = \tau(n)$, verified this for $n <$

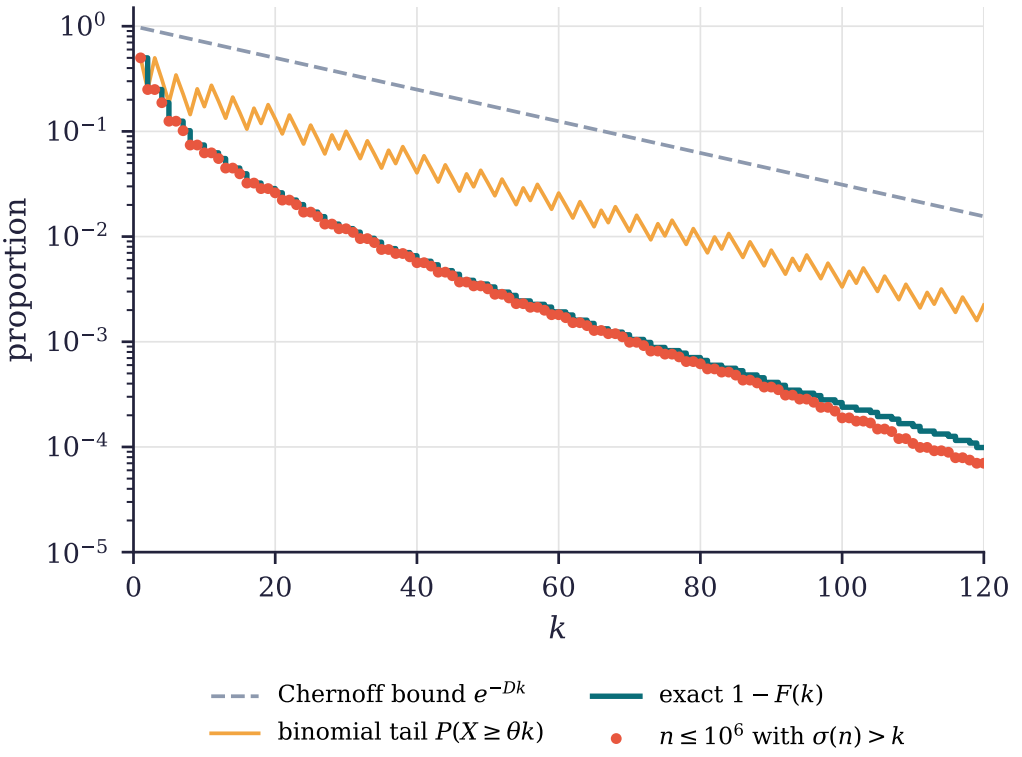


FIGURE 2.6. Decay of the proportion of residues that have not stopped. Dark solid line: the exact density $1 - F(k)$ of $\{\tau > k\}$. Dots: the fraction of $n \leq 10^6$ with $\sigma(n) > k$. Light solid line: the binomial tail used in Step 3 of the proof. Dashed line: the Chernoff bound e^{-Dk}.

250 000, and deduced from an explicit bound that it holds for every n with $\tau(n) < 2592$; see Lagarias [53, Theorem H]. We have also checked it directly for $2 \le n \le 10^6$. The conjecture would rule out nontrivial cycles: if $n \ge 2$ were the least element of a cycle of length k with m odd elements, then $\sigma(n) = \infty$, while $3^m < 2^k$ by Corollary 2.14, so $\tau(n) \le k$. It says nothing directly about unbounded orbits, since a starting value of such an orbit could have $\tau = \infty$ as well. Among $n \le 10^6$ the largest stopping time is $\sigma(626\,331) = 176$; half of all n have $\sigma(n) = 1$ and a quarter have $\sigma(n) = 2$, in agreement with Table 2.2.

Remark 2.28 *(Why density one is not enough).* Theorem 2.26 says that almost every n falls below itself. It does not say that almost every n reaches 1. To descend from n to 1 an orbit must pass through a chain of smaller values each of which must itself have finite stopping time, and the exceptional set, though of density zero, might contain a value on every orbit. Repeating the argument from the value $T^{(\sigma(n))}(n)$ is not legitimate, because that value is not uniformly distributed in any residue class: it is determined by n, and its residues have been used up in computing $\sigma(n)$. This is the gap between ensemble and orbit that recurs throughout the book. Results that go further, from Korec's exponent $\log 3/\log 4$ to Tao's almost bounded values, are described in Chapter 1.

Notes

The parity vector and its bijection with residues are due to Terras [99] and Everett [22]; see [53, Theorem B]. Terras also introduced the coefficient stopping time, proved the facts collected in Proposition 2.23, gave the recursion of Proposition 2.25 with a table of the resulting distribution (his Table A) and the description of the values of τ with a table of them (his Table B), proved that the limiting distribution $F(k)$ exists, as in Steps 1 and 2 of the proof of Theorem 2.26, and, by the central limit theorem, that $F(k) \to 1$, which is the density statement of Theorem 2.26 (further details of the proof are in a later note [100]), and stated the coefficient stopping time conjecture. The density statement was proved independently by Everett [22], who worked with $f(2n) = n$, $f(2n+1) = 3n+2$, which is our accelerated map T. The exponential bound of Theorem 2.26 is Theorem D of Lagarias [53], proved there by bounding the same binomial tail; Lagarias also observes that its exponent cannot be improved. The survey of Lagarias [53] gives a unified account of these results, the 2 adic extension and the conjugacy of Theorem 2.18, and many generalizations; its section on the 2 adic viewpoint is still the best introduction. Bernstein [10] gave the noniterative formula for Q^{-1}, and the rational cycles of the extended map are studied by Lagarias [54]. Wirsching [107] develops the combinatorics of parity words and of the inverse tree in depth. The identity $T^{(j)}(2^j a - 1) = 3^j a - 1$ of Remark 2.4 is classical; see Hercher [38, Lemma 8], who cites Terras [99] for it.

The affine identity (2.4) and the cycle formula (2.8) are classical and appear, in

various notations, in most treatments of cycles; we have followed the notation of our Lean formalization, in which the recursion (2.5) is the inductive step of the proof. The sum (2.6) is a sum of terms $3^j 2^i$ indexed by the parity word, and the question of whether its structure, rather than only its size, can constrain cycles is one of the open problems of Chapter 15.

The Chernoff argument in Step 4 is the simplest route to an explicit rate; the true asymptotics of $1 - F(k)$ are those of the probability that a random walk with negative drift stays positive for k steps. Chapter 3 recovers the same exponential rate for the first descent of the stochastic model, though not the finer asymptotics.

Exercises

Exercise 2.29. Prove the identity $T^{(i)}(2^j a - 1) = 3^i 2^{j-i} a - 1$ of Remark 2.4. Deduce that $T^{(j)}(2^j - 1) = 3^j - 1$, and that for every j there are infinitely many n whose orbits under T begin with exactly j odd steps followed by an even step.

Exercise 2.30. Use Theorem 2.18(iv) to find the 2 adic integer whose parity vector is the periodic word $110110\cdots$, and show that it is -5. Explain the answer in terms of the cycle $-5 \mapsto -7 \mapsto -10 \mapsto -5$. More generally, show that if w is periodic with period k and has m ones in each period, then $Q^{-1}(w) = c(w)/(2^k - 3^m)$ with $c(w)$ the sum (2.6) over one period, in agreement with Corollary 2.14.

Exercise 2.31. For $n \equiv 11 \pmod{32}$ compute $v_5(n)$, $m_5(n)$ and $c_5(n)$, and show that $T^{(5)}(n) = (27n + 23)/32$. Deduce that $\sigma(n) = 5$ for every such n, and compute the natural densities of $\{\sigma = 4\}$ and $\{\sigma = 5\}$.

Exercise 2.32. Let $U = \mathbb{Z}_2 \setminus 2\mathbb{Z}_2$ be the 2 adic units with normalized Haar measure $\mu_U = 2\mu|_U$, and extend the Syracuse map to $U \setminus \{-1/3\}$. Show that for an odd ball $b + 2^k\mathbb{Z}_2$ the preimage under S is the disjoint union over $j \geq 1$ of the balls $3^{-1}(2^j b - 1) + 2^{k+j}\mathbb{Z}_2$, and deduce that S preserves μ_U. Show that under μ_U the halving exponent of S has exactly the Geometric$(\frac{1}{2})$ law.

Exercise 2.33. Let X and θ be as in the proof of Theorem 2.26. Let $w \in \{0,1\}^k$ satisfy $3^{m_k(w)} > 2^k$, and let j be the last index in $\{0, \dots, k-1\}$ at which $m_j(w)\log 3 - j\log 2$ is smallest. Show that the word obtained from w by moving its first j letters to the end is alive, in the sense of Section 2.6.1, and deduce that $\mathbb{P}(X \geq \theta k)/k \leq 1 - F(k) \leq \mathbb{P}(X \geq \theta k)$. Using $\binom{k}{i} \geq 2^{kH(i/k)}/(k+1)$, where $H(x) = -x\log_2 x - (1-x)\log_2(1-x)$, and $D = (1 - H(\theta))\log 2$, show that $\frac{1}{k}\log\bigl(1 - F(k)\bigr) \to -D$, so that the exponent in Theorem 2.26 cannot be improved.

Exercise 2.34. Formulate and prove the analogue of Theorem 2.8 for the standard map T_0. Show that the parity vectors of length k under T_0 are exactly the binary words with no two consecutive ones, that their number is the Fibonacci number Fib_{k+2} (with $\mathrm{Fib}_1 = \mathrm{Fib}_2 = 1$), and that the set of n with a given word w is a single residue class modulo 2^{z+1}, where z is the number of zeros among the first $k-1$

letters of w. Explain why there is no longer a bijection with the residues modulo a fixed power of 2.

Chapter 3

Stochastic Models and Martingale Duality

The previous chapter proved that a uniformly random residue modulo 2^k produces a parity vector of k independent fair coin flips. It is a short step from this theorem to a model: pretend that the parities along an orbit are fair coin flips forever, and study the resulting random walk for $\log x$. The model rests on this theorem of Terras [99] and Everett [22] and was stated as a heuristic by Crandall [18] (Section 3.1 tells its history). Its predictions agree closely with computation (the fitted slope of the total stopping times to within about two percent over the ranges tested, and to within half a percent for the accelerated map below 10^4; their variance to about ten percent), and it is the reason almost everyone who has thought about the problem believes the conjecture.

This chapter develops the model with the tools of martingale theory. The organizing fact is a duality. Under the fair coin model for the accelerated map, the iterates x_n themselves form a martingale, because $\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{3}{2} = 1$ exactly, while their logarithms form a supermartingale with strictly negative drift $\delta^* = \log(\sqrt{3}/2) \approx -0.1438$. The two statements are reconciled by Jensen's inequality, and the drift is exactly the Jensen gap. The heuristic is therefore not a statement about the size of the iterates on average, which in the model does not decrease at all, but about the size of their logarithms. Almost everything else in the chapter follows from this observation: the total stopping time slope $1/|\delta^*| \approx 6.952$ through Wald's identity, the dispersion of total stopping times through the central limit theorem and the moment asymptotics for first passage times, a bound of exactly $1/K$ on the probability of an excursion by a factor K through Ville's maximal inequality, absorption with probability one, and the failure of uniform integrability that prevents the martingale from saying anything about where the orbit ends. We also treat the standard map T_0, whose model parity sequence is not independent but a renewal sequence, and we compute its autocorrelation function exactly: $\rho(h) = (-\frac{1}{2})^h$.

Apart from Propositions 3.2 and 3.10, which are statements about the map, every result in this chapter is a theorem about a probabilistic model, and the model is not the Collatz map. We say so at the start because the point is easy to lose in the middle of a computation. The final section asks what a proof of the conjecture along these lines would have to supply, and the answer, a certificate that transfers the model's conclusions to individual orbits, is the subject of much of the rest of the book.

3.1 The parity heuristic

3.1.1 From Terras to Lagarias and Weiss

Terras [99] and, independently, Everett [22] introduced the parity vector and proved that its first k entries are uniformly distributed when the starting value is uniformly distributed modulo 2^k (Theorem 2.8; see [53, Theorem B]), and both proved that almost all integers have a finite stopping time (the density statement of Theorem 2.26, whose exponential rate is due to Lagarias [53, Theorem D]; see also [100]). Both papers use the coin flip picture to prove theorems about densities of starting values, and both are careful to stay within the range of steps that the residue class actually controls. The heuristic consists in ignoring that limitation.

The heuristic was stated explicitly by Crandall [18]. Lagarias, in his 1985 survey [53, §2.1], recorded it and pointed out that it predicts the growth rate of total stopping times. Lagarias and Weiss [59] then studied two stochastic models in detail, a random walk model for the trajectory of a single integer and a branching random walk model for the tree of preimages, and compared their predictions for total stopping times and maximal excursions with extensive computation. Kontorovich and Sinai [48] proved a structure theorem for a broad family of maps of Collatz type, showing that the logarithm of a trajectory from a random starting value behaves like a random walk with drift; and the survey of Kontorovich and Lagarias [47] places all of this, together with the $5x+1$ problem, where the drift is positive and orbits are expected to diverge, in a common framework.

We shall use three versions of the model, one for each of the maps of Definition 2.1. Throughout the chapter the multiplicative factor $3 + 1/x$ is replaced by 3; we comment on this replacement in Remark 3.3.

Definition 3.1 *(Three stochastic models).*

(a) *Bernoulli model for T.* Let $\xi_1, \xi_2, \dots$ be independent with $\mathbb{P}(\xi_i = \frac{1}{2}) = \mathbb{P}(\xi_i = \frac{3}{2}) = \frac{1}{2}$, and put $x_n = x_0 \xi_1 \cdots \xi_n$. The log increments are $\Delta_i = \log \xi_i$, and $\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$.

(b) *Stride model for S.* Let $K_1, K_2, \dots$ be independent, each with the Geometric$(\frac{1}{2})$ law, so that $\mathbb{P}(K_j = k) = 2^{-k}$ for $k \geq 1$, and put $X_j = X_{j-1} \cdot 3 \cdot 2^{-K_j}$.

(c) *Renewal model for T_0.* Let $(B_n)_{n \geq 0}$ be the Markov chain on $\{0,1\}$ with transition probabilities

$$\mathbb{P}(B_{n+1} = 0 \mid B_n = 1) = 1,$$
$$\mathbb{P}(B_{n+1} = 0 \mid B_n = 0) = \mathbb{P}(B_{n+1} = 1 \mid B_n = 0) = \tfrac{1}{2},$$

and put $x_{n+1} = 3x_n$ if $B_n = 1$ and $x_{n+1} = x_n/2$ if $B_n = 0$. Here B_n is the model counterpart of the parity $b_n = T_0^{(n)}(x_0) \bmod 2$ of an orbit (Lemma 2.3), and $\mathcal{F}_n = \sigma(B_0, \dots, B_n)$.

The three models are consistent with one another in the same way the three maps are. In model (c) an odd step is always followed by a halving, by Lemma 2.3; merging each odd step with the halving that follows it produces a step with factor $3/2$, and the parity after that is again a fair coin, so model (c) with the forced steps deleted is model (a). Grouping the steps of model (a) into runs that begin with a factor $\frac{3}{2}$ and continue with factors $\frac{1}{2}$ until the next $\frac{3}{2}$ produces the factor $3 \cdot 2^{-K}$ with K Geometric$(\frac{1}{2})$, which is model (b). The three models therefore describe one random path measured against three clocks, which tick on average three times, twice and once per stride.

3.1.2 *In what sense the model is exact*

The model is not an analogy chosen for convenience. Over an initial segment of the orbit whose length is controlled by the size of the starting residue class it is exact, up to the replacement of $3 + 1/x$ by 3.

Proposition 3.2 *(Ensemble exactness). Let $k \geq 1$ and let n be uniformly distributed on a complete set of residues modulo 2^k consisting of positive integers. Then the parities of $T^{(j)}(n)$ for $0 \leq j < k$ are independent and fair, and for $j \leq k$*

$$\log T^{(j)}(n) - \log n = \sum_{i=1}^{j} \Delta_i + \sum_{\substack{0 \leq i < j \\ T^{(i)}(n) \text{ odd}}} \log\Big(1 + \frac{1}{3T^{(i)}(n)}\Big),$$

where $\Delta_{i+1} = \log \frac{3}{2}$ if $T^{(i)}(n)$ is odd and $\Delta_{i+1} = -\log 2$ otherwise. Thus the first k log increments of T have exactly the joint law of the increments of the Bernoulli model, apart from correction terms that are positive and at most $1/(3T^{(i)}(n))$ each.

Proof. The first statement is Corollary 2.10. The second is the identity $T(x)/x = \frac{3}{2}(1 + \frac{1}{3x})$ for odd x and $T(x)/x = \frac{1}{2}$ for even x, summed along the orbit. □

Proposition 3.2 is the source of the model's authority and also of its limitation. A starting value n of size about 2^k carries k bits, and the first k steps of its orbit consume them, one bit per step. After that the parities are a deterministic function of information that the orbit itself has generated, and no theorem says that they continue to look like coin flips. Since an orbit from n needs about $\log n/|\delta^*| \approx 4.8 \log_2 n$ steps of T to reach 1 (Proposition 3.8), roughly four fifths of a typical orbit lies beyond the range where the model is a theorem.

Remark 3.3 *(The plus one).* Replacing $3 + 1/x$ by 3 changes each odd step by a factor $1 + 1/(3x)$, so the cumulative effect on $\log x$ along an orbit is at most $\sum 1/(3x_i)$ over its odd steps, which is negligible unless the orbit lingers at small values. The replacement matters in two places only. Near the end of an orbit it decides whether the path lands on the cycle $1 \to 2 \to 1$ under T, which the multiplicative model cannot see. And it is the whole difference between the $3x + 1$

map and its multiplicative caricature, in which an exact return to a previous value would require $3^m = 2^k$; the cycle analysis of Chapter 10 lives entirely in the $+1$. For questions of drift, dispersion and excursion, which are the subject of this chapter, it is harmless.

3.2 Martingale duality

We begin with the accelerated map, where the model is simplest.

Proposition 3.4 *(Martingale duality). In the Bernoulli model for the accelerated map T:*

(i) Multiplicative martingale. $\mathbb{E}[\xi_i] = \frac{1}{2}\cdot\frac{1}{2} + \frac{1}{2}\cdot\frac{3}{2} = 1$, *so (x_n) is a nonnegative martingale with $\mathbb{E}[x_n] = x_0$ for every n.*

(ii) Log supermartingale. $\mathbb{E}[\Delta_i] = \frac{1}{2}\log\frac{3}{2} - \frac{1}{2}\log 2 = \log\frac{\sqrt{3}}{2} = \delta^* \approx -0.1438$, *so $(\log x_n)$ is a strict supermartingale, and $\log x_n - n\delta^*$ is a martingale.*

(iii) Wald root. *The moment generating function of the increment,*

$$M(\theta) = \mathbb{E}\left[e^{\theta\Delta_i}\right] = \tfrac{1}{2}\,2^{-\theta} + \tfrac{1}{2}\left(\tfrac{3}{2}\right)^{\theta},$$

is strictly convex with $M(0) = M(1) = 1$. Its nonzero root θ^* of $M(\theta) = 1$ is $\theta^* = 1$, and x_n^θ is a supermartingale for $0 \le \theta \le 1$ and a submartingale for $\theta \ge 1$ or $\theta \le 0$.

(iv) Jensen gap. $\delta^* = \mathbb{E}[\log \xi_i] - \log \mathbb{E}[\xi_i]$, *so the drift of $\log x_n$ is exactly the Jensen defect of the logarithm at the law of ξ_i.*

Proof. Parts (i) and (ii) are the two displayed computations together with the independence of ξ_{n+1} from $\mathcal{F}_n$: $\mathbb{E}[x_{n+1} \mid \mathcal{F}_n] = x_n\mathbb{E}[\xi_{n+1}] = x_n$ and $\mathbb{E}[\log x_{n+1} \mid \mathcal{F}_n] = \log x_n + \delta^*$. For (iii), $M(0) = 1$ and $M(1) = \frac{1}{4} + \frac{3}{4} = 1$. Since $M''(\theta) = \mathbb{E}[\Delta^2 e^{\theta\Delta}] > 0$, M is strictly convex, so it equals 1 at exactly two points and is below 1 strictly between them and above 1 outside. Finally $\mathbb{E}[x_{n+1}^\theta \mid \mathcal{F}_n] = x_n^\theta M(\theta)$. Part (iv) restates (ii) using $\log \mathbb{E}[\xi_i] = 0$ from (i). □

Parts (i) and (ii) of Proposition 3.4 appear to contradict each other. By (i) the expected size of x_n never changes. By (ii) and the strong law of large numbers, $\log x_n / n \to \delta^* < 0$ almost surely, so almost every path tends to zero exponentially fast, at the rate $e^{n\delta^*} = (\sqrt{3}/2)^n$. Both are true. The mean is held up by rare paths that grow enormously, and the typical path has nothing to do with the mean. This is the familiar behaviour of a product of independent positive factors with mean one, the same mechanism by which a fair multiplicative bet ruins almost every gambler while preserving the expected wealth of the population.

Remark 3.5 *(Size of the Jensen gap).* The factor ξ has mean 1 and variance $\mathbb{E}\xi^2 - 1 = \frac{1}{2}(\frac{1}{4} + \frac{9}{4}) - 1 = \frac{1}{4}$. The second order expansion $\mathbb{E}\log\xi \approx \log\mathbb{E}\xi - \mathrm{Var}(\xi)/(2(\mathbb{E}\xi)^2)$ gives $-\frac{1}{8} = -0.125$, against the exact value $\delta^* = -0.1438$. The two values are

symmetric about 1, so the odd moments of $\xi - 1 = \pm\frac{1}{2}$ vanish and $\mathbb{E}\log\xi = -\sum_{j\geq 1} 4^{-j}/(2j) = \frac{1}{2}\log\frac{3}{4}$; the second order value $-\frac{1}{8}$ is the first term of this series, and the difference comes from the higher even moments. Thus the descent of a Collatz orbit, in the model, is a pure fluctuation effect. A process with the same mean factor and no fluctuation would neither rise nor fall. This is why, in the model for T, any argument that works with x_n directly, or with any quantity linear in x_n, sees no drift at all; we shall meet the same phenomenon from the side of majorization in Chapter 4, where the linear majorizer of log is blind to descent for exactly this reason.

For a random walk $S_n = \sum_{i\leq n}\Delta_i$ with negative drift, the positive root θ^* of $M(\theta) = 1$ governs the probability of large upward excursions: $e^{\theta^* S_n}$ is a martingale, and the maximal inequality applied to it gives the Lundberg bound $\mathbb{P}(\sup_n S_n \geq u) \leq e^{-\theta^* u}$, as we show in Section 3.7. For the Collatz model the root is $\theta^* = 1$ exactly, which is (i) seen from the other side: $e^{S_n} = x_n/x_0$, so the Wald martingale is the orbit itself. The same root appears for the stride model, whose moment generating function $M_S(p) = 3^p/(2^{1+p} - 1)$ also satisfies $M_S(1) = 1$, as we compute in (4.10). It must: whenever a model makes the iterates themselves a martingale, the exponential martingale with parameter one is the identity, and the root is 1 whatever clock is used.

3.3 The standard map and its renewal structure

The fair coin heuristic is usually stated for T, and there is a good reason. Applied naively to T_0 it gives the wrong sign.

Remark 3.6 *(A tempting but false argument).* "Under T_0 the parities are fair coins, so with probability $\frac{1}{2}$ a step multiplies by 3 and with probability $\frac{1}{2}$ it divides by 2. The expected log increment is $\frac{1}{2}\log 3 - \frac{1}{2}\log 2 = \frac{1}{2}\log\frac{3}{2} > 0$, so orbits of T_0 should grow." The error is in the first sentence. Under T_0 an odd step is always followed by an even step (Lemma 2.3), so the parities are not independent and are not fair: in the long run only one step in three is odd. The correct computation, in Proposition 3.7 below, gives the drift $\frac{1}{3}\log\frac{3}{4} < 0$ per step.

The renewal model of Definition 3.1(c) encodes the forced halving. Its transition matrix, with states ordered $(0, 1)$ and columns indexed by the current state, is

$$P = \begin{pmatrix} \frac{1}{2} & 1 \\ \frac{1}{2} & 0 \end{pmatrix},$$

with eigenvalues 1 and $-\frac{1}{2}$. The eigenvalue $-\frac{1}{2}$ will reappear as the base of the autocorrelation function in Section 3.8.

Proposition 3.7 *(Martingale structure of the standard map). In the renewal model for T_0:*

(i) The stationary law of the parity chain is $\mathbb{P}(B_n = 1) = \frac{1}{3}$*, and the fraction of odd steps among the first n converges to* $\frac{1}{3}$ *almost surely from any initial state.*

(ii) (x_n) *is not a martingale:* $\mathbb{E}[x_{n+1} \mid \mathcal{F}_n] = 3x_n$ *on* $\{B_n = 1\}$*. The process* $M_n = 3^{B_n} x_n$ *is a martingale.*

(iii) $(\log x_n)$ *is not a supermartingale, since its increment is* $+\log 3$ *on* $\{B_n = 1\}$*. The process* $\log M_n$ *is a supermartingale, with conditional drift* 0 *on* $\{B_n = 1\}$ *and* δ^* *on* $\{B_n = 0\}$*.*

(iv) Almost surely, $\frac{1}{n} \log x_n \to \frac{1}{3} \log 3 - \frac{2}{3} \log 2 = \frac{1}{3} \log \frac{3}{4} = \frac{2}{3}\delta^* \approx -0.0959$*.*

(v) Let $\tau_1 < \tau_2 < \cdots$ *be the successive times at which* $B_n = 1$*. Then* $X_j = x_{\tau_j}$ *satisfies* $X_{j+1} = 3X_j 2^{-K_{j+1}}$*, where* $K_{j+1} = \tau_{j+1} - \tau_j - 1$ *are independent Geometric*$(\frac{1}{2})$*, and* (X_j) *is a martingale:* $\mathbb{E}[3 \cdot 2^{-K}] = 3 \sum_{k \geq 1} 4^{-k} = 1$*.*

Proof. (i) The stationary vector π of P satisfies $\pi_1 = \frac{1}{2}\pi_0$ and $\pi_0 + \pi_1 = 1$, so $\pi_1 = \frac{1}{3}$. The chain is irreducible and aperiodic (state 0 has a loop), so the ergodic theorem for finite Markov chains gives the almost sure frequency.

(ii) On $\{B_n = 1\}$ we have $x_{n+1} = 3x_n$ and $B_{n+1} = 0$, so $M_{n+1} = 3x_n = M_n$. On $\{B_n = 0\}$ we have $x_{n+1} = x_n/2$ and B_{n+1} is a fair coin, so $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = \frac{x_n}{2} \cdot \frac{1}{2}(1+3) = x_n = M_n$.

(iii) On $\{B_n = 1\}$ the increment $\log M_{n+1} - \log M_n$ is 0. On $\{B_n = 0\}$ it is $-\log 2 + B_{n+1} \log 3$, with conditional mean $-\log 2 + \frac{1}{2} \log 3 = \delta^*$.

(iv) $\log x_n = \log x_0 + a_n \log 3 - (n - a_n) \log 2$, where a_n is the number of odd steps before time n, and $a_n/n \to \frac{1}{3}$ by (i).

(v) Between two successive ones the chain makes a forced step to 0 and then stays at 0 for a Geometric number of further steps with parameter $\frac{1}{2}$ on $\{0, 1, \ldots\}$, so the number K of zeros between ones is Geometric$(\frac{1}{2})$ on $\{1, 2, \ldots\}$, and successive gaps are independent by the Markov property. The value of x is multiplied by 3 at the odd step and by $\frac{1}{2}$ at each of the K even steps. □

The martingale $M_n = 3^{B_n} x_n$ has a simple meaning. When x_n is odd, the next step will triple it with certainty, and M_n books that tripling in advance. With the tripling booked, the process is fair. Deleting the steps on which $\log M_n$ does not move, which are exactly the tripling steps, leaves the Bernoulli walk of Proposition 3.4; the drift per step of T_0 is $\frac{2}{3}\delta^*$ because two thirds of the steps of T_0 are steps of T in disguise and one third are bookkeeping. The three drifts in play, $\frac{1}{3} \log \frac{3}{4}$ per step of T_0, $\frac{1}{2} \log \frac{3}{4}$ per step of T and $\log \frac{3}{4}$ per Syracuse stride, are one quantity measured against the three clocks of Section 3.1.

3.4 Log Lyapunov parameters and Wald's identity

The function $V(x) = \log x$ is the natural Lyapunov candidate, and the model supplies its first two moments.

Proposition 3.8 *(Log Lyapunov parameters). In the Bernoulli model for T, the log increment has mean $\delta^* = \log(\sqrt{3}/2) \approx -0.143841$ and variance*

$$\sigma^2 = \tfrac{1}{2}(\log 2)^2 + \tfrac{1}{2}\big(\log \tfrac{3}{2}\big)^2 - (\delta^*)^2 = \tfrac{1}{4}(\log 3)^2 \approx 0.301737.$$

The ratio $C = \sigma^2/(\delta^*)^2 = (\log 3/\log \frac{4}{3})^2 \approx 14.58$ is the asymptotic index of dispersion of the passage time defined below (see Section 3.5), and the expected passage time grows with slope

$$\gamma_0 = \frac{1}{|\delta^*|} = \frac{2}{\log(4/3)} \approx 6.952.$$

Precisely, if $\tau = \inf\{n \geq 0 : x_n \leq 1\}$ and $x_0 > 1$, then

$$\gamma_0 \log x_0 \;\leq\; \mathbb{E}[\tau] \;<\; \gamma_0\big(\log x_0 + \log 2\big). \tag{3.1}$$

Proof. A random variable taking two values with probability $\frac{1}{2}$ each has variance one quarter of the squared distance between them, and the two values $\log \frac{3}{2}$ and $-\log 2$ differ by $\log 3$. The expression for C follows from $|\delta^*| = \frac{1}{2}\log\frac{4}{3}$. For (3.1), the process $N_n = \log x_n - \log x_0 - n\delta^*$ is a martingale with increments bounded by $\log 3$, so optional stopping at the bounded time $n \wedge \tau$ gives

$$\mathbb{E}[\log x_{n\wedge\tau}] = \log x_0 + \delta^*\,\mathbb{E}[n \wedge \tau].$$

Before time τ we have $\log x > 0$, and x falls at most by a factor 2 in one step, so $\log x_{n\wedge\tau} > -\log 2$. Hence $|\delta^*|\,\mathbb{E}[n\wedge\tau] < \log x_0 + \log 2$, and letting $n \to \infty$ by monotone convergence, $\mathbb{E}[\tau] \leq \gamma_0(\log x_0 + \log 2)$. In particular $\tau < \infty$ almost surely, and since $\log x_{n\wedge\tau}$ is bounded above by $\log x_0 + n \wedge \tau \cdot \log\frac{3}{2}$, dominated convergence applies to the displayed identity and gives Wald's identity $\mathbb{E}[\log x_\tau] = \log x_0 + \delta^*\mathbb{E}[\tau]$. As $\log x_\tau \leq 0$, the lower bound follows, and as $\log x_\tau > -\log 2$ almost surely, the upper bound is strict. □

The proof is an instance of Wald's identity [103]: for a random walk with integrable increments of mean μ and a stopping time with finite mean, $\mathbb{E}[S_\tau] = \mu\,\mathbb{E}[\tau]$. The two sided bound (3.1) pins the slope down to γ_0 with an additive error of at most $\gamma_0 \log 2 \approx 4.8$ steps, coming from the overshoot of the barrier. A finer analysis by renewal theory [23, 28] shows that $\mathbb{E}[\tau] - \gamma_0 \log x_0$ converges to a constant determined by the stationary overshoot distribution. For the slope we need nothing more, but the constant matters when the model is compared with orbits in Section 3.5: the exact law of τ, computed by dynamic programming over the paths of the walk (script `ch3_model_means.py`), puts the excess at about 2.1 steps at every size from $x_0 \approx 10^3$ on, a mean overshoot of about 0.3 below the barrier.

Comparison with the orbits. The total stopping time of T over $2 \leq n \leq 10^4$ has least squares slope 6.920 against $\log n$ (through the origin; script `ch3_stats.py`), which differs from $\gamma_0 = 6.952$ by less than half a percent. The model also predicts

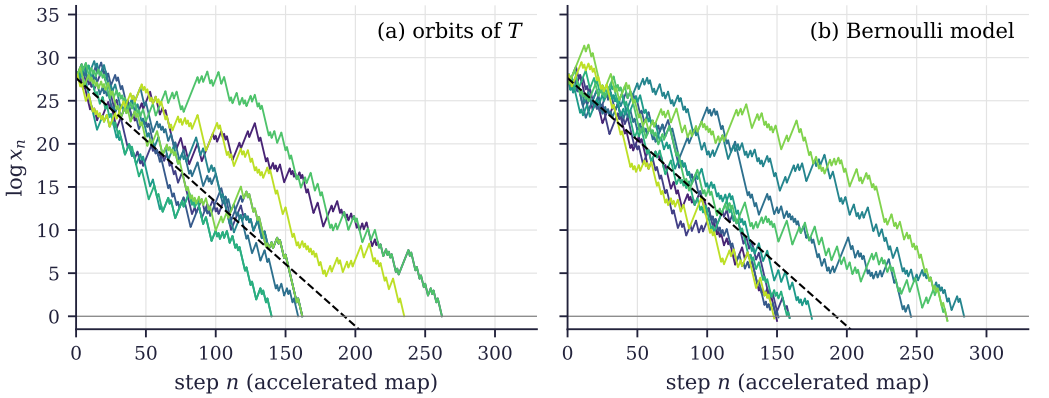


FIGURE 3.1. Left: $\log x_n$ along ten orbits of the accelerated map started at odd integers near 10^{12}. Right: ten paths of the Bernoulli model from the same height. The dashed line is the mean path $\log x_0 + n\delta^*$.

the slope for the standard map: a stride costs on average 3 steps of T_0 and lowers $\log x$ by $\log \frac{4}{3}$, so the slope is $3/\log \frac{4}{3} \approx 10.428$, and the empirical slope over the same range is 10.359. Figure 3.1 compares ten orbits of T started near 10^{12} with ten paths of the Bernoulli model from the same height. The two panels cannot be told apart by eye. The real orbits reached 1 after between 140 and 262 steps, with mean 184.6; the model paths took between 148 and 284 steps, with mean 201.6; the passage time of the model from 10^{12} has mean 194.1, the leading term $\gamma_0 \log 10^{12} \approx 192.1$ plus the overshoot, and standard deviation 53.2, against the leading term 52.9. (Several of the real orbits share their final segments, since orbits of nearby integers merge, which is why some lengths repeat.)

REMARK 3.9 *(The constant* 14.58 *elsewhere in the book).* The number $C = (\log 3/\log \frac{4}{3})^2$ appears again in Chapter 7, as the condition number at which the Switching MM momentum coefficient $\beta = \log(3/2)/\log 2$ is the optimal Nesterov coefficient. This is the same algebra and not a second piece of evidence: with $\beta = \log_2 3 - 1$ one has $(1+\beta)/(1-\beta) = \log 3/\log \frac{4}{3}$ exactly.

The tail of the stopping time. The first time ν at which a model path falls below its starting height, the model counterpart of the stopping time $\sigma(n)$ of the map, satisfies $\mathbb{P}(\nu > n) \le \mathbb{P}(S_n \ge 0) \le \mathbb{E}[e^{\theta S_n}] = M(\theta)^n$ for every $\theta > 0$, by Markov's inequality (this θ is not the θ of Theorem 2.26). The minimum of M is attained at $\theta = \log(\log 2/\log \frac{3}{2})/\log 3 \approx 0.4881$, where $M(\theta) \approx 0.9659$. Thus $\mathbb{P}(\nu > n) \le 0.9659^n$. This is the bound e^{-Dk} of Theorem 2.26, obtained there by a Chernoff argument for the binomial count of odd steps; the two computations are the same optimization in different coordinates, since Proposition 3.2 identifies the binomial count with the number of factors $\frac{3}{2}$ in the model.

TABLE 3.1. Total stopping time $\sigma_\infty(n)$ of the accelerated map for 4000 starting values n drawn independently and uniformly from each range, against the leading terms of the Bernoulli model, $\gamma_0 \log n$ for the mean (a lower bound for the model mean, by (3.1)) and $101.4 \log n$ for the variance, averaged over the sample.

Range of n	Mean	$\gamma_0 \log n$	Variance	$101.4 \log n$	Ratio
$[10^3, 2 \cdot 10^3)$	49.66	50.73	685.3	739.8	0.926
$[10^4, 2 \cdot 10^4)$	65.36	66.69	871.2	972.6	0.896
$[10^5, 2 \cdot 10^5)$	81.18	82.76	1101.8	1206.9	0.913
$[10^6, 2 \cdot 10^6)$	96.90	98.74	1292.5	1440.0	0.898
$[10^7, 2 \cdot 10^7)$	114.33	114.71	1538.3	1672.9	0.920

3.5 THE CENTRAL LIMIT THEOREM FOR TOTAL STOPPING TIMES

Wald's identity gives the mean of τ; the central limit theorem for first passage times gives its fluctuations. For a random walk whose increments have mean $\mu < 0$ and variance σ^2, the first passage time $\tau_u = \inf\{n : S_n \le -u\}$ satisfies, as $u \to \infty$,

$$\frac{\tau_u - u/|\mu|}{\sqrt{\sigma^2 u/|\mu|^3}} \implies N(0,1). \tag{3.2}$$

This is a classical consequence of Anscombe's theorem and the central limit theorem for the walk; see Gut [28] for a complete treatment. A limit in distribution does not by itself give the moments: that $\mathbb{E}[\tau_u] \sim u/|\mu|$ and $\mathrm{Var}(\tau_u) \sim \sigma^2 u/|\mu|^3$ is the classical first passage asymptotics of Heyde [39] and Siegmund [91] (see [28, Theorem III.9.1]). The heuristic is quickly told. At time $t = u/|\mu|$ the walk sits at $-u$ plus a fluctuation of order $\sigma\sqrt{t}$, and closing that gap at speed $|\mu|$ takes an extra time of order $\sigma\sqrt{t}/|\mu|$, whose variance is $\sigma^2 t/\mu^2 = \sigma^2 u/|\mu|^3$.

With $u = \log x_0$, $\mu = \delta^*$ and $\sigma^2 = \frac{1}{4}(\log 3)^2$, the passage time τ_u is the time τ of Proposition 3.8, and the model predicts that it is approximately normal with mean $\gamma_0 \log x_0$ and variance $\sigma^2 \log x_0/|\delta^*|^3 \approx 101.4 \log x_0$. For this model the normal limit with exactly these constants was proved by Borovkov and Pfeifer [13, Theorem 5]. Both constants are fixed by the drift and the variance of a single step, so the prediction has no free parameter. The ratio of the variance to the mean is $C = \sigma^2/(\delta^*)^2 \approx 14.58$, which explains the name index of dispersion in Proposition 3.8. The counterpart of τ for the map is the total stopping time $\sigma_\infty(n)$ of T, the number of steps of T that take n to 1 (in Chapter 1 the same symbol counts steps of T_0). It is a deterministic integer, and what can be compared with the law of τ is its spread over many starting values n.

Table 3.1 tests both predictions on samples of 4000 starting values drawn independently and uniformly, of either parity, from five ranges (consecutive integers would not give independent samples, since their orbits share their tails). The means fall short of $\gamma_0 \log n$ by 0.4 to 1.8 steps, whereas by (3.1) the model

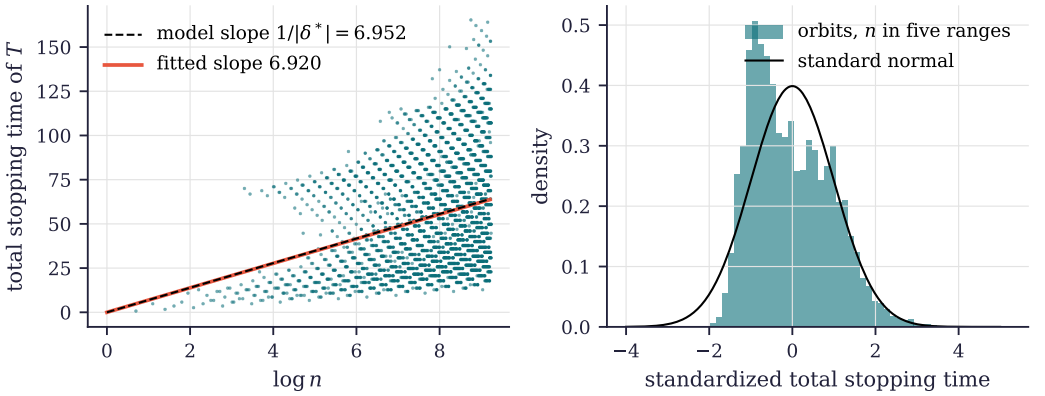


FIGURE 3.2. Left: total stopping time of T against $\log n$ for $2 \le n \le 10^4$, with the model slope γ_0 (dashed) and the fitted slope (solid). Right: standardized total stopping times for 20000 random n from the five ranges of Table 3.1, against the standard normal density.

mean lies above $\gamma_0 \log n$, and the overshoot of the barrier puts it 2.07 steps higher in every range (script `ch3_model_means.py`). So the real orbits are shorter than model paths from the same height, by 2.5 to 3.9 steps in these samples, whose means have standard errors of 0.4 to 0.6, and by 3.5 steps in the exact census at $k = 29$ of Section 3.6; the overshoot does not explain the shortfall but adds to it. It is a boundary effect at the end of the orbit: a real orbit stops on hitting 1 exactly, with no overshoot, and small integers such as the powers of two descend to 1 faster than a random walk from the same height would. (The $+1$ acts in the opposite direction, lengthening orbits by about one step on average, almost all of it in the final descent; see Section 3.6.) The variances are about 0.9 times the leading term $101.4 \log n$ in every range; the exact variance of the model exceeds its leading term by 2 to 4 percent at these sizes, so the variances of the map are 0.87 to 0.90 times those of the model, with no clear trend at this sample size; Section 3.6 locates the deficit exactly. Figure 3.2 shows the regression of total stopping time on $\log n$ and the pooled standardized residuals $(\sigma_\infty(n) - \gamma_0 \log n)/\sqrt{101.4 \log n}$, whose sample variance is 0.909 and whose skewness is 0.57; first passage times of a walk with drift are right skewed at finite u, and the skew decays like $u^{-1/2}$.

The conclusion is that the model gets the dispersion of total stopping times right to about ten percent, erring slightly on the side of too much variability. There is no excess dispersion to explain. This matters because it is easy to construct a wrong explanation for a discrepancy that is not there; we return to this in Remark 3.15.

3.6 Where the missing variance lives

The ten percent deficit in Table 3.1 is small, but it is systematic, and a complete census locates its source. For every $n < 2^{30}$ we computed the total stopping time $\sigma_\infty(n)$ of T and the number $a(n)$ of odd steps on the way to 1 (script `ch3_o4_decomp.py`, which takes about two minutes), and analysed each dyadic block $[2^k, 2^{k+1})$ exactly, not by sampling. The analysis rests on an identity that holds orbit by orbit.

Proposition 3.10 *(Pathwise decomposition). For every $n \geq 2$ whose orbit reaches 1 (as every $n < 2^{30}$ of the census does), with $u = \log n$ and $|\delta^*| = \log(2/\sqrt{3})$,*

$$|\delta^*|\,\sigma_\infty(n) = u + R(n) + Z(n),$$
$$R = \sigma_\infty \log 2 - a \log 3 - u, \qquad Z = \tfrac{1}{2}\log 3\,(2a - \sigma_\infty).$$

Here Z is the centred parity sum, a sum of σ_∞ terms $\pm\frac{1}{2}\log 3$, and $R = \sum_{\text{odd steps}} \log(1 + 1/(3x_i)) \geq 0$ is the total contribution of the $+1$ terms. Splitting $Z = Z_{\mathrm{h}} + Z_{\mathrm{t}}$ into the sums over the first k steps and over the rest, for n uniform in $[2^k, 2^{k+1})$ the head Z_{h} is exactly a sum of k independent fair signs, so that $\operatorname{Var} Z_{\mathrm{h}} = \sigma^2 k$ *with* $\sigma^2 = \frac{1}{4}(\log 3)^2$.

Proof. The identity is algebra: $u + R + Z = \sigma_\infty \log 2 - a \log 3 + a \log 3 - \frac{1}{2}\sigma_\infty \log 3 = \sigma_\infty(\log 2 - \frac{1}{2}\log 3)$. Taking logarithms along the orbit, $0 = \log 1 = u + a \log \frac{3}{2} - (\sigma_\infty - a)\log 2 + \sum_{\mathrm{odd}} \log(1 + 1/(3x_i))$, which identifies R with the sum of the $+1$ corrections. For the head, the block $[2^k, 2^{k+1})$ contains each residue class modulo 2^k exactly once, and by Theorem 2.8 the first k parities are a bijective function of the class; so they are uniformly distributed over $\{0,1\}^k$, that is, independent fair bits. □

In the Bernoulli model Z would be a martingale stopped at τ, with $\mathbb{E}Z^2 = \sigma^2 \mathbb{E}\tau$ by Wald's second identity, and R would be a bounded overshoot; the model variance $\sigma^2 \log n / |\delta^*|^3$ is the leading term of $\operatorname{Var}(Z)/|\delta^*|^2$. Table 3.2 compares each piece with its model value.

Table 3.2. Exact census of the dyadic blocks $[2^k, 2^{k+1})$: mean and variance of σ_∞, the leading term $\sigma^2 \mathbb{E}\log n/|\delta^*|^3$ of the model variance, the head ratio $\operatorname{Var} Z_{\mathrm{h}}/(\sigma^2 k)$, the tail ratio $\operatorname{Var} Z_{\mathrm{t}}/(\sigma^2 \mathbb{E}(\sigma_\infty - k))$, and the additive deficit $D_{\mathrm{v}} = \sigma^2 \mathbb{E}(\sigma_\infty - k) - \operatorname{Var} Z_{\mathrm{t}}$.

k	$\mathbb{E}\sigma_\infty$	$\operatorname{Var}\sigma_\infty$	model	ratio	head	tail	D_{v}
10	50.1	709	742	0.956	1.0000	0.939	0.74
15	73.6	977	1093	0.894	1.0000	0.909	1.61
20	97.7	1308	1445	0.905	1.0000	0.902	2.30
25	121.7	1656	1796	0.922	1.0000	0.914	2.51
29	141.0	1938	2077	0.933	1.0000	0.924	2.56

Three conclusions follow. First, the head is exact, as the proposition requires, and the plus one term R contributes almost nothing to the variance: its variance

is 0.003 in every block from $k = 10$ on, and its covariance with Z is about 0.05. (Its mean, about 0.15 in every block and so contributed almost entirely by the final descent, lengthens orbits by about one step.) The whole deficit lives in the tail parity sum, the part of the orbit beyond the Terras horizon.

Second, the deficit is *additive*, not multiplicative. The quantity D_{v} grows from 0.74 at $k = 10$ and levels off near 2.6 (in units of $\log^2$, about 124 squared steps of T), while the model variance grows linearly in k. So the ratio of observed to model variance first falls, to a minimum near 0.89 around $k = 15$, and then rises again, to 0.933 at $k = 29$. If D_{v} stays bounded, as the census suggests, the ratio tends to 1 like $1 - c/\log n$, and the model is asymptotically exact to second order. At $n \approx 10^{12}$ a saturated deficit of 2.6 would give a ratio of about 0.95.

Third, the deficit sits at the end of the orbit. Splitting each orbit of a random sample at the first time it falls below $Y = 2^{14}$ (script `ch3_o4_end.py`), the final segment from below 2^{14} down to 1 contributes a deficit of 1.94 to 1.98 for $k = 20$, 24 and 28, independent of k, and is uncorrelated with what came before; the middle segment, from the Terras horizon down to 2^{14}, contributes 0.40, 0.62 and 0.64, with its variance ratio rising towards 1 as k grows (0.914, 0.930, 0.952). The final segment is a finite deterministic system: every orbit enters $[1, 2^{14})$ and then follows one of finitely many fixed paths to 1. Those paths are shorter and less variable than a random walk descending from the same heights, because the small integers are funnelled into a few common tails, through 20, 10, 5 and the powers of two. The model treats the last sixty or so steps as fresh coin flips, and they are not.

So the census indicates that the deficit is a bounded boundary effect at the bottom of the orbit, plus a smaller contribution from the passage to the bottom, and that the Bernoulli model is asymptotically right about the variance; a proof that the deficit stays bounded is the first part of open problem O4 of Chapter 15. The third cumulant is less well explained: the observed skewness of σ_∞ is 0.62 at $k = 20$ and 0.67 at $k = 29$, while the inverse Gaussian law of a drifted Brownian first passage, with the model's drift and variance, predicts $3\sigma/\sqrt{|\delta^*|\,\mathbb{E}\log n}$, which is 1.15 and 0.96. The real total stopping times are less skewed than the model's passage times, and we leave the explanation open.

3.7 Excursions: the Ville and Lundberg bound

How high can an orbit climb before it descends? The orbit of 27 under T_0 climbs to 9232, a factor of about 342 (under T its peak is 4616), and such excursions are the most visible feature of the problem. In the model the answer is controlled exactly by the martingale property, through a maximal inequality that goes back to Ville [102].

Proposition 3.11 *(Ville bound). In the Bernoulli model for T, for every $x_0 > 0$ and*

$K \geq 1$,

$$\mathbb{P}\Big(\sup_{n\geq 0} x_n \geq K x_0\Big) \;\leq\; \frac{1}{K}.$$

Equivalently, in terms of the log walk, $\mathbb{P}(\sup_n S_n \geq u) \leq e^{-u}$ for every $u \geq 0$. The same bound holds in the stride model for S, and in the renewal model for T_0 when $B_0 = 0$; when $B_0 = 1$ the bound becomes $3/K$.

Proof. Let $c = Kx_0$ and $\eta = \inf\{n : x_n \geq c\}$. The stopped process $x_{n\wedge\eta}$ is a nonnegative martingale, so for each n

$$x_0 = \mathbb{E}[x_{n\wedge\eta}] \geq \mathbb{E}[x_\eta; \eta \leq n] \geq c\,\mathbb{P}(\eta \leq n).$$

Letting $n \to \infty$ gives $\mathbb{P}(\eta < \infty) \leq x_0/c = 1/K$. For the stride model apply the same argument to the martingale X_j; the supremum over strides is the supremum over odd values, which is the relevant one. For the renewal model apply it to $M_n = 3^{B_n} x_n \geq x_n$, whose initial value is $3^{B_0} x_0$. □

The proposition is the Lundberg inequality of risk theory in the special case $\theta^* = 1$. For a general walk with negative drift the bound reads $\mathbb{P}(\sup_n S_n \geq u) \leq e^{-\theta^* u}$, and the Cramér and Lundberg approximation sharpens it to $\mathbb{P}(\sup_n S_n \geq u) \sim C_{\mathrm{L}} e^{-\theta^* u}$ for a constant $C_{\mathrm{L}} < 1$ that depends on the overshoot over the level [4]. For the Collatz model this says that $K \cdot \mathbb{P}(\sup x_n \geq K x_0)$ should settle to a constant a little below one.

Figure 3.3 tests the bound. For each n with $2 \leq n \leq 10^6$ we computed the maximum of the orbit of n under T and recorded the fraction of n whose maximum is at least Kn, for $K = 2, 4, \ldots, 4096$. The product K times this fraction lies between 0.76 and 0.85 for $K \leq 1024$, comfortably below the bound. A simulation of 10^6 paths of the Bernoulli model gives values between 0.82 and 0.85 over the same range, an estimate of the Cramér and Lundberg constant. The orbits sit a little below the model, in line with the slightly smaller dispersion found in Section 3.5. The drop at $K = 2048$ and $K = 4096$ is an edge effect: at these factors only a few hundred of the million starting values qualify, and the integers below 10^6 are too small for the orbit to have room to make such excursions in the proportion the model predicts.

The exponent 2. The Ville bound explains a prediction of Lagarias and Weiss [59] about the maximal excursion $t(n) = \max_k T^{(k)}(n)$. Treat the orbits of different n as independent model paths. Taking $K = n^{1+\varepsilon}$ gives $\mathbb{P}(t(n) \geq n^{2+\varepsilon}) \leq n^{-1-\varepsilon}$, which is summable, so by the Borel and Cantelli lemma only finitely many n would have $t(n) \geq n^{2+\varepsilon}$. In the other direction the Cramér and Lundberg asymptotics give $\mathbb{P}(t(n) \geq n^{2-\varepsilon}) \approx C_{\mathrm{L}} n^{-1+\varepsilon}$, which is not summable, and independence would give infinitely many such n. The heuristic conclusion is $\limsup_n \log t(n)/\log n = 2$. Over $1000 < n \leq 10^6$ the largest value of $\log t(n)/\log n$ for the accelerated map is 1.9095, attained at $n = 159487$. The heuristic is of course only as good as the

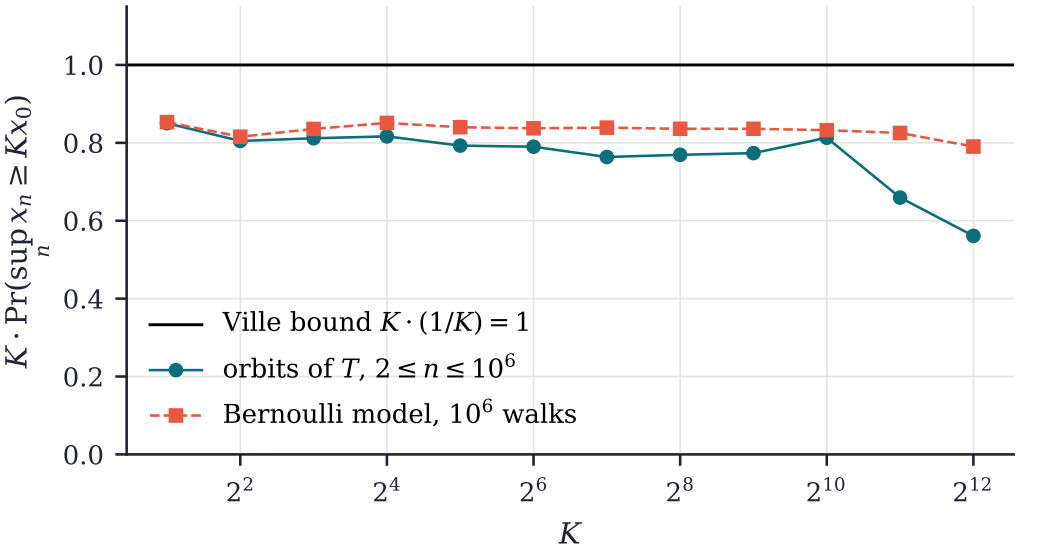


FIGURE 3.3. The quantity $K \cdot \mathbb{P}(\sup_n x_n \geq K x_0)$ for orbits of T with $2 \leq n \leq 10^6$ (circles) and for 10^6 paths of the Bernoulli model (squares). For the model, Ville's inequality (Proposition 3.11) says that it never exceeds 1.

independence assumption, which is false for neighbouring integers whose orbits merge, and there is no proof that $t(n) \leq n^C$ for any constant C.

3.8 Parity autocorrelations

Under the accelerated map the model parities are independent. Under the standard map they are not, and their dependence can be computed exactly.

PROPOSITION 3.12 *(Parity autocorrelation function). In the renewal model for T_0, started from its stationary law, the model parity chain (B_n) is stationary with*

$$\mathbb{P}(B_n = 1) = \frac{1}{1 + \mathbb{E}[K]} = \frac{1}{3}, \qquad \operatorname{Var}(B_n) = \frac{2}{9},$$

and its autocorrelation function is

$$\rho(h) = \operatorname{Corr}(B_n, B_{n+h}) = \left(-\frac{1}{2}\right)^h, \qquad h \geq 0. \tag{3.3}$$

Proof. We give the renewal proof, which explains where the formula comes from; a second proof is in Remark 3.13. By Proposition 3.7(v), the gaps between successive ones of (B_n) are independent copies of $G = 1 + K$, with

$$\mathbb{P}(G = m) = \mathbb{P}(K = m - 1) = 2^{-(m-1)}, \qquad m \geq 2.$$

Hence $\mathbb{E}[G] = 1 + \mathbb{E}[K] = 3$, and by the renewal theorem the stationary probability of a one is $1/\mathbb{E}[G] = \frac{1}{3}$, so $\mathrm{Var}(B_n) = \frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9}$. The generating function of the gap law is

$$F(z) = \sum_{m \geq 2} 2^{-(m-1)} z^m = \frac{z^2}{2 - z}, \qquad |z| < 2.$$

Let $u(h) = \mathbb{P}(B_{n+h} = 1 \mid B_n = 1)$ be the renewal sequence, with $u(0) = 1$. Decomposing according to the time of the first renewal after time n gives the renewal equation $u(h) = \sum_{m=1}^{h} \mathbb{P}(G = m) u(h - m)$ for $h \geq 1$, which in generating functions reads $U(z) = 1 + F(z)U(z)$, where $U(z) = \sum_{h \geq 0} u(h) z^h$. Therefore

$$U(z) = \frac{1}{1 - F(z)} = \frac{2 - z}{2 - z - z^2} = \frac{2 - z}{(1 - z)(2 + z)}.$$

Partial fractions give

$$U(z) = \frac{1/3}{1 - z} + \frac{4/3}{2 + z} = \frac{1/3}{1 - z} + \frac{2/3}{1 + z/2},$$

and expanding both terms as geometric series,

$$u(h) = \frac{1}{3} + \frac{2}{3}\Big(-\frac{1}{2}\Big)^h, \qquad h \geq 0. \tag{3.4}$$

The first values are $u(0) = 1$, $u(1) = 0$, $u(2) = \frac{1}{2}$, $u(3) = \frac{1}{4}$, $u(4) = \frac{3}{8}$. Finally, with $p = \frac{1}{3}$,

$$\begin{aligned} \rho(h) &= \frac{\mathbb{P}(B_n = 1, B_{n+h} = 1) - p^2}{p(1 - p)} = \frac{p\, u(h) - p^2}{p(1 - p)} \\ &= \frac{u(h) - p}{1 - p} = \frac{3}{2}\Big(u(h) - \frac{1}{3}\Big) = \Big(-\frac{1}{2}\Big)^h. \end{aligned}$$

□

The first three values of the renewal sequence carry the arithmetic of a single stride. $u(1) = 0$ is Lemma 2.3: there are never two consecutive odd steps. $u(2) = \frac{1}{2}$ and $u(3) = \frac{1}{4}$ are the probabilities that the halving exponent equals 1 and 2, that is $\mathbb{P}(x \equiv 3 \pmod 4)$ and $\mathbb{P}(x \equiv 1 \pmod 8)$ for odd x (Proposition 2.15). From $h = 4$ on the values involve the joint law of successive halving exponents, beginning with $u(4) = \mathbb{P}(K = 3) + \mathbb{P}(K = 1)^2 = \frac{3}{8}$; that law is exact for random residues over the range of Proposition 3.2, and beyond it the independence of successive exponents is an assumption of the model.

Remark 3.13 *(A second proof).* Since (B_n) is a Markov chain with transition matrix P, the conditional law of B_{n+h} given B_n is given by P^h. Writing $P = \Pi + (-\frac{1}{2})E$ with Π the rank one projection onto the stationary vector and E the complementary spectral projection, $P^h = \Pi + (-\frac{1}{2})^h E$, and the autocorrelation of any function of a reversible two state chain decays like the hth power of

the second eigenvalue. The renewal proof has the advantage of exhibiting the generating function $U(z)$, which is what one needs for the variance computation below.

Corollary 3.14 *(Long run variance of the log path). In the renewal model for T_0, the autocorrelation factor is*

$$\mathcal{A} := 1 + 2\sum_{h\geq 1} \rho(h) = 1 + 2\cdot \frac{-1/2}{1+1/2} = \frac{1}{3},$$

and the variance of the log path satisfies

$$\frac{\operatorname{Var}(\log x_n)}{n} \;\longrightarrow\; \mathcal{A}\cdot \frac{2}{9}(\log 6)^2 = \frac{2}{27}(\log 6)^2 \approx 0.2378.$$

Proof. The log increment at step n is $B_n \log 3 - (1-B_n)\log 2 = B_n \log 6 - \log 2$, a linear function of B_n with slope $\log 6$. For a stationary sequence with summable autocovariances we have $\operatorname{Var}(\sum_{i<n} B_i)/n \to \operatorname{Var}(B)\,\mathcal{A}$. The same limit holds from any initial state: the parities form a two state Markov chain whose second eigenvalue is $-\frac{1}{2}$, so the law of B_i approaches the stationary law geometrically fast, which changes $\operatorname{Var}(\sum_{i<n} B_i)$ by $O(1)$. Here $\operatorname{Var}(B) = \frac{2}{9}$ and $\sum_{h\geq 1}(-\frac{1}{2})^h = -\frac{1}{3}$, so $\mathcal{A} = \frac{1}{3}$ and the limit is $\frac{1}{3}\cdot\frac{2}{9}(\log 6)^2$. As a check, the variance asymptotic of renewal theory [28, Theorem III.9.1] says that the number a_n of renewals before time n has $\operatorname{Var}(a_n)/n \to \operatorname{Var}(G)/\mathbb{E}[G]^3 = 2/27$, since $\operatorname{Var}(G) = \operatorname{Var}(K) = 2$ and $\mathbb{E}[G] = 3$; and $\log x_n = \log x_0 + a_n \log 6 - n\log 2$. The two routes agree. □

Corollary 3.14 has a consequence that runs against a natural intuition.

Remark 3.15 *(A tempting but false argument).* One might argue as follows: "the parities of T_0 are correlated, correlated increments make a random walk more variable than an independent one, and so the parity correlations should inflate the dispersion of total stopping times well beyond what an independent model predicts." Each clause sounds reasonable, and the conclusion is wrong in sign. The correlations are negative at odd lags and alternate, $\rho(h) = (-\frac{1}{2})^h$, and their sum is $-\frac{1}{3}$. The long run variance factor is $\mathcal{A} = \frac{1}{3} < 1$, so the correlations *reduce* the variance of the log path by a factor of three relative to an independent sequence with the same marginal law. Negative lag one correlation is exactly what one expects from a forced halving after every tripling: an up step is always cancelled in part by the next step. Two further facts close the question. First, measured on the clock of T, where the model parities are independent, the reduction disappears and what remains is the variance σ^2 of Proposition 3.8; the factor $\frac{1}{3}$ and the deletion of forced steps are two descriptions of the same thing. So $\mathcal{A}$ concerns only the log path of T_0 counted in its own steps, and parity correlations play no part in the dispersion law of Section 3.5, which is stated on the clock of T. Second, as Table 3.1 shows, there is no excess dispersion of total stopping times to

explain: the observed variance is about 0.9 of the model value. Any mechanism that predicts an inflation of the variance by a large factor, from parity correlations or from anything else, is contradicted by the data.

3.8.1 The empirical check

Figure 3.4 compares (3.3) with two empirical autocorrelation functions for the parity sequence b_j of T_0, both estimated with the mean and variance pooled over all recorded steps: if $\hat{p}$ is the fraction of odd steps, the estimate at lag h is $(\overline{b_j b_{j+h}} - \hat{p}^2)/(\hat{p}(1-\hat{p}))$, where the bar is the average over all recorded pairs of steps h apart. The first is a scan over small starting values: orbits started at 27 and then restarted at a random odd number below 10^4 each time they reach 1, for a total of 4×10^5 steps. These orbits are not truncated but followed all the way to 1. The fraction of odd steps is 0.3351 and the first six lags are

$$-0.5040, \quad +0.2482, \quad -0.1530, \quad +0.0822, \quad -0.0032, \quad -0.0139,$$

against the predicted values $(-\frac{1}{2})^h$, that is -0.5, $+0.25$, -0.125, $+0.0625$, -0.0313 and $+0.0156$. The first two lags agree (at lag one this is forced: no two odd steps are consecutive, so the estimate is $-\hat{p}/(1-\hat{p})$ whatever the orbits). From lag three on the agreement is visibly worse, and lag six even has the wrong sign. Lag three still reflects a single stride: the halving exponent 2, which has probability $u(3) = \frac{1}{4}$ in the model, occurs in only 23.3 percent of the strides of the scan. We regard this as a genuine feature of short orbits and not as sampling noise. Small starting values have few bits, so most of each orbit lies outside the range of Proposition 3.2, and the scan includes the final descent of every orbit through the powers of two and the cycle $4 \to 2 \to 1$, whose parity pattern is far from stationary.

The second check uses large starting values and truncated orbits. For 4000 random integers n between 2^{200} and 2^{208}, of either parity and with $n \bmod 2^{200}$ uniformly distributed, we recorded the parities of steps 50 to 299 of T_0, far above the trivial cycle. By Proposition 3.2 all but about one percent of these steps lie inside the range in which the parities of T are exactly independent and fair, so apart from that one percent and from sampling error the model is a theorem here. The fraction of odd steps is 0.3330, and the autocorrelations are -0.4993, $+0.2497$, -0.1252, $+0.0629$, -0.0320, $+0.0155$ for $h = 1, \ldots, 6$, in agreement with $(-\frac{1}{2})^h$ to the precision of the sample. The renewal formula is exact for an ensemble of starting values over the initial segment of their orbits; along the late part of short orbits it is a prediction, and the prediction is visibly imperfect from lag three on.

3.9 Absorption and the failure of uniform integrability

We now come to the two results that locate the obstruction precisely. The first says that the model settles its own version of the conjecture. The second says why the martingale of Proposition 3.4 cannot be used to settle it, even in the model.

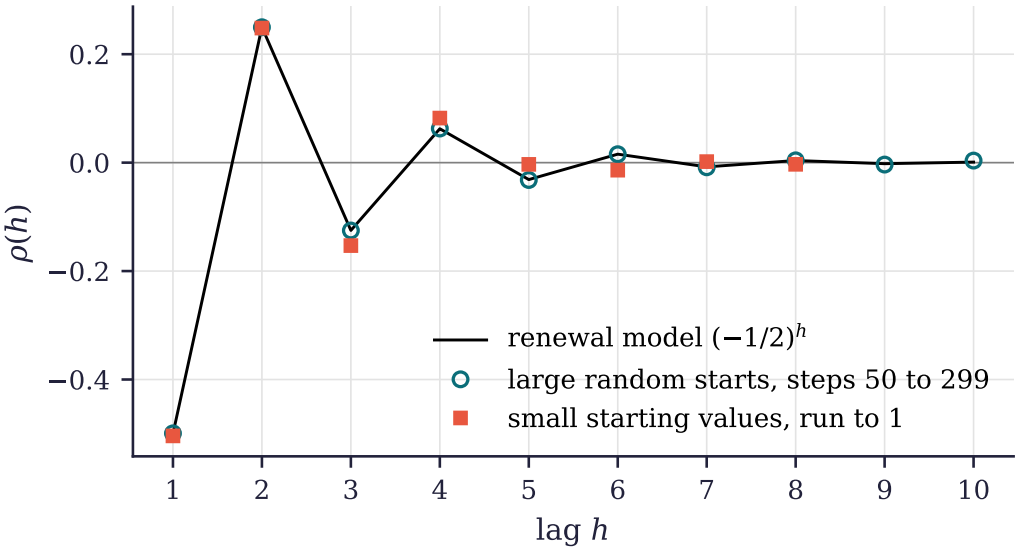


FIGURE 3.4. Autocorrelation of the parity sequence of the standard map, estimated with the pooled mean and variance. Solid line: the renewal model $(-\frac{1}{2})^h$. Circles: 4000 large random starting values of either parity, steps 50 to 299. Squares: small odd starting values, orbits run to 1.

PROPOSITION 3.16 *(The model absorbs almost surely). Let $\Delta_1, \Delta_2, \ldots$ be independent and identically distributed and integrable with mean $\mu < 0$, and put $\log x_n = \log x_0 + \sum_{i \le n} \Delta_i$ with $x_0 > 1$. Let $\tau = \inf\{n : x_n \le 1\}$. Then $\mathbb{E}[\tau] < \infty$; in particular*

$$\mathbb{P}\big(x_n > 1 \text{ for all } n\big) = 0.$$

If moreover $\Delta_i \ge -L$ almost surely, then $\mathbb{E}[\tau] < (\log x_0 + L)/|\mu|$.

Proof. Suppose first that $\Delta_i \ge -L$. The process $N_n = \log x_n - n\mu$ is a martingale, and optional stopping at the bounded time $n \wedge \tau$ gives

$$\mathbb{E}[\log x_{n\wedge\tau}] = \log x_0 + \mu\, \mathbb{E}[n \wedge \tau].$$

Before time τ the value $\log x$ is positive, and at time τ it is at least $\log x_{\tau-1} - L > -L$. Hence $\log x_{n\wedge\tau} > -L$, and $|\mu|\,\mathbb{E}[n \wedge \tau] < \log x_0 + L$. Monotone convergence as $n \to \infty$ gives $|\mu|\,\mathbb{E}[\tau] \le \log x_0 + L$. So τ is integrable, Wald's identity $\mathbb{E}[\log x_\tau] = \log x_0 + \mu\,\mathbb{E}[\tau]$ holds, and since $\log x_\tau > -L$ almost surely the bound is strict.

In general, put $\Delta_i' = \max(\Delta_i, -L)$. As $L \to \infty$, $\mathbb{E}[\Delta_i']$ decreases to μ by monotone convergence, so we may fix L with $\mu' = \mathbb{E}[\Delta_i'] < 0$. The walk built from the Δ_i' lies above the original walk, so its passage time τ' below 0 satisfies $\tau \le \tau'$, and by the first part $\mathbb{E}[\tau'] < (\log x_0 + L)/|\mu'|$. In either case τ is integrable and therefore finite almost surely. □

For the Bernoulli model $L = \log 2$ and the bound is the upper half of (3.1). For the stride model the downward steps $\log 3 - K \log 2$ are unbounded, and the truncation in the second half of the proof is needed.

The proposition is the strong law of large numbers in martingale dress, and the dress is the point. Its only inputs are a negative mean and independence of the increments from the past, which is what makes N_n a martingale. For the Collatz map the increments are deterministic functions of the current iterate, there is no filtration with respect to which $\log x_n - n\delta^*$ is a martingale, and Proposition 3.16 has no analogue. The Cesàro liminf conjecture of Chapter 11 asks for a deterministic substitute for the part of this argument that excludes escape to infinity; it says nothing about cycles.

Proposition 3.17 *(Failure of uniform integrability). In the Bernoulli model for T with $x_0 > 1$, let $\tau = \inf\{n : x_n \le 1\}$. Then $\tau < \infty$ almost surely, the stopped process $(x_{n\wedge\tau})_{n\ge 0}$ is a nonnegative martingale converging almost surely to $x_\tau \le 1$, and it is not uniformly integrable. More precisely, for every n,*

$$\mathbb{E}\big[x_n; \tau > n\big] \;\ge\; x_0 - 1, \qquad \textit{while} \qquad \mathbb{P}(\tau > n) \to 0.$$

Proof. Almost sure finiteness of τ is Proposition 3.16. The stopped process of a martingale is a martingale, so $\mathbb{E}[x_{n\wedge\tau}] = x_0$ for every n, and

$$x_0 = \mathbb{E}[x_\tau; \tau \le n] + \mathbb{E}[x_n; \tau > n] \le \mathbb{P}(\tau \le n) + \mathbb{E}[x_n; \tau > n],$$

which gives the first inequality. If the stopped process were uniformly integrable, then $x_{n\wedge\tau} \to x_\tau$ in L^1, and $x_0 = \lim \mathbb{E}[x_{n\wedge\tau}] = \mathbb{E}[x_\tau] \le 1$, a contradiction. Equivalently, the expectations $\mathbb{E}[x_n; \tau > n]$ stay above $x_0 - 1 > 0$ on events whose probabilities tend to zero, which is exactly the failure of uniform integrability. □

The two propositions are consistent and together they locate the obstruction. The martingale (x_n) has constant mean x_0 while almost every path is absorbed. The mass that keeps the mean up escapes along the rare paths that have not yet been absorbed, and these paths carry, at every time n, an expected value of at least $x_0 - 1$. Taking logarithms converts the martingale into a supermartingale with drift, which is why Proposition 3.16 works for $\log x$ and cannot be run for x. The power functions x^θ with $0 < \theta < 1$ sit in between: by Proposition 3.4(iii) they are supermartingales with geometric drift $M(\theta) < 1$, and they are the model counterpart of the power majorizers of Chapter 4.

Remark 3.18 *(What a certificate would have to be).* Suppose we tried to turn the model into a proof. In a probabilistic setting one would look for a uniformly integrable martingale Y_n that converges to the indicator of absorption; then $Y_0 = \mathbb{P}(\tau < \infty)$, and showing $Y_0 = 1$ is the statement to be proved. Such a martingale always exists, namely $Y_n = \mathbb{P}(\tau < \infty \mid \mathcal{F}_n)$, but writing it down requires knowing the absorption probability, which is the thing to be proved; in the model it is

identically 1 by Proposition 3.16, so the certificate is trivial and uninformative. In the model the only martingales of the form x_n^θ are those with $\theta = 0$ and $\theta = 1$, the roots of $M(\theta) = 1$ in Proposition 3.4, and neither carries information about termination.

For the deterministic map the analogue of the conditional probability is the function $h(x) = 1$ if the orbit of x reaches 1 and $h(x) = 0$ otherwise. It is invariant, $h(T(x)) = h(x)$, and the conjecture is the assertion $h \equiv 1$. A useful certificate would be a computable function $g \geq 0$ that decreases along orbits by a definite amount away from 1, and Chapter 4 shows that no such function that increases with x exists, whatever fixed number of strides is allowed for the decrease (Theorem 4.8 and Corollary 4.15). What the model offers instead is decrease in expectation, and expectation requires a measure. The only measures for which the Collatz parities are provably independent are the uniform measures on residue classes of Proposition 3.2, and they control a single orbit for only as many steps as the residue class has bits. A certificate along these lines would therefore have to show that the information an orbit generates as it proceeds is, in a quantitative sense, as good as fresh randomness. This is the content of the equidistribution questions of Chapter 11, and it is where every known approach stops.

Notes

Section 3.1 tells the history of the model; the survey of Kontorovich and Lagarias [47] is the best general reference for stochastic models of the $3x + 1$ and $5x + 1$ problems. The presentation of the martingale duality of Proposition 3.4, the martingale $3^{B_n} x_n$ for the standard map and the renewal computation of $\rho(h) = (-\frac{1}{2})^h$ and $\mathcal{A} = \frac{1}{3}$ is ours, but all of these are elementary, and the closed form is the autocorrelation of a two state Markov chain; the renewal structure of the model parity sequence is implicit in all treatments that use the Geometric law of the halving exponent. The dispersion law of Section 3.5 is the classical first passage asymptotics of Heyde [39] and Siegmund [91]; for this model Borovkov and Pfeifer [13, Theorem 5] proved the normal limit with these constants.

For martingales, optional stopping and the convergence theorems we have used, Williams [105] is a concise and elegant source. Wald's identity is from Wald [103]; the maximal inequality for nonnegative supermartingales appears in Ville's thesis [102], which also introduced the word martingale into probability. Renewal theory, the renewal central limit theorem and the ladder variables behind the Cramér and Lundberg approximation are in Feller [23]; Gut [28] treats first passage times and Anscombe's theorem; Asmussen and Albrecher [4] is the standard reference for Lundberg inequalities in risk theory, where the Wald root is called the adjustment coefficient.

All numerical results in this chapter were computed by the authors; the scripts are `ch3_stats.py`, `ch3_stopping.py`, `ch3_figures.py`,

`ch3_o4_decomp.py`, `ch3_o4_census.py` and `ch3_o4_end.py` in the code appendix; the exact moments of the model passage time come from `ch3_model_means.py`, which uses `ch3_o4_model.py`.

Exercises

Exercise 3.19. Repeat the computations of Proposition 3.4 for the $5x+1$ map, with accelerated form $x \mapsto (5x+1)/2$ for odd x. Show that the log drift under the fair coin model is $\frac{1}{2}\log\frac{5}{4} > 0$, that the factors no longer have mean one, and find the nonzero root of the moment generating function. What does the sign of the root say about excursions?

Exercise 3.20. In the stride model, let τ be the first stride at which $X_j \le 1$. Use Wald's identity to show that $\mathbb{E}[\tau] = \log X_0 / \log\frac{4}{3} + O(1)$, and reconcile this with the slope γ_0 of Proposition 3.8 by counting steps of T per stride.

Exercise 3.21. Let the halving exponent have a general law on $\{1,2,\dots\}$ with generating function $\phi(z) = \mathbb{E}[z^K]$. Show that the renewal sequence of the parity chain of T_0 has generating function $U(z) = 1/(1 - z\phi(z))$, and verify (3.4) for $\phi(z) = z/(2-z)$. For which laws of K is $\rho(2) > 0$?

Exercise 3.22. In the Bernoulli model, let η be the first time $x_n \ge K x_0$ and ζ the first time $x_n \le \varepsilon$, where $0 < \varepsilon < x_0$. Apply optional stopping to the bounded martingale $x_{n\wedge\eta\wedge\zeta}$, using the fact that a level is overshot by at most a factor $\frac{3}{2}$, to show that

$$\frac{2}{3K} \;\le\; \mathbb{P}\Big(\sup_n x_n \ge K x_0\Big) \;\le\; \frac{1}{K}.$$

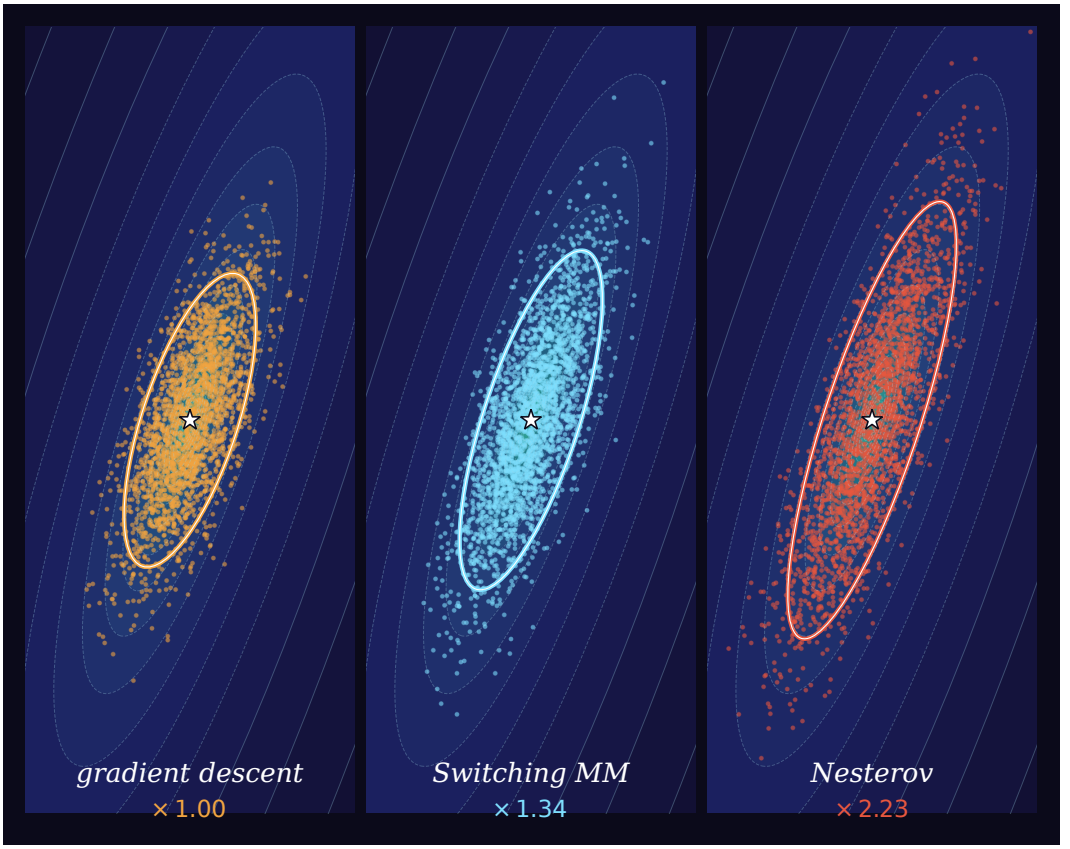
Thus the Ville bound is sharp up to a constant factor.

Exercise 3.23. In the renewal model for T_0, let τ be the first time $x_n \le 1$, and suppose that $x_0 > 1$ and $3^{B_0} x_0 > 2$, that is, $x_0 > 2$ or $B_0 = 1$. Show that the stopped martingale $M_{n\wedge\tau}$, with $M_n = 3^{B_n} x_n$, is not uniformly integrable, and that $\mathbb{E}[M_n; \tau > n] \ge 3^{B_0} x_0 - 2$ for every n while $\mathbb{P}(\tau > n) \to 0$. (Hint: the step into τ is a halving, so B_τ is a fair coin independent of the past.) What happens when $B_0 = 0$ and $1 < x_0 \le 2$?

Exercise 3.24. Write a program that estimates the Cramér and Lundberg constant $C_{\mathrm{L}} = \lim_{K\to\infty} K\,\mathbb{P}(\sup_n x_n \ge K x_0)$ for the Bernoulli model, and compare your estimate with the values near 0.84 reported in Section 3.7.

Part II

The Optimization Reading



Stationary noise clouds of gradient descent, Switching MM and Nesterov ($\beta = \log_2 3 - 1$, step $1/L$) on a quadratic with condition number 20 and noisy gradients; ellipses are two standard deviation contours of the exact stationary covariances, and the numbers give the stationary variance along the slow eigenvector relative to gradient descent (1.36 and 2.41 in the limit of infinite condition number).

Chapter 4

Lyapunov Functions and the Limits of Descent

The heuristic that has accompanied the $3x + 1$ problem since the work of Terras [99], Crandall [18] and Lagarias and Weiss [59] is, at bottom, a statement about a potential function. The quantity $\log x$ goes down on average along a Collatz orbit, and a quantity that goes down on average, and cannot go below zero, ought eventually to reach its floor. In dynamical systems and optimization this is the language of Lyapunov functions, and the purpose of this chapter is to take that language seriously: to say precisely what a Lyapunov certificate for the Collatz map would be, to prove what can be proved about the natural candidates, and to show, with complete proofs, why every candidate of the most natural kind must fail.

The chapter has a positive half and a negative half. The positive half is the stride descent theorem, Theorem 4.7, which describes exactly when a single Syracuse stride decreases $\log x$ and by how much, together with an exact analysis of two stride compositions. The negative half consists of three obstructions. No single stride Lyapunov function that is monotone in the size of x can certify descent (Theorem 4.8); no monotone certificate built from any fixed number J of strides can do so either, because ascending strides can be chained to arbitrary length (Proposition 4.14 and Corollary 4.15); and the natural majorization minimization construction for log yields only a tangent line bound, which holds for any next iterate and whose surrogate drifts upward in the model (Propositions 4.18 and 4.19). The last obstruction has a clean explanation in terms of the martingale duality of Chapter 3, and it points to the family of power majorizers, which repair the drift but not the pointwise failure.

Throughout, S denotes the Syracuse map of Definition 2.1, $S(x) = (3x+1)/2^k$ with $k = v_2(3x+1)$, acting on odd positive integers, and T_0 and T denote the standard and accelerated maps.

4.1 Lyapunov functions for discrete dynamics

Let $f : X \to X$ be a map on a set X and let $A \subset X$ be a target set. Lyapunov's idea, set out in his 1892 thesis on the stability of motion [65] and carried into the discrete setting by LaSalle and others [61], is to certify that orbits approach A without computing them, by exhibiting a scalar function that decreases along every orbit outside A.

Definition 4.1 *(Lyapunov certificate).* A function $V : X \to \mathbb{R}$ is a *Lyapunov function* for (f, A) if $V(f(x)) \le V(x)$ for every $x \notin A$. It is a *strict* Lyapunov function if $V(f(x)) < V(x)$ for every $x \notin A$, and a *strong* (or geometric) Lyapunov function if $V > 0$ on $X \setminus A$ and there is a constant $\rho < 1$ with $V(f(x)) \le \rho\, V(x)$ for every $x \notin A$.

On a continuum, a strict Lyapunov function need not drive orbits into A; the decrease can become arbitrarily small, and LaSalle's invariance principle is the tool that identifies where such orbits accumulate [61]. The situation is simpler when V takes values in the nonnegative integers, or decreases by at least a fixed amount, because a strictly decreasing sequence of nonnegative integers, or a sequence of reals bounded below that decreases by at least a fixed amount at each step, must stop.

Proposition 4.2 *(Discrete Lyapunov criterion). Let $f : X \to X$ and $A \subset X$. Suppose that either*

(i) V takes values in $\mathbb{N} \cup \{0\}$ and $V(f(x)) < V(x)$ for $x \notin A$; or

(ii) $V \ge 0$ and there is $\varepsilon > 0$ with $V(f(x)) \le V(x) - \varepsilon$ for $x \notin A$; or

(iii) $V \ge c > 0$ on $X \setminus A$ and $V(f(x)) \le \rho V(x)$ for $x \notin A$, with $\rho < 1$.

Then every orbit of f enters A. In case (ii) it does so within $V(x)/\varepsilon$ steps, and in case (iii) within $1 + \log(V(x)/c)/\log(1/\rho)$ steps.

Proof. Suppose that the first $n \ge 1$ points $x, f(x), \dots, f^{(n-1)}(x)$ of the orbit lie outside A. In case (i) V takes the $n+1$ distinct values $V(x) > V(f(x)) > \cdots > V(f^{(n)}(x))$, all in $\{0, 1, \dots, V(x)\}$, so $n \le V(x)$. In case (ii), $0 \le V(f^{(n)}(x)) \le V(x) - n\varepsilon$, so $n \le V(x)/\varepsilon$. In case (iii) the bound $V \ge c$ is known only outside A, so $c \le V(f^{(n-1)}(x)) \le \rho^{n-1} V(x)$ and $n \le 1 + \log(V(x)/c)/\log(1/\rho)$. In each case n is bounded, so the orbit enters A, and the number of steps it takes obeys the stated bound. □

The Collatz conjecture is equivalent to the existence of such a certificate, and this is worth recording at the outset because it shows that existence is not the issue.

Remark 4.3 *(Certificates exist if and only if the conjecture holds).* Let X be the odd positive integers, $f = S$ and $A = \{1\}$. If every Syracuse orbit reaches 1, let $\sigma_S(x)$ be the number of strides it takes. Then $V = \sigma_S$ satisfies case (i) of Proposition 4.2, since $\sigma_S(S(x)) = \sigma_S(x) - 1$, and $W = 2^{\sigma_S}$ satisfies case (iii) with $c = 2$ and $\rho = \frac{1}{2}$. Conversely any certificate of the kind in Proposition 4.2 proves that every orbit reaches 1. The difficulty is therefore never whether a Lyapunov function exists but whether one can be written down without knowing the answer. The functions σ_S and 2^{σ_S} are wildly irregular in x; the candidates one can actually analyse are

smooth increasing functions of the size of x, and those are the ones this chapter shows to be inadequate.

4.2 Stochastic Lyapunov functions and Foster's criterion

The stochastic analogue of a Lyapunov function replaces pointwise decrease by decrease in conditional expectation. The idea goes back to Foster [25], who used it to decide positive recurrence of countable Markov chains arising in queueing, and it is the organizing principle of the monograph of Meyn and Tweedie [69], where it appears in many forms under the name of drift conditions.

Theorem 4.4 *(Foster's criterion). Let (X_n) be an irreducible Markov chain on a countable set E, let $V : E \to [0,\infty)$, let $C \subset E$ be finite and $\varepsilon > 0$. Suppose*

$$\mathbb{E}\big[V(X_1) \mid X_0 = x\big] \le V(x) - \varepsilon \qquad (x \notin C),$$
$$\mathbb{E}\big[V(X_1) \mid X_0 = x\big] < \infty \qquad (x \in C).$$

Then the chain is positive recurrent. Moreover, if τ_C is the hitting time of C, then $\mathbb{E}_x[\tau_C] \le V(x)/\varepsilon$ for every $x \notin C$.

The positive recurrence statement is Foster's theorem [25]; see Meyn and Tweedie [69] for the general theory. The hitting time bound is elementary and we prove it, because it is exactly the stochastic counterpart of case (ii) of Proposition 4.2.

Proof of the hitting time bound. Fix $x \notin C$ and put $M_n = V(X_{n\wedge\tau_C}) + \varepsilon\,(n \wedge \tau_C)$. On the event $\{\tau_C > n\}$ the drift condition gives $\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] \le V(X_n) - \varepsilon + \varepsilon(n+1) = M_n$, and on $\{\tau_C \le n\}$ we have $M_{n+1} = M_n$. So (M_n) is a nonnegative supermartingale and $\varepsilon\,\mathbb{E}_x[n \wedge \tau_C] \le \mathbb{E}_x[M_n] \le M_0 = V(x)$. Letting $n \to \infty$ and using monotone convergence gives $\mathbb{E}_x[\tau_C] \le V(x)/\varepsilon$. □

The proof uses neither the countability of E, nor irreducibility, nor the finiteness of C, so it applies to the stride model of Chapter 3; but it does use $V \ge 0$, through the nonnegativity of (M_n), and this is where the obvious candidate fails. In the stride model the successive log increments $\Delta_j = \log 3 - k_j \log 2$ are independent, with k_j distributed as Geometric$(\frac{1}{2})$ (Definition 3.1; Proposition 2.15 is the arithmetic source of this law), so that $\mathbb{E}[\Delta_j] = \log 3 - 2\log 2 = \log(3/4) \approx -0.2877$. Write x_j for the state after j strides, and let τ be the first stride with $\log x_\tau \le 0$. Taking $V = \log x$ and $C = \{\log x \le 0\}$, the drift condition holds with $\varepsilon = |\log(3/4)|$, but at time τ the value of V is $-O$, where $O = -\log x_\tau \ge 0$ is the overshoot of the last stride below 0, and $O > 0$ with positive probability. So the argument fails, and the bound $\mathbb{E}_{x_0}[\tau] \le \log x_0/|\log(3/4)|$ that it would give is false: by the next proposition, $\log x_0/|\log(3/4)|$ is in fact a strict lower bound for $\mathbb{E}_{x_0}[\tau]$. The overshoot has to be paid for.

Proposition 4.5 *(Passage time of the stride model). In the stride model, let $x_0 > 1$, and let τ and O be as above. Then $\log 2 \le \mathbb{E}[O] < 2\log 2$, and*

$$\frac{\log x_0 + \log 2}{|\log(3/4)|} \;\le\; \mathbb{E}_{x_0}[\tau] \;<\; \frac{\log x_0 + 2\log 2}{|\log(3/4)|}. \tag{4.1}$$

Proof. By Proposition 3.16, τ is integrable, so Wald's identity applies to the walk $\log x_j = \log x_0 + \Delta_1 + \cdots + \Delta_j$ and gives $\mathbb{E}[\log x_\tau] = \log x_0 + \log(3/4)\,\mathbb{E}_{x_0}[\tau]$, that is,

$$\mathbb{E}_{x_0}[\tau] = \frac{\log x_0 + \mathbb{E}[O]}{|\log(3/4)|}.$$

To bound $\mathbb{E}[O]$, suppose that $\log x_{j-1} = y > 0$. Stride j ends at or below 0 exactly when $y + \log 3 - k_j \log 2 \le 0$, that is, when $k_j \ge m$ with $m = \lceil (y + \log 3)/\log 2 \rceil$, and then $O = (k_j - m)\log 2 + r$ with $r = m\log 2 - y - \log 3 \in [0, \log 2)$. Since k_j is independent of the first $j-1$ strides and the Geometric law has no memory, given $k_j \ge m$ the excess $k_j - m$ has the law of $k_j - 1$, whose mean is 1. So, given the first $j-1$ strides and $\tau = j$, the conditional mean of O is $\log 2 + r$, which lies in $[\log 2, 2\log 2)$. Averaging over j and over the first $j-1$ strides, with total weight $\mathbb{P}(\tau < \infty) = 1$, gives $\log 2 \le \mathbb{E}[O] < 2\log 2$, and (4.1) follows. □

The model therefore needs about $3.476 \log x_0$ strides, and the overshoot adds between $\log 2/|\log(3/4)| \approx 2.41$ and 4.82 more. At $\log x_0 = 10$ the bounds are 37.17 and 39.58, against 34.76 for the leading term alone, and an exact computation (script `ch4_exact.py`) gives $\mathbb{E}[\tau] \approx 38.19$ and $\mathbb{E}[O] \approx 0.99$, in agreement with a simulation of 2×10^5 paths. Since a stride of the accelerated map contains on average two steps, the leading term is the slope $\gamma_0 = 1/|\delta^*| \approx 6.952$ of Proposition 3.8, measured in strides instead of steps, and (4.1) is the stride counterpart of (3.1): in both, $\log x_0$ divided by the drift is a lower bound, and the overshoot accounts for the rest. In the model, then, $\log x$ becomes a Foster function once it is shifted to pay for the overshoot: the function equal to $\log x + 2\log 2$ off C and to 0 on C is nonnegative and satisfies the drift condition with $\varepsilon = |\log(3/4)|$, and the hitting time bound gives the upper half of (4.1) again, with $\le$ in place of $<$ (Exercise 4.25). The model's version of the conjecture follows, as Proposition 3.16 already showed by a martingale argument. The whole of this chapter is an investigation of what survives when the expectation is removed.

4.3 The conjecture as a descent statement

The connection between the conjecture and the potential $V(x) = \log x$ is not only heuristic. Recall (Definition 2.22) that the *stopping time* $\sigma(n)$ of $n > 1$ is the least $k \ge 1$ with $T^{(k)}(n) < n$, or ∞ if there is none; it is finite exactly when some $T_0^{(j)}(n) < n$, because the iterates of T are those of T_0 with the values $3m+1$ omitted, and such a value, being larger than the odd m before it, is never the first

iterate below n. Terras [99] and, independently, Everett [22] proved that almost every n has a finite stopping time (Theorem 2.26); the conjecture is exactly the statement that every n does.

PROPOSITION 4.6 *(The conjecture as eventual descent). The following are equivalent.*

(i) Every positive integer eventually reaches 1 *under* T_0.

(ii) For every integer $n > 1$ *there is* $j \geq 1$ *with* $\log T_0^{(j)}(n) < \log n$.

Proof. If (i) holds and $n > 1$, the orbit of n reaches $1 < n$, so (ii) holds. Conversely, assume (ii) and suppose some integer fails to reach 1; let n be the least such. Then $n > 1$ (since $1 \to 4 \to 2 \to 1$), and by (ii) some iterate $m = T_0^{(j)}(n)$ satisfies $m < n$. By minimality m reaches 1, hence so does n, a contradiction. □

So the conjecture asserts that $V = \log x$ is an *eventual* Lyapunov function: it need not decrease at every step, or at every stride, but from every starting point it eventually drops below its initial value. What one would like is to replace "eventually" by a fixed horizon, or to replace $\log x$ by a function that decreases at every step. The rest of the chapter shows that within the class of functions increasing in x neither replacement is possible.

4.4 The stride descent theorem

The basic computation concerns one Syracuse stride. For odd x and $k = v_2(3x + 1) \geq 1$,

$$\log S(x) - \log x = \log\Big(3 + \frac{1}{x}\Big) - k \log 2. \tag{4.2}$$

The first term lies in $(\log 3, \log(10/3)]$ for odd $x \geq 3$; the second is a positive integer multiple of $\log 2$. Since $\log 2 < \log 3 < \log 4$, the sign of the increment is decided entirely by whether $k = 1$ or $k \geq 2$.

THEOREM 4.7 *(Stride descent). Let* $x \geq 3$ *be odd and* $k = v_2(3x + 1)$.

(i) $\log S(x) - \log x = \log(3 + 1/x) - k \log 2$.

(ii) If $k \geq 2$, *then* $S(x) < x$,

$$\log S(x) - \log x \;\leq\; \log\Big(\frac{3}{4} + \frac{1}{4x}\Big) \;<\; 0,$$

and $\log S(x) - \log x \leq \log(7/9)$, *with equality exactly at* $x = 9$.

(iii) If $k = 1$, *which happens exactly when* $x \equiv 3 \pmod 4$, *then* $S(x) = (3x + 1)/2 > x$, *and*

$$\log \frac{3}{2} \;<\; \log S(x) - \log x = \log\Big(\frac{3}{2} + \frac{1}{2x}\Big) \;\leq\; \log \frac{5}{3},$$

with equality on the right exactly at $x = 3$.

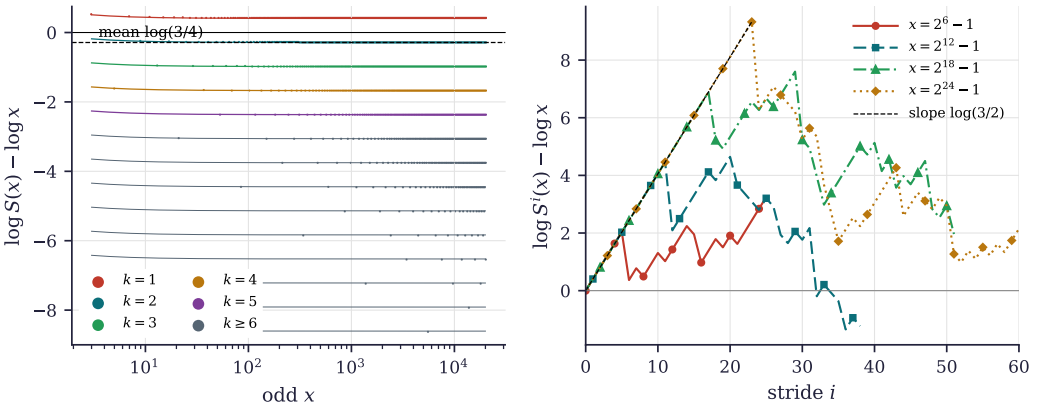


FIGURE 4.1. Left: the stride increment $\log S(x) - \log x$ for every odd $x \leq 20000$, coloured by $k = v_2(3x+1)$. Each point lies on the curve $\log(3 + 1/x) - k\log 2$ of its k (thin lines), and only $k = 1$ lies above zero; the dashed line is the model mean $\log(3/4)$. Right: the log trajectory $\log S^i(x) - \log x$ for $x = 2^j - 1$ with $j = 6, 12, 18, 24$. Each climbs in a straight run of $j - 1$ ascending strides, close to the black dashed line of slope $\log(3/2)$, before the first descent (Proposition 4.14).

Proof. Part (i) is (4.2). For (ii), $k \geq 2$ gives $S(x) \leq (3x+1)/4$, which is less than x because $x > 1$; taking logarithms, $\log S(x) - \log x \leq \log((3x+1)/(4x)) = \log(3/4 + 1/(4x)) < 0$. If $k = 2$ then $3x + 1 \equiv 4 \pmod 8$, that is $x \equiv 1 \pmod 8$, so $x \geq 9$ and $\log(3/4 + 1/(4x)) \leq \log(7/9)$, with equality exactly at $x = 9$; if $k \geq 3$ then $\log S(x) - \log x \leq \log(3/8 + 1/(8x)) \leq \log(5/12) < \log(7/9)$. For (iii), $k = 1$ means $3x + 1 \equiv 2 \pmod 4$, that is $3x \equiv 1 \pmod 4$, that is $x \equiv 3 \pmod 4$. Then $S(x) = (3x+1)/2 > x$, and $\log(S(x)/x) = \log(3/2 + 1/(2x))$ is strictly decreasing in x, exceeds $\log(3/2)$ for all x, and equals $\log(5/3)$ at $x = 3$. □

Two features of the theorem deserve comment. First, the x dependent bound in (ii) is $\log(3/4) + O(1/x)$, not $\log(3/4)$: when $k = 2$ the increment is $\log(3/4 + 1/(4x))$, which is slightly *larger* than $\log(3/4)$. The number $\log(3/4)$ is the mean of the increment under the Geometric$(\frac{1}{2})$ law, not a pointwise bound, and it is easy to conflate the two. Second, the theorem splits the odd integers exactly in half. The residue class $x \equiv 3 \pmod 4$ ascends, and its complement $x \equiv 1 \pmod 4$ descends; within the descending half, $k = j$ exactly on a single residue class modulo 2^{j+1}, of relative density $2^{-(j-1)}$ among the descending integers. The left panel of Figure 4.1 shows the increments for all odd $x \leq 20000$, which fall on the curves $\log(3 + 1/x) - k\log 2$, one for each value of k.

Case (iii) is the overshoot, or in the language of Chapter 5 the momentum injection; case (ii) is the correction. Each occurs for half of all odd integers, and the mean increment under the model is $\frac{1}{2}\log\frac{3}{2} + \sum_{j\geq 2} 2^{-j}(\log 3 - j\log 2) = \log\frac{3}{4}$, in agreement with the Foster drift of Section 4.2.

4.5 No monotone strong Lyapunov function for a single stride

Theorem 4.7 already contains the first obstruction. If half of all odd integers ascend, then no function that increases with x can decrease along every stride.

Theorem 4.8 *(Impossibility of a monotone single stride strong Lyapunov function). Let W be a strictly increasing function from the odd positive integers to $(0,\infty)$, and let $\rho \le 1$. Then $W(S(x)) > \rho\, W(x)$ for every $x \equiv 3 \pmod 4$. In particular no strictly increasing W is a strict or strong Lyapunov function for $(S,\{1\})$, and the set of odd integers on which it fails contains the residue class* 3 mod 4, *which has natural density $\frac{1}{2}$ among the odd integers; for $\rho = 1$ it is exactly this class.*

Proof. For $x \equiv 3 \pmod 4$, Theorem 4.7(iii) gives $S(x) > x$, so $W(S(x)) > W(x) \ge \rho W(x)$. The residue class 3 mod 4 contains half of the odd integers. For $\rho = 1$ the inequality $W(S(x)) \le W(x)$ holds at every $x \equiv 1 \pmod 4$ with $x \ge 5$, since there $S(x) < x$ by Theorem 4.7(ii). □

The same argument covers every nondecreasing W with values in $(0,\infty)$, since then $W(S(x)) \ge W(x) > \rho W(x)$ on the class $x \equiv 3 \pmod 4$ when $\rho < 1$, and $W(S(x)) < W(x)$ fails there when $\rho = 1$; and no nonincreasing W works either, since $S(5) = 1 < 5$ gives $W(S(5)) \ge W(5)$. So no monotone W is a strict or strong Lyapunov function for $(S,\{1\})$.

The proof is two lines, and the theorem is correspondingly robust. It does not depend on the form of W, only on its monotonicity, and it does not depend on any probabilistic model. It applies to $W = \log x$, to $W = x^p$ for every $p > 0$, to $W = (\log x)^2$ on $x \ge 3$, and to any composite of these with an increasing function. Nor is it a boundary effect: the failures are not confined to small x or to a thin set, but occupy a full residue class.

Remark 4.9 *(Rate of ascent).* On the ascending class the increment is $\log(S(x)/x) = \log(3/2 + 1/(2x))$, which decreases to $\log(3/2) \approx 0.405$ as $x \to \infty$. So the absolute ascent per stride is bounded below by $\log(3/2)$ uniformly, while the relative ascent

$$\frac{\log S(x)}{\log x} = 1 + \frac{\log(3/2)}{\log x} + O\Big(\frac{1}{x \log x}\Big)$$

tends to 1 from above. A reparametrization $W = \phi(\log x)$ with ϕ increasing converts this into $W(S(x))/W(x) = \phi(\log x + \log(3/2) + o(1))/\phi(\log x)$, which exceeds 1 for every increasing ϕ; no choice of ϕ pushes the ratio below 1, and the most one can arrange is that it tends to 1, as it does for $\phi(u) = u^2$.

Remark 4.10 *(Monotonicity is the essential hypothesis).* Theorem 4.8 is sharp in the following sense. If one drops monotonicity, then by Remark 4.3 a strong Lyapunov function bounded away from zero exists if and only if the conjecture is true; if it is, the function $2^{\sigma_S(x)}$ is one. The theorem therefore draws a precise line. Certificates

that increase with the size of x are impossible; certificates that see the arithmetic of x are possible in principle but are, at present, equivalent to the conjecture. Any viable certificate must depend on x through more than its magnitude, for instance through its residues modulo powers of 2 and 3, which is where Chapters 2 and 3 and Chapter 11 look for structure.

4.6 Two stride compositions

A natural response to Theorem 4.8 is to look at several strides at once. Every ascent is followed sooner or later by a descent, and one might hope that over two strides the descents always win. They do not, and the two stride case can be analysed completely.

For odd x write $k_1 = v_2(3x+1)$, $y = S(x)$ and $k_2 = v_2(3y+1)$. Expanding $S(y) = (3y+1)/2^{k_2}$ gives

$$S^{(2)}(x) = \frac{9x + 3 + 2^{k_1}}{2^{k_1+k_2}}. \tag{4.3}$$

Lemma 4.11 *(Two stride residue classes). For integers $a, b \geq 1$, the set of odd x with $k_1 = a$ and $k_2 = b$ is a single residue class modulo 2^{a+b+1}. Consequently it has natural density $2^{-(a+b)}$ among the odd integers.*

Proof. The condition $k_1 = a$ says $3x + 1 \equiv 2^a \pmod{2^{a+1}}$, and since 3 is invertible modulo 2^{a+1} this is a single class $x \equiv x_0 \pmod{2^{a+1}}$. Write $x = x_0 + 2^{a+1}t$. Then $S(x) = (3x_0+1)/2^a + 6t = y_0 + 2 \cdot 3t$ with y_0 odd. As t runs over a complete residue system modulo 2^b, so does $3t$, and hence $S(x)$ runs over all odd residues modulo 2^{b+1}, each exactly once. The condition $k_2 = b$ selects one odd residue class of $S(x)$ modulo 2^{b+1}, hence one class of t modulo 2^b, hence one class of x modulo 2^{a+b+1}. There are 2^{a+b} odd classes modulo 2^{a+b+1}, which gives the density. □

The lemma is the two stride case of the parity vector bijection of Theorem 2.8, written in the language of strides.

Proposition 4.12 *(Exact two stride behaviour). Let $x \geq 3$ be odd.*

(i) $S^{(2)}(x) < x$ *if and only if* $k_1 + k_2 \geq 4$.

(ii) The odd integers with a two stride descent have natural density exactly $\frac{1}{2}$ among the odd integers.

(iii) $\max_{x \geq 3 \text{ odd}} \bigl(\log S^{(2)}(x) - \log x\bigr) = \log \frac{17}{7} \approx 0.8873$, *attained only at $x = 7$, where $S(7) = 11$ and $S^{(2)}(7) = 17$.*

Proof. Put $s = k_1 + k_2 \geq 2$. If $s \leq 3$ then $2^{k_1} \geq 2$ and (4.3) gives $S^{(2)}(x) \geq (9x+5)/8 > x$. If $s \geq 4$ then $2^{k_1} \leq 2^{s-1}$, so it suffices to check $9x + 3 + 2^{s-1} < 2^s x$, that is $(2^s - 9)x > 3 + 2^{s-1}$. For $s = 4$ this reads $7x > 11$, true for $x \geq 3$; for $s \geq 5$

the left side is at least $3(2^s - 9)$ and $3 \cdot 2^s - 27 - 3 - 2^{s-1} = \frac{5}{2}2^s - 30 > 0$. This proves (i).

For (ii), by (i) and Lemma 4.11 the ascending set is the union of the classes $(k_1, k_2) \in \{(1,1),(1,2),(2,1)\}$, of total density $2^{-2} + 2^{-3} + 2^{-3} = \frac{1}{2}$.

For (iii), if $(k_1, k_2) = (1,1)$ then $S^{(2)}(x)/x = (9x+5)/(4x)$, which decreases in x. The condition $k_1 = 1$ means $x \equiv 3 \pmod 4$, and then $k_2 = 1$ means $(3x+1)/2 \equiv 3 \pmod 4$, that is $x \equiv 7 \pmod 8$. The least such x is 7, giving the ratio $68/28 = 17/7$. If $s = 3$ the ratio is at most $(9x+7)/(8x) \le 34/24 < 17/7$, and if $s \ge 4$ the ratio is below 1. □

Under the Geometric$(\frac{1}{2})$ model the two stride increment is $\Delta_1 + \Delta_2$ with mean $2\log(3/4) \approx -0.5754$, and the probability that $k_1 + k_2 \le 3$ is $\frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}$, so the model reproduces part (ii) exactly. The arithmetic agrees with it at every scale we have examined. Over the odd x with $3 \le x \le 20001$ the script `ch4_ch5_numbers.py` (see Appendix B) finds the two stride increment negative for exactly half of the 10000 starting values, with mean increment -0.574681 against the theoretical $2\log(3/4) = -0.575364$, and largest ascent 0.8873 at $x = 7$. Extending the scan to all odd x with $3 \le x < 2 \times 10^6$ gives a descending fraction of 0.4999995 and mean -0.575359; the residual discrepancy from $\frac{1}{2}$ is the edge effect of truncating the residue classes of Lemma 4.11.

Averaging over two strides therefore doubles the mean descent without reducing the proportion of ascending starting points at all. The left panel of Figure 4.2 shows the same phenomenon for the two step composition of the accelerated map, where the split is by $x \bmod 4$: the class $x \equiv 1 \pmod 4$ falls on the curve $\log(3/4 + 1/(4x))$ and the class $x \equiv 3 \pmod 4$ on $\log(9/4 + 5/(4x))$, about 0.81 above zero. The right panel follows whole orbits of T in the phase coordinates $(\log x_n, \log x_{n+1})$: every step lands on the lower branch $\log x_{n+1} = \log x_n - \log 2$ or just above the upper branch $\log x_{n+1} = \log x_n + \log(3/2)$, and the orbit of 27, for example, takes the upper branch 41 times and the lower one only 29 times, yet descends, because a halving moves $\log x$ by more than an odd step does.

4.7 The log squared candidate

A second natural response to Theorem 4.8 is to change the potential. Quadratic potentials are the default in optimization, and the obvious candidate is $W(x) = (\log x)^2$. Its behaviour illustrates, more sharply than any other example, the difference between descent in expectation and descent at every point.

Write $u = \log x$ and $\Delta = \log S(x) - \log x$. Then, identically,

$$(\log S(x))^2 - (\log x)^2 = (u + \Delta)^2 - u^2 = 2u\,\Delta + \Delta^2. \tag{4.4}$$

Now take expectations under the stride model, in which $\Delta = \log 3 - k \log 2$ with k Geometric$(\frac{1}{2})$ and independent of u. Then $\mathbb{E}[\Delta] = \log(3/4)$ and $\mathrm{Var}(\Delta) =$

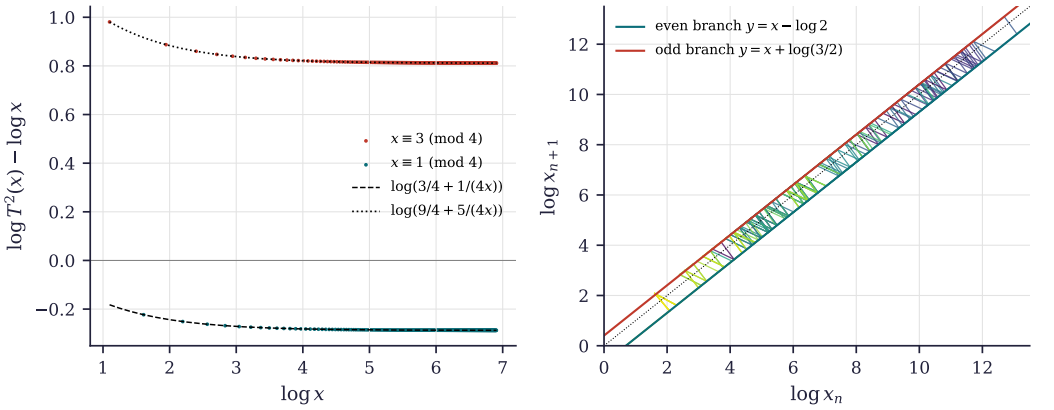


FIGURE 4.2. Left: two step increment $\log T^2(x) - \log x$ of the accelerated map for odd $x \le 1000$. The class $x \equiv 1 \pmod 4$ descends, on the curve $\log(3/4 + 1/(4x))$, which lies above its limit $\log(3/4)$; the class $x \equiv 3 \pmod 4$ ascends, on the curve $\log(9/4 + 5/(4x))$. Right: phase portrait in log coordinates for the accelerated orbits of 27, 97, 255, 703 and 6171, coloured by the fraction of the orbit elapsed. Every step lands on one of the two branches $\log x_{n+1} = \log x_n - \log 2$ and $\log x_{n+1} \approx \log x_n + \log(3/2)$.

$(\log 2)^2 \operatorname{Var}(k) = 2(\log 2)^2$, so $\mathbb{E}[\Delta^2] = 2(\log 2)^2 + (\log(3/4))^2 \approx 1.0437$ and

$$\begin{aligned} \mathbb{E}\left[(\log S(x))^2 - (\log x)^2\right] &= 2\log(3/4)\,\log x + \mathbb{E}[\Delta^2] \\ &\approx -0.5754\,\log x + 1.0437. \end{aligned} \tag{4.5}$$

This is negative as soon as $\log x > \mathbb{E}[\Delta^2]/(2|\log(3/4)|) \approx 1.814$, that is for $x \ge 7$, and the expected decrease grows without bound in $\log x$. For J strides the same computation gives $2J\log(3/4)\log x + \mathbb{E}[(\Delta_1 + \cdots + \Delta_J)^2]$, negative for $\log x \gtrsim 1.958$ when $J = 2$.

One might read (4.5) as saying that $(\log x)^2$ is a "super Lyapunov" function whose descent accelerates as the orbit grows. That reading is wrong in two separate ways.

REMARK 4.13 *(A tempting but false argument).* First, (4.5) is a statement about the model, not about the map. For an individual $x \equiv 3 \pmod 4$ the increment Δ is positive, and (4.4) gives $(\log S(x))^2 - (\log x)^2 = 2u\Delta + \Delta^2 > 0$: the potential increases, and by *more* the larger x is. The expected decrease and the pointwise increase both grow linearly in $\log x$; the first averages the increase on the ascending half with the decreases on the descending half, which are larger on average. Since $(\log x)^2$ is increasing on $x \ge 1$, Theorem 4.8 applies to it directly, and Corollary 4.15 below extends the failure to any fixed number of strides.

Second, even in the model the growth of the absolute decrease is an artefact of

the scale of W. The relative change is

$$\frac{\mathbb{E}[W(S(x))]}{W(x)} - 1 = \frac{2\log(3/4)}{\log x} + \frac{\mathbb{E}[\Delta^2]}{(\log x)^2},$$

which tends to 0. Measured as a contraction factor, which is what a strong Lyapunov function requires, $(\log x)^2$ contracts *less* as x grows, not more. There is no constant $\rho < 1$ with $\mathbb{E}[W(S(x))] \le \rho W(x)$ for all large x even in the model.

The lesson generalizes. For any potential of the form $\phi(\log x)$ with ϕ smooth and increasing, a Taylor expansion gives an expected change $\phi'(u)\log(3/4) + \frac{1}{2}\phi''(u)\mathbb{E}[\Delta^2] + \cdots$, and one can choose ϕ to make this negative or to make it grow. None of these choices touches the pointwise behaviour, which is governed by the sign of Δ alone.

4.8 Multi stride certificates and chained ascents

The last hope for a monotone certificate is to fix a number of strides J and ask for $W(S^{(J)}(x)) \le \rho\, W(x)$ for all odd $x \ge 3$. For $J = 2$ this fails by Proposition 4.12. The following proposition shows that it fails for every J, for an explicit and simple reason: ascending strides can be chained to any length, and the starting points that produce long chains are completely explicit.

Proposition 4.14 *(Chained ascents). Let x be odd and write $x + 1 = 2^j q$ with q odd, so that $j = v_2(x+1) \ge 1$. Then for $0 \le i \le j-1$,*

$$S^{(i)}(x) = 3^i\, 2^{j-i}\, q - 1, \tag{4.6}$$

the first $j-1$ strides from x all have $k = 1$ and ascend, and the jth stride has $k \ge 2$ and, if $x \ge 3$, descends. In other words, the length of the initial run of ascending strides from x is exactly $v_2(x+1) - 1$.

Proof. We prove (4.6) by induction on i; the case $i = 0$ is the definition of j and q. Suppose (4.6) holds for some $i \le j-2$, and put $z = S^{(i)}(x) = 3^i 2^{j-i} q - 1$. Since $j - i \ge 2$, $z \equiv -1 \equiv 3 \pmod 4$, so by Theorem 4.7(iii) the next stride has $k = 1$ and

$$S(z) = \frac{3z+1}{2} = \frac{3^{i+1}2^{j-i}q - 2}{2} = 3^{i+1}2^{j-i-1}q - 1,$$

which is (4.6) for $i+1$. This shows that strides $1, \dots, j-1$ have $k = 1$. For the jth stride, $z = S^{(j-1)}(x) = 2 \cdot 3^{j-1}q - 1$ with $3^{j-1}q$ odd, so $z \equiv 1 \pmod 4$ and the stride has $k \ge 2$, hence, since $z \ge 3$ when $x \ge 3$, descends by Theorem 4.7(ii) (for $x = 1$ the stride is $S(1) = 1$). □

Formula (4.6) is the identity of Remark 2.4 written for S, and it is classical: see Hercher [38, Lemma 8], who cites Terras [99] for it. For example, $x = 7 = 2^3 - 1$ gives $j = 3$, $q = 1$, and indeed $S(7) = 11 = 3 \cdot 2^2 - 1$, $S^{(2)}(7) = 17 = 9 \cdot 2 - 1$,

after which $S(17) = 13$ descends. The integer $x = 2^{24} - 1$ climbs through 23 consecutive ascents, multiplying by nearly $3/2$ each time, before its first descent; the right panel of Figure 4.1 shows this and three shorter chains.

Corollary 4.15 *(No fixed horizon monotone certificate). Fix $J \geq 1$, $\rho \leq 1$ and a strictly increasing W from the odd positive integers to $(0, \infty)$. Then $W(S^{(J)}(x)) > \rho\, W(x)$ for every odd $x \equiv -1 \pmod{2^{J+1}}$, that is for every $x = 2^{J+1}q - 1$ with $q \geq 1$. This set is infinite and has natural density 2^{-J} among the odd integers. Moreover, along it,*

$$\frac{S^{(J)}(x)}{x} \longrightarrow \left(\frac{3}{2}\right)^{J} \qquad (q \to \infty).$$

Proof. If $x \equiv -1 \pmod{2^{J+1}}$ then $v_2(x+1) \geq J+1$, so by Proposition 4.14 the first J strides all ascend and $S^{(J)}(x) > x$; monotonicity of W finishes the argument. The residue class -1 mod 2^{J+1} is one of the 2^J odd classes modulo 2^{J+1}. For the limit, write $x + 1 = 2^j q'$ with $j \geq J+1$ and q' odd; then (4.6) with $i = J$ gives $S^{(J)}(x)/x = (3^J 2^{j-J} q' - 1)/(2^j q' - 1)$, which tends to $(3/2)^J$ as x grows within the class. □

The corollary is the precise content of the slogan that "chaining ascending strides defeats every monotone multi stride certificate". It says more than that each monotone fixed horizon certificate fails somewhere: it fails on a full residue class, and on that class $\log x$ grows over the horizon by an amount approaching $J \log(3/2)$, which is unbounded as J grows. The horizon cannot be chosen to outrun the chains, because the chains are as long as the 2 adic valuation of $x + 1$, and that valuation is unbounded. As after Theorem 4.8, nondecreasing W are covered by the same argument, and nonincreasing W fail at $x = 5$, since $S^{(J)}(5) = 1$ for every $J \geq 1$.

Remark 4.16 *(How the obstruction is distributed).* The initial run length $v_2(x+1) - 1$ has, among odd x, natural density $2^{-(r+1)}$ of being equal to r, so long chains are rare but not negligible in any fixed proportion: a fraction 2^{-J} of odd integers defeats every J stride certificate that increases with x. This is the same Geometric law as the halving exponent k in Proposition 2.15, and for the same reason; both are statements about the 2 adic digits of x. It is also the reason the obstruction is not merely an artefact of small numbers. The class $x \equiv -1 \pmod{2^{J+1}}$ contains integers of every size.

4.9 Majorization minimization for the log potential

We turn now from Lyapunov functions to a second piece of optimization technology, the majorization minimization (MM) principle of Lange, Hunter and Yang [60] and Hunter and Lange [40], and ask whether it can supply a descent certificate that the direct approach cannot. The answer is no, but the reason is illuminating and connects back to the martingale duality of Proposition 3.4.

Recall the principle. To minimize an objective V, one constructs at the current point x_n a surrogate $g(\cdot \mid x_n)$ satisfying *domination*, $g(y \mid x_n) \geq V(y)$ for all y, and *tangency*, $g(x_n \mid x_n) = V(x_n)$. Any move that does not increase the surrogate then does not increase the objective, since $V(x_{n+1}) \leq g(x_{n+1} \mid x_n) \leq g(x_n \mid x_n) = V(x_n)$. More generally, for any move, the change in the objective is bounded by the change in the surrogate.

4.9.1 The linear majorizer

Take the objective to be $V(x) = \log x$, which on the positive integers is minimized exactly at $x = 1$. Since log is concave, its tangent line at any $x_n > 0$ lies above it:

$$\log y \;\leq\; g(y \mid x_n) := \log x_n + \frac{y - x_n}{x_n}, \qquad y > 0, \tag{4.7}$$

with equality at $y = x_n$. Both defining properties hold, so g is a majorizer in the sense of the MM literature.

Remark 4.17 *(A pitfall: majorizing $-\log$).* It is easy to set the problem up the wrong way round. If one takes the objective to be $-\log x$ and asserts that its tangent line majorizes it, the inequality is reversed: $-\log$ is convex, so its tangent lies *below* it and is a minorizer, and nothing can be concluded about descent. The formulation also has the wrong target, since $-\log x$ is minimized at $x = \infty$, not at $x = 1$. With $V = \log$ as in (4.7), both the direction of the inequality and the location of the minimum are correct.

Domination and tangency give the following bound: the true objective improves by at least as much as the surrogate. It holds at every step, in both parities.

Proposition 4.18 *(Tangent line bound for T_0). For every $x \in \mathbb{N}$,*

$$V(T_0(x)) - V(x) \;\leq\; g(T_0(x) \mid x) - g(x \mid x) = \begin{cases} -\frac{1}{2}, & x \text{ even}, \\ 2 + \frac{1}{x}, & x \text{ odd}. \end{cases}$$

Proof. Apply (4.7) with $x_n = x$ and $y = T_0(x)$, and use $g(x \mid x) = V(x)$. For even x the surrogate increment is $(x/2 - x)/x = -\frac{1}{2}$, while the true increment is $-\log 2 \approx -0.693$. For odd x the surrogate increment is $(3x + 1 - x)/x = 2 + 1/x$, while the true increment is $\log(3 + 1/x) \leq \log(10/3) \approx 1.204$ for $x \geq 3$. □

The inequality uses only domination and tangency, so it holds for any next iterate whatever; $T_0(x)$ is not obtained by minimizing $g(\cdot \mid x)$, and the Collatz map is not an MM iteration in the usual sense. Worse, the bound certifies nothing.

Proposition 4.19 *(The linear majorizer cannot certify descent). Under the stochastic model for the standard map, in which the parities form the renewal chain (B_n) of*

Proposition 3.12 and a step is odd with stationary probability $\frac{1}{3}$, the mean surrogate increment per step of T_0 is

$$\tfrac{1}{3} \cdot 2 + \tfrac{2}{3} \cdot \left(-\tfrac{1}{2}\right) = +\tfrac{1}{3} \;>\; 0,$$

up to the $O(1/x)$ term, whereas the mean true increment per step is $\frac{1}{3}\log 3 - \frac{2}{3}\log 2 = \frac{1}{3}\log(3/4) \approx -0.0959 < 0$. Per Syracuse stride, the mean surrogate increment is $+1$ and the mean true increment is $\log(3/4) \approx -0.2877$.

Proof. The per step values follow from Proposition 4.18 and the stationary parity frequencies. A stride consists of one odd step followed by k even steps, so the surrogate increments summed over a stride equal $2 + 1/x - k/2$, with mean $2 - \mathbb{E}[k]/2 = 1$ up to $O(1/x)$; the true increment over the stride is (4.2), with mean $\log 3 - 2\log 2$ in the model. □

A word on levels, since three drifts are now in play. The figure $\frac{1}{3}\log(3/4)$ is the drift per step of T_0; the drift per step of the accelerated map T is $\delta^* = \log(\sqrt{3}/2) = \frac{1}{2}\log(3/4) \approx -0.1438$; and the drift per Syracuse stride is $\log(3/4)$. They are the same quantity measured against clocks that tick on average three times, twice and once per stride. The comparison in Proposition 4.19 is between quantities measured on the same clock, and the sign disagreement does not depend on which clock is used.

In the model, therefore, the surrogate drifts upward while the potential drifts downward, and iterating the majorization will never produce a descent certificate. The reason is visible in (4.7): the tangent line is an accurate upper bound only for y close to x_n, and the odd step multiplies by three. At a multiplicative jump of that size the linear bound gives 2 where the logarithm gives $\log 3 \approx 1.10$, an overestimate by a factor of about 1.8, which is more than enough to swamp the halvings.

4.9.2 Why the linear surrogate sees no drift

There is a cleaner way to say what goes wrong, which explains why no tinkering with the linear surrogate can help. At the level of strides, the surrogate increment of (4.7) is

$$g(S(x) \mid x) - g(x \mid x) = \frac{S(x)}{x} - 1 = \frac{3 + 1/x}{2^k} - 1.$$

Under the Geometric($\frac{1}{2}$) law, $\mathbb{E}[2^{-k}] = \sum_{j \geq 1} 4^{-j} = \frac{1}{3}$, so the mean surrogate increment per stride, computed directly at the stride level, is

$$\mathbb{E}\Big[\frac{S(x)}{x}\Big] - 1 = \frac{3 + 1/x}{3} - 1 = \frac{1}{3x}. \tag{4.8}$$

Up to the vanishing term $1/(3x)$, the stride ratio $S(x)/x$ has mean exactly one: the Syracuse iterates form a multiplicative martingale in the model, just as the

accelerated iterates do in Proposition 3.4. (The value $+1$ in Proposition 4.19 is larger because there the majorizer is recentred at every intermediate step, and each recentring adds a further Jensen gap.) A linear surrogate measures the expected change of x itself, which is zero to first order, and is therefore structurally blind to the descent of $\log x$, which is the Jensen defect $\mathbb{E}[\log(S(x)/x)] - \log \mathbb{E}[S(x)/x] \approx \log(3/4)$. This is the martingale duality of Chapter 3 seen through the MM lens: any surrogate that is affine in x inherits the drift of the martingale, which is nil.

4.9.3 Power majorizers and the Wald root

If the linear surrogate is too loose under multiplication, one can ask for a surrogate that is tighter under multiplication. A natural family is obtained from the inequality $\log r \le (r^p - 1)/p$, valid for $r > 0$ and $p > 0$ with equality at $r = 1$. For $0 < p \le 1$ define

$$g_p(y \mid x) = \log x + \frac{(y/x)^p - 1}{p}. \tag{4.9}$$

Each g_p dominates $\log y$ and is tangent to it at $y = x$, so each is an MM majorizer of $V = \log$; the case $p = 1$ is the linear majorizer (4.7), and as $p \downarrow 0$ the surrogate tends to $\log y$ itself. Its stride increment is $((S(x)/x)^p - 1)/p$, and its mean under the stride model involves the moment generating function of the log increment $\Delta = \log 3 - k \log 2$:

$$M(p) := \mathbb{E}\left[e^{p\Delta}\right] = 3^p \sum_{j \ge 1} 2^{-j} 2^{-pj} = \frac{3^p}{2^{1+p} - 1}. \tag{4.10}$$

Proposition 4.20 *(Power majorizers). Under the stride model, the mean stride increment of the surrogate* (4.9)*, ignoring the* $1/x$ *correction, is* $(M(p) - 1)/p$*, and:*

(i) $M(0) = M(1) = 1$*, and* $M(p) < 1$ *for* $0 < p < 1$*;*

(ii) $p \mapsto (M(p) - 1)/p$ *is increasing on* $(0, 1]$*, tends to* $\mathbb{E}[\Delta] = \log(3/4)$ *as* $p \downarrow 0$*, and equals* 0 *at* $p = 1$*.*

In particular every power majorizer with $0 < p < 1$ *has strictly negative mean surrogate increment, and the linear majorizer* $p = 1$ *sits exactly on the boundary.*

Proof. From (4.10), $M(0) = 1/(2 - 1) = 1$ and $M(1) = 3/(4 - 1) = 1$. A moment generating function of a nondegenerate random variable is strictly convex, so $M < 1$ strictly between its two roots of $M = 1$. For (ii), $(M(p) - 1)/p$ is the slope of the chord of the convex function M from 0 to p, which is nondecreasing in p and tends to $M'(0) = \mathbb{E}[\Delta]$ as $p \downarrow 0$; strict convexity makes it increasing. □

Numerically, $M(\frac{1}{2}) = \sqrt{3}/(2\sqrt{2} - 1) \approx 0.9473$, so the surrogate $g_{1/2}$ has mean stride increment about -0.105, against -0.181 for $p = \frac{1}{4}$ and -0.047 for $p = \frac{3}{4}$. The family interpolates monotonically between the true drift $\log(3/4)$ at $p = 0$ and the martingale drift 0 at $p = 1$. The point $p = 1$ is the nonzero root of

$M(p) = 1$, the *Wald root* of the stride walk; it is the stride counterpart of the Wald root 1 of the accelerated map (Proposition 3.4), and it is not a coincidence that it falls exactly on the linear majorizer. The equation $M(p) = 1$ is the statement that $e^{p \log x} = x^p$ is a martingale, and at $p = 1$ that is the martingale property of x itself. Including the exact correction, $M_x(p) = (3 + 1/x)^p / (2^{1+p} - 1)$, so $M_x(1) = 1 + 1/(3x)$ as in (4.8), and for each $p < 1$ the mean surrogate increment is negative for all sufficiently large x.

Equivalently, for $0 < p < 1$ the function $W_p(x) = x^p$ is a supermartingale along strides in the model, $\mathbb{E}[W_p(S(x))] \approx M(p)\, W_p(x)$ with $M(p) < 1$: in the model it is even a geometric Foster function.

4.9.4 What a multiplicatively tight majorizer would need

Proposition 4.20 shows that tightening the majorizer under multiplication does exactly what one would hope at the level of the model: as soon as the surrogate curves below the tangent line, its mean increment turns negative. It also shows what it cannot do. Every g_p is an increasing function of y, so the surrogate increment is positive whenever $S(x) > x$, that is on the whole residue class $x \equiv 3 \pmod 4$; and $W_p = x^p$ is strictly increasing, so Theorem 4.8 and Corollary 4.15 apply to it without change. A majorizer that certified descent pointwise would have to satisfy three requirements at once.

(a) It must dominate log and be tangent to it, so that surrogate descent implies true descent.

(b) It must be tight enough under multiplication that its drift in the model is negative; by Proposition 4.20 this means curving below the linear majorizer, as g_p does for $p < 1$.

(c) Its increment must be negative at every odd x, including those with $k = 1$, and over every horizon, including the chains of Proposition 4.14.

Requirements (a) and (b) are compatible and are met by the power family. Requirement (c) is incompatible with (a) for every surrogate, because on the ascending class $\log S(x) > \log x$, and domination and tangency then force the surrogate increment up to at least $\log S(x) - \log x > 0$, as in Proposition 4.18. No majorizer of log can meet all three; a certificate of this kind would have to majorize a potential other than $\log x$, one that depends on the arithmetic of x and not merely on its magnitude, and by Remark 4.10 constructing one is, as far as anyone knows, as hard as the conjecture. We do not know of one, and we regard the search for an MM certificate of this kind as a reformulation of the problem rather than a route around it.

Notes

Lyapunov's 1892 memoir on the stability of motion, available in English translation [65], is the origin of the method; for discrete systems, including the invariance principle, see LaSalle [61]. Foster's criterion appeared in Foster [25]; the systematic theory of drift conditions, including geometric drift and its relation to geometric ergodicity, is in Meyn and Tweedie [69], whose second edition (Cambridge University Press, 2009) adds a new prologue and, by Meyn, commentary and an epilogue on later developments. The equivalence of the conjecture with finiteness of all stopping times (Proposition 4.6) is folklore and is implicit in Terras [99]; see also the survey of Lagarias [53] and the collection [55]. The stride descent theorem is a direct computation from the definition of S, and the impossibility of a monotone certificate at any fixed stride depth follows in one line from the chained ascent formula (Proposition 4.14), which is the classical identity $T^{(J)}(a2^J-1)=a3^J-1$ for the accelerated map written for S; see Hercher [38, Lemma 8], who cites Terras [99] for it. The observation that $x\equiv -1 \pmod{2^j}$ produces long ascending runs appears in many expositions of the Terras parity vector. The exact two stride analysis (Proposition 4.12) and the power majorizers (Proposition 4.20) are presented here for the first time in this form. For the MM principle see Lange, Hunter and Yang [60], Hunter and Lange [40] and, for incremental versions, Mairal [66]. The role of Wald's identity, of the overshoot (Proposition 4.5) and of the root of the moment generating function in random walks with negative drift is standard; in the Collatz setting the corresponding computations for the accelerated map are in Chapter 3.

Exercises

Exercise 4.21. Show that for odd x and $j\geq 1$, $v_2(3x+1)=j$ exactly when x lies in a single residue class modulo 2^{j+1}, and find this class for $j=1,2,3,4$. Deduce that $v_2(3x+1)=1$ if and only if $x\equiv 3 \pmod 4$.

Exercise 4.22. Prove (4.3). The two stride ascending set of Proposition 4.12 is a union of residue classes modulo 16; find them, and check that there are exactly four of them among the eight odd classes.

Exercise 4.23. Let $x\equiv 3 \pmod 4$ and put $j=v_2(x+1)$, so that $j\geq 2$. Use Proposition 4.14 to show that $\log S^{(j-1)}(x)-\log x > (j-1)\log(3/2)$. Conclude that for every $B>0$ there is an odd x and a number of strides J with $\log S^{(J)}(x)-\log x > B$, so that no bound on the cumulative ascent holds uniformly in the horizon.

Exercise 4.24. Using the Geometric$(\tfrac12)$ law, compute $M(p)=\mathbb{E}[e^{p\Delta}]$ for the stride increment and verify (4.10). Repeat the computation for the accelerated map with the Bernoulli$(\tfrac12)$ model of Proposition 3.4, and show that its nonzero Wald root

is also 1. Explain why both roots must equal 1 whenever the iterates form a multiplicative martingale.

Exercise 4.25. In the stride model, let τ be the first stride at which $\log x \le 0$, and $O \ge 0$ the overshoot below 0 at time τ, as in Proposition 4.5. Explain where the optional stopping argument in the proof of Theorem 4.4 breaks down for $V = \log x$, and why the bound $\log x_0 / |\log(3/4)|$ that it would give is never valid. For $s \ge 0$ let $V_s = \log x + s$ when $\log x > 0$ and $V_s = 0$ when $\log x \le 0$. Show that V_s satisfies the drift condition with $\varepsilon = |\log(3/4)|$ at every x with $\log x > 0$ if and only if $s \ge 2\log 2$, and deduce from the hitting time bound that $\mathbb{E}_{x_0}[\tau] \le (\log x_0 + 2\log 2)/|\log(3/4)|$, the upper half of (4.1) with $\le$ in place of $<$.

Exercise 4.26. For the accelerated map T and the linear majorizer (4.7), show that the surrogate increment is $\frac{1}{2} + \frac{1}{2x}$ for odd x and $-\frac{1}{2}$ for even x, so that under the Bernoulli$(\frac{1}{2})$ model its mean is $1/(4x)$. Relate this to the martingale property in Proposition 3.4.

Chapter 5

Majorization and Momentum

Chapter 4 ended with a negative verdict on descent. The potential $\log x$ decreases on average along a Collatz orbit, but it does not decrease along every stride, and no increasing function of x, however it is reparametrized or majorized, can repair this. Half of all odd integers ascend in a single stride, and ascending strides can be chained to any length. An optimizer who met an iteration with these properties would not conclude that it fails to converge. She would conclude that it is not a descent method, and she would reach for the other great family of first order methods, the momentum methods, whose iterates are allowed to overshoot and whose convergence is certified by a potential that is not monotone in the objective.

This chapter takes that step. We review the two classical momentum methods, Polyak's heavy ball method [80] and Nesterov's accelerated gradient method [72], with their rates on strongly convex quadratics and the optimal coefficient $(\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)$, and we recall the continuous time view of Su, Boyd and Candès [95] and Shi, Du, Jordan and Su [90], in which momentum is a damped oscillator and the certificate of convergence is an energy that includes a kinetic term. We then read the Collatz orbit in the same terms: the odd step is a momentum injection that overshoots, and the run of halvings that follows is the correction. Matching the two pictures produces a small set of constants, and the main business of the chapter is to say precisely which of them mean something and which do not.

Three conclusions emerge. The first is a set of stride constants (Definition 5.3) whose ratio $\kappa \approx 1.776$, the largest ascent of a stride over its mean descent, is a ratio of step sizes and not a condition number (Remark 5.4). The second is a cautionary tale: mixing constants defined at different levels of the iteration produces spurious agreements (Remark 5.5). The third is a single number, $\beta = \log_2 3 - 1 = \log(3/2)/\log 2$, that turns up as an ascent to descent ratio, as a definitional rewriting of a pair of drift constants, as the excess over one of the Cesàro threshold of Chapter 11, and as a ratio of Kullback and Leibler divergences. These are one fact seen four ways, and the number becomes the momentum coefficient of Chapter 7. Parrondo's paradox supplies an analogy along the way.

5.1 Momentum methods

Let $f : \mathbb{R}^d \to \mathbb{R}$ be μ strongly convex with L Lipschitz gradient, and write $\kappa = L/\mu \geq 1$ for its condition number. The model case, and the only one we need for

rates, is the quadratic

$$f(x) = \tfrac{1}{2}(x - x^*)^\top A (x - x^*), \qquad \mu I \preceq A \preceq L I. \tag{5.1}$$

Gradient descent $x_{t+1} = x_t - \alpha \nabla f(x_t)$ acts on the component of the error $e_t = x_t - x^*$ along an eigenvector with eigenvalue λ by multiplication by $1 - \alpha\lambda$. With $\alpha = 1/L$ the worst component, $\lambda = \mu$, contracts by $1 - 1/\kappa$ per step; with the best constant step $\alpha = 2/(L + \mu)$ the worst contraction is $(\kappa - 1)/(\kappa + 1)$. Either way the number of iterations needed to reduce the error by a factor ε is of order $\kappa \log(1/\varepsilon)$. Momentum reduces this to order $\sqrt{\kappa}\log(1/\varepsilon)$, and the square root is the whole point.

5.1.1 Heavy ball

Polyak [80] proposed adding to the gradient step a fraction of the previous displacement,

$$x_{t+1} = x_t - \alpha \nabla f(x_t) + \beta (x_t - x_{t-1}), \tag{5.2}$$

and showed that on the quadratic (5.1) the choice

$$\alpha = \frac{4}{(\sqrt{L} + \sqrt{\mu})^2}, \qquad \beta = \Big(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \Big)^2 \tag{5.3}$$

makes the spectral radius of the two step recursion equal to $(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$ [81], which matches the lower bound of Nesterov for first order methods on strongly convex quadratics in high dimension [71]. The guarantee is a statement about quadratics, and more generally about a neighbourhood of a nondegenerate minimum. It does not extend globally: Lessard, Recht and Packard [63] exhibit a strongly convex function on which heavy ball with the tuning (5.3) fails to converge, settling instead into a limit cycle. Momentum that is too aggressive for the global geometry can produce a limit cycle, a phrase the reader of Chapter 10 will recognize.

5.1.2 Nesterov acceleration

Nesterov [72] moved the gradient evaluation to an extrapolated point,

$$y_t = x_t + \beta_t (x_t - x_{t-1}), \qquad x_{t+1} = y_t - \alpha \nabla f(y_t). \tag{5.4}$$

For merely convex f and a momentum sequence β_t that tends to one like $1 - 3/t$, with $\alpha = 1/L$, the method achieves $f(x_t) - f^* = O(L\|x_0 - x^*\|^2 / t^2)$, against $O(1/t)$ for gradient descent, and this order cannot be improved by any first order method in the worst case [71]. For strongly convex f one uses a constant coefficient. The next proposition gives the optimal constant on quadratics with a complete proof, since it is the formula that Chapter 7 sets out to avoid.

PROPOSITION 5.1 *(Optimal constant momentum on quadratics). Let f be the quadratic (5.1) with spectrum in $[\mu, L]$, and run (5.4) with $\alpha = 1/L$ and constant $\beta \in [0,1)$. On the eigendirection with eigenvalue λ, put $a = \lambda/L \in [1/\kappa, 1]$. Then the error obeys a linear two step recursion with characteristic polynomial*

$$p_a(z) = z^2 - (1-a)(1+\beta)z + (1-a)\beta. \tag{5.5}$$

Let $r(\beta) = \max_{a\in[1/\kappa,1]} \rho_a(\beta)$, where ρ_a is the largest modulus of a root of p_a. Then r is minimized at

$$\beta^{\mathrm{opt}}(\kappa) = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}, \qquad \text{with} \qquad r\big(\beta^{\mathrm{opt}}\big) = 1 - \frac{1}{\sqrt{\kappa}}. \tag{5.6}$$

Proof. On the eigendirection, $y_t = (1+\beta)e_t - \beta e_{t-1}$ and $e_{t+1} = (1-a)y_t$, which gives (5.5). Write $c = 1-a$. The roots of p_a are real when $\Delta = c(1+\beta)^2 - 4\beta \geq 0$ and complex conjugate with modulus $\sqrt{c\beta}$ otherwise.

First fix $a = 1/\kappa$, so $c = 1 - 1/\kappa$, and regard Δ as a quadratic in β. Its roots have product one, and a direct substitution shows that β^{opt} is one of them: $(1+\beta^{\mathrm{opt}})^2 = 4\kappa/(\sqrt{\kappa}+1)^2$, so $c(1+\beta^{\mathrm{opt}})^2 = 4(\kappa-1)/(\sqrt{\kappa}+1)^2 = 4\beta^{\mathrm{opt}}$. Hence $\Delta < 0$ for $\beta \in (\beta^{\mathrm{opt}}, 1)$, where $\rho_{1/\kappa} = \sqrt{c\beta}$ is increasing in β; and $\Delta \geq 0$ for $\beta \leq \beta^{\mathrm{opt}}$, where the larger root z_+ lies below 1 (because $p_a(1) = a > 0$ and the vertex $c(1+\beta)/2$ is below 1), and there $\partial p_a/\partial\beta = c(1-z_+) > 0$ while $\partial p_a/\partial z = \sqrt{c\Delta} \geq 0$, so z_+ is nonincreasing in β by implicit differentiation. Thus $\rho_{1/\kappa}$ is minimized at β^{opt}, where $p_{1/\kappa}$ has the double root

$$\sqrt{c\beta^{\mathrm{opt}}} = \sqrt{\frac{\kappa-1}{\kappa}\cdot\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}} = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}} = 1 - \frac{1}{\sqrt{\kappa}}.$$

Since $r(\beta) \geq \rho_{1/\kappa}(\beta)$, this is a lower bound for r. It is attained: at $\beta = \beta^{\mathrm{opt}}$ and $a > 1/\kappa$ the discriminant is smaller than at $a = 1/\kappa$, hence negative, and the modulus $\sqrt{(1-a)\beta^{\mathrm{opt}}}$ is smaller than $1 - 1/\sqrt{\kappa}$. □

We checked the proposition numerically by minimizing the worst case spectral radius over a grid of β for $\kappa = 4$, 14.58 and 100 (script `ch5_nag_check.py`); the minimizers agree with (5.6) to the grid resolution and the minimal radii are 0.500, 0.738 and 0.900, as $1 - 1/\sqrt{\kappa}$ predicts. Note the shape of the optimum. The slowest direction is tuned to critical damping, a double root, and every faster direction is underdamped and oscillates. Accelerated methods converge by spiralling, and their objective values need not decrease monotonically; the characteristic ripples of Nesterov's method and the restart schemes designed to remove them are discussed by O'Donoghue and Candès [75].

5.2 The continuous time view

The heavy ball update is a discretization of a damped oscillator. Take the equation

$$\ddot{X} + \gamma\dot{X} + \nabla f(X) = 0 \tag{5.7}$$

and replace the derivatives by differences with time step h: $(x_{t+1} - 2x_t + x_{t-1})/h^2 + \gamma(x_t - x_{t-1})/h + \nabla f(x_t) = 0$. Solving for x_{t+1} gives exactly (5.2) with

$$\alpha = h^2, \qquad \beta = 1 - \gamma h.$$

The momentum coefficient is one minus the damping per step. Critical damping of the slowest mode of a quadratic, $\gamma = 2\sqrt{\mu}$, with $h = 1/\sqrt{L}$, gives $\beta = 1 - 2/\sqrt{\kappa}$, which agrees with $\beta^{\text{opt}} = 1 - 2/(\sqrt{\kappa}+1)$ to leading order in $1/\sqrt{\kappa}$. So the optimal coefficient of Proposition 5.1 is, asymptotically, the statement that the continuous system should be critically damped in its slowest direction.

The certificate of convergence for (5.7) is the energy

$$\begin{aligned} \mathcal{E}(t) &= f(X(t)) - f^* + \tfrac{1}{2}\|\dot{X}(t)\|^2, \\ \dot{\mathcal{E}} &= \dot{X}^\top\big(\nabla f(X) + \ddot{X}\big) = -\gamma\|\dot{X}\|^2 \le 0. \end{aligned} \tag{5.8}$$

The energy decreases, but the objective $f(X(t))$ alone need not: potential energy is converted to kinetic energy and back, and f can rise while the ball climbs the far wall of the valley. This is the structural feature that Chapter 4 lacked. There every candidate was an increasing function of the current position x, and Theorem 4.8 showed that no such function can decrease along every stride. For momentum methods the Lyapunov function lives on the phase space of positions and velocities, and the systematic construction of such functions for accelerated methods is carried out by Wilson, Recht and Jordan [106]. The Collatz analogue of a velocity would be some memory of the recent parity pattern, which is precisely the arithmetic information that Remark 4.10 says a successful certificate must use.

For Nesterov's method Su, Boyd and Candès [95] found the limiting equation in the convex case, with $\beta_t = (t-1)/(t+2)$, step $s \to 0$ and time $t\sqrt{s}$:

$$\ddot{X} + \frac{3}{t}\dot{X} + \nabla f(X) = 0, \qquad f(X(t)) - f^* \le \frac{2\|x_0 - x^*\|^2}{t^2}. \tag{5.9}$$

The damping $3/t$ vanishes as $t \to \infty$, which is why the method keeps oscillating. The constant 3 is not arbitrary: for the family $\ddot{X} + (r/t)\dot{X} + \nabla f(X) = 0$ Su, Boyd and Candès identify $r = 3$ as a threshold at which the qualitative behaviour changes. In the strongly convex case with constant momentum both heavy ball and Nesterov have the same low resolution limit, $\ddot{X} + 2\sqrt{\mu}\,\dot{X} + \nabla f(X) = 0$, and so the limit cannot explain why Nesterov's method is globally accelerated while heavy ball is not. Shi, Du, Jordan and Su [90] resolved this by keeping terms of order $\sqrt{s}$. Their high resolution equations are

$$\text{heavy ball:} \quad \ddot{X} + 2\sqrt{\mu}\,\dot{X} + (1 + \sqrt{\mu s})\nabla f(X) = 0, \tag{5.10}$$

$$\text{Nesterov:} \quad \ddot{X} + 2\sqrt{\mu}\,\dot{X} + \sqrt{s}\,\nabla^2 f(X)\dot{X} + (1 + \sqrt{\mu s})\nabla f(X) = 0. \tag{5.11}$$

The extra term $\sqrt{s}\,\nabla^2 f(X)\dot{X}$ is a gradient correction: it damps the motion more strongly where the curvature is large, and it is the continuous trace of evaluating

the gradient at the extrapolated point rather than at the current one. In the language we shall use for the Collatz map, the extrapolation overshoots and the gradient correction at the overshot point pulls back.

5.3 The Collatz orbit as overshoot and correction

Under the standard map T_0 an odd x goes to $3x+1$, which is even, so every odd step is followed by at least one halving (Lemma 2.3). On the logarithmic scale the orbit therefore alternates between a single upward jump of $\log(3+1/x) \approx \log 3$ and a run of $k = v_2(3x+1) \geq 1$ downward jumps of $\log 2$ each. Grouping the two gives the Syracuse stride, whose increment $\log(3+1/x) - k\log 2$ was analysed exactly in Theorem 4.7. The picture has the shape of the one in the previous section, with the roles labelled:

- the odd step is a *momentum injection*: it moves the potential in the wrong direction by a fixed amount, independently of how far the orbit is from its target;
- the halvings are the *correction*: each lowers the potential by the same amount, and their number is decided by the arithmetic of $3x+1$, not by the size of the overshoot.

The second point is where the analogy is weakest and also where it is most suggestive. In a momentum method the correction is computed from the gradient at the overshot point, so it responds to the overshoot. In the Collatz map the number of halvings is $v_2(3x+1)$, which depends only on the residue of x modulo a power of 2, and in the model, where successive halving exponents are independent draws from the Geometric$(\frac{1}{2})$ law of Proposition 2.15, it is independent of the past. The correction is blind, and the conjecture asserts that blind corrections nevertheless dominate along every orbit.

Example 5.2 *(The orbit of 27 as injections and corrections).* The orbit of 27 under T_0 reaches 1 after 111 steps, of which 41 are odd steps and 70 are halvings, and it climbs as high as 9232 on the way. Grouped into strides it consists of 41 Syracuse strides with halving exponents

$$1,2,1,1,1,1,2,2,1,2,1,1,2,1,1,1,2,3,1,1,2,$$
$$1,2,1,1,1,1,1,3,1,1,1,4,2,2,4,3,1,1,5,4.$$

Of the 41 strides, 24 have $k=1$ and so ascend by Theorem 4.7: more than half of the corrections are too small to undo their injection, and the longest run of consecutive ascending strides has length 5. The orbit descends anyway because the remaining 17 strides carry large exponents. The average number of halvings per odd step is $\bar{k} = 70/41 = 1.7073$, and the orbit satisfies $70\log 2 = 48.520 > 41\log 3 + \log 27 = 48.339$, the inequality that Proposition 11.3 forces on every

orbit that reaches 1. The margin is small: about 0.18 on the logarithmic scale out of 48.5. (Script `ch5_numbers.py`.)

5.4 Stride constants

An accelerated method converges at a rate governed by the ratio of the largest to the smallest curvature. The stride descent theorem has the same two sided shape, an ascent in case (iii) and a descent in case (ii), so it is natural to form the corresponding ratio. The quantities must be defined at the same level of the iteration, and the stride is the natural level, since it is the smallest unit that contains both an injection and its correction.

Definition 5.3 *(Stride constants).* For the Syracuse map S on odd integers $x \geq 3$ put

$$L := \sup_{x \text{ odd},\ x \geq 3} \big(\log S(x) - \log x\big) = \log \tfrac{5}{3} = 0.5108,$$
$$\mu := -\mathbb{E}\big[\log S(x) - \log x\big] = \big|\log \tfrac{3}{4}\big| = 0.2877,$$

where the expectation is under the Geometric($\frac{1}{2}$) law of $k = v_2(3x+1)$ with the correction $\log(1 + 1/(3x))$ neglected. The *stride ratio* is $\kappa := L/\mu = 1.776$.

The supremum is attained at $x = 3$, where $S(3) = 5$, by Theorem 4.7(iii); the mean was computed in Section 4.4. To four decimal places, $\kappa = 1.7757$, and the *Nesterov style rate* $1 - \kappa^{-1/2}$ that one may form from it by analogy with Proposition 5.1 is 0.2496. If one insisted on the formula of Proposition 5.1, the optimal coefficient at this κ would be $(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1) = 0.143$.

Remark 5.4 *(What this is and is not).* L is the largest ascent that a single stride can produce and μ is the mean descent per stride. Both are increments of the potential $\log x$ at the same level of the iteration, which is what makes their ratio dimensionless and meaningful as a comparison of step sizes. Neither is a curvature bound. L is not the Lipschitz constant of a gradient and μ is not a strong convexity modulus, because there is no gradient and no convex objective: the Collatz map does not minimize anything by evaluating derivatives. Consequently κ is a ratio of increments rather than a condition number, and the rate $1 - \kappa^{-1/2}$ is an analogy with no theorem behind it. No result in this book uses it. We report it because it is the natural dimensional comparison and because it fixes an order of magnitude: the largest net ascent of a stride is less than twice the mean net descent per stride.

The number is also sensitive to conventions that have no optimization meaning. The supremum L is attained only at $x = 3$; removing that single point moves it to $\log(11/7)$, attained at $x = 7$, and the ratio to 1.571 (rate 0.202). Using instead the asymptotic ascent $\log(3/2)$ of a stride with $k = 1$ and large x gives

$\log(3/2)/\log(4/3) = 1.409$ and a rate 0.158. A genuine condition number does not move when a single point is removed from the domain.

Remark 5.5 *(A pitfall: mixing levels).* One might take for L the ascent of the bare odd step $x \mapsto 3x+1$, whose largest value over odd $x \geq 3$ is $\log(10/3) = 1.204$, while keeping the per stride mean descent $\mu = |\log(3/4)|$. The ratio is then $\log(10/3)/|\log(3/4)| = 4.185$ and the Nesterov style rate is $1 - 4.185^{-1/2} = 0.511$. This number is close to 0.52, the modulus of the second eigenvalue of the chain on residues modulo 27 of Remark 11.10 when its states are restricted to the nine odd representatives of the units in $[1, 27)$. An agreement between a rate derived from the potential and a rate derived from the residue dynamics would be remarkable if it were real.

It is not real. The mixed ratio compares an ascent defined at the level of a single step of T_0 with a descent defined at the level of a stride, and it corresponds to no consistent picture of the iteration; with the levels matched as in Definition 5.3 the rate is 0.250 and there is nothing to match. The value 0.52 is an artefact of a different kind. Since 27 is odd, every residue class modulo 27 contains integers of both parities, so the nine states are not a set of odd residues; the chain leaves them with positive probability, and conditioning on a set that is not invariant produces spurious eigenvalues. The model chain on all eighteen units, with independent Geometric($\frac{1}{2}$) halving exponents, is exactly stationary after three strides, by an identity of Tao [96, eq. (1.21)], and so every eigenvalue other than 1 is zero (Lemma 11.8). The same lesson applies to the matched rate itself: its value 0.2496 lies within 0.0005 of $1 - e^{-\mu} = 1/4$, the fractional decrease per stride of the geometric mean of x in the model, and this closeness is equally accidental, since $\kappa^{-1/2}$ and $e^{-\mu}$ are unrelated functions of the constants. With a handful of constants of order one and a tolerance of a few percent, coincidences are to be expected, and none of them should be reported as a finding.

5.5 Per step and per macro step constants

A second pair of constants is natural if one looks at the accelerated map $T(x) = x/2$ or $(3x+1)/2$ of Definition 2.1 rather than at strides. Every step of T changes $\log x$ by $-\log 2$ if x is even and by $\log(3/2 + 1/(2x)) \in (\log(3/2), \log 2]$ if x is odd, with $\log 2$ attained only at $x = 1$; over odd $x \geq 3$ the largest ascent is $\log(5/3)$, at $x = 3$, which is the stride constant L, since $T(3) = S(3) = 5$. Put

$$L_* := \log 2 = 0.6931, \quad \mu_* := \left|\log \tfrac{3}{4}\right| = 0.2877, \quad \kappa_* := \frac{L_*}{\mu_*} = 2.409. \tag{5.12}$$

Here L_* is the size of one halving, and also the largest one step ascent of T, but only because $x = 1$ is admitted; over the odd $x \geq 3$ used in the other rows of Table 5.1 the ratio would be κ itself. Read as an ascent, L_* is a per step extremal quantity, while μ_* is the mean descent per Syracuse macro step under

TABLE 5.1. Ratios of an ascent constant to a descent constant at different levels of the iteration. Only the first row compares quantities defined at the same level. The rate column is $1-\kappa^{-1/2}$ in each case and has no theorem behind it. The ascent $\log(10/3)$ is the largest over odd $x \geq 3$; at $x = 1$ the step $1 \mapsto 4$ ascends by $\log 4$. The ascent $\log 2$ is attained only at $x = 1$; over odd $x \geq 3$ a step of T ascends by at most $\log(5/3)$, as in the first row.

Ascent	Level	Descent	κ	$1-\kappa^{-1/2}$
$\log(5/3)$	stride	$\lvert\log(3/4)\rvert$ per stride	1.776	0.250
$\log 2$	step of T	$\lvert\log(3/4)\rvert$ per stride	2.409	0.356
$\log(10/3)$	step of T_0	$\lvert\log(3/4)\rvert$ per stride	4.185	0.511

the Geometric$(\frac{1}{2})$ law, a per macro step expectation. The ratio κ_* therefore pairs quantities defined at different levels, as in Remark 5.5, but its excess over κ comes entirely from the point $x = 1$. Table 5.1 collects the three ratios side by side.

The pair (L_*, μ_*) has one virtue, which is also a trap. Since $\log(3/4) = \log 3 - 2\log 2$,

$$1 - \frac{\mu_*}{L_*} = 1 - \frac{2\log 2 - \log 3}{\log 2} = \frac{\log 3 - \log 2}{\log 2} = \frac{\log(3/2)}{\log 2} =: \beta. \tag{5.13}$$

The left side has the form $1 - 1/\kappa_*$, and an optimizer will recognize in it a familiar way of writing a momentum coefficient in terms of a condition number. One might read (5.13) as a discovery, namely that the Collatz map selects its own momentum coefficient. It is nothing of the kind. It is a definitional identity, a rearrangement of $\log(3/4) = \log 3 - 2\log 2$, and it would hold for any pair of constants so defined; Remark 7.3 returns to it from the optimizer's side. Equivalently $1/(1-\beta) = \kappa_*$, and when this quantity reappears in Chapter 8 as the slow mode noise amplification of every step momentum at coefficient β, it is the same algebra again.

REMARK 5.6 *(Another definitional coincidence).* By Proposition 5.1, a fixed coefficient β is optimal for exactly one condition number, $\kappa = \big((1+\beta)/(1-\beta)\big)^2$. For $\beta = \log(3/2)/\log 2$ we have $(1+\beta)/(1-\beta) = \log 3/\log(4/3)$, so this condition number is

$$\Big(\frac{\log 3}{\log(4/3)}\Big)^2 = 14.58.$$

The same number appears in Chapter 3 as the constant $C = \sigma^2/(\delta^*)^2$ that sets the scale of the central limit theorem for stopping times. Again the coincidence is algebraic. Under the Bernoulli model a step of T is a fair choice between $\log(3/2)$ and $-\log 2$, so its standard deviation is half the gap, $\sigma = \frac{1}{2}\log 3$, and its mean is $\delta^* = \frac{1}{2}\log(3/4)$; the ratio $\sigma/|\delta^*|$ is $\log 3/\log(4/3) = (1+\beta)/(1-\beta)$ by the same rearrangement as (5.13). Two quantities built from $\log 2$ and $\log 3$ in the same way are equal; nothing more follows.

5.6 One number in four places

Set aside the condition number language, which does not survive scrutiny, and look at the number itself:

$$\beta = \frac{\log(3/2)}{\log 2} = \log_2 3 - 1 = 0.584963\ldots \tag{5.14}$$

It is the fractional part of $\log_2 3$, whose continued fraction expansion governs the cycle problem of Chapter 10. It appears in four guises in this book, and they are a single fact.

As an ascent to descent ratio. An odd step of the accelerated map T raises the potential by $\log(3/2) + O(1/x)$, and a halving lowers it by $\log 2$. Measured in units of one halving, the overshoot of an injection is β, asymptotically. In the standard map T_0 the odd step has not yet absorbed its first halving, and its overshoot in the same units is $\log_2 3 = 1 + \beta$.

As a Cesàro threshold. Along an orbit of odd iterates $y_0, y_1, \ldots$ with halving exponents k_j, the average $\bar{k}_N$ is the number of halvings per odd step. Every orbit that reaches 1 from $x_0 > 1$ has $\bar{k}_N > \log_2 3$ at the stride N at which it reaches 1 (Proposition 11.3), and an unbounded orbit has $\limsup \bar{k}_N \le \log_2 3$ (Corollary 11.4). So the threshold for halvings per odd step is exactly the overshoot of one injection measured in halvings,

$$\bar{k} > 1 + \beta,$$

which is the statement that corrections outweigh injections on average, counted in the same units. Under the Geometric$(\frac{1}{2})$ law $\mathbb{E}k = 2$, and the margin is $2 - (1+\beta) = 1 - \beta = \mu_*/L_* = 0.415$, a margin that describes the model, in which k is evaluated at a uniformly random odd residue; along an unbounded orbit the identity leaves no margin at all. The orbit of 27 in Example 5.2 has $\bar{k} = 1.707$, above 1.585, as every orbit that reaches 1 must be, and well below the model mean. Chapter 11 reduces the divergence half of the conjecture to a lower bound on $\liminf \bar{k}_N$ of exactly this form (Conjecture 11.6).

As a rewriting of drift constants. This is (5.13), and we have said what it is worth: the per macro step drift $\log(3/4)$ equals $(\beta - 1)\log 2$, so the drift is negative because $\beta < 1$, and $\beta < 1$ is the statement $3 < 4$.

As a ratio of divergences. Let U_j be the uniform law on a set of j points, with $U_1 \subset U_2 \subset U_3$ nested in the sense of supports. Then $D_{\mathrm{KL}}(U_j \,\|\, U_{j+1}) = \log\big((j+1)/j\big)$, since each of the j atoms of U_j contributes $\frac{1}{j}\log\big((j+1)/j\big)$, and therefore

$$\frac{D_{\mathrm{KL}}(U_2 \,\|\, U_3)}{D_{\mathrm{KL}}(U_1 \,\|\, U_2)} = \frac{\log(3/2)}{\log 2} = \beta. \tag{5.15}$$

TABLE 5.2. Long run gain per play in the capital dependent Parrondo games. Random mixture: A or B chosen by a fair coin at each play. Periodic patterns are repeated indefinitely. Exact stationary computations; a simulation of 2×10^6 plays of AABB at $\epsilon = 0.005$ gave 0.0138.

Game	$\epsilon = 0$	$\epsilon = 0.005$
A alone	0	-0.0100
B alone	0	-0.0087
random mixture of A and B	$18/709 = 0.0254$	0.0157
periodic AABB	0.0245	0.0147
periodic ABB	0.0679	0.0574

Lemma 7.2 states this formally and discusses its Sanov reading; here we note only how it connects to the other three guises. An odd step of T multiplies by $3/2$ and a halving divides by 2, so the potential moves by $\log(3/2)$ and by $\log 2$. These are exactly the divergences in (5.15), the information lost in compressing three equally likely states to two and two to one, and their ratio is the ratio of the two moves. The integers 2 and 3 do all the work in every guise, which is why the four agree.

5.7 PARRONDO'S PARADOX

The Collatz iteration alternates two moves with opposite effects on the potential, and which move is made is decided by the state, through its parity. A well known family of games has the same architecture and a surprising conclusion. Parrondo's games, devised by Parrondo and published by Harmer and Abbott [33, 34], show that two games, each losing when played alone, can combine to make a winning one.

The capital dependent version runs as follows. A player's capital changes by ± 1 at each play. In game A a coin with winning probability $\frac{1}{2} - \epsilon$ is tossed. In game B the coin depends on the capital: if the capital is a multiple of 3 the winning probability is $\frac{1}{10} - \epsilon$, and otherwise it is $\frac{3}{4} - \epsilon$. Game A is plainly losing for $\epsilon > 0$. Game B is a Markov chain on the capital modulo 3, and its long run gain per play is $\sum_s \pi_s(2p_s - 1)$, where π is the stationary law of the residue and p_s the winning probability in state s. We computed these quantities exactly (script `ch5_parrondo.py`); Table 5.2 gives the results.

The mechanism is visible in game B at $\epsilon = 0$. If the residue of the capital modulo 3 were uniformly distributed, the gain per play would be $\frac{1}{3}(2 \cdot \frac{1}{10} - 1) + \frac{2}{3}(2 \cdot \frac{3}{4} - 1) = \frac{1}{15} > 0$. But the residue is not uniform: the bad state 0 has stationary probability $\pi_0 = 5/13$ rather than $1/3$, and the gain is exactly $\frac{5}{13}(-\frac{4}{5}) + \frac{8}{13} \cdot \frac{1}{2} = 0$. The state dependent rule steers the chain towards the state in which it loses. Interleaving game A randomizes the capital, pushes the residue law back towards uniform ($\pi_0 = 245/709$ for the fair mixture), and lets the favourable states of B

pay off.

For the Collatz map the analogy reads as follows. The ascent move (the odd step) and the descent move (a halving) are the two games; the first loses for a player whose aim is to reach 1, the second wins, and the switching between them is dictated by the state through its residues modulo powers of 2. The heuristic that the orbit descends computes the long run drift under the assumption that these residues are uniformly distributed, which is what the Geometric$(\frac{1}{2})$ law expresses. Parrondo's games show, with exact arithmetic, how far that kind of computation can be from the truth when the switching rule is coupled to the state: the uniform residue heuristic says $+1/15$ for game B and the truth is 0. An orbit that diverged would be an orbit whose state dependent switching steered it, in the manner of game B, into the residues where halvings are scarce, so that $\bar{k}$ stayed at or below $1+\beta$. The statement that this cannot happen is a statement about the distribution of residues along deterministic orbits, which is the subject of the Cesàro liminf conjecture, Conjecture 11.6; the residue chains of Chapter 11 answer the corresponding question for the model, in which the halving exponents are drawn independently, and not for a single orbit.

The analogy has limits. In Parrondo's games both components lose and the combination wins; in the Collatz map one move ascends and one descends, and there is no paradox in their combination descending. What carries over is the lesson: the long run drift of a state switched system is a property of the stationary law of the switching state, not of the components. The optimizer of Chapter 7, which alternates a noise amplifying momentum step with a plain gradient step, is governed by the same lesson in a benign form.

5.8 Towards Switching MM

Proposition 5.1 makes the dependence of momentum on curvature explicit: the optimal constant coefficient $(\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)$ is an increasing function of the condition number, near zero for well conditioned problems and near one for badly conditioned ones. Any fixed coefficient is therefore optimal for exactly one κ. Figure 5.1 plots the optimal coefficient against κ together with the two fixed values that will matter in Chapters 7 and 8: the Collatz value $\beta = 0.585$ and the machine learning default 0.9. Solving $(\sqrt{\kappa}-1)/(\sqrt{\kappa}+1) = \beta$ gives $\kappa = ((1+\beta)/(1-\beta))^2$, so the crossings are at $\kappa = 14.58$ for $\beta = 0.585$ and at exactly $\kappa = 361$ for $\beta = 0.9$ (script `ch4_ch5_numbers.py`). The three ratios of Table 5.1 are marked for reference only; all three lie in the region where the optimal coefficient is small, but since none of them is a condition number (Remark 5.4), this says only that the largest ascent is a small multiple of the mean descent.

The figure states the practitioner's dilemma that opens Chapter 7: the optimal coefficient needs κ, which is usually unknown. The alternative pursued there is to fix the coefficient in advance at the number (5.14), the overshoot of an injection measured in corrections. The figure also shows the price. The value

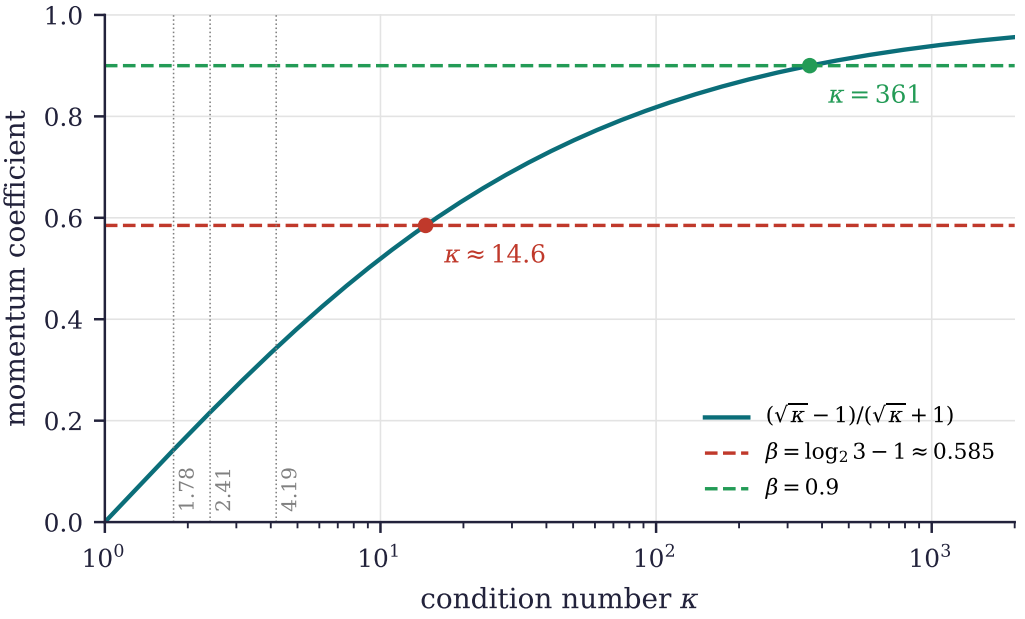


FIGURE 5.1. The optimal constant Nesterov coefficient $(\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)$ on κ conditioned quadratics, with the fixed values $\beta = \log_2 3 - 1 \approx 0.585$ and $\beta = 0.9$. The crossings, at $\kappa = ((1+\beta)/(1-\beta))^2$, are $\kappa \approx 14.6$ and $\kappa = 361$. Dotted vertical lines mark the ratios 1.78, 2.41 and 4.19 of Table 5.1.

0.585 is the right amount of momentum only near $\kappa = 14.6$ and is too little for harder problems; Chapter 7 shows that on quadratics the resulting method is not accelerated at all, and Chapter 8 locates its real advantage elsewhere, in robustness to noisy gradients. The second ingredient taken from the arithmetic is the parity pattern: the Collatz orbit injects only on odd steps, and Switching MM extrapolates only on odd iterations. In the language of this chapter, it is a momentum method whose overshoots are always followed by a correction, which is the one feature of the Collatz orbit that Lemma 2.3 guarantees. Chapter 6 first presses the majorization idea as far as it will go on the Collatz map itself; Chapters 7 and 8 then develop the optimizer, exactly on quadratics and under noise; and Chapter 9, at the end of Part II, places both objects, the Collatz recursion and the momentum methods, in the common framework of nonlinear difference equations.

NOTES

The heavy ball method is due to Polyak [80], whose book [81] remains the clearest treatment of momentum and Chebyshev methods on quadratics. Nesterov's method appeared in [72]; the lower bounds showing its optimality among first

order methods, and the strongly convex version with constant momentum, are in his lectures [71]. The failure of heavy ball to converge globally on a strongly convex function, and a systematic method for certifying first order algorithms by small semidefinite programs, are in Lessard, Recht and Packard [63]. The differential equation (5.9) is from Su, Boyd and Candès [95], and the high resolution equations (5.10) and (5.11) from Shi, Du, Jordan and Su [90]; Lyapunov functions for accelerated methods in continuous and discrete time are developed systematically by Wilson, Recht and Jordan [106]. Restarting schemes that suppress the oscillations of accelerated methods are studied by O'Donoghue and Candès [75].

The stride constants, the per step and per macro step constants and the identity (5.13) are elementary; we present them with the level of the iteration at which each is defined made explicit, which is the point of Remarks 5.4 and 5.5. Proposition 5.1 is classical; the proof given here, by locating the double root, is the standard one. Parrondo's games were devised by Parrondo as a pedagogical illustration of the flashing Brownian ratchet; the capital dependent games used here were published by Harmer and Abbott [33, 34], games whose rules depend on the history of play rather than on the capital were introduced by Parrondo, Harmer and Abbott [77], and [35] reviews both; the observation that game B fails exactly because its residue law is not uniform is the reason we find the analogy with Collatz parity switching useful. For the Kullback and Leibler divergence see any text on information theory; nothing beyond its definition is used.

Exercises

Exercise 5.7. For the heavy ball recursion (5.2) on a scalar quadratic with curvature λ, show that the characteristic polynomial is $z^2 - (1 + \beta - \alpha\lambda)z + \beta$. With the tuning (5.3), show that both endpoints $\lambda = \mu$ and $\lambda = L$ give double roots of modulus $(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$, and that every interior λ gives complex roots of modulus $\sqrt{\beta}$, which is the same number.

Exercise 5.8. Verify the energy identity (5.8). For the quadratic $f(x) = \frac{1}{2}\lambda x^2$ and $\gamma < 2\sqrt{\lambda}$, show that $f(X(t))$ is not monotone even though $\mathcal{E}(t)$ is. Find the analogous discrete energy for (5.2) on a scalar quadratic, that is, a positive definite quadratic form in (x_t, x_{t-1}) that decreases along the iteration whenever the spectral radius is less than one.

Exercise 5.9. Compute the stride ratio and the Nesterov style rate when L is defined as the supremum of the stride increment over odd $x \geq X$, for $X = 3, 7, 11, \ldots$, and show that the ratio decreases to $\log(3/2)/\log(4/3) = \beta/(1-\beta)$ as $X \to \infty$. Explain why this dependence on X is evidence that the stride ratio is not a condition number.

Exercise 5.10. Let $a, b > 0$ with $a < b$ and define $\beta(a, b) = \log(1 + a)/\log(1 + b)$,

the ascent to descent ratio of a map that multiplies by $1+a$ on one parity class and divides by $1+b$ on the other. Show that $1-\big(\log(1+b)-\log(1+a)\big)/\log(1+b) = \beta(a,b)$ identically, and deduce that the identity (5.13) holds for every map of this type. For which a and b is $\beta(a,b)$ also a ratio $D_{\mathrm{KL}}(U_m \,\|\, U_{m+1})/D_{\mathrm{KL}}(U_n \,\|\, U_{n+1})$ of nested uniform divergences?

Exercise 5.11. For the capital dependent Parrondo game B with winning probabilities p_0 in state 0 and p_1 in states 1 and 2 (capital modulo 3), show that the stationary probability of state 0 is

$$\pi_0 = \frac{1 - p_1 + p_1^2}{3 - p_0 - 2p_1 + 2p_0 p_1 + p_1^2}.$$

Hence verify that $\pi_0 = 5/13$ and the gain is 0 for $p_0 = \frac{1}{10}$, $p_1 = \frac{3}{4}$, and that the fair mixture with game A, which has $p_0 = \frac{3}{10}$ and $p_1 = \frac{5}{8}$, has $\pi_0 = 245/709$ and gain $18/709$.

Exercise 5.12. For an odd starting value $x_0 > 1$ whose orbit reaches 1 after N strides, show that the number of ascending strides ($k_j = 1$) can exceed $N/2$, as it does for $x_0 = 27$. Using Proposition 11.3, find a lower bound on the average of k_j over the descending strides in terms of the fraction of ascending strides, and evaluate it for the orbit of 27.

Chapter 6

Majorization Pressed Harder

Chapter 4 showed that the linear majorizer of $\log x$ cannot certify descent. This chapter asks what the majorization idea can do for the Collatz map when it is pushed as far as it will go. We obtain four things. First, the natural majorizer of a Syracuse stride is multiplicative, not linear; summed around a cycle, its bound *is* the cycle inequality on which Chapter 10 builds, and its minorizer is the window lemma of that chapter. The cycle half of the conjecture becomes the question of how tight this majorizer can be made along a closed orbit, a reformulation rather than a method. Second, a potential that corrects $\log x$ by a bounded function of residues is impossible, and the reason is a negative cycle condition from shortest path theory: the residue graph of the map contains, for every modulus, cycles of positive mean weight, among them the images of the cycles of T on the negative integers. Third, any pointwise potential must assign a surplus to integers with a run of trailing ones that grows linearly in the length of the run. Fourth, majorizing over a whole run of ascents, a block, gives an exact description of the dynamics, a renewal process under the model, in which the descent of a block is decided by a single 2 adic valuation, and the resulting block potentials fail at a computable fraction of odd integers that no choice of weight brings below 0.143. Together the failures locate the conjecture where Chapter 13 will also put it, in the integrality of n.

6.1 The multiplicative majorizer

The tangent line majorizer of Chapter 4 bounds $\log y$ by a linear function of y, which is tight for y near x and poor at $y = 3x + 1$. The Syracuse map is multiplicative, and the right surrogate is multiplicative too.

Proposition 6.1 *(Stride majorizer). For odd $x \geq 1$ with $k = v_2(3x+1)$,*

$$\begin{aligned} \log S(x) &= \log x + \log 3 - k \log 2 + \varepsilon(x), \\ 0 < \varepsilon(x) &= \log\Big(1 + \frac{1}{3x}\Big) \leq \frac{1}{3x}, \end{aligned} \tag{6.1}$$

so that $g(x) := \log x + \log 3 - k \log 2 + 1/(3x)$ bounds $\log S(x)$ from above, with gap $1/(3x) - \varepsilon(x) < 1/(18x^2)$, and the surrogate decrease $\log x - g(x) = k \log 2 - \log 3 - 1/(3x)$ is a lower bound for the true decrease $\log x - \log S(x)$.

Proof. $3x+1 = 3x(1+1/(3x))$, and $u - u^2/2 < \log(1+u) \le u$ for $u > 0$. □

This is the surrogate bound in multiplicative form, not a descent property: the surrogate decrease is negative whenever $k = 1$, and no minimization step is involved, so the Syracuse map is not an MM iteration in the usual sense, any more than T_0 is. What the multiplicative form gains over the tangent line bound of Proposition 4.18 is accuracy: the surrogate decrease is exactly computable from the halving exponent, and it underestimates the true decrease by less than $1/(18x^2)$. The surrogate misses the true decrease only by this second order gap; what separates the map from the model decrease $k\log 2 - \log 3$ are the terms $\varepsilon(x) \approx 1/(3x)$, which the majorization keeps up to that gap, and those are the entire content of the cycle half.

Proposition 6.2 *(The cycle inequality from the stride majorizer). Let $x_1, \dots, x_r$ be the odd elements of a cycle of S with halving exponents $k_1, \dots, k_r$ and $s = \sum k_i$. Summing* (6.1) *around the cycle gives*

$$0 = \sum_{i=1}^{r} \bigl(\log S(x_i) - \log x_i\bigr) = r\log 3 - s\log 2 + \sum_{i=1}^{r} \varepsilon(x_i),$$

so $\Lambda = s\log 2 - r\log 3 = \sum_i \varepsilon(x_i) \in (0, r/(3M))$ with $M = \min_i x_i$, which is the cycle identity that Chapter 10 proves as Lemma 10.1. The surrogate decrease around the cycle, $\Lambda - \sum 1/(3x_i)$, is negative, and the true decrease is zero: a cycle is exactly an orbit along which the multiplicative majorizer is too pessimistic by the whole accumulated gap.

The companion inequality is a *minorization*. The affine identity of Chapter 2, applied to the s steps of T, r of them odd, that return an element x of the cycle to itself, writes $x = c_s(x)/(2^s - 3^r)$, and the trivial lower bound $c_s(x) \ge 3^r - 2^r$ gives the lower end of the window in the window lemma that Chapter 10 proves (Lemma 10.21, with $m = r$ and $k = s$). So the two halves of the window are the majorizer and minorizer of the same quantity, the total gap $\sum \varepsilon(x_i)$, and part (i) of Corollary 10.22 there records that an improvement of the upper end by the ratio of the two ends, a factor of about $r/3$ when Λ is small, for every r and s would close the window and exclude every nontrivial cycle.

6.2 Residue potentials and negative cycles

For the divergence half the natural MM refinement is to correct the potential by a function of the residue of x. Theorem 4.8 rules out monotone potentials, but $V(x) = \log x + \varphi(x \bmod 2^d)$ is not monotone, and one might hope to choose φ so that V decreases at every step of T for all large x. The question has an exact answer, and it is negative for every d.

Fix d and consider the directed graph G_d on the residues modulo 2^d in which a has an edge to each residue a' with $a' \equiv T(a) \pmod{2^{d-1}}$, the two lifts of $T(a)$,

with weight $w(a) = \log(3/2)$ if a is odd and $w(a) = -\log 2$ if a is even. For large $x \equiv a$ the step $x \mapsto T(x)$ changes $\log x$ by $w(a) + O(1/x)$ and moves the residue along an edge of G_d.

THEOREM 6.3 *(Residue potentials).*

(i) There is a function φ on $\mathbb{Z}/2^d\mathbb{Z}$ and X such that $V(T(x)) < V(x)$ for all $x \geq X$, with $V = \log x + \varphi(x \bmod 2^d)$, if and only if every cycle of G_d has negative total weight.

(ii) For every $d \geq 1$ the graph G_d contains cycles of positive weight: the loop at -1, of weight $\log(3/2)$, and the images of the cycles of T on the negative integers, whose weights are $r\log 3 - s\log 2 > 0$ for a cycle with r odd steps among s. The maximum cycle mean of G_d is exactly $\log(3/2)$ for every d.

(iii) Consequently no potential $\log x + \varphi(x \bmod 2^d)$ with bounded φ decreases at every step for all large x, for any d, and the same holds for potentials over any fixed number J of steps.

Proof. (i) Suppose such φ exists and let $a_0 \to a_1 \to \cdots \to a_\ell = a_0$ be a cycle of G_d. By the theorem of Terras and Everett (Theorem 2.8) the residue class of x modulo $2^{d+\ell}$ determines $T^i(x) \bmod 2^d$ for $i \leq \ell$, and the two lifts of $x \bmod 2^{d+i}$ to 2^{d+i+1} send $T^{i+1}(x) \bmod 2^d$ to the two lifts of $T^{i+1}(x) \bmod 2^{d-1}$; so every walk of length ℓ in G_d is realized by a residue class modulo $2^{d+\ell}$, which contains integers $x \geq 2^\ell X$, so that $T^i(x) \geq X$ for $i < \ell$. Along the corresponding ℓ steps the actual increments are $w(a_i)$ plus a positive correction at odd steps, and φ returns to its starting value, so $V(T^\ell(x)) - V(x) \geq \sum_i w(a_i)$; strict decrease forces $\sum_i w(a_i) < 0$. Conversely, if every cycle has negative weight then, G_d being finite, there is $\eta > 0$ such that every cycle still has negative weight for the shifted weights $w + \eta$. Let $\varphi(a)$ be the maximum weight, for the shifted weights, of a walk starting at a (finite because no cycle has nonnegative shifted weight, and at least 0 by the empty walk). Then $\varphi(a) \geq w(a) + \eta + \varphi(a')$ for every edge (a, a'), so $V(T(x)) - V(x) = w(a) + \varphi(a') - \varphi(a) + O(1/x) \leq -\eta + O(1/x) < 0$ for $x \geq X$.

(ii) The residue $-1 \bmod 2^d$ is odd and $T(-1) = -1$, so G_d has a loop at -1 of weight $\log(3/2)$. A cycle of T on negative integers with r odd and $s - r$ even steps maps to a closed walk in G_d of weight $r\log(3/2) - (s-r)\log 2 = r\log 3 - s\log 2$, which is positive because $3^r > 2^s$ for negative cycles: for an element $x < 0$ of the cycle the affine identity (2.4) gives $(2^s - 3^r)x = c_s(x) > 0$. For the maximum cycle mean, every edge weight is at most $\log(3/2)$, so no cycle has mean above $\log(3/2)$, and the loop attains it. We also computed the maximum cycle mean of G_d by Karp's algorithm for $2 \leq d \leq 10$ and found $\log(3/2)$ in every case (script `ch5c_mm.py`).

(iii) is (i) and (ii), and the J step version follows by applying (i) to the graph of J step walks, whose cycles are closed walks of G_d and include the loop at -1 traversed J times. □

Figure 6.1(a) draws G_5 with three of its positive cycles, the loop at -1 and the images of the cycles of T through -5 and -17, whose means are 0.405, 0.039 and 0.006. In all G_5 has 91 simple cycles of positive weight among its 30 176 simple cycles (script `ch5c_mm.py`). By Theorem 2.8 a closed walk in G_d is determined by its parity word, and it is the trace of the periodic point of T with that word; so the positive cycles are the traces of cycles of T on negative rationals with odd denominator, such as $-19/11 \mapsto -23/11 \mapsto -29/11 \mapsto -38/11$, whose trace is the cycle $7 \to 27 \to 9 \to 14 \to 7$, and the three drawn are the traces of the cycles through negative integers. The theorem turns the obstruction that Chapter 13 will describe, the point -1 of the exceptional set E of 2 adic integers with infinite coefficient stopping time (Definition 2.22), into the language of optimization: the residue graph has a positive cycle, so no potential on residues can be a Lyapunov function, exactly as a shortest path problem with a negative cycle, here for the edge lengths $-w$, has no feasible potential. The cycles of T on the negative integers are not a curiosity here; they are the simplest witnesses of why residue arithmetic cannot certify the positive integers. A certificate would have to beat the loop at -1 at every modulus, and the loop has mean $\log(3/2)$ whatever the modulus.

6.3 What a pointwise potential must read

Residues modulo 2^d read d bits of x. The chains of Proposition 4.14 show that the number of bits a potential must read is unbounded, and the following proposition makes the requirement quantitative.

Proposition 6.4 *(Surplus along chains). Suppose* $V(x) = \log x + \varphi(x)$ *satisfies* $V(S(x)) < V(x)$ *for every odd* $x \geq 3$. *Then for every* $j \geq 2$ *and odd* $q \geq 1$,

$$\varphi(2^j q - 1) - \varphi(2 \cdot 3^{j-1} q - 1) \;>\; (j-1) \log \frac{3}{2}.$$

In particular φ *is unbounded above or below, and* $\sup \varphi - \inf \varphi \geq (j-1)\log(3/2)$ *for every* j.

Proof. By Proposition 4.14 the first $j-1$ strides from $x = 2^j q - 1$ are $x_i = 3^i 2^{j-i} q - 1$, each with $k = 1$, and $\log(x_{i+1}/x_i) = \log\big((3 \cdot 2^{j-i-1} q \cdot 3^i - 1)/(3^i 2^{j-i} q - 1)\big) > \log(3/2)$, since $2(3^{i+1} 2^{j-i-1} q - 1) > 3(3^i 2^{j-i} q - 1)$. Summing the strict decreases $V(x_{i+1}) < V(x_i)$ over $i = 0, \ldots, j-2$ gives $\varphi(x_{j-1}) - \varphi(x_0) < -\sum_i \log(x_{i+1}/x_i) < -(j-1)\log(3/2)$. □

So a pointwise potential must give the integer $2^j q - 1$, whose binary expansion ends in j ones, a surplus of at least $(j-1)\log(3/2)$ over the integer $2 \cdot 3^{j-1} q - 1$ it becomes after its run of ascents. The surplus has to grow without bound with the number of trailing ones, and trailing ones are the simplest unbounded feature of the binary expansion. A potential that reads only boundedly many bits cannot do this, which is Theorem 6.3 again; a potential that reads the run length can, and the next section shows what happens when it does.

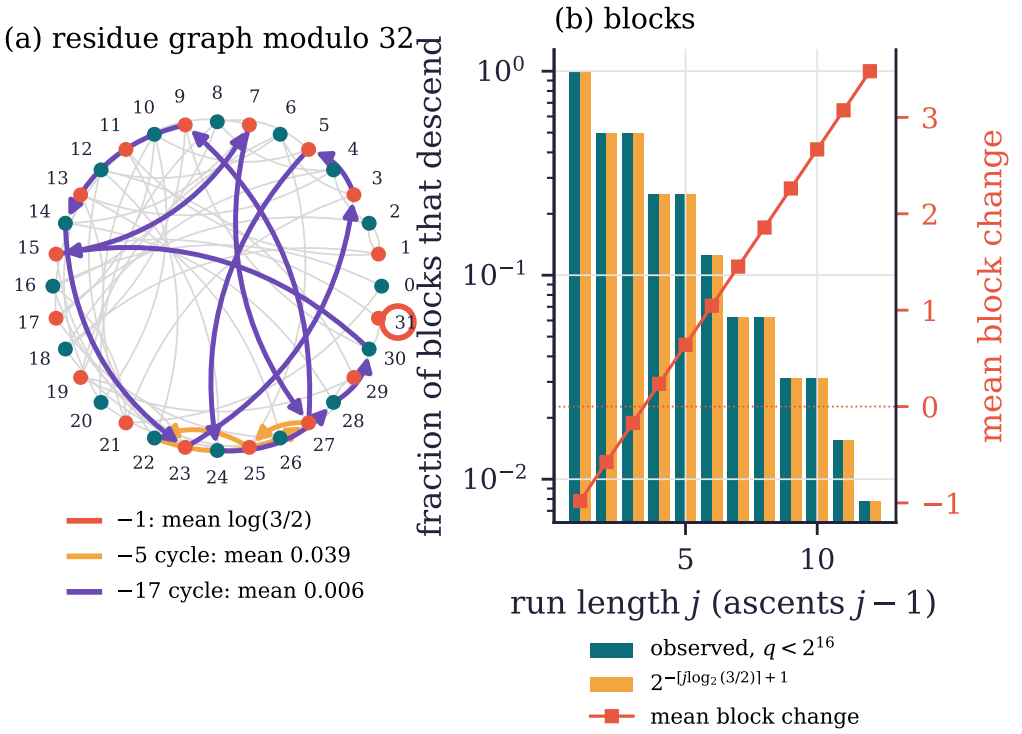


FIGURE 6.1. (a) The residue graph G_5 modulo 32, its odd and even residues in two shades and its edges in grey, and three of its 91 simple cycles of positive weight, the images of the cycles of T on the negative integers, highlighted with their mean weights. (b) Blocks: the fraction of blocks of run length j that descend, from an exact scan over $q < 2^{16}$ and from the formula $2^{-\lceil j\log_2(3/2)\rceil+1}$ of Proposition 6.5, with the mean block change $(j-1)\log(3/2)+\log 3-3\log 2$ under the model.

6.4 Block majorization

Let $x = 2^j q - 1$ with q odd and $j \geq 1$, so that $j - 1 = v_2(x+1) - 1$ is the number of ascending strides that follow. Majorize not over one stride but over the whole *block*: the $j-1$ ascents and the descent that ends them.

PROPOSITION 6.5 *(Block map). For $x = 2^j q - 1$ with q odd,*

$$S^{(j)}(x) = \frac{3^j q - 1}{2^v}, \qquad v = v_2(3^j q - 1) \geq 1, \tag{6.2}$$

and the block descends, $S^{(j)}(x) < x$, if and only if $v > j\log_2(3/2)$ (for $x \geq 3$). Under the model in which q is uniform on odd residues, v is Geometric$(\frac{1}{2})$, the fraction of blocks of run length j that descend is $2^{-\lceil j\log_2(3/2)\rceil+1}$, and the mean change of $\log x$ over the block is $(j-1)\log(3/2)+\log 3-3\log 2+O(1/x)$, which is negative for $j \leq 3$ and positive for $j \geq 4$.

Proof. By Proposition 4.14, $S^{(j-1)}(x) = 2 \cdot 3^{j-1}q - 1$, which is $\equiv 1 \pmod 4$; the next stride is $(3(2 \cdot 3^{j-1}q - 1) + 1)/2^k = (2 \cdot 3^j q - 2)/2^k = (3^j q - 1)/2^{k-1}$ with $k - 1 = v_2(3^j q - 1)$. Comparing with $x = 2^j q - 1$, write $3^j q - 1 = 2^v u$ with $u = S^{(j)}(x)$ odd; then $q(2^{j+v} - 3^j) = 2^v(2^j q - u) - 1$, and $2^j q - u$ is odd. If $2^{j+v} < 3^j$ the left side is negative, so $u > 2^j q > x$ and the block ascends. If $2^{j+v} > 3^j$ the left side is positive, so $2^j q - u$ is a positive odd integer, and $S^{(j)}(x) < x$ unless $2^j q - u = 1$, that is, unless $S^{(j)}(x) = x$ and x lies on a cycle consisting of a single block. By the theorem of Steiner [94] (see also [53, Theorem J]) the only such cycle is the trivial one, so for $x \geq 3$ the block descends exactly when $2^{j+v} > 3^j$, that is, when $v > j \log_2(3/2)$; the exact scan in `ch5c_mm.py` agrees for every $q < 2^{16}$ and $j \leq 12$. For uniform odd q, $3^j q - 1$ is uniform over even residues, so $\mathbb{P}(v \geq i) = 2^{-(i-1)}$. The threshold is the least integer v with $v > j \log_2(3/2)$, namely $\lceil j \log_2(3/2) \rceil$ when that is not an integer, which it never is for $j \geq 1$ since $\log_2(3/2)$ is irrational; this gives the fraction. The mean change is $(j-1)\log(3/2)$ from the ascents plus $\log 3 - \mathbb{E}[k] \log 2$ with $\mathbb{E}[k] = \mathbb{E}[v] + 1 = 3$ from the descent. □

The block is the natural MM unit: it is the shortest horizon over which the surrogate can be made exact, since within a run of ascents the halving exponents are forced to be 1 and the only random quantity is the valuation v at the end. Figure 6.1(b) shows the descent fractions, which the scan reproduces exactly for $j \geq 2$ (at $j = 1$ it gives $1 - 2^{-15}$, because it includes the fixed point $x = 1$), and the mean block change, which crosses zero between $j = 3$ and $j = 4$. Blocks of run length four or more, that is, with three or more ascents, are, on average, net ascents. Their rarity, 2^{-j} for run length j, is what makes the overall drift negative: averaging over j gives $\sum_j 2^{-j}((j-1)\log(3/2) + \log 3 - 3\log 2) = 2\log(3/4)$, two strides' worth of the Syracuse drift, as it must.

Block potentials. The block suggests the potential $V_c(x) = \log x + c\, v_2(x+1)$, which pays in advance for the coming ascents.

Proposition 6.6 *(Run length potential). Let $V_c(x) = \log x + c\, v_2(x+1)$ with $c > \log(3/2)$. Then V_c decreases at every ascending stride from $x > 1/(2(e^c - 3/2))$, hence at every ascending stride when $c > \log(5/3)$ and at all but finitely many when $c \leq \log(5/3)$. The descending strides at which it fails to decrease are exactly those $x \mapsto y$, with k halvings, that land on a y with $v_2(y+1) - 1 \geq (k\log 2 - \log 3 - \varepsilon(x))/c$. Under the model the fraction of odd x at which V_c fails is*

$$p(c) = \sum_{k \geq 2} 2^{-k}\, 2^{-\lceil (k\log 2 - \log 3)/c \rceil},$$

which is 0.2087 *at $c = 1$ and decreases to* 0.1431 *as $c \downarrow \log(3/2)$, and V_c fails at a positive fraction of odd integers for every c.*

Proof. At an ascending stride from x with $v_2(x+1) = j \geq 2$, by (4.6) $v_2(S(x) + 1) = j - 1$ and $S(x) = (3x+1)/2$, so V_c changes by $\log(3/2 + 1/(2x)) - c$, which

is negative exactly when $x > 1/(2(e^c - 3/2))$. Ascending strides start at $x \geq 3$, so this covers all of them when $c > \log(5/3)$; for $c \leq \log(5/3)$ the stride $3 \mapsto 5$ fails, and for $c \leq \log(11/7)$ also $7 \mapsto 11$ (the scan in `ch5c_mm.py` lists the ascending failures). At a descending stride with $k \geq 2$ halvings, $v_2(x+1) = 1$ and the change is $\log 3 - k \log 2 + \varepsilon(x) + c(v_2(y+1) - 1)$, which gives the stated criterion. Under the model k is Geometric$(\frac{1}{2})$, the descending strides being those with $k \geq 2$, and $v_2(y+1)$ is an independent Geometric$(\frac{1}{2})$, so that $\mathbb{P}(v_2(y+1) - 1 \geq t) = 2^{-t}$, and summing over k gives $p(c)$; the scan over odd $x < 2^{20}$ gives 0.2087 at $c = 1$, matching the model to four digits (Figure 6.2). Positivity of $p(c)$ is clear from the $k = 2$ term. The integers fail at a positive fraction as well: if $x \equiv 9 \pmod{16}$ then $k = 2$, $v_2(x+1) = 1$ and $y = (3x+1)/4 \equiv 3 \pmod 4$, so $v_2(y+1) \geq 2$ and V_c changes by more than $\log(3/4) + c = c - \log(4/3) > 0$, and these x are one eighth of the odd integers. (For $c \leq \log(3/2)$ the ascending strides, half of the odd integers, fail instead, by the first step of the proof.) □

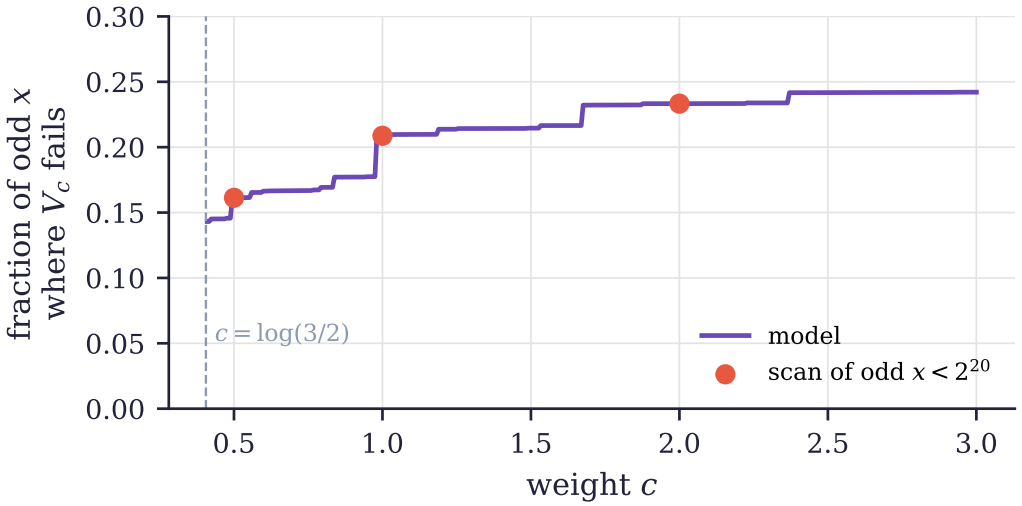


FIGURE 6.2. The fraction of odd integers at which the run length potential $V_c = \log x + c\, v_2(x+1)$ fails to decrease over one stride, as a function of the weight c: the model formula of Proposition 6.6 and an exact scan of odd $x < 2^{20}$. The infimum, 0.143, is approached as c decreases to $\log(3/2)$; below that value the ascending strides fail instead.

The run length potential does precisely what Proposition 6.4 demands, and it moves the obstruction rather than removing it. For $c > \log(5/3)$ it succeeds on every ascent, and fails on the descents that land on an integer whose next run of ascents is long enough to outweigh the halvings just gained. One can pay for that run too, with a potential that reads the run after the next descent, and then the failures move to the landing after that; the regress ends only with the potential that reads the whole orbit to the first descent below x, which is the stopping

time, and whose finiteness is the conjecture. This is the sense in which the MM approach locates the problem: each potential in this regress is a truncation of the stopping time at some depth of the binary expansion, and the conjecture asks, in effect, that the stopping time they truncate be finite.

6.5 Majorization in expectation

There is one setting in which the majorization works perfectly. Under the stride model of Definition 3.1, take expectations in (6.1): with k Geometric($\frac{1}{2}$) the surrogate decrease has mean $2\log 2 - \log 3 - 1/(3x)$, which is $|\log(3/4)|$ up to $O(1/x)$, so $\log x$ is a strict supermartingale and Proposition 3.16 gives absorption. Majorization in expectation is exactly the supermartingale method, and it proves the model's version of the conjecture in three lines. The pathwise version fails at the half of all odd integers where $k = 1$, and the preceding sections show that no repair by residues, by run lengths, or by any bounded depth of the expansion restores it. The difference between majorization in expectation and pathwise majorization is the difference between the model and the map, and it is the whole of the divergence half.

6.6 What the approach offers

We can now state exactly what majorization offers the Collatz problem.

For the cycle half it offers a reformulation: the cycle inequality is the bound of the multiplicative majorizer summed around the cycle, the window lemma is its minorizer, and the cycle half is the question of the tightness of that majorizer along closed orbits. This is a clean statement and nothing more; the arithmetic that would close the window is not supplied by majorization.

For the divergence half it offers three impossibility results, each with a proof: no residue potential (Theorem 6.3), because the negative cycles of T are positive cycles of every residue graph; no potential $\log x + \varphi(x)$ with bounded φ at all (Proposition 6.4); and no block potential of any weight (Proposition 6.6). And it offers one structural fact: the block decomposition is exact, each block's fate is decided by the single valuation $v_2(3^j q - 1)$, and under the model the blocks form a renewal process; the conjecture in descent form is the statement that the sequence of blocks, started at any n, reaches a block whose descent takes the orbit below n.

The conclusion is that majorization, pressed as hard as we are able to press it, converts the conjecture into an equivalent termination statement with an explicit and simple structure, and proves that the truncations of that statement examined here, to boundedly many bits of x or to a single run of ascents, are false. What it does not do is produce the unbounded depth argument. We believe that this is the right place to be stuck, because, as Chapter 13 will show, it is also where the automated termination provers and the 2 adic analysis of Proposition 13.1 are

stuck, and three independent routes ending at the same wall is some evidence that the wall is real.

Notes

The negative cycle criterion for the existence of potentials is the Bellman [9] and Ford [24] condition of shortest path theory, and the maximum cycle mean is computed by Karp's algorithm [42]; both are in any text on combinatorial optimization. The observation that the negative integer cycles obstruct residue based certificates is, in the language of 2 adic integers, the statement that -1, -5 and -17 lie in the exceptional set E of Chapter 13 (Proposition 13.1); the cycles themselves are classical, see Lagarias [53]. Its formulation as a positive cycle in the residue graph, the surplus proposition and the run length potential are ours. The block map itself is the classical identity behind Proposition 4.14 followed by one descending stride; Hercher [38, Lemma 8] cites Terras [99] for the identity. A cycle consisting of a single block is called a circuit, and Steiner [94] showed that the trivial cycle is the only circuit [53, Theorem J]; by the identity in the proof of Proposition 6.5, this is exactly the "if" half of the descent criterion for blocks, whose "only if" half is elementary.

Exercises

Exercise 6.7. Verify by hand that the -5 cycle of T maps to a closed walk of weight $2\log 3 - 3\log 2$ in G_d for every $d \geq 2$, and locate its three residues in G_5 in Figure 6.1(a).

Exercise 6.8. Show that the maximum cycle mean of G_d would drop below $\log(3/2)$ if the loop at -1 were deleted, compute the new maximum for $d = 5$, and explain why deleting it is not legitimate.

Exercise 6.9. Every odd $x \geq 3$ is uniquely $2^j q - 1$ with q odd, and the block map sends it to $S^{(j)}(x) = (3^j q - 1)/2^v$. Show that under the model in which the residue of q is uniform, the pair (j', v') of the next block is independent of the current pair, with j' and v' independent Geometric$(\frac{1}{2})$, so that, under the model, the sequence of blocks is a renewal process. Compute the mean number of strides per block and recover the Syracuse drift.

Exercise 6.10. Compute $p(c)$ of Proposition 6.6 for $c = 2$ and compare with the scan value 0.2333. Why does the failure fraction increase with c for large c?

Exercise 6.11. Define the depth two potential $V_c^{(2)}(x) = \log x + c\, v_2(x+1) + c\, v_2(y+1)$, where y is the first element of the orbit after the next descent. Show that it decreases at every stride except descents whose second landing has a long run, and compute the model fraction of failures. Describe the general depth h potential and its failure set.

Chapter 7

Switching MM

The previous chapters read the Collatz iteration as an optimization method: the odd step is an injection that raises the potential $\log x$, the halvings are the corrective descent, and Chapter 5 made precise how far the analogy with accelerated gradient methods can be pushed before it breaks. In this chapter and the next we run the analogy in the opposite direction. We take two pieces of the arithmetic, a single number and a parity pattern, and build from them an ordinary first order method for smooth optimization. The method is called Switching MM, abbreviated SMM. It uses the momentum coefficient

$$\beta \;=\; \frac{\log(3/2)}{\log 2} \;=\; \log_2 3 - 1 \;\approx\; 0.585,$$

and it applies the momentum only on odd iterations.

We want to be clear from the start about the logical status of the Collatz material here. The arithmetic motivates the constant and the schedule; it supplies no convergence theorem, and nothing in these two chapters bears on the Collatz conjecture. What the arithmetic does supply is a reproducible default that is fixed before any problem is seen. Whether that default is a good one is a question about optimization, and we answer it with optimization tools: an exact analysis on quadratics in this chapter, an exact noise analysis in Chapter 8, and the experiments reported there.

This chapter rests on two results, Lemma 7.5 and Theorem 7.6. For every differentiable f, two consecutive SMM steps starting from an even index are exactly two gradient steps, of lengths $(1+\beta)\alpha$ and α; on a quadratic the order of the two steps is immaterial, and the odd iterates lie on a cyclic gradient path as well. Switching MM is therefore a cyclic step size method in disguise, and on quadratics its behaviour can be read off from a polynomial of degree two. This reframing explains most of what the experiments of Chapter 8 show, including several things that a reading of SMM as thinned momentum would not predict. Chapter 9 uses the same identity to prove that, with step $1/L$, $\mathrm{SMM}_{\mathrm{det}}$ decreases f on alternate iterates for every L smooth f and contracts the distance to the minimizer at an explicit linear rate when f is also strongly convex (Proposition 9.5 and Theorem 9.9).

7.1 The practitioner's problem

Consider the unconstrained problem

$$\min_{x \in \mathbb{R}^d} f(x), \tag{7.1}$$

where f is differentiable and L smooth, meaning that ∇f is Lipschitz with constant L. Some of the methods we compare against also use a strong convexity modulus $\mu > 0$. Throughout we write α for the base step size, usually $\alpha = 1/L$.

Momentum methods add to the gradient step a multiple of the previous displacement. Polyak's heavy ball method [80] reads

$$x_{t+1} = x_t - \alpha \nabla f(x_t) + \beta (x_t - x_{t-1}), \tag{7.2}$$

and Nesterov's accelerated gradient method [72], in the constant momentum form we shall use, reads

$$y_t = x_t + \beta (x_t - x_{t-1}), \qquad x_{t+1} = y_t - \alpha \nabla f(y_t). \tag{7.3}$$

The difference is only where the gradient is evaluated. On a μ strongly convex, L smooth function with condition number $\kappa = L/\mu$, the classical tunings choose β from κ: for Nesterov's method with $\alpha = 1/L$ the textbook choice is

$$\beta^{\mathrm{opt}}(\kappa) = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}, \tag{7.4}$$

and heavy ball uses the square of this ratio together with a modified step $4/(\sqrt{L} + \sqrt{\mu})^2$ [81]. On quadratics these tunings deliver the accelerated iteration count of order $\sqrt{\kappa}\log(1/\varepsilon)$ instead of the order $\kappa \log(1/\varepsilon)$ of gradient descent. For Nesterov's method the same order holds on every μ strongly convex, L smooth function [71]; for heavy ball it does not, since Lessard, Recht and Packard [63] give a strongly convex function on which the tuned method settles into a limit cycle (Section 5.1).

The difficulty is that (7.4) moves with the instance. A practitioner who solves a family of related problems, say a sequence of regularized regressions along a path of penalty parameters, or the same model refitted as data arrive, faces two unattractive options.

1. *Retune per instance.* Estimate μ each time and set β by (7.4). This buys near optimal speed, but estimating the smallest eigenvalue of a Hessian is often as expensive as the optimization itself, and a poor estimate can make matters worse than no momentum at all.

2. *Use the default.* Set $\beta = 0.9$, the value that has become conventional in machine learning, for instance as the first moment coefficient of Adam [46]. This avoids tuning, but a high fixed momentum overshoots on well conditioned problems and, as Chapter 8 shows in detail, amplifies gradient noise by a factor $1/(1 - \beta) = 10$ in the slow directions.

The approach of this chapter is a third option: *fix the momentum coefficient in advance by a rule* that uses no information about the instance, and then study what the resulting method does and does not achieve.

7.2 Fixing the coefficient in advance

We give two derivations of the same number. Neither uses μ, κ, the dimension, or any property of f, which is the point; both rest on the two logarithms $\log 2$ and $\log(3/2)$.

7.2.1 The Collatz derivation

Recall from Chapter 2 the accelerated map $T(n) = n/2$ for even n and $T(n) = (3n+1)/2$ for odd n. On the logarithmic scale its increments take two kinds of value,

$$\log T(n) - \log n = \begin{cases} -\log 2, & n \text{ even}, \\ \log\left(\frac{3}{2} + \frac{1}{2n}\right) \in (0, \log 2), & n \text{ odd},\ n \geq 3, \end{cases}$$

so every halving lowers the potential by exactly $\log 2$ while every odd step raises it by slightly more than $\log(3/2)$. Grouping an odd step with the run of halvings that follows it gives the Syracuse map $S(n) = (3n+1)/2^k$ with $k = v_2(3n+1) \geq 1$, and the increment of the potential across this macro step obeys an exact identity.

Lemma 7.1 *(Log displacement identity). For every odd $n \geq 1$, with $k = v_2(3n+1)$,*

$$\log S(n) - \log n \;=\; \log 3 - k \log 2 + \varepsilon_n, \qquad \varepsilon_n = \log\Big(1 + \frac{1}{3n}\Big), \tag{7.5}$$

and $0 < \varepsilon_n \leq \log(4/3)$, with equality on the right only at $n = 1$. Moreover $\varepsilon_n < 1/(3n)$, so $\varepsilon_n \to 0$.

Proof. Write $3n + 1 = 3n(1 + 1/(3n))$, take logarithms and subtract $k \log 2$; this is (7.5). The function $n \mapsto \log(1 + 1/(3n))$ is positive and strictly decreasing on $n \geq 1$, so its maximum over odd n is attained at $n = 1$, where it equals $\log(4/3)$. The bound $\log(1+u) < u$ for $u > 0$ gives the last statement. □

The identity separates the macro step into a fixed ascent $\log 3$, a variable descent $k \log 2$, and a vanishing correction. In the accelerated map the ascent is split as $\log 3 = \log(3/2) + \log 2$: one halving is absorbed into the odd step itself, and the net ascent of the odd step tends to $\log(3/2)$ as $n \to \infty$. The asymptotic ratio of the magnitude of one ascent to the magnitude of one descent is therefore

$$\beta \;=\; \frac{\log(3/2)}{\log 2} \;=\; \log_2 3 - 1 \;=\; 0.584963\ldots \tag{7.6}$$

This is the number that parameterizes the two primitive moves of the parity switching rule, and it depends on nothing else. In the language of Chapter 5,

the odd step is the injection that overshoots and the halving is the correction, and β is the size of the overshoot measured in units of the correction. Under the Geometric($\frac{1}{2}$) law of Proposition 2.15 the mean macro step increment is $\log 3 - 2\log 2 = \log(3/4) < 0$, so the corrections win on average; the identity $\log(3/4) = (\beta - 1)\log 2$ expresses this as the statement $\beta < 1$.

7.2.2 The Kullback and Leibler derivation

The second derivation comes from elementary information theory. Let $A_1 \subset A_2 \subset A_3$ be nested sets with $|A_j| = j$, and let U_j be the uniform distribution on A_j, viewed as a distribution on A_3.

LEMMA 7.2 *(A ratio of divergences).* $D_{\mathrm{KL}}(U_1 \,\|\, U_2) = \log 2$ *and* $D_{\mathrm{KL}}(U_2 \,\|\, U_3) = \log(3/2)$. *Consequently*

$$\beta \;=\; \frac{D_{\mathrm{KL}}(U_2 \,\|\, U_3)}{D_{\mathrm{KL}}(U_1 \,\|\, U_2)}. \tag{7.7}$$

Proof. Since U_j is supported inside the support of U_{j+1}, we have $D_{\mathrm{KL}}(U_j \,\|\, U_{j+1}) = \sum_{a \in A_j} \frac{1}{j} \log\big(\frac{1/j}{1/(j+1)}\big) = \log \frac{j+1}{j}$. Take $j = 1$ and $j = 2$. □

A divergence $D_{\mathrm{KL}}(P \,\|\, Q)$ is the exponential rate at which an empirical distribution drawn from Q comes to look like P, by Sanov's theorem. Read this way, β is the ratio of the information rates of two elementary compressions, from three equally likely states to two and from two to one. We invoke no specific large deviation bound and no result below depends on this reading. We include it because it shows the same number from another side, not as independent evidence that the constant is natural: every ratio $\log \frac{m+1}{m} / \log \frac{n+1}{n}$ is a ratio of divergences of nested uniform laws, and, as Section 5.6 explains, the integers 2 and 3 do all the work in both derivations, so their agreement is forced.

7.2.3 What the derivations do not show

Two remarks guard against reading more into the derivations than is there.

REMARK 7.3 *(A definitional identity).* Put $L_* = \log 2$, the size of one halving and also the largest one step ascent of the potential under the accelerated map (attained only at $n = 1$), and $\mu_* = |\log(3/4)|$, the mean descent per macro step under the Geometric($\frac{1}{2}$) law. Then

$$\beta = \frac{L_* - \mu_*}{L_*} = 1 - \frac{\mu_*}{L_*},$$

which looks like the familiar expression of a momentum coefficient in terms of a condition number. It is not a discovery: it follows at once from $\log(3/4) = \log 3 - 2\log 2$ and the definition (7.6). The ratio $\kappa_* = L_*/\mu_* = 2.409$ is not a condition number: it compares a per step extremal quantity with a per macro step expectation, and both are increments of the potential, not curvature bounds,

ALGORITHM 7.1. $\mathrm{SMM}_{\mathrm{det}}$: Switching MM with deterministic parity injection

Input: ∇f, step size $\alpha = 1/L$, initial point $x_0 \in \mathbb{R}^d$, horizon T; fixed momentum $\beta = \log(3/2)/\log 2$

$x_{-1} \leftarrow x_0$
for $t = 0, 1, \ldots, T-1$ *do*
  if t is odd *then*
    $y_t \leftarrow x_t + \beta(x_t - x_{t-1})$ ▷ injection: extrapolate
    $x_{t+1} \leftarrow y_t - \alpha\nabla f(y_t)$
  else
    $x_{t+1} \leftarrow x_t - \alpha\nabla f(x_t)$ ▷ descent: plain gradient step
  end if
end for
return x_T

as Remark 5.4 explains for the closely related stride constants of Definition 5.3. No rate follows from it. The coincidence that $1/(1-\beta) = \kappa_*$ will reappear in Chapter 8 as the noise amplification of every step momentum at this β; it is the same algebra, not new evidence.

REMARK 7.4 *(Structure is not optimality).* A structural derivation makes a default reproducible and removes one tuning decision. It does not make the default optimal for any particular problem. By (7.4), the value 0.585 is the optimal Nesterov coefficient only at $\kappa = \big((1+\beta)/(1-\beta)\big)^2 = 14.58$; it is too small for more ill conditioned problems and too large for better conditioned ones.

7.3 THE ALGORITHM

The second ingredient taken from the arithmetic is the parity pattern. Under the standard map an odd step is always followed by a halving (Lemma 2.3), so an injection is never followed directly by another injection, although it may be followed by several corrections. Switching MM imposes the simplest pattern with this property, strict alternation, which in an orbit is a run of strides with $k = 1$, and it does so with the iteration counter: on odd iterations it extrapolates by $\beta(x_t - x_{t-1})$ and takes a gradient step from the extrapolated point, exactly as in (7.3); on even iterations it takes a plain gradient step. Algorithm 7.1 is the formal method, which we denote $\mathrm{SMM}_{\mathrm{det}}$.

We also consider an adaptive variant, $\mathrm{SMM}_{\mathrm{state}}$ (Algorithm 7.2), which injects momentum only when the most recent reduction of a merit gap was inadequate. It needs two extra inputs, a reference value $f_{\mathrm{ref}} \le \inf f$ and a threshold $\rho_{\mathrm{thr}} \in (0,1)$. We take $\rho_{\mathrm{thr}} = 0.356$, which is $1 - \kappa_*^{-1/2}$ for the ratio $\kappa_* = 2.409$ of Remark 7.3, the number in the rate column of Table 5.1; it is carried over from the Collatz constants untuned, and like that column it has no theorem behind it. We report

ALGORITHM 7.2. $\mathrm{SMM}_{\mathrm{state}}$: adaptive injection variant

Input: f, ∇f, $\alpha = 1/L$, x_0, T, reference value $f_{\mathrm{ref}} \le \inf f$, threshold $\rho_{\mathrm{thr}} \in (0,1)$, small $\epsilon > 0$; $\beta = \log(3/2)/\log 2$

$x_{-1} \leftarrow x_0$; $\quad g_{\mathrm{prev}} \leftarrow \max(f(x_0) - f_{\mathrm{ref}}, \epsilon)$
for $t = 0, 1, \ldots, T-1$ *do*
  $g_t \leftarrow \max(f(x_t) - f_{\mathrm{ref}}, \epsilon)$ ▷ nonnegative merit gap
  if $t = 0$ *or* $g_t / g_{\mathrm{prev}} > \rho_{\mathrm{thr}}$ *then*
    $y_t \leftarrow x_t + \beta(x_t - x_{t-1})$; $\quad x_{t+1} \leftarrow y_t - \alpha \nabla f(y_t)$ ▷ progress too slow: inject
  else
    $x_{t+1} \leftarrow x_t - \alpha \nabla f(x_t)$ ▷ progress adequate: descend
  end if
  $g_{\mathrm{prev}} \leftarrow g_t$
end for
return x_T

$\mathrm{SMM}_{\mathrm{state}}$ as an empirical probe only; the one result that covers it is the analysis of arbitrary switching on quadratics in Chapter 9, where Corollary 9.8 proves, conditionally on Conjecture 9.7, that every switching rule is stable. The claim that SMM has no tuned momentum coefficient attaches to $\mathrm{SMM}_{\mathrm{det}}$, and all the theory below concerns $\mathrm{SMM}_{\mathrm{det}}$.

7.3.1 Relation to heavy ball and Nesterov

The injection step of $\mathrm{SMM}_{\mathrm{det}}$ is the Nesterov update (7.3) with fixed momentum β, and the descent step is gradient descent. Equivalently, $\mathrm{SMM}_{\mathrm{det}}$ is fixed momentum Nesterov at $\beta = 0.585$ with the extrapolation switched off on every other iteration. Heavy ball (7.2) injects at every iteration and evaluates the gradient before extrapolating; when we speak of every step momentum we mean either of (7.2) and (7.3) applied at every iteration with the same fixed β.

The restriction to injection on alternate steps is unusual. Every classical inertial method injects at every step, and Section 7.6 shows that in the noiseless setting the restriction costs a constant factor in speed. The reason for keeping it is the noise analysis of Chapter 8: alternation caps the amplification of gradient noise uniformly in β.

Figure 7.1 shows the mechanism on a tilted two dimensional quadratic with $\kappa = 8$. The injection steps of $\mathrm{SMM}_{\mathrm{det}}$ push the iterate along the valley floor, and the objective decreases faster than under gradient descent. After 28 steps from $x_0 = (1.3, 0.9)$ the objective is 8.5×10^{-5} under gradient descent and 7.4×10^{-6} under $\mathrm{SMM}_{\mathrm{det}}$. The adaptive rule of $\mathrm{SMM}_{\mathrm{state}}$ fires on 24 of the 28 steps on this instance, because the progress per step along the slow direction rarely beats the threshold ρ_{thr}, and it behaves almost like every step Nesterov: it ends

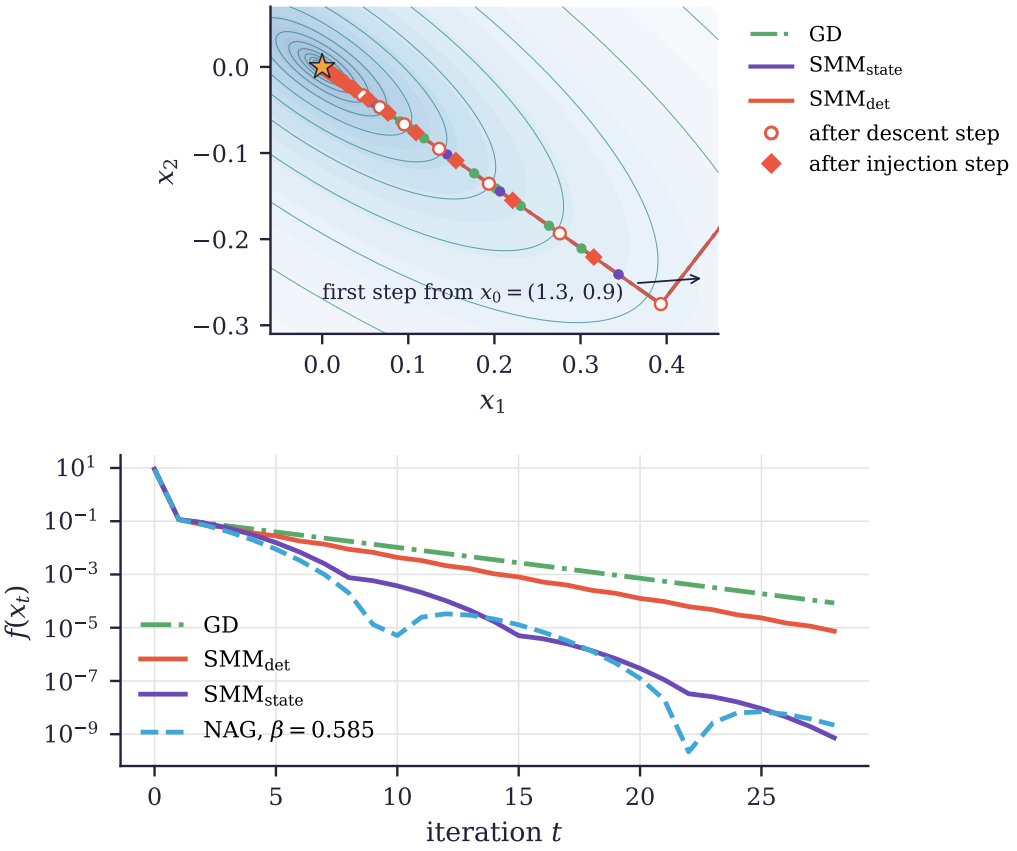


FIGURE 7.1. SMM$_{\text{det}}$ on a tilted two dimensional quadratic with $\kappa = 8$ and step $1/L$. Top: contours and the first iterates of gradient descent, SMM$_{\text{det}}$ and SMM$_{\text{state}}$; open circles mark the points reached after descent steps and filled diamonds those after injection steps. Bottom: objective value on a log scale over 28 iterations, with every step Nesterov at the same β for comparison.

at 7.3×10^{-10}, against 2.2×10^{-9} for every step Nesterov at the same β (script `ch6_trajectory.py`). In the noiseless setting more injection is better; Chapter 8 explains why the conclusion reverses under noise.

7.3.2 *What the method needs*

SMM$_{\text{det}}$ requires the smoothness constant L, for the step size, and nothing else. Heavy ball and Nesterov in their textbook strongly convex tunings need both L and μ. The fixed momentum baselines of Chapter 8 need L and a chosen β, and the point of the structural derivation is that SMM does not leave that choice to the user. Memory cost is one extra vector, the previous iterate, the same as for heavy ball and Nesterov and less than for adaptive methods such as Adam, which keep two moment vectors.

TABLE 7.1. Candidate fixed momentum coefficients. The column κ gives the condition number at which β equals the optimal Nesterov coefficient; the last column is the slow mode variance amplification of every step momentum relative to gradient descent.

β	Origin	κ where optimal	$1/(1-\beta)$
0.500	round number	9.0	2.00
0.585	$\log_2(3/2)$, Collatz and KL ratio	14.6	2.41
0.667	$2/3$	25.0	3.00
0.900	machine learning default [46]	361	10.0

7.4 Candidate fixed momenta

The value 0.585 is one of several natural fixed coefficients. Table 7.1 lists the four that the experiments of Chapter 8 compare, with the condition number at which each would be the optimal Nesterov coefficient (7.4), namely $\kappa = \big((1+\beta)/(1-\beta)\big)^2$, and the slow mode noise amplification $1/(1-\beta)$ of every step momentum derived in Chapter 8.

None of these is uniformly best, and the experiments confirm it: 0.585 does well at small κ, 0.667 is best on a mildly ill conditioned logistic regression, and 0.9 wins on strongly ill conditioned quadratics. What distinguishes 0.585 is not performance but provenance: it is the only entry that the arithmetic of the Collatz map fixes, rather than a round number or a convention.

7.5 Related work

Fixed momentum methods. SMM belongs to the tradition of inertial first order methods that runs from Polyak's heavy ball [80] through Nesterov's accelerated gradient [72]; Polyak's monograph [81] remains the clearest source for their analysis on quadratics. The ordinary differential equation view of Su, Boyd and Candès [95] and the high resolution refinement of Shi, Du, Jordan and Su [90] explain acceleration as the interplay of an inertial term and a gradient correction. That picture describes every step momentum with β tending to 1 as the step shrinks. SMM has no inertial limit: whatever the value of $\beta < 1$, its continuous time limit is a first order gradient flow, because two of its steps are two gradient steps (Lemma 7.5) and it carries one gradient of memory rather than a velocity (Proposition 9.11).

Low tuning design. A recurring theme in modern optimization is the search for methods whose defaults work across instances. Iterate averaging in stochastic approximation [82, 83, 89] and adaptive moment methods such as Adam [46] are two successful examples. SMM is adjacent in spirit, but takes its fixed coefficient from a structural source instead of from an adaptive estimate.

Arithmetic structure in optimization. Number theoretic structure enters

optimization in several established ways. Fibonacci search, which Kiefer [45] proved minimax optimal for unimodal functions, uses ratios of Fibonacci numbers, and in the limit the golden ratio, to place function evaluations. Quasi Monte Carlo methods [73] use low discrepancy sequences and lattice rules built from modular arithmetic to derandomize integration and stochastic optimization. In both cases arithmetic decides *where to evaluate*. SMM differs in where the arithmetic enters: it decides *how much inertial memory to retain*, and when.

Switching systems. Alternating between two linear updates is a switched linear system, and the stability of such systems under arbitrary or periodic switching is the subject of Liberzon's monograph [64]. Periodic switching is the benign case: stability is decided by the spectral radius of the product of the matrices over one period, and Section 7.6 computes that product in closed form.

Majorization and minimization. The descent half of SMM is a majorization minimization step in the sense of Lange, Hunter and Yang [60] and Hunter and Lange [40]: for L smooth f the gradient step minimizes the quadratic upper bound $f(x_t) + \nabla f(x_t)^\top (x - x_t) + \frac{L}{2}\|x - x_t\|^2$. Mairal [66] develops incremental versions of this principle for large scale learning. At the level of the optimizer the MM reading supplies the descent inequality behind Proposition 9.5: $\mathrm{SMM}_{\mathrm{det}}$ is the momentum MM scheme of Section 9.3, with the majorization point extrapolated on odd iterations. In the Collatz setting of Chapter 4 the tangent line of $\log x$ bounds the increment of the potential at every step of T_0, but the map is not an MM iteration in the usual sense, and Proposition 4.19 shows that the bound cannot certify descent. That reading of the Collatz map is the origin of the name Switching MM.

Step size schedules. The connection that turns out to matter most is invisible in the momentum reading of the method. Varying the step size periodically in gradient descent on quadratics goes back to Richardson [86]; Young [110] showed how to choose the steps from the roots of Chebyshev polynomials, and Golub and Varga [27] connected these schedules to second order iterations of heavy ball type. Much more recently Altschuler and Parrilo [2] showed that carefully designed nonconstant step size schedules, the silver stepsize schedule, accelerate gradient descent on smooth convex problems beyond what any constant step achieves. Theorem 7.6 places SMM in this family.

7.6 Switching MM is a cyclic step size method

We now analyse $\mathrm{SMM}_{\mathrm{det}}$ exactly. Write $G_h(x) = x - h\nabla f(x)$ for a gradient step of length h. The starting observation is an identity valid for every differentiable f.

Lemma 7.5 *(The extrapolated point is a long gradient step). Let t be even. Then $x_{t+1} = G_\alpha(x_t)$, and the extrapolated point at the following odd iteration is*

$$y_{t+1} = x_{t+1} + \beta(x_{t+1} - x_t) = x_t - (1+\beta)\alpha\nabla f(x_t) = G_{(1+\beta)\alpha}(x_t). \tag{7.8}$$

Consequently $x_{t+2} = G_\alpha\big(G_{(1+\beta)\alpha}(x_t)\big)$*, and in the form that exhibits the momentum,*

$$x_{t+2} = x_{t+1} - \beta\alpha\nabla f(x_t) - \alpha\nabla f(y_{t+1}). \tag{7.9}$$

Proof. At the even iteration t the update is a plain gradient step, so $x_{t+1} - x_t = -\alpha\nabla f(x_t)$. Substituting this displacement into the extrapolation gives $y_{t+1} = x_{t+1} - \beta\alpha\nabla f(x_t) = x_t - (1+\beta)\alpha\nabla f(x_t)$. The odd update is $x_{t+2} = y_{t+1} - \alpha\nabla f(y_{t+1})$, which gives both displayed forms. □

So the even iterates $x_0, x_2, x_4, \dots$ of $\mathrm{SMM}_{\mathrm{det}}$ are *exactly* every second iterate of gradient descent with the periodic step sizes $(1+\beta)\alpha, \alpha, (1+\beta)\alpha, \alpha, \dots$, whose intermediate iterates are the extrapolated points y_{t+1}, for any f. The momentum buffer of $\mathrm{SMM}_{\mathrm{det}}$ never holds anything except the last gradient, so the extrapolation simply lengthens that gradient step by the factor $1+\beta$. The odd iterates x_{t+1} are an observation point off this path. On a quadratic they lie on a second cyclic path, because gradient steps commute.

Theorem 7.6 *(Two step equivalence on quadratics). Let* $f(x) = \frac{1}{2}x^\top Ax - b^\top x$ *with* A *symmetric positive definite and minimizer* $x^* = A^{-1}b$*, let* $\alpha > 0$ *and* $\beta \geq 0$*, and let* (x_t) *be generated by Algorithm 7.1 with step* α*. For every even* $t \geq 0$*,*

$$x_{t+2} = G_{(1+\beta)\alpha}(x_{t+1}) = G_{(1+\beta)\alpha}\big(G_\alpha(x_t)\big), \tag{7.10}$$

that is, two $\mathrm{SMM}_{\mathrm{det}}$ *steps from an even index are a gradient step of length* α *followed by a gradient step of length* $(1+\beta)\alpha$*, and every iterate of* $\mathrm{SMM}_{\mathrm{det}}$ *is an iterate of this cyclic schedule. In terms of the error* $e_t = x_t - x^*$*,*

$$e_{t+2} = (I - \alpha A)\big(I - (1+\beta)\alpha A\big)e_t, \qquad t \text{ even}. \tag{7.11}$$

Proof. For a quadratic, $\nabla f(x) = A(x - x^*)$, so a gradient step of any length h acts on the error linearly: $G_h(x) - x^* = (I - hA)(x - x^*)$. By Lemma 7.5, for even t,

$$e_{t+1} = (I - \alpha A)e_t, \qquad y_{t+1} - x^* = \big(I - (1+\beta)\alpha A\big)e_t,$$
$$e_{t+2} = (I - \alpha A)(y_{t+1} - x^*),$$

and therefore $e_{t+2} = (I - \alpha A)(I - (1+\beta)\alpha A)e_t$, which is (7.11). The matrices $I - \alpha A$ and $I - (1+\beta)\alpha A$ are both polynomials in A, hence commute, so also

$$e_{t+2} = \big(I - (1+\beta)\alpha A\big)(I - \alpha A)e_t = \big(I - (1+\beta)\alpha A\big)e_{t+1},$$

which says $x_{t+2} = G_{(1+\beta)\alpha}(x_{t+1})$. Combined with $x_{t+1} = G_\alpha(x_t)$ this is (7.10). By induction on even t, starting from x_0, every iterate of $\mathrm{SMM}_{\mathrm{det}}$ is an iterate of gradient descent with steps $\alpha, (1+\beta)\alpha, \alpha, (1+\beta)\alpha, \dots$. □

The same fact can be seen through the state space description that the switching systems literature would use. Restricted to an eigenvector of A with eigenvalue

λ, and writing $a = \alpha\lambda$, the state (e_t, e_{t-1}) evolves by the descent matrix D on even steps and the Nesterov matrix N on odd steps, where

$$D = \begin{pmatrix} 1-a & 0 \\ 1 & 0 \end{pmatrix}, \qquad N = \begin{pmatrix} (1-a)(1+\beta) & -(1-a)\beta \\ 1 & 0 \end{pmatrix}.$$

The product over one period is

$$ND = \begin{pmatrix} (1-a)(1-(1+\beta)a) & 0 \\ 1-a & 0 \end{pmatrix},$$

a matrix of rank one with eigenvalues 0 and $(1-a)(1-(1+\beta)a)$. The zero eigenvalue records that the descent step forgets e_{t-1}, which is why the period map involves only e_t.

7.6.1 Consequences

Write

$$\rho(a) = (1-a)\big(1-(1+\beta)a\big) = 1-(2+\beta)a+(1+\beta)a^2 \tag{7.12}$$

for the two step multiplier on an eigendirection with $a = \alpha\lambda$.

Corollary 7.7 *(Two step contraction). Under the hypotheses of Theorem 7.6, for even t,*

$$\|e_{t+2}\| \le \max_{\lambda\in\operatorname{spec}(A)} |\rho(\alpha\lambda)|\ \|e_t\|,$$

with equality for a suitable e_t. The per step contraction factor on an eigendirection is $|\rho(a)|^{1/2}$.

Proof. The matrix in (7.11) is symmetric with eigenvalues $\rho(\alpha\lambda)$, so its operator norm is the largest of their absolute values, attained on the corresponding eigenvector. □

Corollary 7.8 *(Stability). Let $0 \le \beta < 1$. Then $|\rho(a)| < 1$ if and only if $0 < a < (2+\beta)/(1+\beta)$. In particular* $\mathrm{SMM}_{\det}$ *converges linearly on every strongly convex quadratic whenever $\alpha \le 1/L$, and in fact whenever $\alpha L < (2+\beta)/(1+\beta)$. For all $a > 0$, $\rho(a) \ge 1-(2+\beta)^2/(4(1+\beta))$, a minimum attained at $a = (2+\beta)/(2(1+\beta))$.*

Proof. From (7.12), $\rho(a) < 1$ if and only if $a\big((1+\beta)a-(2+\beta)\big) < 0$, that is $0 < a < (2+\beta)/(1+\beta)$. The condition $\rho(a) > -1$ reads $(1+\beta)a^2-(2+\beta)a+2 > 0$; the discriminant of this quadratic in a is $(2+\beta)^2-8(1+\beta) = \beta^2-4\beta-4$, which is negative for $0 \le \beta < 1$, so the inequality holds for every a. The minimum of the convex quadratic ρ is at its vertex. □

At $\beta = \log_2 3 - 1$ the stability limit is $(2+\beta)/(1+\beta) = 1+\log_3 2 = 1.631$ and the minimum of ρ is -0.054, attained at $a = 0.815$. The multiplier vanishes at $a = 1/(1+\beta) = 0.631$ and at $a = 1$: those eigendirections are annihilated in one period.

Corollary 7.9 *(Slow modes). As $a \to 0$, $\rho(a) = 1 - (2+\beta)a + O(a^2)$, and the per step contraction factor is $1 - \frac{2+\beta}{2}a + O(a^2)$. Thus in the slow directions* $\mathrm{SMM}_{\mathrm{det}}$ *behaves like gradient descent with the effective step*

$$m_{\mathrm{SMM}}\,\alpha, \qquad m_{\mathrm{SMM}} = \frac{2+\beta}{2} = 1.2925 \quad \text{at } \beta = \log_2 3 - 1.$$

Proof. Expand (7.12) and take the square root. □

The effective step is just the average of the two step lengths α and $(1+\beta)\alpha$, as it must be for a cyclic schedule in the slow modes. For comparison, every step momentum at fixed β, heavy ball or Nesterov, has in the overdamped slow regime a real dominant root $1 - a/(1-\beta) + O(a^2)$, so its effective step multiplier is

$$m_{\mathrm{HB}} = m_{\mathrm{NAG}} = \frac{1}{1-\beta} = 2.409 \quad \text{at } \beta = \log_2 3 - 1,$$

and $m = 10$ at $\beta = 0.9$. On a κ conditioned quadratic with $\alpha = 1/L$ the slowest mode has $a = 1/\kappa$, so the iteration count of $\mathrm{SMM}_{\mathrm{det}}$ is of order κ/m_{SMM}: SMM is *not* an accelerated method. It improves the constant of gradient descent by about 29 percent and is slower than every step momentum at the same β by the factor $m_{\mathrm{NAG}}/m_{\mathrm{SMM}} = 2.409/1.2925 = 1.86$ in the slow modes. The frequency ablation of Chapter 8 measures 32.85 against 16.45 iterations at $\kappa = 10$, a ratio of 2.00, close to this prediction. At $\kappa = 100$ it reports 293 against 167, a ratio of 1.75, but that parity count is censored by the horizon $T = 300$ of the experiment: 14 of the 20 parity runs have not reached the tolerance by T, and continued to convergence the parity schedule needs 318 iterations, a ratio of 1.90. A rerun of the same experiment with 200 replicates (script `ch6_ablation.py`) gives nearly the same censored means and leaves about two thirds of the parity runs short of the tolerance at T; run to convergence, the parity schedule needs about 315 iterations against 166, again a ratio of 1.90, close to the prediction.

7.6.2 Comparison with every step Nesterov

Figure 7.2 plots the per step contraction factor against a for gradient descent, every step Nesterov at $\beta = 0.585$ and $\beta = 0.9$, and $\mathrm{SMM}_{\mathrm{det}}$. Selected values are collected in Table 7.2. Every step Nesterov has characteristic polynomial $z^2 - (1-a)(1+\beta)z + (1-a)\beta$; its roots are complex, with modulus $\sqrt{(1-a)\beta}$, as soon as $(1-a)(1+\beta)^2 < 4\beta$, which at $\beta = 0.585$ means $a > 0.069$. In that underdamped region the comparison with $\mathrm{SMM}_{\mathrm{det}}$ is explicit: $|\rho(a)|^{1/2} < \sqrt{(1-a)\beta'}$ for Nesterov with coefficient β' if and only if $1 - (1+\beta)a < \beta'$, that is

$$a > \frac{1-\beta'}{1+\beta},$$

when $(1+\beta)a \le 1$; when $(1+\beta)a > 1$ the condition is $(1+\beta)a - 1 < \beta'$, which for $\beta' = \beta$ holds for every $a < 1$. With $\beta' = \beta = 0.585$ the crossover is at $a = 0.262$,

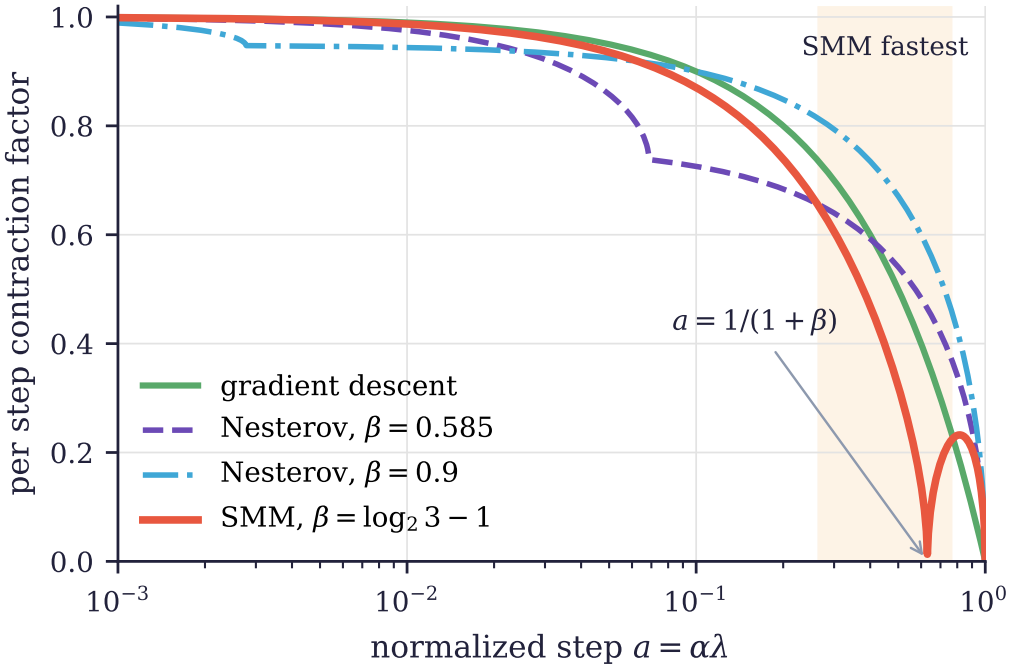


FIGURE 7.2. Per step contraction factor on an eigendirection against $a = \alpha\lambda$ (log scale) for gradient descent, every step Nesterov at $\beta = 0.585$ and $\beta = 0.9$, and $\mathrm{SMM}_{\mathrm{det}}$. For $\mathrm{SMM}_{\mathrm{det}}$ the factor is $|(1-a)(1-(1+\beta)a)|^{1/2}$; it vanishes at $a = 1/(1+\beta)$ and $a = 1$. SMM contracts fastest of the four for $0.262 < a < 0.774$ and is beaten by every step momentum in the slow modes.

and with $\beta' = 0.9$ at $a = 0.063$. Against gradient descent, $\mathrm{SMM}_{\mathrm{det}}$ contracts faster for every $a < 2/(2+\beta) = 0.774$.

The picture is consistent with the experiments. On very well conditioned problems every $a = \alpha\lambda$ lies in $[1/\kappa, 1]$, and the slowest mode $a = 1/\kappa$ falls in the window $0.262 < a < 0.774$ where $\mathrm{SMM}_{\mathrm{det}}$ contracts fastest; indeed at $\kappa = 2$ it is the fastest method tested. As κ grows the slow modes dominate and every step momentum takes over. Nothing here depends on the Collatz origin of β; it is the algebra of the polynomial ρ.

7.6.3 Chebyshev steps and silver steps

Theorem 7.6 invites comparison with the best cyclic schedule of period two. On a quadratic with spectrum in $[\mu, L]$, the two step polynomial $(1-h_1\lambda)(1-h_2\lambda)$ that minimizes the maximum modulus over $[\mu, L]$ is the scaled Chebyshev polynomial of degree two; its roots are

$$\lambda_{\pm} = \frac{L+\mu}{2} \pm \frac{L-\mu}{2\sqrt{2}},$$

TABLE 7.2. Per step contraction factor on an eigendirection, $\beta = 0.585$ unless stated. Values computed from the characteristic roots.

$a = \alpha\lambda$	GD	NAG, $\beta = 0.585$	NAG, $\beta = 0.9$	$\mathrm{SMM}_{\mathrm{det}}$
0.01	0.9900	0.9754	0.9439	0.9871
0.04	0.9600	0.8920	0.9295	0.9482
0.10	0.9000	0.7256	0.9000	0.8703
0.25	0.7500	0.6624	0.8216	0.6729
0.50	0.5000	0.5408	0.6708	0.3221

so $h_\pm = 1/\lambda_\pm$, and the worst case two step factor is $1/T_2\big((\kappa+1)/(\kappa-1)\big)$ [27, 110]. As $\kappa \to \infty$ the ratio of the two steps tends to $(1+1/\sqrt{2})/(1-1/\sqrt{2}) = (\sqrt{2}+1)^2 = 5.83$, the long step approaches $6.83/L$, and the per step factor is $1 - 4/\kappa + O(\kappa^{-2})$. We computed, for example, a per step factor of 0.709 at $\kappa = 10$ and 0.9615 at $\kappa = 100$, against 0.870 and 0.9871 for $\mathrm{SMM}_{\mathrm{det}}$ with $\alpha = 1/L$. SMM is thus a very conservative member of the period two family: its step ratio $1 + \beta = 1.585$ is fixed and its long step is short, so that the scheme is stable with no knowledge of μ. The Chebyshev schedule needs μ to place its roots and is fragile when the estimate is wrong; this is the cyclic step size version of the retuning problem of Section 7.1.

Periodic Chebyshev schedules of a fixed period p cannot change the order of the iteration count: since $T_p(1 + 2/\kappa) = 1 + 2p^2/\kappa + O(\kappa^{-2})$, the per step factor is $1 - 2p/\kappa + O(\kappa^{-2})$. Altschuler and Parrilo [2] showed that suitably designed nonperiodic schedules, built recursively and involving the silver ratio $1 + \sqrt{2}$, do accelerate gradient descent for smooth convex optimization. We mention this only to locate SMM: it is a fixed period two schedule and inherits the order κ iteration count of gradient descent, with a better constant.

7.6.4 Nonquadratic objectives

For a general f, Lemma 7.5 still says that the even iterates follow the cyclic schedule exactly, but with the long step *first*: $x_{t+2} = G_\alpha(G_{(1+\beta)\alpha}(x_t))$. The second statement of Theorem 7.6, that $x_{t+2} = G_{(1+\beta)\alpha}(x_{t+1})$ so that the odd iterates also lie on a cyclic gradient path, used commutation and is only approximately true. The defect is of third order in the step size.

PROPOSITION 7.10 *(Order of the defect). Suppose $\nabla^2 f$ is Lipschitz with constant M. For even t, with $h = x_{t+1} - x_t = -\alpha\nabla f(x_t)$,*

$$x_{t+2} - G_{(1+\beta)\alpha}(x_{t+1}) = \alpha\big[(1+\beta)\nabla f(x_{t+1}) - \beta\nabla f(x_{t+1} - h) - \nabla f(x_{t+1} + \beta h)\big],$$

and

$$\left\| x_{t+2} - G_{(1+\beta)\alpha}(x_{t+1}) \right\| \le \frac{M}{2}\, \beta(1+\beta)\, \alpha^3\, \|\nabla f(x_t)\|^2.$$

The bracket vanishes identically when ∇f is affine.

Proof. By (7.9), $x_{t+2} = x_{t+1} - \beta\alpha\nabla f(x_t) - \alpha\nabla f(y_{t+1})$, and $x_t = x_{t+1} - h$, $y_{t+1} = x_{t+1} + \beta h$. Subtracting $G_{(1+\beta)\alpha}(x_{t+1}) = x_{t+1} - (1+\beta)\alpha\nabla f(x_{t+1})$ gives the displayed identity. Write $u = x_{t+1}$ and $H = \nabla^2 f(u)$. Taylor's theorem with the Lipschitz Hessian gives $\nabla f(u-h) = \nabla f(u) - Hh + r_1$ and $\nabla f(u+\beta h) = \nabla f(u) + \beta Hh + r_2$ with $\|r_1\| \le \frac{M}{2}\|h\|^2$ and $\|r_2\| \le \frac{M}{2}\beta^2\|h\|^2$. The bracket equals $-\beta r_1 - r_2$, whose norm is at most $\frac{M}{2}(\beta+\beta^2)\|h\|^2$. Multiply by α and use $\|h\| = \alpha\|\nabla f(x_t)\|$. □

We checked the exact statement numerically on a regularized logistic regression in five dimensions: the even iterates of $\mathrm{SMM}_{\mathrm{det}}$ agree with the cyclic schedule $(1+\beta)\alpha, \alpha$ to 10^{-16}, while the reversed order agrees only up to a defect that shrinks with the gradient, from 0.13 at the first period to 2×10^{-8} by the fourth. For the purposes of this book the point is simple: on smooth problems SMM is a period two step size schedule with steps $(1+\beta)\alpha$ and α, observed at two different points of each period. For μ strongly convex and L smooth f the same identity gives the linear rate of Theorem 9.9 in Chapter 9.

Remark 7.11 *(What the reframing changes).* A natural first reading describes the parity schedule as momentum that is active half the time, a buffer that is thinned. Lemma 7.5 shows that the buffer is never more than one gradient deep, so SMM carries no long memory at all. This is the structural reason why its noise amplification, computed in Chapter 8, stays bounded however close β is to one: a cyclic step size method can only amplify noise through its step lengths, and these are bounded by $(1+\beta)\alpha < 2\alpha$.

Notes

The two derivations of β are ours; the log displacement identity is Theorem 4.7 of Chapter 4 in Syracuse form, and the Kullback and Leibler reading is elementary. The coefficient $\log(3/2)/\log 2 = \log_2 3 - 1$ is the fractional part of $\log_2 3$, whose continued fraction is the subject of Chapter 10; the present chapter uses only its numerical value.

Polyak's book [81] treats heavy ball and Chebyshev iterations on quadratics side by side, and is the best place to see why a period two cyclic step size method and a momentum method are close relatives. For switched linear systems see Liberzon [64]; for MM algorithms, Lange, Hunter and Yang [60] and the tutorial [40].

Theorem 7.6, Lemma 7.5 and Proposition 7.10 are, as far as we know, new in this book, and we regard them as the correct way to think about the method.

Exercises

Exercise 7.12. Show that $D_{\mathrm{KL}}(U_j \,\|\, U_{j+1}) = \log\big((j+1)/j\big)$ for nested uniform distributions, and that $D_{\mathrm{KL}}(U_{j+1} \,\|\, U_j) = \infty$. Show that

$$\log_3 2 = D_{\mathrm{KL}}(U_1 \,\|\, U_2)/D_{\mathrm{KL}}(U_1 \,\|\, U_3) = 1/(1+\beta),$$

and that $D_{\mathrm{KL}}(U_1 \,\|\, U_3) = D_{\mathrm{KL}}(U_1 \,\|\, U_2) + D_{\mathrm{KL}}(U_2 \,\|\, U_3)$ is not one of the consecutive divergences. Write the stability limit $1 + \log_3 2$ of Corollary 7.8 in terms of β alone, and explain why its expression in divergences carries no information about stability.

Exercise 7.13. For odd n with $v_2(3n+1) = 1$, show that $\log S(n) - \log n$ decreases to $\log(3/2)$ and is at most $\log(5/3)$ for $n \geq 3$. Relate the excess over $\log(3/2)$ to ε_n in Lemma 7.1.

Exercise 7.14. Let $0 \leq \beta < 1$, $\beta' \in (0,1)$ and $0 < a < 1$. Show that every step Nesterov with coefficient β' is underdamped on an eigendirection if and only if $(1-a)(1+\beta')^2 < 4\beta'$, and that in this regime $\mathrm{SMM}_{\mathrm{det}}$ has the smaller per step contraction factor if and only if $|1-(1+\beta)a| < \beta'$, that is $(1-\beta')/(1+\beta) < a < (1+\beta')/(1+\beta)$. Verify the crossovers 0.262 and 0.063 quoted in the text, and show that for $\beta' \geq \beta$ the upper limit is not reached by any $a < 1$.

Exercise 7.15. Replace the parity schedule by injection every third step: descend at $t \equiv 0, 1 \pmod 3$ and inject at $t \equiv 2 \pmod 3$. Show that, as in Lemma 7.5, the extrapolated point is a gradient step of length $(1+\beta)\alpha$ from the previous iterate for every differentiable f, so that the iterates at $t \equiv 0 \pmod 3$ follow gradient descent with the periodic steps $\alpha, (1+\beta)\alpha, \alpha$, and that the slow mode effective step multiplier is $(3+\beta)/3$. What happens if injection occurs at two consecutive steps? Show that the second extrapolated point then involves gradients at two different points.

Exercise 7.16. Take $f(x) = x^4/4$ on $\mathbb{R}$. Starting from $x_0 = 1$ with $\alpha = 1/3$ and $\beta = \log_2 3 - 1$, compute two steps of $\mathrm{SMM}_{\mathrm{det}}$ and compare x_2 with $G_\alpha(G_{(1+\beta)\alpha}(x_0))$ and with $G_{(1+\beta)\alpha}(G_\alpha(x_0))$. Check the bound of Proposition 7.10 with $M = 6 \sup |x|$ over the relevant interval.

Exercise 7.17. Show that the period two schedule minimizing $\max_{\lambda \in [\mu, L]} |(1-h_1\lambda)(1-h_2\lambda)|$ has $h_2/h_1 \to (\sqrt{2}+1)^2$ as $\kappa \to \infty$, and that its worst case per step factor is $1 - 4/\kappa + O(\kappa^{-2})$. For which κ is the SMM ratio $h_2/h_1 = 1+\beta$ optimal?

Chapter 8

Noise, Variance and Experiments

Chapter 7 showed that in the noiseless setting the parity schedule of Switching MM is not an acceleration device. On a quadratic, $\mathrm{SMM}_{\mathrm{det}}$ is a period two step size schedule with steps α and $(1+\beta)\alpha$ (Theorem 7.6); in the slow directions it improves the constant of gradient descent by the factor $m_{\mathrm{SMM}} = (2+\beta)/2$ and is beaten by every step momentum, whose factor is $1/(1-\beta)$. If SMM has a reason to exist, that reason must lie elsewhere. A natural hope is that the parity schedule acts as a *variance filter*: momentum recycles gradient shocks through its memory, and a method that uses its memory only half the time ought to recycle less noise.

This chapter puts that hope to an exact test. After recalling the classical AR(2) analysis of every step momentum under additive gradient noise, we prove the corresponding exact result for $\mathrm{SMM}_{\mathrm{det}}$, Theorem 8.4. In the slow modes SMM amplifies noise, relative to gradient descent at the same step, by $(2+2\beta+\beta^2)/(2+\beta)$: this is 1.359 at the Collatz value of β and never exceeds $5/3$, whereas every step momentum amplifies by the unbounded factor $1/(1-\beta)$. Compared at matched speed in the slow modes rather than at matched step, SMM has no advantage over gradient descent in those modes, although it is quieter in the fast ones; it is quieter than every step momentum largely because it moves more cautiously. Its genuine virtue is robustness: its noise behaviour hardly depends on β, which is the precise content of the claim that SMM is a low tuning method. The second half of the chapter presents experiments with SMM, rerun for this book, and reconciles them quantitatively with the theory.

8.1 A model of gradient noise

We work with the strongly convex quadratic $f(x) = \frac{1}{2}x^\top Ax - b^\top x$, A symmetric positive definite with eigenvalues in $[\mu, L]$, and suppose that the algorithm does not see ∇f but a noisy version

$$g(x) = \nabla f(x) + \xi, \tag{8.1}$$

where a fresh ξ is drawn at every call, independent of the past, with mean zero and covariance $\sigma^2 I$. This is the simplest model of a stochastic gradient; it ignores the state dependence and anisotropy of minibatch noise, but isolates how a method turns independent shocks into fluctuations of the iterate. Every method we consider decouples along the eigenvectors of A. Fix an eigenvalue λ, write

$a = \alpha\lambda$ for the normalized step, e_t for the component of $x_t - x^*$ and ξ_t for the component of the noise drawn at iteration t; the ξ_t are independent with mean zero and variance σ^2. With $\alpha = 1/L$, a ranges over $[1/\kappa, 1]$. Each scalar error has a variance that converges to a limit, the *stationary variance*, and since

$$\mathbb{E}\big[f(x_t) - f(x^*)\big] = \tfrac{1}{2}\sum_{\lambda} \lambda\, \mathbb{E}[e_t(\lambda)^2], \tag{8.2}$$

these limits determine the noise floor that remains after the deterministic transient has died away.

For gradient descent, $e_{t+1} = (1-a)e_t - \alpha\xi_t$, an autoregression of order one, and the stationary variance is

$$V_{\mathrm{GD}}(a) = \frac{\alpha^2\sigma^2}{1-(1-a)^2} = \frac{\alpha^2\sigma^2}{a(2-a)}, \qquad 0 < a < 2. \tag{8.3}$$

In the slow modes $V_{\mathrm{GD}} \approx \alpha^2\sigma^2/(2a) = \alpha\sigma^2/(2\lambda)$: the floor is proportional to the step. A longer step is faster and noisier in exactly compensating amounts, and every comparison below must account for this.

8.2 Every step momentum as an autoregression

8.2.1 The stationary variance of an AR(2) process

The classical tool is an elementary lemma on autoregressions of order two; see Brockwell and Davis [14] for the general theory.

Lemma 8.1 *(Stationary variance of an AR(2) process). Let $e_{t+1} = a_1 e_t + a_2 e_{t-1} + \varepsilon_t$ with ε_t independent, mean zero, variance s^2, and ε_t independent of (e_t, e_{t-1}). Suppose both roots of $z^2 - a_1 z - a_2$ lie in the open unit disc, which holds if and only if*

$$|a_2| < 1, \qquad 1 - a_1 - a_2 > 0, \qquad 1 + a_1 - a_2 > 0. \tag{8.4}$$

Then $\mathrm{Var}(e_t)$ *converges, from any initial condition with finite second moments, to*

$$\gamma_0 = \frac{(1-a_2)\, s^2}{(1+a_2)\big((1-a_2)^2 - a_1^2\big)}. \tag{8.5}$$

Proof. In state form the covariance of (e_t, e_{t-1}) obeys an affine recursion whose linear part has spectral radius equal to the square of that of the companion matrix, which is less than one; so it converges geometrically to its unique fixed point. In stationarity let $\gamma_k = \mathrm{Cov}(e_t, e_{t-k})$. Multiplying the recursion by e_t, by e_{t-1} and by e_{t+1} and taking expectations gives the Yule and Walker equations

$$\gamma_1 = a_1\gamma_0 + a_2\gamma_1, \qquad \gamma_2 = a_1\gamma_1 + a_2\gamma_0, \qquad \gamma_0 = a_1\gamma_1 + a_2\gamma_2 + s^2.$$

The first gives $\gamma_1 = a_1\gamma_0/(1-a_2)$. Substituting the second into the third, $\gamma_0(1 - a_2^2) = a_1(1+a_2)\gamma_1 + s^2 = a_1^2(1+a_2)\gamma_0/(1-a_2) + s^2$. Multiplying by $1-a_2$ and collecting terms gives $\gamma_0(1+a_2)\big((1-a_2)^2 - a_1^2\big) = (1-a_2)s^2$, which is (8.5). The conditions (8.4) are the classical stability triangle for a real quadratic. □

8.2.2 Heavy ball

The noisy heavy ball iteration $x_{t+1} = x_t - \alpha g(x_t) + \beta(x_t - x_{t-1})$ gives, on an eigendirection,

$$e_{t+1} = (1 + \beta - a)\, e_t - \beta\, e_{t-1} - \alpha\xi_t, \tag{8.6}$$

an AR(2) process with $a_1 = 1 + \beta - a$, $a_2 = -\beta$ and $s^2 = \alpha^2\sigma^2$.

Proposition 8.2 *(Heavy ball under additive noise). Let* $0 \le \beta < 1$ *and* $0 < a < 2(1+\beta)$. *The stationary variance of* (8.6) *is*

$$V_{\mathrm{HB}}(a) = \frac{\alpha^2\sigma^2\,(1+\beta)}{(1-\beta)\,a\,\big(2(1+\beta)-a\big)}. \tag{8.7}$$

As $a \to 0$, $V_{\mathrm{HB}}(a) \sim \alpha^2\sigma^2/\big(2a(1-\beta)\big)$, *so* $V_{\mathrm{HB}}/V_{\mathrm{GD}} \to 1/(1-\beta)$.

Proof. The conditions (8.4) read $\beta < 1$, $1 - a_1 - a_2 = a > 0$ and $1 + a_1 - a_2 = 2(1+\beta) - a > 0$, which are the hypotheses. In (8.5), $1 - a_2 = 1 + \beta$, $1 + a_2 = 1 - \beta$, and

$$(1 - a_2)^2 - a_1^2 = (1+\beta)^2 - (1+\beta-a)^2 = a\big(2(1+\beta) - a\big).$$

This is (8.7). For small a the last factor is $2(1+\beta) + O(a)$, the factors $1 + \beta$ cancel, and the limit follows on dividing by $V_{\mathrm{GD}} \sim \alpha^2\sigma^2/(2a)$. □

The factor $1/(1-\beta)$ is 2.409 at $\beta = \log_2 3 - 1$, 3 at $\beta = 2/3$ and 10 at $\beta = 0.9$. Its origin is easy to see. In the slow regime the characteristic polynomial has a fast root near β and a slow root $1 - a/(1-\beta) + O(a^2)$. A shock $\alpha\xi_t$ enters the velocity, and by the time the fast root has dissipated it the position has moved by $\alpha\xi_t(1 + \beta + \beta^2 + \cdots) = \alpha\xi_t/(1-\beta)$. Momentum thus multiplies the effective step by $m = 1/(1-\beta)$, speeding the slow mode up by m, and multiplies each shock by m, hence its variance by m^2; the net effect on the floor is $m^2/m = m$. We return to this bookkeeping in Section 8.4.

8.2.3 Nesterov

Constant momentum Nesterov evaluates the noisy gradient at the extrapolated point: $y_t = x_t + \beta(x_t - x_{t-1})$, $x_{t+1} = y_t - \alpha g(y_t)$. On an eigendirection,

$$e_{t+1} = (1-a)(1+\beta)\, e_t - (1-a)\beta\, e_{t-1} - \alpha\xi_t. \tag{8.8}$$

Proposition 8.3 *(Nesterov under additive noise). Let* $0 \le \beta < 1$ *and* $0 < a \le 1$. *The stationary variance of* (8.8) *is*

$$V_{\mathrm{NAG}}(a) = \frac{\alpha^2\sigma^2\,(1+\beta-a\beta)}{a\,(1-\beta+a\beta)\,\big(2(1+\beta)-a(1+2\beta)\big)}. \tag{8.9}$$

As $a \to 0$, $V_{\mathrm{NAG}}/V_{\mathrm{GD}} \to 1/(1-\beta)$, *the same slow mode amplification as heavy ball; at* $a = 1$, $V_{\mathrm{NAG}} = V_{\mathrm{GD}} = \alpha^2\sigma^2$.

Proof. Here $a_1 = (1-a)(1+\beta)$ and $a_2 = -(1-a)\beta$, so $|a_2| < 1$, $1 - a_1 - a_2 = 1 - (1-a) = a > 0$ and $1 + a_1 - a_2 = 1 + (1-a)(1+2\beta) > 0$; the lemma applies. Now $1 - a_2 = 1 + \beta - a\beta$, $1 + a_2 = 1 - \beta + a\beta$, and

$$(1-a_2)^2 - a_1^2 = (1 - a_2 - a_1)(1 - a_2 + a_1) = a\big(2(1+\beta) - a(1+2\beta)\big)$$

since $1 - a_2 + a_1 = 1 + \beta - a\beta + (1-a)(1+\beta) = 2(1+\beta) - a(1+2\beta)$. Substituting in (8.5) gives (8.9), and both limits are immediate. □

Heavy ball and Nesterov agree in the slow modes and differ for larger a: at $a = 1$ the Nesterov error is just the last shock, while heavy ball still carries momentum.

8.3 The exact variance of Switching MM

For $\mathrm{SMM}_{\mathrm{det}}$ the coefficients alternate with the parity of t, so the error is not a stationary AR(2) process. The way through is Lemma 7.5, which holds for any gradient oracle and therefore for the noisy one: starting from an even index, the extrapolated point is a single long step from x_t using the *same* noisy gradient,

$$y_{t+1} = x_{t+1} + \beta(x_{t+1} - x_t) = x_t - (1+\beta)\alpha\, g(x_t).$$

The noise drawn at the even step is therefore used twice, once in the descent step and once more, scaled by β, through the extrapolation. Two steps of SMM consume two independent shocks with weights $(1+\beta)\alpha$ and α, exactly as noisy gradient descent with the cyclic steps $(1+\beta)\alpha$ and α would.

Theorem 8.4 *(Stationary variance of* $\mathrm{SMM}_{\mathrm{det}}$*). Let f be the quadratic of Section 8.1, with the noisy oracle* (8.1)*, let $0 \le \beta < 1$, and fix an eigendirection with $a = \alpha\lambda \in (0, 1]$. Let*

$$\rho = (1-a)\big(1 - (1+\beta)a\big).$$

(i) For every even t the error of $\mathrm{SMM}_{\mathrm{det}}$ *satisfies the exact two step recursion*

$$e_{t+2} = \rho\, e_t + w_t, \qquad w_t = -(1-a)(1+\beta)\alpha\,\xi_t - \alpha\,\xi_{t+1}, \tag{8.10}$$

where the w_t, t even, are independent, independent of e_t, with mean zero and $\mathrm{Var}(w_t) = \alpha^2\sigma^2\big[(1-a)^2(1+\beta)^2 + 1\big]$.

(ii) As $t \to \infty$ through even values, $\mathrm{Var}(e_t)$ *converges to*

$$V_{\mathrm{even}} = \frac{\alpha^2\sigma^2\big[(1-a)^2(1+\beta)^2 + 1\big]}{1 - \rho^2}, \tag{8.11}$$

and through odd values to

$$V_{\mathrm{odd}} = (1-a)^2\, V_{\mathrm{even}} + \alpha^2\sigma^2. \tag{8.12}$$

(iii) The phase averaged stationary variance $V_{\mathrm{SMM}} = \frac{1}{2}(V_{\mathrm{even}} + V_{\mathrm{odd}})$ *is the rational function*

$$\frac{V_{\mathrm{SMM}}(a)}{\alpha^2\sigma^2} = \frac{Q(a,\beta)}{1-\rho^2}, \tag{8.13}$$

with

$$\begin{aligned} Q(a,\beta) &= (2+2\beta+\beta^2) - (1+\beta)(2+3\beta)\,a \\ &\quad + (1+\beta)(1+3\beta)\,a^2 - \beta(1+\beta)\,a^3, \\ 1-\rho^2 &= a\big(2+\beta-(1+\beta)a\big)\big(2-(2+\beta)a+(1+\beta)a^2\big). \end{aligned}$$

Proof. (i) Let t be even. The descent step gives $e_{t+1} = (1-a)e_t - \alpha\xi_t$, so $e_{t+1} - e_t = -ae_t - \alpha\xi_t$ and the extrapolated point has error

$$e_{t+1} + \beta(e_{t+1} - e_t) = \big(1-(1+\beta)a\big)e_t - (1+\beta)\alpha\xi_t.$$

The injection step evaluates a noisy gradient at this point, with fresh noise ξ_{t+1}, and multiplies the error by $1-a$:

$$e_{t+2} = (1-a)\big[\big(1-(1+\beta)a\big)e_t - (1+\beta)\alpha\xi_t\big] - \alpha\xi_{t+1},$$

which is (8.10). The error e_t is a function of e_0 and of $\xi_0, \dots, \xi_{t-1}$, while w_t is a function of ξ_t and ξ_{t+1}; for distinct even t the pairs (ξ_t, ξ_{t+1}) are disjoint. This gives the independence claims, and the variance of w_t is the sum of the variances of its two independent terms.

(ii) By (i), $\mathrm{Var}(e_{t+2}) = \rho^2\,\mathrm{Var}(e_t) + \mathrm{Var}(w_t)$ for even t. By Corollary 7.8, $|\rho| < 1$ whenever $0 < a < (2+\beta)/(1+\beta)$, in particular for $a \in (0,1]$, so this affine recursion converges geometrically to its fixed point (8.11). For odd times, $e_{t+1} = (1-a)e_t - \alpha\xi_t$ with ξ_t independent of e_t, so $\mathrm{Var}(e_{t+1}) = (1-a)^2\,\mathrm{Var}(e_t) + \alpha^2\sigma^2$, and (8.12) follows by letting $t \to \infty$ through even values.

(iii) We have $V_{\mathrm{SMM}} = \frac{1}{2}\big[(1+(1-a)^2)V_{\mathrm{even}} + \alpha^2\sigma^2\big]$. The factorization of $1-\rho^2$ comes from $1-\rho = a(2+\beta-(1+\beta)a)$ and $1+\rho = 2-(2+\beta)a+(1+\beta)a^2$. Over the common denominator $1-\rho^2$ the numerator is $\frac{1}{2}\big[(1+(1-a)^2)\big((1-a)^2(1+\beta)^2+1\big)+1-\rho^2\big]$, and expanding this polynomial in a gives $Q(a,\beta)$. We checked the expansion with a computer algebra system, which confirms that (8.13) agrees identically with the definition of V_{SMM}. □

The even subsequence is thus an autoregression of order one: the second state variable of every step momentum has disappeared, because the descent step forgets e_{t-1}, and the noise of the even step is amplified by $1+\beta < 2$ and no more.

8.3.1 The slow mode limit

Corollary 8.5 *(Slow mode amplification). As* $a \to 0$*, both* aV_{even} *and* aV_{odd} *converge to*

$$\alpha^2\sigma^2\,\frac{(1+\beta)^2+1}{2(2+\beta)} = \alpha^2\sigma^2\,\frac{1+\beta+\beta^2/2}{2+\beta}.$$

Consequently

$$\lim_{a\to 0}\frac{V_{\mathrm{SMM}}(a)}{V_{\mathrm{GD}}(a)} = V(\beta) := \frac{2+2\beta+\beta^2}{2+\beta} = \beta + \frac{2}{2+\beta}. \tag{8.14}$$

Proof. From (8.11), $1-\rho^2 = a(2+\beta)\cdot 2 + O(a^2)$ and the numerator tends to $\alpha^2\sigma^2\big[(1+\beta)^2+1\big]$, which gives the limit for aV_{even}. By (8.12), $aV_{\mathrm{odd}} = (1-a)^2 aV_{\mathrm{even}} + a\alpha^2\sigma^2$ has the same limit. Since $(1+\beta)^2+1 = 2+2\beta+\beta^2$ and $aV_{\mathrm{GD}} \to \alpha^2\sigma^2/2$, the ratio tends to $(2+2\beta+\beta^2)/(2+\beta)$. Finally $(2+\beta)\beta + 2 = 2+2\beta+\beta^2$. □

The formula has a transparent reading in terms of the cyclic step size picture. Write the two steps of a period as $h_1\alpha$ and $h_2\alpha$ with $h_1 = 1+\beta$ and $h_2 = 1$. In the slow modes one period contracts the error by $1-(h_1+h_2)a$ and injects noise of variance $(h_1^2+h_2^2)\alpha^2\sigma^2$, so the stationary variance is approximately $(h_1^2+h_2^2)\alpha^2\sigma^2/\big(2(h_1+h_2)a\big)$. Dividing by the gradient descent value $\alpha^2\sigma^2/(2a)$ gives

$$V(\beta) = \frac{(h_1^2+h_2^2)/2}{(h_1+h_2)/2} = \frac{\text{mean square step}}{\text{mean step}}. \tag{8.15}$$

For every step momentum the same bookkeeping, with effective step $m = 1/(1-\beta)$ and shocks multiplied by m, gave $m^2/m = m$. The difference between the two methods is that SMM never lengthens any single step by more than the factor $1+\beta < 2$.

Proposition 8.6 *(Uniform bound in β). The function $V(\beta) = \beta + 2/(2+\beta)$ is strictly increasing on $[0,1]$, with $V(0) = 1$ and $V(1) = 5/3$. Hence*

$$1 \le V(\beta) < \tfrac{5}{3} \qquad \textit{for all } 0 \le \beta < 1,$$

whereas the every step amplification $1/(1-\beta)$ tends to infinity as $\beta \to 1$.

Proof. $V'(\beta) = 1 - 2/(2+\beta)^2$, and $(2+\beta)^2 \ge 4 > 2$ for $\beta \ge 0$, so $V' > 0$. The values at the endpoints are $0+1 = 1$ and $1+2/3 = 5/3$. □

At $\beta = \log_2 3 - 1$ the slow mode amplification is $V = 1.359$, against 2.409 for every step momentum at the same coefficient. At $\beta = 0.9$ it is 1.590, against 10. The slow modes are the worst case, as the next result shows: the bound $V(\beta)$ holds on every eigendirection, not only in the limit.

Proposition 8.7 *(The slow mode bound is uniform in a). For $0 \le \beta < 1$ and every $a \in (0,1]$,*

$$V_{\mathrm{GD}}(a) \le V_{\mathrm{SMM}}(a) \le V(\beta)\, V_{\mathrm{GD}}(a) < \tfrac{5}{3}\, V_{\mathrm{GD}}(a),$$

with $V_{\mathrm{SMM}}(a) = V_{\mathrm{GD}}(a)$ only when $\beta = 0$ or $a = 1$, and $V_{\mathrm{SMM}}(a) = V(\beta)V_{\mathrm{GD}}(a)$ only when $\beta = 0$.

Proof. Write $\rho = (1-a)(1-(1+\beta)a)$ as before. A direct computation from (8.3) and (8.13), which we verified by computer algebra, gives

$$\frac{V(\beta)V_{\mathrm{GD}}(a) - V_{\mathrm{SMM}}(a)}{\alpha^2\sigma^2} = \frac{\beta(1+\beta)\,\big(-P(a,\beta)\big)}{(2-a)(2+\beta)\big(2+\beta-(1+\beta)a\big)(1+\rho)},$$

where

$$P(a,\beta) = (2+\beta)a^3 - (7+2\beta-\beta^2)a^2 + (10+\beta-2\beta^2)a - (6-\beta^2).$$

For $a \in (0,1]$ all four factors of the denominator are positive: $2-a>0$, $2+\beta>0$, $2+\beta-(1+\beta)a \geq 1 > 0$, and $1+\rho>0$ by Corollary 7.8. It remains to show $P<0$. Since $(2+\beta)a^3 \leq (2+\beta)a^2$ for $a \leq 1$,

$$P(a,\beta) \leq q(a) := -(5+\beta-\beta^2)a^2 + (10+\beta-2\beta^2)a - (6-\beta^2).$$

The quadratic q has negative leading coefficient, because $5+\beta-\beta^2>0$ on $[0,1]$, and discriminant

$$(10+\beta-2\beta^2)^2 - 4(5+\beta-\beta^2)(6-\beta^2) = 5\beta^2 - 4\beta - 20,$$

which is at most $5-20<0$ on $[0,1]$. Hence $q<0$ for every real a, so $P<0$, and the displayed difference is nonnegative, and positive when $\beta>0$. The last inequality is Proposition 8.6. For the first, the same computation gives

$$\frac{V_{\mathrm{SMM}}(a) - V_{\mathrm{GD}}(a)}{\alpha^2\sigma^2} = \frac{\beta(1+\beta)(1-a)^2\big((1-a)^2+1\big)}{a(2-a)\big(2+\beta-(1+\beta)a\big)(1+\rho)},$$

whose denominator is positive for $a \in (0,1]$, as above, and whose numerator is nonnegative and vanishes only when $\beta=0$ or $a=1$ (script `ch7_extra.py`). □

Remark 8.8 *(A tempting but incomplete argument).* A natural first argument for the parity schedule runs as follows. Since the momentum term is active on only half of the iterations, the slow mode amplification $1/(1-\beta)$ should be roughly halved, to 1.20 at $\beta = 0.585$. This is close to the exact 1.359 but not equal to it. At $\beta = 0.9$ it predicts 5, against the exact 1.590; as $\beta \to 1$ it predicts an unbounded amplification, against the exact bound $5/3$. The heuristic treats the parity schedule as a thinned version of a long memory, whereas Lemma 7.5 shows that SMM has no long memory at all: its buffer never holds more than the last gradient. What the schedule controls is not how often noise enters a buffer but how long each step is, and the correct amplification is the ratio (8.15) of mean square to mean step length.

8.4 What the variance buys: efficiency at matched speed

A smaller noise floor at a fixed step size is not by itself an advantage. Any method can lower its floor by moving more slowly, and gradient descent with a shorter step is the simplest example. One fair comparison is between methods run at the same speed in the slow modes, which determine the iteration count. They do not determine the floor of the objective: by (8.2) an eigendirection contributes $\frac{1}{2}\lambda V_{\mathrm{GD}} = \alpha\sigma^2/\big(2(2-a)\big)$ for gradient descent, between $\alpha\sigma^2/4$ and $\alpha\sigma^2/2$, so every eigendirection contributes to the floor on the same scale. The comparison below concerns each slow eigendirection, not the total floor. Recall from Corollary 7.9 and the discussion following it the effective step multipliers

$$m_{\mathrm{GD}} = 1, \qquad m_{\mathrm{HB}} = m_{\mathrm{NAG}} = \frac{1}{1-\beta}, \qquad m_{\mathrm{SMM}} = \frac{2+\beta}{2}.$$

Gradient descent with step $m\alpha$ has slow mode variance $m\alpha^2\sigma^2/(2a)$, that is m times the floor at step α. So the natural figure of merit is the amplification divided by the speed multiplier, the *noise floor at matched speed*.

Proposition 8.9 *(Noise floor at matched speed). In the slow mode limit, the noise floor at matched speed relative to gradient descent is* 1 *for heavy ball and for Nesterov, and for* $\mathrm{SMM}_{\mathrm{det}}$ *it is*

$$\frac{V(\beta)}{m_{\mathrm{SMM}}} = \frac{2(2+2\beta+\beta^2)}{(2+\beta)^2} = 1 + \frac{\beta^2}{(2+\beta)^2} \;\geq\; 1.$$

It equals 1.051 *at* $\beta = \log_2 3 - 1$ *and increases to* $10/9$ *as* $\beta \to 1$. *More generally, for any period two schedule with steps* $h_1\alpha$ *and* $h_2\alpha$ *the ratio is* $\frac{1}{2}(h_1^2+h_2^2)\big/\big(\frac{1}{2}(h_1+h_2)\big)^2 \geq 1$, *with equality only for equal steps.*

Proof. For heavy ball and Nesterov the amplification and the speed multiplier are both $1/(1-\beta)$ (Propositions 8.2 and 8.3). For SMM, divide (8.14) by $(2+\beta)/2$ and use $2(2+2\beta+\beta^2) = (2+\beta)^2 + \beta^2$. The function $\beta^2/(2+\beta)^2$ is increasing with value $1/9$ at $\beta = 1$. The general statement is the inequality between the mean square and the square of the mean, which follows from (8.15). □

This is what the variance theory says. In the slow modes of a quadratic, gradient descent, heavy ball and Nesterov all sit on the same efficiency frontier, trading speed for noise one for one. SMM sits slightly *behind* that frontier, by five percent at the Collatz coefficient, because a cyclic schedule with unequal steps wastes a little variance relative to a constant schedule of the same mean. Gradient descent with step $(1+\beta/2)\alpha$ would be as fast as SMM in the slow modes and slightly quieter in them, but noisier in the fast modes, where its longer step costs more than the parity schedule does (Section 8.10). So the lower floor of SMM at a fixed α, compared with every step momentum, comes largely from its smaller effective step, 1.29α against 2.41α; it is not a free lunch.

What then is the virtue of SMM? It is the uniformity of Propositions 8.6 and 8.7. For every step momentum the cost of a badly chosen large β is unbounded: the floor at step $1/L$ is multiplied by $1/(1-\beta)$, which is 10 at the popular default and 100 at $\beta = 0.99$. For SMM the floor at step $1/L$ lies between that of gradient descent and $5/3$ of it for *every* $\beta \in [0,1)$, and on every eigendirection. Equally, the speed multiplier lies in $[1, 3/2)$. Switching from $\beta = 0.585$ to $\beta = 0.9$ changes the slow mode floor of every step momentum by a factor 4.15 and that of SMM by 17 percent. SMM is a conservative method: it can neither gain much nor lose much by the choice of β, and it is stable for every $\beta < 1$ at the step $1/L$ (Corollary 7.8). That is the precise sense in which it is a low tuning default.

Remark 8.10 *(Fixed step versus fixed budget).* In practice methods are compared at the same $\alpha = 1/L$ and the same number of iterations, which is the protocol of the experiments below. Under that protocol a method that is both faster and noisier can lose on the final gap, and a method that is slower and quieter can win, without either being more efficient in the sense of Proposition 8.9. The experiments therefore measure a mixture of speed and floor. The theory tells us how to unmix them.

8.5 Numerical illustration and simulation check

Table 8.1 gives the exact ratio of the stationary variance of each method to that of gradient descent at the same a, computed from (8.3), (8.7), (8.9) and (8.13). The SMM ratio stays below $V(\beta)$, as Proposition 8.7 asserts, and falls to 1 at $a = 1$; the Nesterov ratio falls from $1/(1-\beta)$ to 1; the heavy ball ratio decreases only moderately, from $1/(1-\beta)$ to $(1+\beta)/\big((1-\beta)(1+2\beta)\big)$ at $a = 1$, which is 1.760 at $\beta = \log_2 3 - 1$, about 30 percent below the slow mode value.

Figure 8.1 plots the two slow mode amplification curves against β, with simulated ratios from 4000 independent chains of the scalar recursions at $a = 0.01$; at the five values of β shown, the simulated and exact ratios differ by at most 0.006 for SMM and 0.004 for Nesterov (script `ch7_variance.py`). A single long chain of $\mathrm{SMM}_{\mathrm{det}}$ at $a = 0.02$ and $\beta = \log_2 3 - 1$, run for 2×10^6 steps with unit noise, gave a sample variance of 34.16, against the exact phase averaged value 34.11 and the slow mode approximation $V(\beta)/(2a) = 33.97$ (script `smm_sim.py`).

8.6 Beyond period two: the gain weight theorem

Proposition 8.9 compared $\mathrm{SMM}_{\mathrm{det}}$ with gradient descent at matched speed and found a small penalty, $1 + \beta^2/(2+\beta)^2$. Is that penalty an accident of the particular schedule, or would some other pattern of injections, of period three or four, or with injection on several residues, beat gradient descent? This is problem P6 of Chapter 15. The answer is no, and the reason is a single structural fact: in any first order method the noise enters exactly where the gradient enters.

TABLE 8.1. Stationary variance relative to gradient descent at the same normalized step $a = \alpha\lambda$, from the exact formulas. The last row gives the slow mode limits $1/(1-\beta)$ and $V(\beta)$.

	$\beta = 0.585$			$\beta = 0.9$		
a	NAG	HB	$\mathrm{SMM}_{\mathrm{det}}$	NAG	HB	$\mathrm{SMM}_{\mathrm{det}}$
0.001	2.406	2.409	1.358	9.908	9.998	1.589
0.01	2.372	2.405	1.355	9.153	9.976	1.584
0.04	2.264	2.391	1.343	7.284	9.904	1.568
0.1	2.074	2.364	1.319	5.142	9.757	1.535
0.25	1.707	2.289	1.256	2.909	9.366	1.445
0.5	1.314	2.145	1.146	1.648	8.636	1.267
1	1.000	1.760	1.000	1.000	6.786	1.000
$a \to 0$	2.409	2.409	1.359	10.00	10.00	1.590

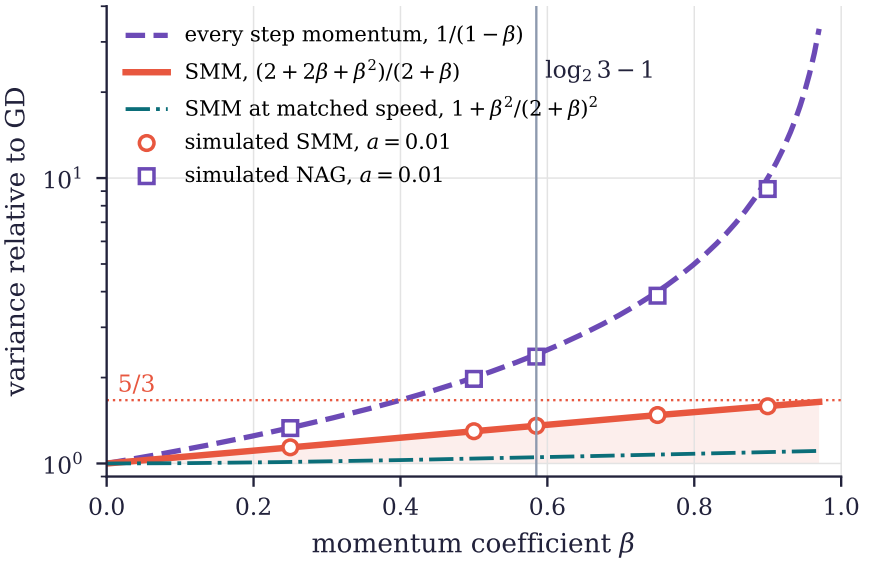


FIGURE 8.1. Slow mode variance amplification relative to gradient descent at the same step, as a function of the momentum coefficient (log scale). Dashed: every step momentum, $1/(1-\beta)$. Solid: $\mathrm{SMM}_{\mathrm{det}}$, $(2+2\beta+\beta^2)/(2+\beta)$, bounded by $5/3$ (dotted); the shaded band lies between 1 and this curve. Dash dotted: $\mathrm{SMM}_{\mathrm{det}}$ at matched speed, $1+\beta^2/(2+\beta)^2$ (Proposition 8.9). Markers: simulated ratios at $a = 0.01$. The vertical line marks $\beta = \log_2 3 - 1$.

Setting. Fix a period p and a memory length K. On one eigendirection of a quadratic, with $a = \alpha\lambda$ and state $z_t = (e_t, \dots, e_{t-K+1})$, a *linear periodic first order method* evaluates at step t (with residue $j = t \bmod p$) a gradient at an extrapolated point $y_t = \gamma_j \cdot z_t$ and updates

$$e_{t+1} = \delta_j \cdot z_t - h_j\big(a\, y_t + \alpha\, \xi_t\big), \tag{8.16}$$

with fixed vectors $\gamma_j, \delta_j \in \mathbb{R}^K$ and step multipliers $h_j > 0$, where ξ_t is the gradient noise. Consistency requires that constants are preserved when the gradient vanishes, $\delta_j \cdot \iota = \gamma_j \cdot \iota = 1$, where $\iota = (1, \dots, 1)$. Gradient descent, heavy ball, Nesterov at every step, $\mathrm{SMM}_{\mathrm{det}}$, cyclic step sizes, and momentum applied on any subset of the residues modulo p are all of this form. Write $B(a) = B_0 + aB_1 + O(a^2)$ for the product of the p step matrices over one period, and assume the method is *stable at* $a = 0$: the eigenvalue 1 of B_0, with right eigenvector ι, is simple, and all other eigenvalues of B_0 lie strictly inside the unit disc.

Definition 8.11 *(Gains).* Let w be the left eigenvector of B_0 for the eigenvalue 1, normalized so that $w \cdot \iota = 1$. The *gain* c_i of the gradient evaluated at residue i is the total long run effect, at $a = 0$, of a unit injection $-h_i$ into e at that step, measured in the slow coordinate $w \cdot z$: $c_i = h_i\, w \cdot \Pi_i u_1$, where u_1 is the first unit vector and Π_i the product of the step matrices at $a = 0$ from the step after i to the end of the period. Put $M = \sum_{i=1}^{p} c_i$.

The proof of part (i) of the theorem below shows that, whatever the sign of M, the period product has the eigenvalue $1 - Ma + O(a^2)$, so $M \geq 0$ is necessary for stability at small $a > 0$. We assume $M > 0$; this holds whenever all gains are positive, as for every method listed above with $\beta \in [0, 1)$.

For gradient descent $c = 1$; for every step momentum $c = 1/(1-\beta)$, since a unit impulse is carried forward with weights $1, \beta, \beta^2, \dots$; for $\mathrm{SMM}_{\mathrm{det}}$ the gains are $(1+\beta,\, 1)$, because the gradient taken at a descent step also enters the next extrapolation.

Theorem 8.12 *(Gain weight theorem). For a linear periodic first order method* (8.16) *that is stable at* $a = 0$ *and has* $M > 0$*, with uncorrelated noise of variance* σ^2*, as* $a \to 0$*:*

(i) the per step contraction of the slow mode is $1 - (M/p)\, a + O(a^2)$*;*

(ii) the phase averaged stationary variance of e_t *is* $\alpha^2\sigma^2\, \dfrac{\sum_i c_i^2}{2Ma}\, (1 + o(1))$*;*

(iii) consequently the noise floor relative to gradient descent run at the same slow mode speed, step $(M/p)\alpha$*, is*

$$\mathcal{F} = \frac{p \sum_{i=1}^{p} c_i^2}{\big(\sum_{i=1}^{p} c_i\big)^2} \;\geq\; 1, \tag{8.17}$$

with equality if and only if all gains are equal.

Proof. Let $s_t = w \cdot z_t$ at the start of each period. Since w is a left eigenvector of B_0, s is unchanged by B_0, and over one period

$$s_{t+p} = s_t + a\, w \cdot B_1 z_t - \alpha \sum_{i=1}^{p} c_i\, \xi_{t+i-1} + O(a\alpha)\, \|\xi\|,$$

where the noise coefficients are the gains by Definition 8.11. The key observation is that the deterministic term has the same coefficients. The gradient enters (8.16) through $h_j(a\, y_t + \alpha\xi_t)$, so replacing $\alpha\xi_t$ by $a\, y_t$ shows that $a\, w \cdot B_1 z_t = -a \sum_i c_i\, y_{t+i-1} + O(a^2)\|z\|$. On the slow manifold $z \approx s\iota$ we have $y \approx s$, because $\gamma_j \cdot \iota = 1$, and therefore $w \cdot B_1 \iota = -M$. Simple eigenvalue perturbation now gives the eigenvalue $1 - Ma + O(a^2)$ of $B(a)$ for one period, which is (i).

Decompose $z_t = s_t \iota + f_t$ with $w \cdot f_t = 0$. The fast part f obeys a recursion whose matrix is a perturbation of B_0 restricted to the complement of ι, which has spectral radius below 1, so f has stationary variance $O(\alpha^2\sigma^2)$ uniformly in small a. The slow part satisfies $s_{t+p} = (1 - Ma)s_t + \eta_t + r_t$, where $\eta_t = -\alpha \sum_i c_i \xi_{t+i-1}$ has variance $\alpha^2\sigma^2 \sum_i c_i^2$ and is independent across periods, and the remainder r_t collects terms of order $a\, f_t$ and $a\alpha\, \xi$. Hence $\mathrm{Var}(s) = \alpha^2\sigma^2 \sum_i c_i^2 / (1 - (1 - Ma)^2)\,(1 + o(1)) = \alpha^2\sigma^2 \sum_i c_i^2 / (2Ma)\,(1 + o(1))$. Since $e_t = s_t + (f_t)_1$ and $|\mathrm{Cov}(s_t, (f_t)_1)| \le (\mathrm{Var}\, s\ \mathrm{Var}\, f)^{1/2} = O(\alpha^2\sigma^2 a^{-1/2})$, the variance of e_t at every phase is $\mathrm{Var}(s)(1 + o(1))$, which gives (ii).

Gradient descent with step $h\alpha$ has $c = h$, $M = h$ and variance $\alpha^2\sigma^2 h/(2a)$, and its slow mode contraction per step is $1 - ha$. Matching speeds means $h = M/p$, and the ratio of (ii) to $\alpha^2\sigma^2(M/p)/(2a)$ is (8.17). The inequality is Cauchy and Schwarz, $(\sum c_i)^2 \le p \sum c_i^2$, with equality exactly when the c_i are equal. □

Corollary 8.13 *(No schedule beats gradient descent in the slow mode). Among all linear periodic first order methods that are stable at $a = 0$ and have $M > 0$, none has a lower slow mode noise floor than gradient descent at matched speed. Every step heavy ball and every step Nesterov attain equality, $\mathcal{F} = 1$, which recovers the equivalence of momentum SGD and SGD with a rescaled step at small step sizes [111]. Switching MM has $\mathcal{F} = 2((1+\beta)^2 + 1)/(2+\beta)^2 = 1 + \beta^2/(2+\beta)^2$, recovering Proposition 8.9.*

The theorem answers P6 negatively, and it also explains what the parity schedule buys. Momentum in this regime is never free and never harmful at matched speed: it simply buys speed at the price of noise in exact proportion, provided all gradients receive the same weight. A schedule that weights gradients unequally pays a Cauchy and Schwarz penalty. $\mathrm{SMM}_{\mathrm{det}}$ weights its two gradients in the ratio $1 + \beta$ to 1, which is why its penalty is small; a cyclic step schedule $(1, 3)$ weights them 1 to 3 and pays $\mathcal{F} = 5/4$. What parity switching achieves is not a better tradeoff but a *bounded* one: the speed multiplier $M/p = (2 + \beta)/2$ never exceeds $3/2$ and the penalty $\mathcal{F} = 1 + \beta^2/(2+\beta)^2$ never exceeds $10/9$, so the floor, $\mathcal{F}M/p = V(\beta)$, never exceeds $5/3$ of gradient descent at the same step, whatever β.

TABLE 8.2. Gains, predicted floor factor $\mathcal{F}$ of (8.17), and the exact matched speed ratio from the periodic Lyapunov equation at $a = 10^{-2}$ and $a = 10^{-4}$. "$p = 3$, NAG at $\{2\}$" injects Nesterov momentum only at residue 2 modulo 3, and so on; $\beta_0 = \log_2 3 - 1$.

Schedule	gains c_i	$\mathcal{F}$	$a = 10^{-2}$	$a = 10^{-4}$
gradient descent	1	1	1.0000	1.0000
NAG every step, β_0	2.409	1	0.9773	0.9998
NAG every step, 0.9	10	1	0.8739	0.9986
$\mathrm{SMM}_{\mathrm{det}}$, β_0	1.585, 1	1.0512	1.0466	1.0512
$\mathrm{SMM}_{\mathrm{det}}$, 0.9	1.9, 1	1.0963	1.0902	1.0963
cyclic steps $(1,3)$	1, 3	1.2500	1.2570	1.2501
$p = 3$, NAG $\{2\}$, β_0	1, 1.585, 1	1.0532	1.0500	1.0532
$p = 3$, HB $\{1,2\}$, 0.9	2.71, 1.9, 1	1.1395	1.1377	1.1395
$p = 4$, NAG $\{1,2,3\}$, β_0	2.13, 1.93, 1.58, 1	1.0663	1.0561	1.0662
$p = 4$, NAG $\{2,3\}$, 0.9	1, 2.71, 1.9, 1	1.1859	1.1770	1.1859

Table 8.2 checks the theorem against exact solutions of the periodic Lyapunov equation for a range of schedules, and Figure 8.2 shows the convergence of the exact matched speed ratio to $\mathcal{F}$ as $a \to 0$ (script `ch7_gainweight.py`).

8.7 The best period two ratio

On quadratics $\mathrm{SMM}_{\mathrm{det}}$ is the cyclic schedule with steps $h_1 = \alpha$ and $h_2 = (1+\beta)\alpha$ (Theorem 7.6); in this section $h_1 \le h_2$ denote absolute steps, not multiples of α. Problem S3 asks which ratio $r = h_2/h_1$ is best when both the deterministic rate and the noise floor matter. For period two schedules both quantities depend on the ratio through a single number,

$$G(r) = \frac{(1+r)^2}{4r} \ge 1,$$

and the tradeoff can be solved exactly.

PROPOSITION 8.14 *(Rate and noise for period two steps). Let the spectrum be $[\mu, L]$ with $\kappa = L/\mu$, and consider cyclic gradient steps h_1, h_2 with ratio $r = h_2/h_1 \ge 1$.*

(i) The slow mode noise floor at matched speed is $N = 2(h_1^2 + h_2^2)/(h_1+h_2)^2 = 2 - 1/G(r)$.

(ii) For fixed r with $G(r) \le G_{\mathrm{ch}}(\kappa) = 2(\kappa+1)^2/(\kappa^2+6\kappa+1)$, the smallest possible worst case two step contraction over $[\mu, L]$ is

$$R^*(r) = 1 - \frac{4\kappa\, G(r)}{(\kappa+1)^2},$$

attained when the steps are scaled so that $h_1 + h_2 = 4G(r)/(\mu+L)$. For $G(r) > G_{\mathrm{ch}}(\kappa)$ the smallest worst case contraction exceeds that of the Chebyshev ratio.

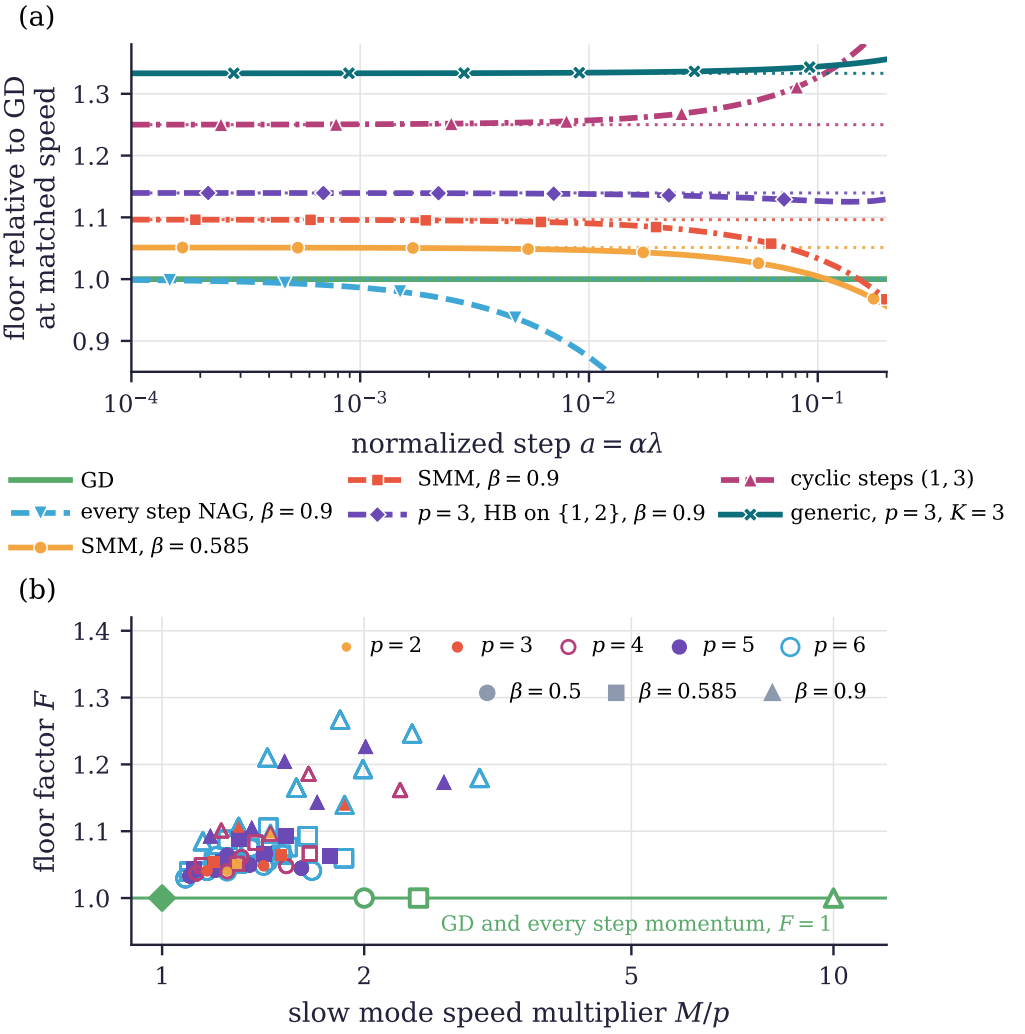


FIGURE 8.2. The gain weight theorem. (a) Exact noise floor relative to gradient descent at matched speed, from the periodic Lyapunov equation, against $a = \alpha\lambda$, for six of the schedules of Table 8.2 and for a period three schedule with three term memory and unrelated coefficients ($\mathcal{F} = 1.333$); dotted lines are the limits $\mathcal{F}$. (b) The floor factor $\mathcal{F}$ against the speed multiplier M/p for all injection subsets of periods 2 to 6 (larger markers for longer periods) and three values of β; only gradient descent and every step momentum reach $\mathcal{F} = 1$.

(iii) Consequently the Pareto frontier between rate and noise floor is $R = 1 - 4\kappa/((2-N)(\kappa+1)^2)$ *for* $1 \le N \le N_{\mathrm{ch}}(\kappa)$*, where* $N_{\mathrm{ch}} = (3\kappa^2+2\kappa+3)/(2(\kappa+1)^2)$ *is the noise floor of the two step Chebyshev schedule, whose ratio tends to* $(1+\sqrt{2})^2 = 5.83$ *and whose floor tends to* $3/2$ *as* $\kappa \to \infty$*.*

Proof. (i) is (8.17) with $c = (h_1, h_2)$, and $2(h_1^2+h_2^2)/(h_1+h_2)^2 = 2 - 4h_1h_2/(h_1+h_2)^2 = 2 - 1/G$. For (ii), the two step polynomial $\Phi(\lambda) = (1-h_1\lambda)(1-h_2\lambda)$ satisfies $\Phi(0) = 1$ and has minimum $1-G$ at $\lambda_m = (h_1+h_2)/(2h_1h_2)$, so it can be written as $\Phi(\lambda) = 1 - G + G(\lambda/\lambda_m - 1)^2$. Its maximum over $[\mu, L]$ is at an endpoint, and $\max\big((\mu/\lambda_m - 1)^2, (L/\lambda_m - 1)^2\big)$ is minimized over λ_m by $\lambda_m = (\mu+L)/2$, with value $((\kappa-1)/(\kappa+1))^2$. Hence the endpoint maximum is at least $1 - G + G((\kappa-1)/(\kappa+1))^2 = 1 - 4\kappa G/(\kappa+1)^2$, with equality at $\lambda_m = (\mu+L)/2$, i.e. $h_1 + h_2 = 4G/(\mu+L)$. At this scaling the interior minimum contributes $|1-G| = G - 1$, and $G - 1 \le 1 - 4\kappa G/(\kappa+1)^2$ exactly when $G \le G_{\mathrm{ch}}$, the two values being equal at the Chebyshev point $G = G_{\mathrm{ch}}$; since the endpoint bound holds for every scaling, this proves (ii) for such G. For $G > G_{\mathrm{ch}}$ no scaling produces the two step Chebyshev polynomial [27, 110], which is the unique minimizer of the worst case over $[\mu, L]$ among quadratic polynomials with value 1 at 0, so the worst case exceeds that of the Chebyshev ratio. (iii) follows by eliminating $G = 1/(2-N)$ for $G \le G_{\mathrm{ch}}$; beyond the Chebyshev point a larger ratio is worse than the Chebyshev ratio in both criteria. □

Every ratio between 1 and the Chebyshev ratio $r_{\mathrm{ch}}(\kappa)$ is therefore on the frontier once its steps are scaled correctly, larger ratios being dominated by the Chebyshev ratio in both criteria, and the question becomes where on the frontier one wants to be. The SMM ratio $1+\beta$ gives $G = (2+\beta)^2/(4(1+\beta)) = 1.0540$ and $N = 1.0512$: a 5.4 percent gain in the rate exponent over optimally tuned gradient descent for a 5.1 percent rise in the noise floor. The Chebyshev ratio gains up to a factor 2 in the exponent but pays a floor up to 1.5. The optimal scaling for the SMM ratio is $\alpha L = 4G\kappa/((2+\beta)(\kappa+1))$, which tends to $(2+\beta)/(1+\beta) = 1.631$; at the conventional $\alpha = 1/L$ the method is not on the frontier, and its per step factor at $\kappa = 100$ is 0.9871 against 0.9791 when optimally scaled. Figure 8.3 draws the frontier. The SMM ratio coincides with the Chebyshev ratio only at $\kappa_0 = 1.94$; for larger κ it sits close to the gradient descent end of the frontier, which is the noise conservative end.

8.8 Multiplicative noise

Additive noise is the simplest model. In least squares, and more generally in stochastic gradient methods with sampled data, the noise also multiplies the error: the sampled curvature fluctuates. Problem S2 asks whether the conclusions survive. Consider on one eigendirection the oracle

$$g(y) = (\lambda + \eta_t)\, e_y + \xi_t,$$

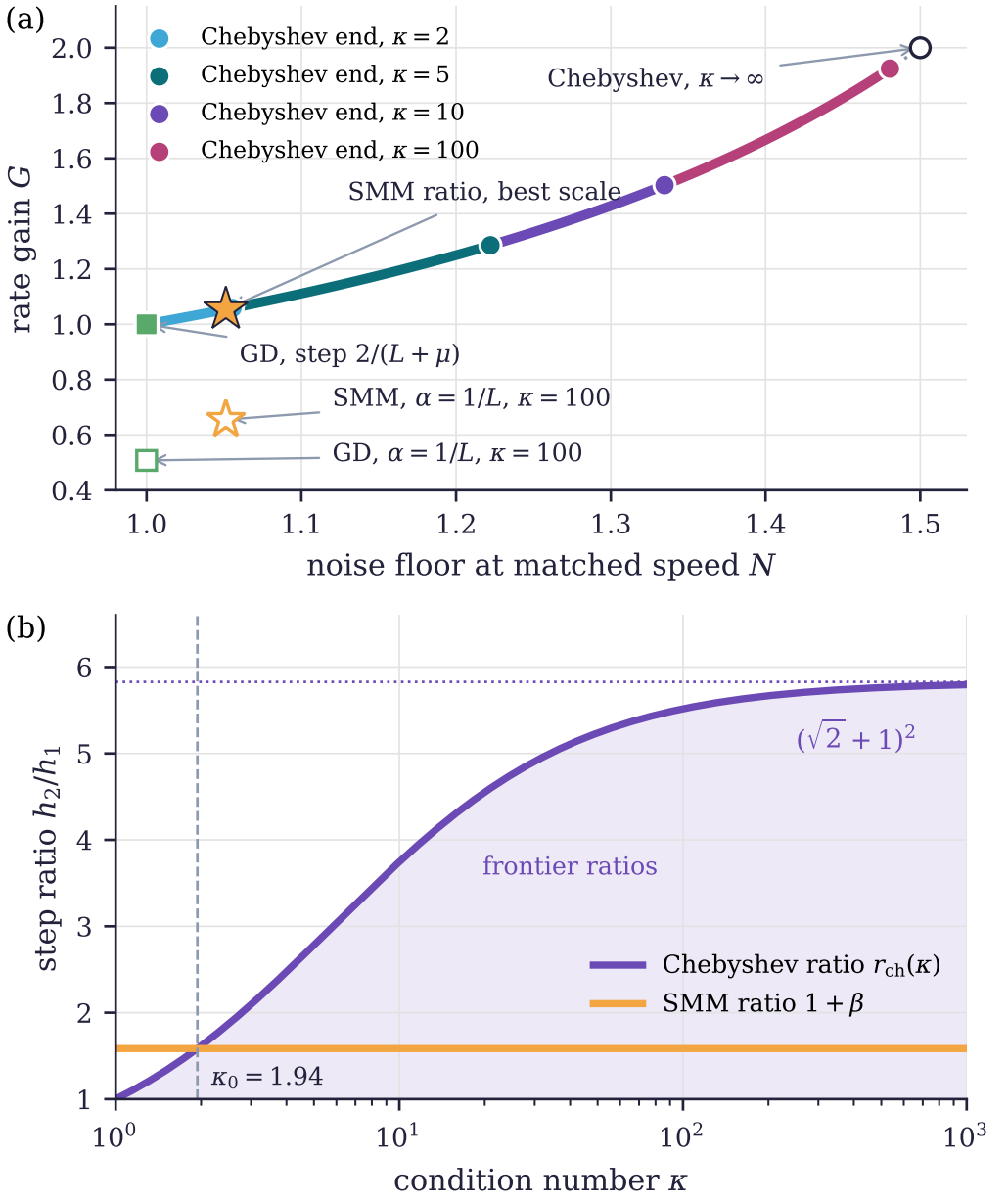


FIGURE 8.3. (a) The rate gain G against the noise floor N at matched speed for period two schedules. By Proposition 8.14(i) all frontiers lie on the curve $G = 1/(2-N)$ and differ only in where they end, at the Chebyshev schedule for each κ. The star is the SMM ratio at its best scale, the open markers SMM and gradient descent at $\alpha = 1/L$ and $\kappa = 100$, and the filled square gradient descent at step $2/(L+\mu)$. The frontier ends for $\kappa = 2, 5, 10, 100$ lie in that order along the curve: the first under the star, the others at the dots. (b) The Chebyshev ratio $r_{\mathrm{ch}}(\kappa)$, rising to $(1+\sqrt{2})^2$, and the SMM ratio $1+\beta$; they cross at $\kappa_0 = 1.94$.

with η_t, ξ_t independent of the past, mean zero, uncorrelated with each other, $\mathrm{Var}(\alpha\eta_t) = \omega^2$ and $\mathrm{Var}\,\xi_t = \sigma^2$. Two scalings are natural: the least squares scaling $\omega^2 = \vartheta a^2$, in which the relative curvature noise $\vartheta = \mathrm{Var}(x^2)/(\mathbb{E}x^2)^2$ is fixed ($\vartheta = 2$ for a Gaussian regressor), and the diffusive scaling $\omega^2 = \epsilon a$.

PROPOSITION 8.15 *(Multiplicative noise). For* $\mathrm{SMM}_{\mathrm{det}}$ *the second moments obey a closed linear recursion: with* $\nu_1 = (1-a)^2 + \omega^2$ *and* $\nu_2 = (1-(1+\beta)a)^2 + (1+\beta)^2\omega^2$, *the stationary second moment at even times is* $\alpha^2\sigma^2(\nu_1(1+\beta)^2+1)/(1-\nu_1\nu_2)$, *finite if and only if* $\nu_1\nu_2 < 1$. *As* $a \to 0$*:*

(i) under the least squares scaling the slow mode ratio to gradient descent is unchanged, $(2+2\beta+\beta^2)/(2+\beta)$, *for every* ϑ*;*

(ii) under the diffusive scaling, for any linear periodic method that is stable at $a = 0$ *and has* $M > 0$, *with gains* c_i, $a\,\mathbb{E}e^2 \to \alpha^2\sigma^2\sum_i c_i^2/(2M - \epsilon\sum_i c_i^2)$, *so the method is mean square stable for small* a *exactly when* $\epsilon < \epsilon^* = 2M/\sum_i c_i^2$.

Proof. For $\mathrm{SMM}_{\mathrm{det}}$ the two step recursion of Theorem 8.4 holds with each factor $(1-ha)$ replaced by $(1-h(a+\alpha\eta))$; squaring and taking expectations, the cross terms vanish by independence and the factors ν_1, ν_2 appear, which gives the closed form and the stability condition. Statement (i) follows because $\omega^2 = \vartheta a^2$ changes ν_1, ν_2 only at order a^2, which does not affect the leading term of $1 - \nu_1\nu_2$. For (ii), repeat the slow coordinate argument of Theorem 8.12: the multiplicative noise contributes to s a term $-\alpha\sum_i c_i\eta_{t+i}s$ per period, of variance $\epsilon a\sum_i c_i^2 s^2$, so the stationary second moment satisfies $\mathbb{E}s^2 = (1 - 2Ma + \epsilon a\sum c_i^2)\mathbb{E}s^2 + \alpha^2\sigma^2\sum c_i^2 + o(a)\mathbb{E}s^2$, which gives the stated limit and threshold. □

The threshold $\epsilon^* = 2M/\sum c_i^2$ is a clean measure of robustness to curvature noise. It is 2 for gradient descent, $2(1-\beta)$ for every step momentum, 0.830 at $\beta = \beta_0 = \log_2 3 - 1$ and 0.2 at $\beta = 0.9$, and $2(2+\beta)/((1+\beta)^2+1)$ for $\mathrm{SMM}_{\mathrm{det}}$, which is 1.472 at β_0 and 1.258 at $\beta = 0.9$ and never falls below $6/5$. So under curvature noise the parity schedule is markedly more robust than every step momentum, by a factor that grows without bound as $\beta \to 1$. At matched slow mode speed the comparison changes, as it did for the variance: gradient descent with step $(M/p)\alpha$ has $\epsilon^* = 2p/M$, and the ratio of the two thresholds is $M^2/(p\sum_i c_i^2) = 1/\mathcal{F} \le 1$. Away from the slow mode the picture is more nuanced. The uniform bound $5/3$ does not survive large multiplicative noise at large a: at $\beta = \beta_0$ $\mathrm{SMM}_{\mathrm{det}}$ remains mean square stable at $a = 1$ only for $\vartheta < 0.566$, and above that it is not mean square stable at $a = 1$, where gradient descent still is for $\vartheta < 1$; but every step Nesterov loses stability earlier at every ϑ we examined (Figure 8.4, script `ch7_multiplicative.py`, with Monte Carlo checks agreeing with the closed forms to three digits).

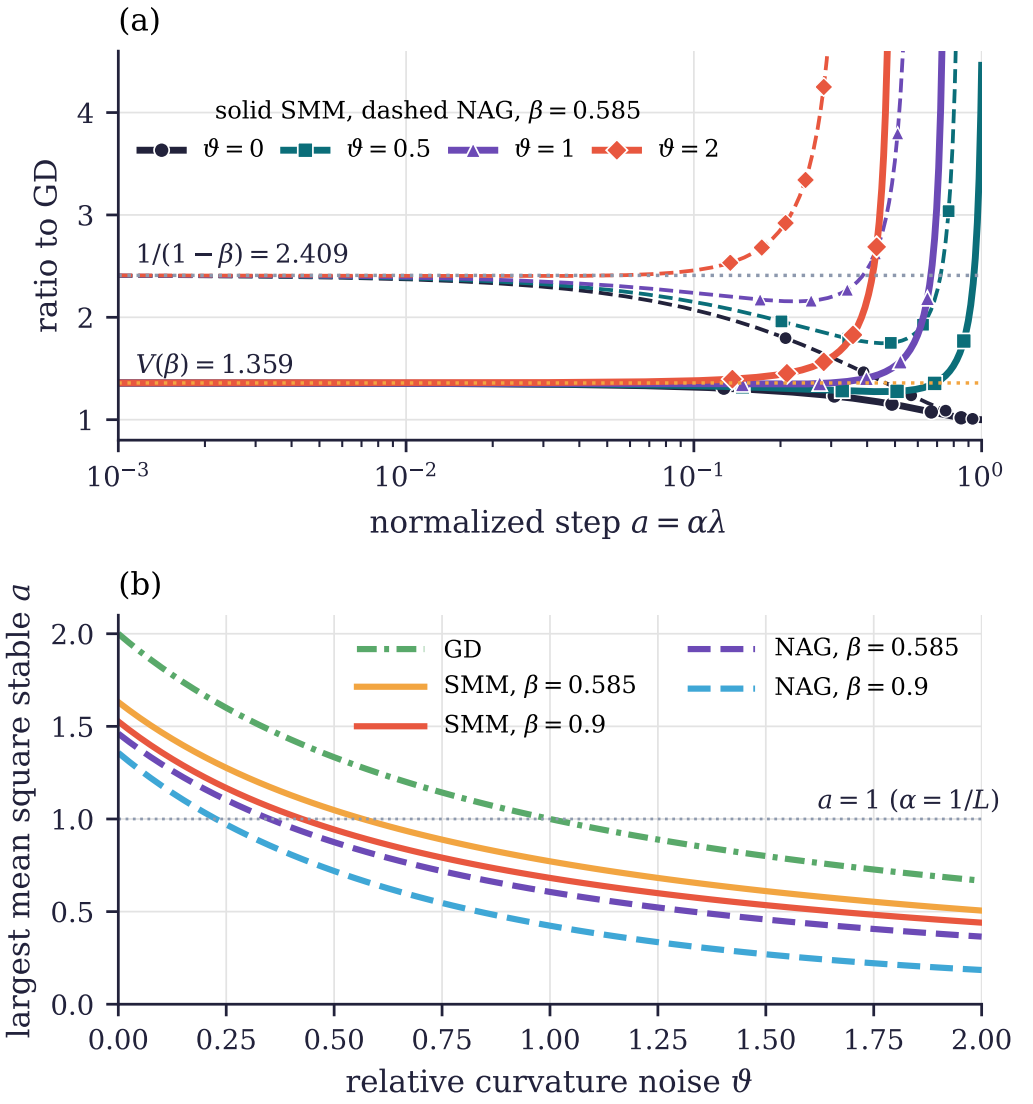


FIGURE 8.4. Multiplicative noise under the least squares scaling. (a) Stationary second moment relative to gradient descent against a, for $\mathrm{SMM}_{\mathrm{det}}$ (solid) and every step Nesterov (dashed) at $\beta = 0.585$ and several ϑ; the slow mode limits $V(\beta) = 1.359$ and $1/(1-\beta) = 2.409$ are marked. (b) The largest mean square stable a as a function of ϑ; $\mathrm{SMM}_{\mathrm{det}}$ is stable on a larger range than every step Nesterov at the same β.

8.9 The experiments

We now describe experiments with SMM and then reconcile them with the exact theory in Section 8.10. The experiments were designed before that theory was available. The numbers and figures below come from a rerun for this book: the script `ch7_experiments.py` runs the experiments, and the scripts `ch7_experiments_fig.py` and `ch7_noisefig.py` draw the figures.

8.9.1 Setup and baselines

SMM_{det} and, where indicated, $\text{SMM}_{\text{state}}$ (Algorithms 7.1 and 7.2) were compared with gradient descent at step $1/L$; with heavy ball and Nesterov at their textbook strongly convex tunings [71, 81], which need μ (heavy ball with $\alpha = 4/(\sqrt{L} + \sqrt{\mu})^2$ and $\beta = \left((\sqrt{L} - \sqrt{\mu})/(\sqrt{L} + \sqrt{\mu})\right)^2$, Nesterov with $\alpha = 1/L$ and $\beta = (\sqrt{L} - \sqrt{\mu})/(\sqrt{L} + \sqrt{\mu})$); and with fixed momentum heavy ball and Nesterov at the four candidates $\beta \in \{0.5, 0.585, 0.667, 0.9\}$ of Table 7.1. The fixed momentum row at $\beta = 0.585$ is the every step analogue of SMM_{det} and isolates the parity schedule. Adam [46] at its defaults ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, step $1/L$) served as a reference from a different family. Every method received L. Fixed momentum Nesterov is the update $y_t = x_t + \beta(x_t - x_{t-1})$, $x_{t+1} = y_t - \alpha\nabla f(y_t)$ at every step, and fixed heavy ball the update of Section 8.2, both with $\alpha = 1/L$; the frequency ablation adds injection at random steps, each step injecting independently with probability $\frac{1}{2}$. $\text{SMM}_{\text{state}}$ runs Algorithm 7.2 with $\rho_{\text{thr}} = 0.356$, reference value $f_{\text{ref}} = f^*$, the minimum of the objective, and $\epsilon = 10^{-12}$.

The test problems were random quadratics $f(x) = \frac{1}{2}x^\top Ax$ in dimension $d = 25$, with eigenvalues geometrically spaced in $[1, \kappa]$, a random orthogonal eigenbasis and x_0 drawn from $N(0, I)$; and ℓ_2 regularized logistic regression on the Breast Cancer Wisconsin data [108] ($n = 569$ samples, 30 standardized features, no intercept, penalty 0.05), with initial weights drawn from $N(0, 0.1^2 I)$; there $\mu = 0.05$ and $L = \lambda_{\max}(X^\top X/n)/4 + 0.05 = 3.370$ for the $n \times 30$ data matrix X, so that $\kappa = L/\mu \approx 67.4$. The count of a run is the first iteration at which the objective gap falls below the tolerance, and a run that has not reached the tolerance within the horizon T counts as $T + 1$. Means and standard errors are over R seeded replicates: $R = 20$ for the quadratic and logistic experiments, $R = 30$ for the noise sweep and the minibatch test.

8.9.2 Quadratic condition number sweep and frequency ablation

Figure 8.5 reports the mean number of iterations to reach an objective gap below 10^{-4} for $\kappa \in \{2, 5, 10, 25, 50, 100\}$, with horizon $T = 300$. At $\kappa = 2$, SMM_{det} is the fastest method tested, with 4.85 iterations on average. At $\kappa = 5$ only every step Nesterov at $\beta = 0.5$ and at $\beta = 0.585$ is clearly faster (11.85 and 12.6 iterations against 15.95); Nesterov at $\beta = 0.667$ ties with SMM_{det}, and every step heavy ball at these coefficients is slower. For $\kappa \geq 10$ every step heavy ball and Nesterov at

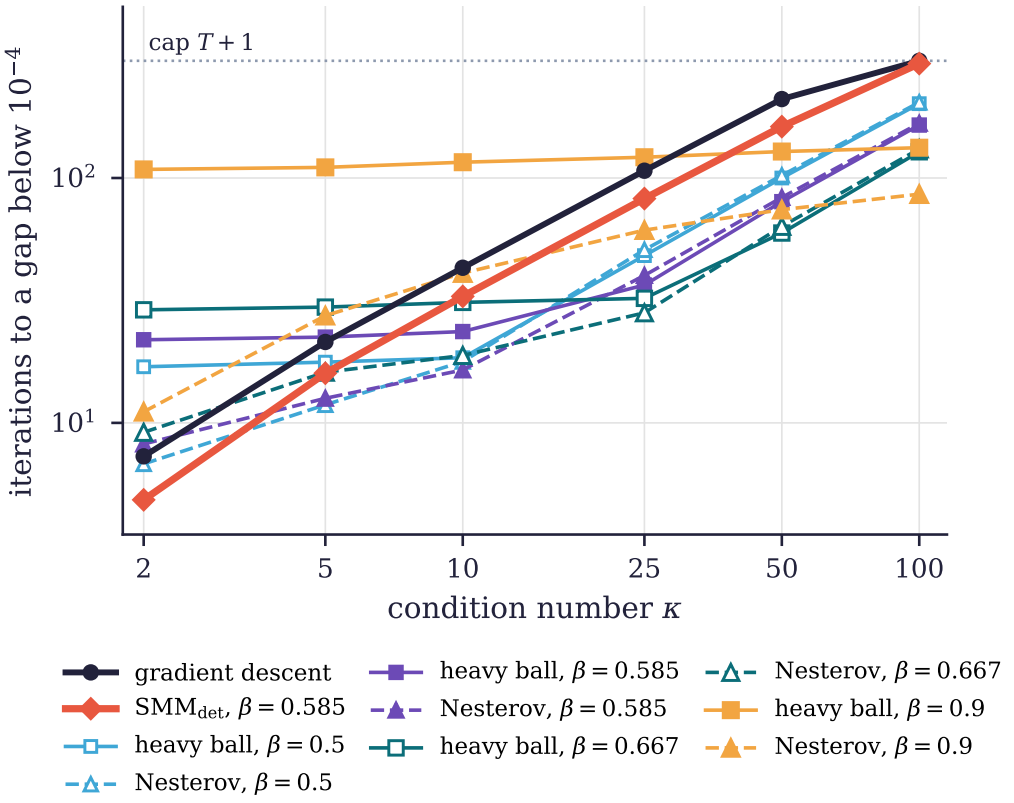


FIGURE 8.5. Fixed momentum baselines on random quadratics ($d = 25$, $R = 20$, $T = 300$): mean iterations to an objective gap below 10^{-4} against the condition number. Heavy ball (squares, solid) and Nesterov (triangles, dashed) at the four candidate coefficients, gradient descent, and $\mathrm{SMM}_{\mathrm{det}}$. A run that has not reached the tolerance by T counts as $T + 1 = 301$ (dotted line); at $\kappa = 100$ this cap censors 19 of the 20 gradient descent runs and 14 of the 20 $\mathrm{SMM}_{\mathrm{det}}$ runs.

$\beta \in \{0.585, 0.667\}$ beat it, by factors from about 1.1 to 2.0 at $\kappa = 10$ and from 2.1 to 2.9 at $\kappa = 25$. At $\kappa = 50$ the coefficient 0.667 is fastest in both forms, and only at $\kappa = 100$ is fixed Nesterov at $\beta = 0.9$ fastest. The nearly flat stretches of the fixed heavy ball curves reflect underdamping: when every normalized step exceeds $(1 - \sqrt{\beta})^2$, which holds for $\kappa < (1 - \sqrt{\beta})^{-2}$ (about 12, 18, 30 and 380 for the four candidates), both characteristic roots have modulus $\sqrt{\beta}$ whatever a, so the iteration count is set by β rather than by the conditioning.

Figure 8.6 isolates the schedule at $\beta = 0.585$: injection at every step, at odd steps ($\mathrm{SMM}_{\mathrm{det}}$), at random steps with probability $\frac{1}{2}$, and never. Without noise, parity does not help. Every step injection needs 16.45 iterations at $\kappa = 10$ against 32.85 for parity, a ratio of 2.00, and 167 against 293 at $\kappa = 100$; there 14 of the 20 parity runs reach the horizon, so the parity count is censored, and continued

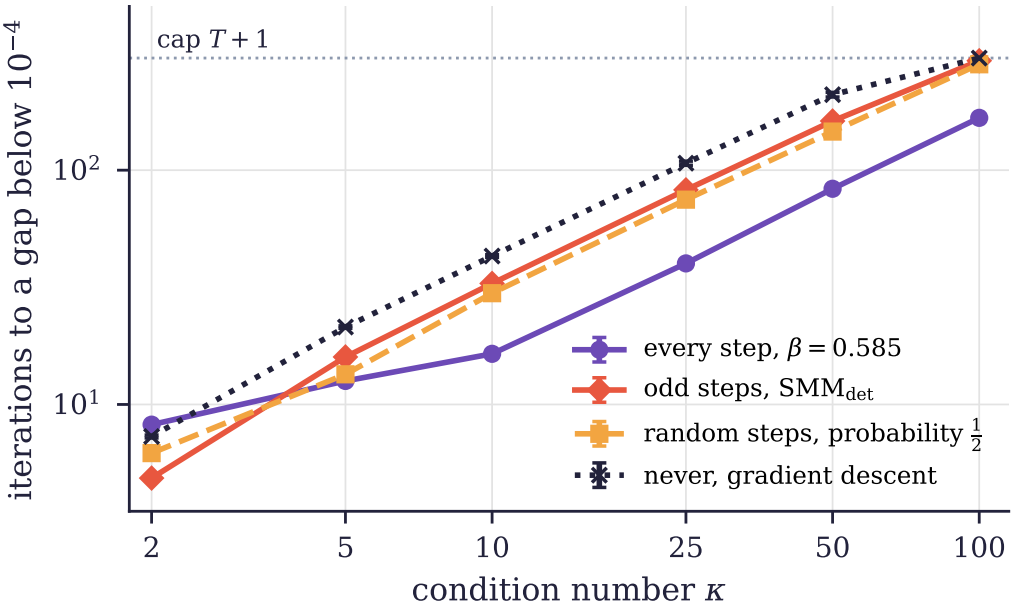


FIGURE 8.6. Frequency ablation on random quadratics at fixed $\beta = 0.585$ ($d = 25$, $R = 20$, $T = 300$): injection at every step, at odd steps ($\mathrm{SMM}_{\mathrm{det}}$), at random steps with probability $\frac{1}{2}$, and never (gradient descent). Bars are standard errors. A run that has not reached the tolerance by T counts as 301 (dotted line), which censors most parity and gradient descent runs and half of the random injection runs at $\kappa = 100$.

to convergence (the script runs every instance for 3000 steps) the parity schedule needs 318 iterations, a ratio of 1.90. Random injection lies between the two, closer to parity for $\kappa \geq 10$. Only at $\kappa = 2$, where all normalized steps are large and the period two polynomial $\rho(a)$ is small, does parity win. The ratios 2.00 and 1.90 are close to the slow mode prediction $m_{\mathrm{NAG}}/m_{\mathrm{SMM}} = 1.86$ of Corollary 7.9 and the discussion following it; Chapter 7 makes the same check with more replicates.

8.9.3 Noise robustness

The central experiment adds Gaussian noise of standard deviation $\sigma \in \{0, 0.05, 0.10, 0.20, 0.40\}$ to the gradients of random quadratics with $\kappa = 25$, with the same noise draws for every method, and records the objective gap after $T = 300$ steps (Figure 8.7 and Table 8.3). At every positive noise level $\mathrm{SMM}_{\mathrm{det}}$ has the lowest final gap among the inertial methods: at $\sigma = 0.1$, 3.7×10^{-3} against 5.0×10^{-3} for every step Nesterov momentum at the same β and 1.1×10^{-2} at $\beta = 0.9$. Across the sweep every step Nesterov at $\beta = 0.585$ is 29 to 36 percent worse than $\mathrm{SMM}_{\mathrm{det}}$, and $\beta = 0.9$ about 2.5 to 3 times worse; heavy ball at the same coefficients is worse still. Gradient descent has the lowest gap of all at each positive σ, and $\mathrm{SMM}_{\mathrm{det}}$ stays within 27 percent of it.

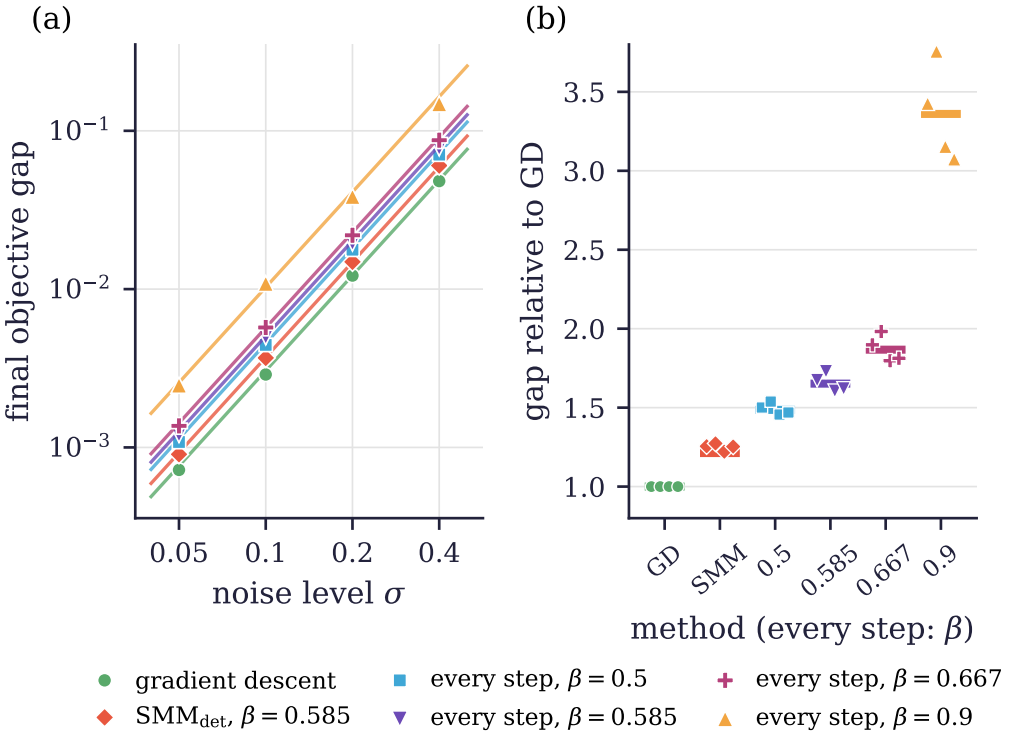


FIGURE 8.7. The noise robustness experiment of Table 8.3. (a) Final objective gap after $T = 300$ steps against the noise level (points), with the stationary predictions of (8.3), Theorem 8.4 and Proposition 8.3 summed over the spectrum (lines). (b) The gap relative to gradient descent for each method at the four positive noise levels (points), with the predicted ratios marked (bars). The every step rows use the Nesterov form.

TABLE 8.3. Noise robustness on random quadratics with $\kappa = 25$ ($d = 25$, $T = 300$, $R = 30$): final objective gap, lower is better. A dagger marks the lowest gap among inertial methods at each positive noise level; gradient descent, which has no momentum, is lower still. "every, β" denotes Nesterov momentum β injected at every step, and "HB, β" heavy ball.

Method	$\sigma = 0$	0.05	0.10	0.20	0.40
GD	1.8×10^{-11}	7.2×10^{-4}	2.9×10^{-3}	1.2×10^{-2}	4.8×10^{-2}
every, 0.5	≈ 0	1.1×10^{-3}	4.4×10^{-3}	1.8×10^{-2}	7.1×10^{-2}
every, 0.585	≈ 0	1.2×10^{-3}	5.0×10^{-3}	2.0×10^{-2}	7.8×10^{-2}
$\mathrm{SMM}_{\mathrm{det}}$, parity	1.1×10^{-14}	9.0×10^{-4} †	3.7×10^{-3} †	1.5×10^{-2} †	6.0×10^{-2} †
every, 0.667	≈ 0	1.4×10^{-3}	5.7×10^{-3}	2.2×10^{-2}	8.7×10^{-2}
every, 0.9	≈ 0	2.5×10^{-3}	1.1×10^{-2}	3.8×10^{-2}	1.5×10^{-1}
HB, 0.585	≈ 0	1.5×10^{-3}	6.7×10^{-3}	2.6×10^{-2}	1.1×10^{-1}
HB, 0.9	1.2×10^{-12}	6.3×10^{-3}	2.8×10^{-2}	1.1×10^{-1}	3.9×10^{-1}

TABLE 8.4. Logistic regression on the Breast Cancer Wisconsin data (penalty 0.05, $\kappa \approx 67.4$, $T = 600$, $R = 20$): iterations to a gap below 10^{-4} (mean and standard error), observed speedup over gradient descent, and the slow mode prediction of Section 8.10. An asterisk marks methods underdamped in the slowest mode.

Method	Iterations	Speedup	Predicted
GD	128.8 ± 1.5	1	1
HB, $\beta = 0.5$	63.1 ± 0.7	2.04	2.08
HB, $\beta = 0.585$	51.6 ± 0.6	2.50	2.58
HB, $\beta = 0.667$	40.3 ± 0.4	3.20	3.41
HB, $\beta = 0.9$	77.0 ± 1.0	1.67	3.52^*
NAG, $\beta = 0.5$	64.1 ± 0.7	2.01	2.05
NAG, $\beta = 0.585$	52.8 ± 0.6	2.44	2.52
NAG, $\beta = 0.667$	41.9 ± 0.5	3.08	3.28
NAG, $\beta = 0.9$	50.3 ± 0.3	2.56	4.02^*
$\mathrm{SMM}_{\mathrm{det}}$, $\beta = 0.585$	99.7 ± 1.2	1.29	1.30

8.9.4 Logistic regression and Adam

On the regularized logistic regression (Table 8.4, horizon $T = 600$), fixed momentum $\beta = 0.667$ is fastest in both forms, with about 41 iterations to a gap below 10^{-4}; the Collatz coefficient at every step needs about 52, $\mathrm{SMM}_{\mathrm{det}}$ about 100 and gradient descent 129. Without noise, every step injection at a moderate fixed β is the better choice. Figure 8.8 adds the tuned methods, which use μ and reach machine precision far sooner (19.3 and 29.0 iterations to the tolerance for heavy ball and Nesterov), and $\mathrm{SMM}_{\mathrm{state}}$, whose rule fires at almost every step on this problem (599.5 of the 600 steps on average), so that it follows every step Nesterov at $\beta = 0.585$, with 53.0 iterations. Adam, with 75.9 iterations, lies between fixed Nesterov at $\beta = 0.9$ (50.3) and $\mathrm{SMM}_{\mathrm{det}}$. Adam keeps two moment vectors per coordinate where SMM, heavy ball and Nesterov keep one extra vector; it needs neither μ nor a tuned coefficient, but on this problem every fixed momentum method except heavy ball at $\beta = 0.9$ is faster.

8.9.5 Minibatch stochastic gradients

A more realistic test replaces the full logistic gradient by minibatch gradients with $|B| \in \{16, 64, 256\}$, the batches drawn without replacement and shared by all methods, keeping the step $1/L$ (Table 8.5 and Figure 8.9). At $|B| = 16$ every method that injects at every step, including tuned Nesterov, reaches the tolerance 10^{-3} late or not at all: tuned Nesterov and Nesterov at $\beta = 0.9$ in none of the 30 runs, Nesterov at $\beta = 0.585$ after 273 iterations on average, with 2 runs that never do. $\mathrm{SMM}_{\mathrm{det}}$ is the only inertial method that reaches it reliably, in 135 iterations, and gradient descent, with 118, is faster still. As the batch grows the deterministic

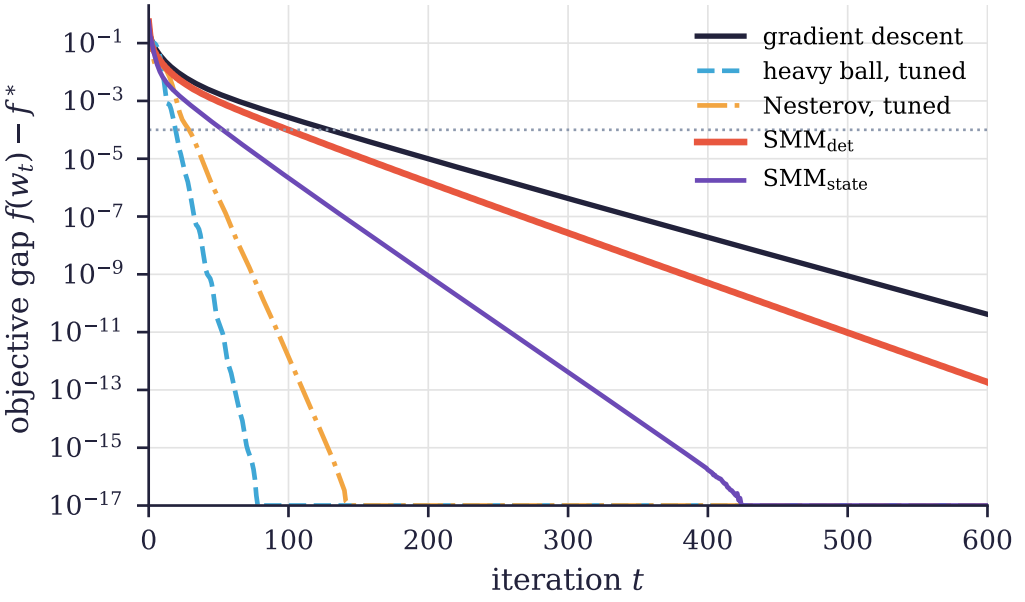


FIGURE 8.8. Logistic regression (penalty 0.05, $\kappa \approx 67$, $T = 600$, mean of $R = 20$ runs): objective gap against iteration for gradient descent, tuned heavy ball and Nesterov, SMM_{det} and $\text{SMM}_{\text{state}}$; the dotted line is the tolerance 10^{-4}. $\text{SMM}_{\text{state}}$ injects at almost every step here and follows every step Nesterov at $\beta = 0.585$.

TABLE 8.5. Minibatch stochastic gradients on the logistic regression: iterations to a gap below 10^{-3} (mean and standard error, $T = 600$, $R = 30$; a run that does not reach the tolerance within T counts as 601). A dagger marks the best inertial method at each batch size; "cap" marks entries in which no run reached the tolerance.

Method	$\|B\| = 16$	$\|B\| = 64$	$\|B\| = 256$
GD	118.2 ± 6.3	77.7 ± 1.5	67.6 ± 1.1
NAG, tuned	601 ± 0 (cap)	$36.5 \pm 2.3^{\dagger}$	$21.2 \pm 0.2^{\dagger}$
NAG, $\beta = 0.585$	272.7 ± 30.7	40.6 ± 1.3	29.3 ± 0.5
NAG, $\beta = 0.9$	601 ± 0 (cap)	113.5 ± 14.7	36.5 ± 0.8
$\mathrm{SMM}_{\mathrm{det}}$, $\beta = 0.585$	$135.2 \pm 11.2^{\dagger}$	62.7 ± 1.7	53.1 ± 0.9

ranking returns: at $|B| = 256$ tuned Nesterov is fastest with 21 iterations. Among the every step Nesterov variants the default $\beta = 0.9$ is never faster than the others: at $|B| = 16$ it ties with tuned Nesterov, neither reaching the tolerance, and at $|B| = 64$ and 256 it is the slowest of the three.

In summary, read on their own, the experiments say the following: without noise, every step Nesterov momentum beats $\mathrm{SMM}_{\mathrm{det}}$ at the same coefficient for $\kappa \geq 5$, and every step heavy ball for $\kappa \geq 10$; with noise, $\mathrm{SMM}_{\mathrm{det}}$ has the lowest final gap among inertial methods and copes best with the smallest minibatch; and 0.585 is a reasonable default among fixed momenta but not a universally optimal one, and nothing in the theory of this chapter singles it out.

8.10 Reconciling the experiments with the exact theory

The noise floors. The $\kappa = 25$ experiment has $\alpha = 1/25$ and normalized steps $a_j = \lambda_j/25 \in [0.04, 1]$. After $T = 300$ steps the deterministic transient has decayed by at least $0.96^{300} < 10^{-5}$, so by (8.2) the final gap is essentially $\frac{1}{2}\sum_j \lambda_j V(a_j)$, with V the exact stationary variance of the method. For gradient descent this sum is $0.3055\,\sigma^2$, that is 7.6×10^{-4}, 3.05×10^{-3}, 1.22×10^{-2} and 4.89×10^{-2} at the four noise levels, against the measured 7.2×10^{-4}, 2.9×10^{-3}, 1.22×10^{-2} and 4.8×10^{-2}. Table 8.6 does the same for the ratios to gradient descent. The every step rows of Table 8.3 use the Nesterov form of every step momentum, and its predictions match the data closely at all four coefficients; so do the much larger heavy ball predictions for the two heavy ball rows. The predicted ratio of every step Nesterov at $\beta = 0.585$ to $\mathrm{SMM}_{\mathrm{det}}$ is $1.65/1.21 = 1.36$, matching the observed 29 to 36 percent, and the predicted excess of $\mathrm{SMM}_{\mathrm{det}}$ over gradient descent is 21 percent, against the observed 22 to 27 percent.

Why gradient descent wins the noise table. That gradient descent has the lowest final gap at every positive σ is forced by the theory. At a fixed step every inertial method considered has stationary variance at least that of gradient descent on

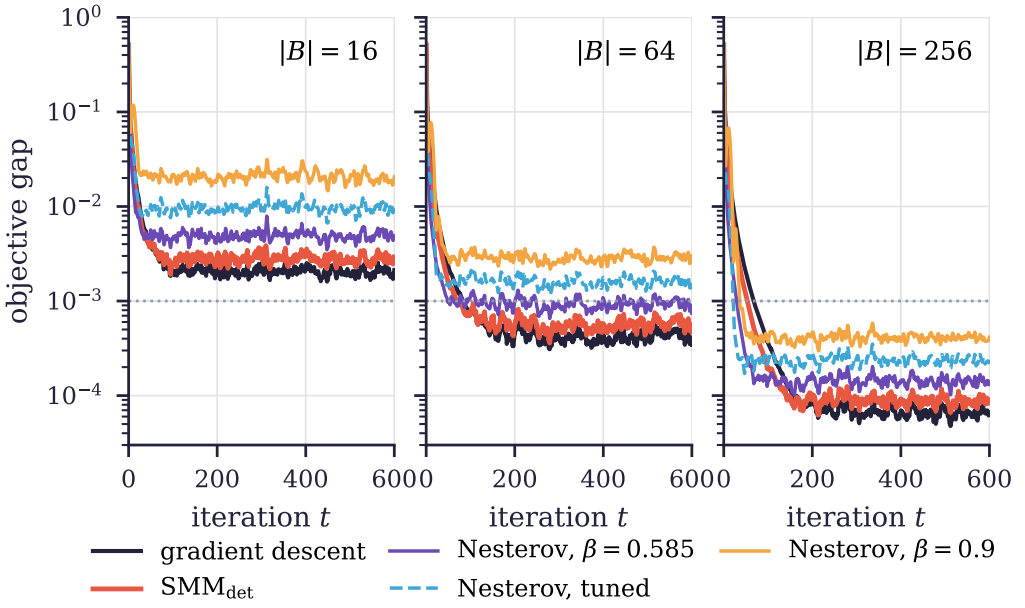


FIGURE 8.9. Minibatch stochastic gradients on the logistic regression ($T = 600$, mean of $R = 30$ runs; the dotted line is the tolerance 10^{-3}). In every panel the plateaus are ordered from gradient descent (lowest) through $\mathrm{SMM}_{\mathrm{det}}$, Nesterov at $\beta = 0.585$ and tuned Nesterov to Nesterov at $\beta = 0.9$ (highest).

TABLE 8.6. Final gap relative to gradient descent in the $\kappa = 25$ noise experiment: prediction from the exact stationary variances summed over the spectrum, and the range of measured ratios over the four positive noise levels of Table 8.3.

Method	Predicted	Measured
$\mathrm{SMM}_{\mathrm{det}}$, $\beta = 0.585$	1.21	1.22 to 1.27
NAG, $\beta = 0.5$	1.49	1.46 to 1.54
NAG, $\beta = 0.585$	1.65	1.61 to 1.73
NAG, $\beta = 0.667$	1.87	1.80 to 1.98
NAG, $\beta = 0.9$	3.36	3.07 to 3.76
HB, $\beta = 0.585$	2.21	2.14 to 2.31
HB, $\beta = 0.9$	8.96	8.02 to 9.86

every eigendirection with $a \leq 1$: for $\mathrm{SMM}_{\mathrm{det}}$ by Proposition 8.7; for Nesterov because $V_{\mathrm{NAG}} - V_{\mathrm{GD}} = 2\beta(1+\beta)(1-a)^2\alpha^2\sigma^2/\big(a(2-a)(1-\beta+a\beta)(2(1+\beta) - a(1+2\beta))\big) \geq 0$, an identity checked by `ch7_extra.py`; and for heavy ball because its ratio $(1+\beta)(2-a)/\big((1-\beta)(2(1+\beta)-a)\big)$ is at least 1 whenever $a \leq 1+\beta$. The dagger in Table 8.3 is therefore to be read with the gradient descent row: $\mathrm{SMM}_{\mathrm{det}}$ gives up little of the quietness of gradient descent while being faster than it without noise, and every step momentum gives up much more.

The minibatch test. The noise of a minibatch gradient is neither additive nor isotropic, but the plateaus of Figure 8.9 are ordered exactly as the slow mode amplifications: gradient descent (1), $\mathrm{SMM}_{\mathrm{det}}$ (1.36), Nesterov at $\beta = 0.585$ (2.41), tuned Nesterov, whose coefficient at $\kappa = 67.4$ is 0.783 with $1/(1-\beta) = 4.60$, and Nesterov at $\beta = 0.9$ (10). The iteration counts mix this ordering with speed (Remark 8.10). At $|B| = 16$ the mean plateaus of all methods lie near or above 10^{-3}, so the count is a first passage time of a fluctuating gap below a level close to the floor; the lower the floor, the sooner the crossing, which favours gradient descent and $\mathrm{SMM}_{\mathrm{det}}$. At $|B| = 256$ all floors lie well below the tolerance and speed decides.

The deterministic speedups. In the slowest mode of the logistic regression, $a = 1/67.4$, the predicted speedup over gradient descent is $\log r / \log(1-a)$, where r is the per step factor of the method: the spectral radius of its companion matrix, or $|\rho(a)|^{1/2}$ for $\mathrm{SMM}_{\mathrm{det}}$. For $\mathrm{SMM}_{\mathrm{det}}$ the prediction, 1.296, agrees within 0.3 percent with the observed ratio of the mean counts, $128.80/99.65 = 1.293$; for the overdamped heavy ball and Nesterov runs, $\beta \leq 0.667$, prediction and observation agree within seven percent, the prediction always slightly optimistic, as expected of a count that includes the initial transient. Only the underdamped runs at $\beta = 0.9$ deviate, since their oscillatory convergence is not governed by the asymptotic rate alone.

In short, the exact theory confirms the qualitative findings, explains their magnitudes, and adjusts one reading: in the slow modes SMM is not more efficient than gradient descent under noise, although over the whole spectrum of the $\kappa = 25$ experiment its floor, 1.21 times that of gradient descent at the same step, is below the 1.42 of gradient descent run at the same slow mode speed; it is much quieter than every step momentum at the same β, and its noise behaviour is almost insensitive to β.

8.11 Practical guidance and limitations

If gradients are exact and μ is available, a tuned accelerated method is the right tool; SMM is not accelerated. If gradients are exact and μ is unknown, every step momentum at a moderate fixed coefficient such as 0.585 or 0.667 is faster than SMM by a factor near $m_{\mathrm{NAG}}/m_{\mathrm{SMM}}$. If gradients are noisy at an unknown level, $\mathrm{SMM}_{\mathrm{det}}$ is a safe default: its floor is within 5/3 of that of gradient descent at

the same step for every $\beta < 1$ and on every eigendirection, it is stable at step $1/L$ for every $\beta < 1$, and at the Collatz coefficient it is about 29 percent faster than gradient descent in the slow modes; if only the floor at a given slow mode speed matters, gradient descent with a slightly longer step is slightly better in the slow eigendirections but worse over the whole spectrum (1.42 against 1.21 in the $\kappa = 25$ experiment). If per coordinate preconditioning is affordable, Adam is an alternative that needs no μ, although in the logistic experiment it was slower than every step momentum at a moderate coefficient. Iterate averaging [83] or a decreasing step lowers the floor of any of these methods; our comparisons are for a constant step.

The exact noise theory covers quadratics, with additive noise independent across iterations (Theorems 8.4 and 8.12) and, for $\mathrm{SMM}_{\mathrm{det}}$, multiplicative noise (Proposition 8.15). For nonquadratic objectives the noiseless theory is Proposition 7.10 together with two results of Chapter 9, which follows this chapter: $\mathrm{SMM}_{\mathrm{det}}$ decreases f on alternate iterates for every L smooth f (Proposition 9.5), and it converges linearly, at an explicit rate in terms of μ and L, for every μ strongly convex and L smooth f (Theorem 9.9). Under noise we have no theory beyond quadratics, and for minibatch noise only the qualitative agreement above. The experiments are small, in dimensions 25 and 30. For $\mathrm{SMM}_{\mathrm{state}}$ the only theory is the stability on quadratics of every switching rule, which Corollary 9.8 proves conditionally on Conjecture 9.7. Nothing here bears on the Collatz conjecture: the arithmetic supplied a number and a pattern, and what they do as an optimizer is ordinary linear algebra.

Notes

The AR(2) analysis of heavy ball under additive noise is classical and appears in many forms; Polyak's book [81] treats momentum on quadratics in detail, and the time series background is in Brockwell and Davis [14]. That constant momentum does not improve the efficiency of stochastic gradient descent in the small step regime, because the effective step and the noise both scale by $1/(1-\beta)$, was established in generality by Yuan, Ying and Sayed [111]; Kidambi, Netrapalli, Jain and Kakade [44] showed that heavy ball and Nesterov with stochastic gradients need not improve on stochastic gradient descent even on least squares. Proposition 8.9 is the period two analogue for SMM.

Theorem 8.4, Proposition 8.6, Proposition 8.7 and Proposition 8.9 are new in this book. The gain weight theorem (Theorem 8.12), Proposition 8.14 and Proposition 8.15 are also proved here; in the equal gain case the gain weight theorem recovers, in the slow mode, the equivalence of Yuan, Ying and Sayed [111]. The experiments of Section 8.9, the reconciliation of Section 8.10, the predicted columns of Tables 8.4 and 8.6, and the corresponding computations were done for this book with the scripts listed in Appendix B. Iterate averaging in stochastic approximation is due independently to Ruppert [89] and to Polyak [82]; Polyak

and Juditsky [83] give the standard analysis.

Exercises

Exercise 8.16. Show that the stationary variance of the AR(2) process of Lemma 8.1 can be written as $s^2(1-a_2)/\big((1+a_2)(1-a_1-a_2)(1+a_1-a_2)\big)$, and use this form to rederive Propositions 8.2 and 8.3 in one line each. Compute γ_1/γ_0 for heavy ball in the slow mode limit and interpret it.

Exercise 8.17. Let $p \geq 1$ and consider noisy gradient descent on a scalar quadratic with the periodic steps $h_1\alpha, \dots, h_p\alpha$, each step using a fresh noisy gradient. Show that the stationary variance at every phase satisfies $aV \to \alpha^2\sigma^2 \sum_j h_j^2 / \big(2\sum_j h_j\big)$ as $a \to 0$. Deduce that the noise floor at matched speed relative to constant steps is $p\sum_j h_j^2/(\sum_j h_j)^2 \geq 1$. Apply this to injection every third step, for which $(h_1, h_2, h_3) = (1+\beta, 1, 1)$ (compare the exercise on this schedule in Chapter 7).

Exercise 8.18. Compute the exact odd phase variance V_{odd} of Theorem 8.4 at $a = 1/(1+\beta)$ and at $a = 1$, where $\rho = 0$, and explain the answers directly from (8.10). Show that $V_{\mathrm{even}} \geq V_{\mathrm{odd}}$ if and only if $a(2-a)V_{\mathrm{even}} \geq \alpha^2\sigma^2$, and decide for which $a \in (0,1]$ this holds.

Exercise 8.19. Using (8.2) and the exact formulas, write a short program that reproduces the predicted column of Table 8.6. Then find the constant step $h\alpha$ for gradient descent that matches the slow mode speed of $\mathrm{SMM}_{\mathrm{det}}$ at $\beta = 0.585$, and compute its predicted final gap in the same experiment. Compare it with that of $\mathrm{SMM}_{\mathrm{det}}$ and explain the result in terms of the fast eigendirections.

Exercise 8.20. For every step Nesterov with fixed β, show that $V_{\mathrm{NAG}}(a)/V_{\mathrm{GD}}(a)$ is decreasing in a on $(0,1]$ and that it is increasing in β. Deduce that at fixed $a \in (0,1]$ its supremum over $\beta < 1$ is $(2-a)^2/\big(a(4-3a)\big)$, which is 1.8 at $a = 1/2$ but tends to infinity as $a \to 0$. Contrast this with the bound of Proposition 8.7, which is uniform in a and β together.

Chapter 9

Nonlinear Difference Equations and Momentum MM

The two halves of this book meet in a single mathematical object: the nonlinear difference equation. The Collatz map is one, driven by the parity of the state. Every first order optimizer is one too, and the momentum methods of Chapter 5 are second order difference equations whose coefficients may switch with the iteration counter. This chapter develops the common theory and proves three results, one of them conditionally on a conjecture that we support by computation.

First, the inverse Collatz tree satisfies a system of difference inequalities, in the spirit of Krasikov and Lagarias, from which we derive and certify explicit growth exponents: for every n not divisible by 3 and not on a cycle, at least $c\,X^{0.477}$ integers below Xn have orbits passing through n. Second, we compute exactly, using a rank one structure, the joint spectral radius of the switched linear system that describes Switching MM on a quadratic. It is below one at every parameter value we computed, and if a transient bound that we state as a conjecture holds it is below one for all of them, so that *every* switching rule, including adaptive and state dependent ones, is exponentially stable. Third, for any μ strongly convex and L smooth function, not necessarily quadratic, two steps of $\mathrm{SMM}_{\mathrm{det}}$ contract the distance to the minimizer by an explicit factor, which is the exact worst case over the class: explicit one dimensional functions attain it, and performance estimation confirms it. This settles open problem S1 of Chapter 15 for the distance; for function values the matching bound is exact only above a threshold condition number. We end with continuous time limits, where a backward error analysis shows that the implicit regularization of Switching MM carries the same Cauchy and Schwarz penalty as its noise floor.

9.1 The Collatz recursion as a switched system

Write $u_n = \log x_n$ for the logarithm of the nth iterate of the accelerated map T. Then

$$\begin{aligned} u_{n+1} &= u_n + g_{b_n}(u_n), \\ g_0(u) = -\log 2, &\qquad g_1(u) = \log\Big(\tfrac{3}{2} + \tfrac{1}{2}e^{-u}\Big), \end{aligned} \tag{9.1}$$

where the switching signal $b_n = x_n \bmod 2$ selects one of two vector fields. In the language of control theory (9.1) is a *switched system* with two modes [64]. Mode

0 is a constant descent; mode 1 is an ascent of size slightly above $\log(3/2)$, larger when u is small. Figure 9.1 shows both branches as traced by real orbits.

Two features distinguish (9.1) from the switched systems of engineering. The first is that the switching is *state dependent* in an arithmetic, not a geometric, way: b_n is a function of $x_n = e^{u_n}$, but of its parity, which changes on scales of size e^{-u_n} in u. The second is that the system is unstable under *arbitrary* switching: the signal $b_n \equiv 1$ gives $u_n \geq u_0 + n\log(3/2)$. Stability, if it holds, must come from the particular signal the arithmetic produces. The following elementary statement makes the threshold precise.

Proposition 9.1 *(The switching frequency decides). Let p_N be the proportion of odd steps among the first N steps of the orbit of x_0 under T. Then*

$$\begin{aligned} N\big(p_N \log 3 - \log 2\big) &\leq u_N - u_0 \\ &\leq N\big(p_N \log 3 - \log 2\big) + \sum_{n<N,\ b_n=1} \frac{1}{3x_n}. \end{aligned}$$

In particular an orbit along which $\liminf_N p_N > 1/\log_2 3 = 0.6309$ is unbounded, and an orbit along which $\limsup_N p_N < 1/\log_2 3 - 1/(3Y\log 3)$ cannot stay above Y at every step.

Proof. Each odd step contributes $\log(3/2 + 1/(2x_n)) = \log(3/2) + \log(1 + 1/(3x_n))$, which lies between $\log(3/2)$ and $\log(3/2) + 1/(3x_n)$, and each even step contributes $-\log 2$. Summing, $u_N - u_0$ lies between $Np_N\log(3/2) - N(1-p_N)\log 2 = N(p_N\log 3 - \log 2)$ and that quantity plus $\sum 1/(3x_n)$ over the odd steps. The first consequence follows from the lower bound. For the second, if $x_n \geq Y$ for all n then the correction is at most $Np_N/(3Y) \leq N/(3Y)$, and the upper bound tends to $-\infty$ under the stated condition, contradicting $x_n \geq Y$. □

The proposition is the Cesàro identity of Chapter 11 written for T rather than for the Syracuse map: the threshold $1/\log_2 3$ for the proportion of odd steps corresponds to the threshold $\log_2 3$ for halvings per odd step. In the language of switched systems, the divergence half of the conjecture asserts that the state dependent switching signal never dwells in mode 1 with frequency at or above $1/\log_2 3$ for long. Theorem 4.8 and Corollary 4.15 say that no strictly increasing function of x decreases along every stride, an odd step together with the halvings that follow it, nor along every block of J strides for any fixed J, so no such function, used as a Lyapunov function, can certify this; the certificate, if there is one, must read the arithmetic of the signal (Remark 4.10).

9.2 Difference inequalities for the inverse tree

The deepest unconditional lower bounds on how many integers reach 1 come from a different difference equation, one satisfied by counting functions on the inverse

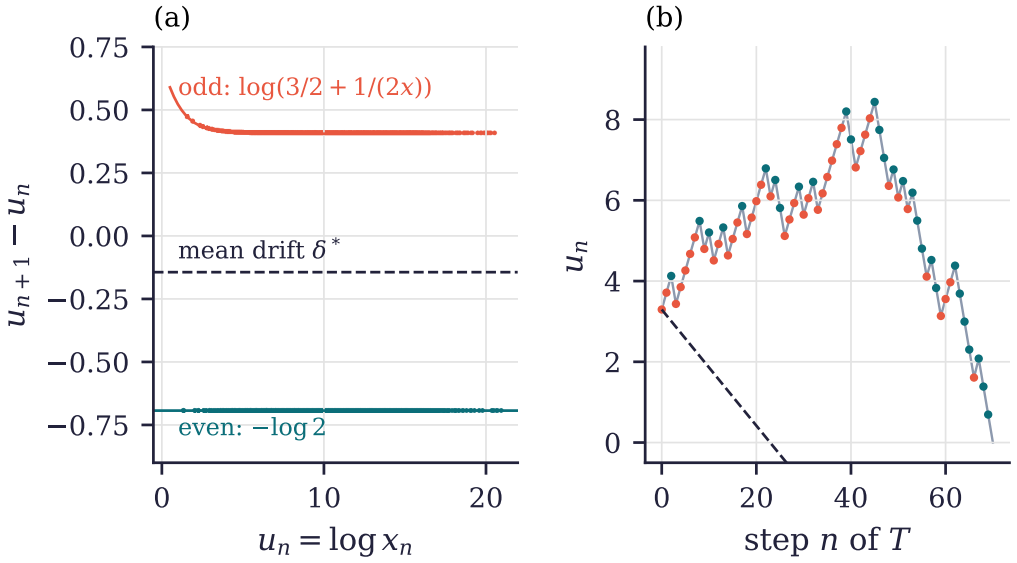


FIGURE 9.1. The Collatz recursion as a switched system in $u = \log x$. (a) Increments $u_{n+1} - u_n$ against u_n along 60 orbits started between 10^6 and 10^7; odd steps follow the curve $\log(3/2 + 1/(2x))$, even steps the constant $-\log 2$, and the dashed line is the mean drift $\delta^* = \log(\sqrt{3}/2)$ of the Bernoulli model. (b) The orbit of 27 under T, shaded by mode, with the drift line $\log 27 + n\delta^*$; every odd step rises and every even step falls.

tree. Krasikov [50] introduced the method, obtaining the exponent $3/7$ from a system of inequalities modulo 9; Applegate and Lagarias [3] reached 0.81 by solving larger systems with linear programming, and Krasikov and Lagarias [51] enlarged the systems further and reached the exponent 0.84: for large x, at least $x^{0.84}$ integers up to x have orbits that reach 1. We develop a simple version of the method here, complete with proofs, which already gives an exponent above $3/7$ and shows clearly where the difference inequalities come from. Accounts of the full method are in [51, 55].

The inverse tree. Under T every n has the preimage $2n$, and also the preimage $(2n-1)/3$ when $n \equiv 2 \pmod 3$. Starting from a root n, the *doubling chain* $n, 2n, 4n, \dots$ lies in its inverse tree, and an element $p = 2^j n$ of the chain with $j \geq 1$ has the odd preimage

$$c_j(n) = \frac{2^{j+1}n - 1}{3} \qquad \text{when } 2^j n \equiv 2 \pmod 3.$$

If moreover $2^j n \equiv 2$ or $8 \pmod 9$, then $c_j(n) \not\equiv 0 \pmod 3$ and $c_j(n)$ has odd preimages of its own; we call such j *fertile*. For $n \not\equiv 0 \pmod 3$ let

$$\pi_n(X) = \#\{m \leq Xn : \text{ the orbit of } m \text{ under } T \text{ passes through } n\}.$$

Lemma 9.2 *(Difference inequality). Let $n \not\equiv 0 \pmod 3$ be an integer that does not lie on a cycle of T. Then for every $X \geq 1$,*

$$\pi_n(X) \;\geq \sum_{j \text{ fertile},\ 2^{j+1} \leq 3X} \pi_{c_j(n)}\Big(\frac{3X}{2^{j+1}}\Big). \tag{9.2}$$

Proof. If m is counted in $\pi_{c_j}(3X/2^{j+1})$ then its orbit passes through c_j, hence through $T(c_j) = 2^j n$ and then through n, and $m \leq (3X/2^{j+1})c_j = Xn - X/2^{j+1} < Xn$. So each term counts integers that are also counted by $\pi_n(X)$. The sets for different fertile j are disjoint: the orbit of m reaches n for the first time along a unique path, which passes through exactly one element of the doubling chain immediately after an odd step, namely $2^j n = T(c_j)$; since n is not on a cycle, the orbit cannot reach n, leave it, and return through another branch. □

The inequality (9.2) is a difference inequality, linear in the counting functions, in the scale variable $\log X$: it bounds a counting function at scale X by the same family of functions at the smaller scales $3X/2^{j+1}$, with an index $c_j(n)$ that moves through residue classes. The residue class of $c_j(n)$ modulo 3^k is determined by n mod 3^{k+1} and j mod $2 \cdot 3^k$, the period of 2 modulo 3^{k+1}. This suggests a power law ansatz with a constant depending on the class.

Theorem 9.3 *(Certified growth exponents). Fix $k \geq 1$ and let $\mathcal{R}_k$ be the residues modulo 3^k prime to 3. For $g > 0$ and a positive vector $(w_r)_{r \in \mathcal{R}_k}$ define*

$$(\Phi_g w)_r = \min_{\ell} \sum_{j \text{ fertile for } \ell} \Big(\frac{3}{2^{j+1}}\Big)^g w_{[c_j(\ell)]},$$

where the minimum is over the three lifts ℓ of r to residues modulo 3^{k+1}, fertility and the class $[c_j(\ell)] \in \mathcal{R}_k$ depend only on ℓ and j, and the sum is over all $j \geq 1$. If $\Phi_g w \geq (1+\eta)w$ componentwise for some $\eta > 0$, then there is $C > 0$ such that

$$\pi_n(X) \;\geq\; C\, w_{[n]}\, X^g \qquad \text{for all } X \geq 1$$

and all $n \not\equiv 0 \pmod 3$ not lying on a cycle of T.

Proof. The sum defining Φ_g converges, since its terms decay like 2^{-jg}; by the periodicity of the classes in j it equals the sum over one period times $1/(1-2^{-Pg})$ with $P = 2 \cdot 3^k$. The terms with $2^{j+1} > 3X$, which are missing from (9.2), have total weight at most $w_{\max} \sum_{2^{j+1} > 3X} (3/2^{j+1})^g \leq w_{\max} X^{-g}/(1-2^{-g})$, which is below $\eta\, w_{\min}$ once $X \geq X_0$ for a suitable X_0. Choose $C = 1/(w_{\max} X_0^g)$, so that the claim holds for $X \leq X_0$ because $\pi_n(X) \geq 1$ (the root itself is counted). For $X > X_0$ argue by strong induction on the scale: every argument $3X/2^{j+1}$ in (9.2) is at most $3X/4 < X$, and the children $c_j(n)$ are again prime to 3 and not on a cycle (an

element of a cycle has its image on the cycle, and $T(c_j) = 2^j n$ leads to n). Hence

$$\pi_n(X) \geq CX^g \sum_{\substack{j \text{ fertile} \\ 2^{j+1} \leq 3X}} \Big(\frac{3}{2^{j+1}}\Big)^g w_{[c_j(n)]}$$
$$\geq CX^g\big((1+\eta)w_{[n]} - \eta w_{\min}\big) \geq CX^g w_{[n]},$$

using that the full sum is at least $(\Phi_g w)_{[n]}$, because the lift of $[n]$ determined by n mod 3^{k+1} is one of the three in the minimum. □

The theorem reduces a lower bound for an infinite family of counting functions to a finite eigenvalue problem: find the largest g for which the monotone, homogeneous operator Φ_g, nonlinear only through the minimum over lifts, has a positive supersolution. We solved it by power iteration and bisection in g, and certified each exponent by exhibiting a vector w with $\min_r(\Phi_g w)_r / w_r > 1$ (script `ch5b_tree.py`), and checked each certificate in exact integer arithmetic, with every power of 2 and 3 rounded down (script `ch5b_tree_exact.py`). The results are

k	2	3	4	5	6	7
classes	6	18	54	162	486	1458
γ_k	0.416	0.435	0.449	0.460	0.470	0.477

with certified margins $1+\eta$ between 1.0002 and 1.0027. The certificate for $k = 7$ still has margin 1.0001 at $g = 0.47705$, so for the root $n = 4$, which reaches 1 and is not on a cycle, the theorem gives $\pi_4(X) \geq c\, X^{0.47705}$ for some $c > 0$, which exceeds $(4X)^{0.477}$ for all large X. This gives the following.

COROLLARY 9.4 *(A counting bound). For all large x, at least $x^{0.477}$ integers $m \leq x$ have orbits that reach* 1.

The exponents increase with the modulus but slowly, and they are bounded by what single doubling chains can produce. A mean field heuristic predicts the limit. On average one third of the j are fertile, and if the class constants were all equal the supersolution condition would read $\frac{1}{3}\sum_{j\geq 1}(3/2^{j+1})^g = 1$, that is $3^{g-1} = 2^g(2^g - 1)$, whose root is $g = 0.4927$. The exact counts of the doubling chain construction approach it: for roots $n = 4, 5, 7$ the number of integers it generates below Xn grows with exponent 0.500, 0.502 and 0.498 over the last decade of X up to 2^{20}, and with exponent 0.492, 0.494 and 0.492 over the last decade up to 2^{36} (script `ch5b_chains.py`), while the true counting functions $\pi_n(X)$ grow with exponent 1.000 (Figure 9.2). The gap between 1/2 and 0.84 comes from a branch that the doubling chain system leaves out. Every element $p \equiv 2 \pmod 3$ of the inverse tree has the odd preimage $(2p-1)/3 < p$, the term $j = 0$; seen from p, its count enters at the larger scale $3X/2$, so the induction on scale in the proof of Theorem 9.3 cannot use it. With this branch the mean field equation becomes $\frac{1}{3}\sum_{j\geq 0}(3/2^{j+1})^g = 1$, that is $3^{g-1} = 2^g - 1$, whose smaller

root is exactly $g = 1$. Krasikov's inequalities keep these terms, which Krasikov and Lagarias call advanced; Applegate and Lagarias truncated them to reach 0.81, and Krasikov and Lagarias reached 0.84 by proving that the linear programming bounds remain valid with the advanced terms present [51]. The elements divisible by 3 play no role: such an element has no odd preimage, so its inverse tree is a single doubling chain. The gap between 0.84 and 1 is open. The exponent 1 would already follow from a positive proportion of integers reaching 1, which is far short of the conjecture.

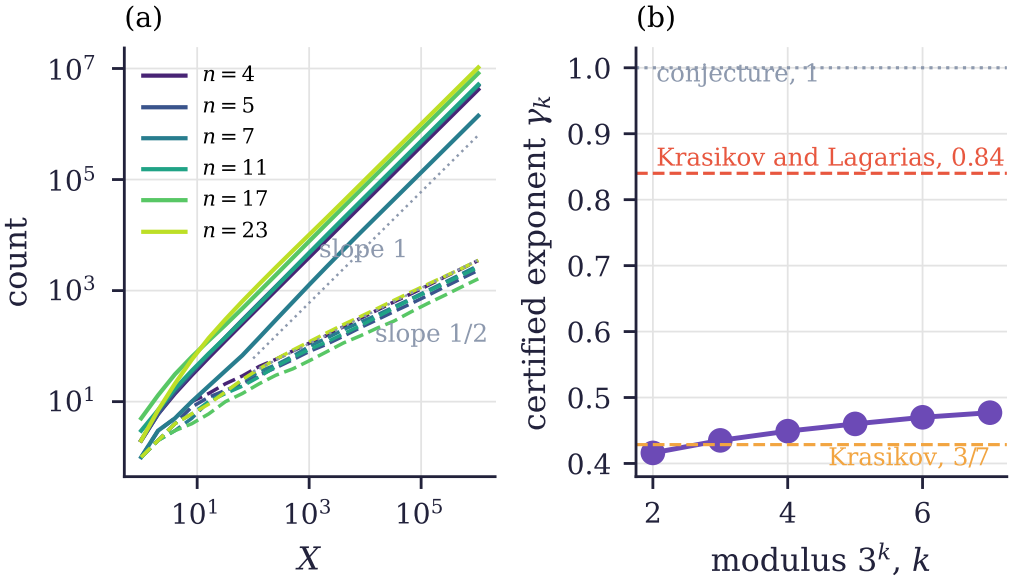


FIGURE 9.2. (a) Exact counting functions $\pi_n(X)$ (solid) and the counts produced by the doubling chain construction (dashed) for several roots n, up to $X = 2^{20}$; the true counts grow linearly, the construction roughly like $X^{1/2}$. (b) Certified exponents γ_k from Theorem 9.3 for moduli 3^k, against Krasikov's $3/7$, the exponent 0.84 of Krasikov and Lagarias, and the conjectured value 1.

9.3 Momentum MM as a difference equation

We now turn to the optimizers. For an L smooth function f the descent lemma says that

$$f(y) \le Q_L(y \mid x) := f(x) + \langle \nabla f(x), y - x \rangle + \tfrac{L}{2}\|y - x\|^2 \tag{9.3}$$

for all x, y, with equality at $y = x$. So $Q_L(\cdot \mid x)$ majorizes f in the sense of Chapter 4, and its minimizer is the gradient step $G_{1/L}(x) = x - \nabla f(x)/L$. Gradient descent is the MM algorithm for this majorizer, and the MM descent property gives $f(G_{1/L}(x)) \le f(x) - \|\nabla f(x)\|^2/(2L)$.

Momentum methods of Nesterov's type majorize at a different point. Given an *anchor* $y_t = x_t + \beta_t(x_t - x_{t-1})$, they set $x_{t+1} = \arg\min_y Q_L(y \mid y_t) = G_{1/L}(y_t)$. With $\beta_t \equiv \beta$ this is Nesterov's method; with $\beta_t = \beta$ for odd t and $\beta_t = 0$ for even t it is $\mathrm{SMM}_{\mathrm{det}}$. We call the general scheme *momentum MM*: an MM iteration whose majorization point is extrapolated on a schedule. Written out, it is the second order nonlinear difference equation

$$x_{t+1} = F_t(x_t, x_{t-1}) := y_t - \tfrac{1}{L}\nabla f(y_t), \qquad y_t = x_t + \beta_t(x_t - x_{t-1}), \tag{9.4}$$

which is periodic in t for Switching MM and autonomous for Nesterov.

Momentum MM gives up the monotonicity of MM: Nesterov's method, like heavy ball, is not a descent method, and its function values oscillate. Switching MM keeps it on alternate iterates, for every smooth function, convex or not.

Proposition 9.5 *(Monotonicity on alternate iterates). Let f be L smooth and bounded below by f_{inf}, and run* $\mathrm{SMM}_{\mathrm{det}}$ *with step $1/L$ and $\beta \in [0, 1)$. For even t,*

$$f(x_{t+2}) \le f(x_t) - \frac{1-\beta^2}{2L}\|\nabla f(x_t)\|^2 - \frac{1}{2L}\|\nabla f(y_{t+1})\|^2.$$

Consequently $\min_{t<2K,\ t \text{ even}} \|\nabla f(x_t)\|^2 \le 2L(f(x_0) - f_{\mathrm{inf}})/((1-\beta^2)K)$.

Proof. By Lemma 7.5, $y_{t+1} = G_{(1+\beta)/L}(x_t)$ and $x_{t+2} = G_{1/L}(y_{t+1})$. For a gradient step of length h the descent lemma gives $f(G_h(x)) \le f(x) - h(1 - Lh/2)\|\nabla f(x)\|^2$. With $h = (1+\beta)/L$ the coefficient is $(1+\beta)(1-\beta)/(2L)$, and with $h = 1/L$ it is $1/(2L)$. Adding the two inequalities gives the first claim, and summing over K periods gives the second. □

The extrapolation is safe because it is a gradient step of length $(1+\beta)/L < 2/L$, inside the range where the descent lemma still guarantees decrease. Momentum at every step has no such interpretation: its anchor carries the whole history of past gradients.

9.4 Stability theory

9.4.1 Lyapunov functions for difference equations

A function $V \ge 0$ is a Lyapunov function for the difference equation $z_{t+1} = F_t(z_t)$ with rate $\rho < 1$ if $V(z_{t+1}) \le \rho V(z_t)$ for all t and z_t; then $V(z_t) \le \rho^t V(z_0)$. For gradient descent on a μ strongly convex, L smooth function, $V = f - f^*$ works with $\rho = 1 - 1/\kappa$. For Nesterov's method with constant momentum $\beta = (\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$ and step $1/L$, Nesterov's estimate sequence construction gives the Lyapunov function $f(x_t) - f^* + \frac{\mu}{2}\|v_t - x^*\|^2$, with v_t an auxiliary sequence that is an affine combination of x_t and x_{t-1}, and the rate $1 - 1/\sqrt{\kappa}$: for every t,

$$f(x_t) - f^* \le \Big(1 - \frac{1}{\sqrt{\kappa}}\Big)^t \Big(f(x_0) - f^* + \frac{\mu}{2}\|x_0 - x^*\|^2\Big),$$

see [71], and [106] for the Lyapunov reading. The point to retain is that the Lyapunov function of an accelerated method must involve two consecutive iterates, because the state of the method is the pair (x_t, x_{t-1}): started at $x_t = x^*$ with $x_{t-1} \neq x^*$ the method in general leaves the minimizer, so no function of x_t alone that vanishes only at x^* can decrease at every step.

9.4.2 Switching and the joint spectral radius

On one eigendirection of a quadratic, with $a = \alpha\lambda$ and state $z_t = (e_t, e_{t-1})$, the two steps of $\mathrm{SMM}_{\mathrm{det}}$ are the matrices

$$D = \begin{pmatrix} 1-a & 0 \\ 1 & 0 \end{pmatrix}, \qquad N = \begin{pmatrix} (1-a)(1+\beta) & -(1-a)\beta \\ 1 & 0 \end{pmatrix},$$

for a descent and an injection respectively. $\mathrm{SMM}_{\mathrm{det}}$ applies them in the periodic order $D, N, D, N, \ldots$, every step Nesterov applies N always, and the adaptive rule $\mathrm{SMM}_{\mathrm{state}}$ of Chapter 7 applies them in an order that depends on the state. What can be said for an *arbitrary* order? The relevant quantity is the joint spectral radius of Rota and Strang [88],

$$\hat{\rho}(\{D, N\}) = \lim_{k \to \infty} \max_{M_i \in \{D,N\}} \|M_k \cdots M_1\|^{1/k},$$

which is the best possible exponential rate valid for every switching sequence; see [41] for the theory. It is in general hard to compute. Here it is not, because D has rank one.

Proposition 9.6 *(Exact joint spectral radius). Write $D = vu^\top$ with $u = (1,0)^\top$ and $v = (1-a, 1)^\top$, and put $\tau_j = u^\top N^j v$, the first coordinate after j injections started from the state left by one descent step. Then*

$$\hat{\rho}(\{D, N\}) = \max\Big(\rho(N),\ \sup_{j \geq 0} |\tau_j|^{1/(j+1)}\Big).$$

Proof. By the theorem of Berger and Wang (see [41]) the joint spectral radius equals $\limsup_k \max \rho(M_k \cdots M_1)^{1/k}$ over products, and by cyclic invariance of the spectrum it suffices to consider products that are either N^k or begin with D. A product beginning with D is, up to cyclic permutation, $DN^{j_1} DN^{j_2} \cdots DN^{j_m}$ with $\sum (j_i + 1) = k$. Since $D = vu^\top$, this product equals $v \left(\prod_i u^\top N^{j_i} v\right)$ times $u^\top$ in the appropriate order, a rank one matrix with the single nonzero eigenvalue $\prod_i \tau_{j_i}$. Hence $\rho^{1/k} = \prod_i |\tau_{j_i}|^{1/k} \leq \max_i |\tau_{j_i}|^{1/(j_i+1)}$, with equality when all j_i are equal. Taking the supremum over products gives the formula. □

The formula needs only the scalar sequence τ_j, which tends to zero geometrically at rate $\rho(N) < 1$, so the maximum in the formula is attained, at a finite j or by $\rho(N)$, and can be computed to any accuracy (script `ch5b_jsr.py`).

The result is striking (Figure 9.3). For every a we computed the joint spectral radius is below 1, and it exceeds the larger of the two pure rates, $\max(\rho(D), \rho(N))$, by at most 0.0056 at $\beta = \log_2 3 - 1$ and 0.0011 at $\beta = 0.9$, over a grid of 400 values of a from 10^{-4} to 0.999. For small a it equals $1 - a$, the rate of gradient descent, attained by the product D alone.

Since $\tau_j \to 0$, the joint spectral radius is below 1 exactly when $|\tau_j| < 1$ for every j. Over a grid of 200 values of $\beta \in [0, 0.999]$ and 300 values of $a \in [10^{-5}, 0.9999]$ we found $\sup_j |\tau_j| = \tau_0 = 1 - a$ in every case, with a quadratic Lyapunov function of N bounding the terms beyond those computed (script `ch5b_transient.py`): the transient of the injection steps never exceeds the first descent. We record this as a conjecture.

Conjecture 9.7 *(Transient bound). For all $a \in (0, 1]$, $\beta \in [0, 1)$ and $j \geq 0$, $|\tau_j| \leq 1 - a$.*

Note that while e stays positive the bound is immediate: with $d_t = e_{t-1} - e_t$ one has $e_{t+1} = (1 - a)(e_t - \beta d_t)$ and $d_{t+1} = a e_t + (1 - a)\beta d_t$, with $e_0 = 1 - a$ and $d_0 = a$. As long as e stays nonnegative, d stays nonnegative and $e_{t+1} \leq (1 - a)e_t$, so e decreases and $|e_t| \leq 1 - a$. The difficulty is the oscillating regime after e changes sign.

Corollary 9.8 *(Every switching rule is stable). Fix $a \in (0, 1)$ and suppose $|\tau_j| < 1$ for all j, as Conjecture 9.7 asserts and as computation confirms on the grid above. Then on a quadratic with spectrum in $(0, L]$ and step $1/L$, every switching sequence between descent and injection steps, whether periodic, random, or chosen adaptively from the state as in* $\mathrm{SMM}_{\mathrm{state}}$*, converges to the minimizer on the eigendirection with $a = \alpha\lambda$, with exponential rate at most $\hat{\rho}(\{D(a), N(a)\}) < 1$.*

This is partial progress on open problem S4 of Chapter 15. At every grid point above, and for all a and β if Conjecture 9.7 holds, adaptive switching cannot destabilize the method. On the grid of 400 values of a it also cannot make the contraction factor more than 0.0056 larger than that of the worse of gradient descent and every step Nesterov at $\beta = \log_2 3 - 1$, which is about one percent per step and 1.8 percent in the number of iterations, and at $\beta = 0.9$ the cost is 0.12 percent per step and 1.1 percent in iterations. Stability alone does not close the noiseless gap to every step momentum that S4 asks about. The Collatz switched system (9.1) is the opposite case: there the joint behaviour under arbitrary switching is unstable, and all the difficulty lies in the switching signal.

9.5 A linear rate beyond quadratics

Theorem 7.6 described $\mathrm{SMM}_{\mathrm{det}}$ exactly on quadratics. For a general smooth strongly convex function the same identity, Lemma 7.5, holds with no change, and it reduces the convergence question to the classical contraction of a single gradient step.

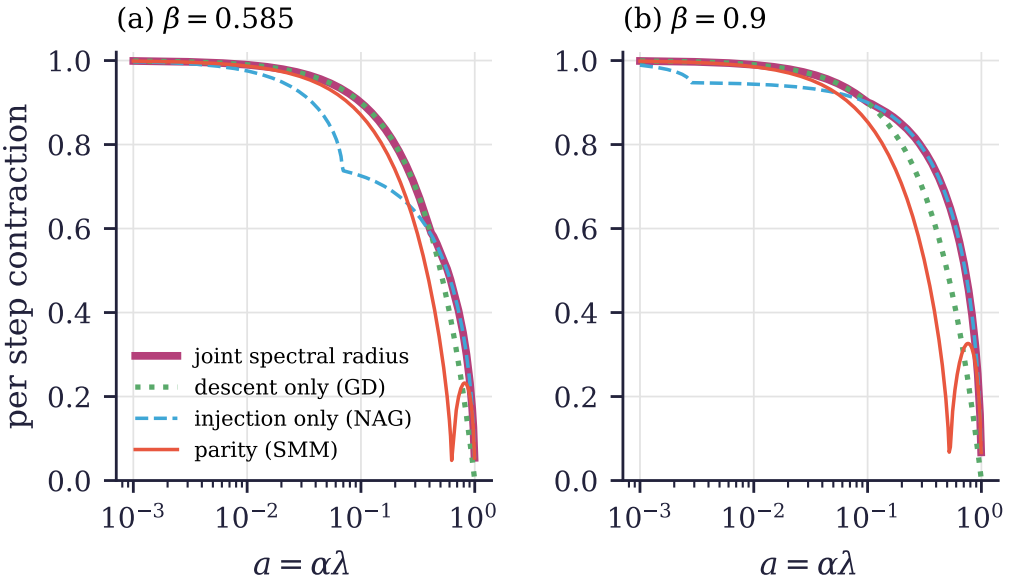


FIGURE 9.3. The joint spectral radius of the descent and injection matrices, computed exactly by Proposition 9.6, against $a = \alpha\lambda$, together with the per step rates of pure descent (gradient descent), pure injection (every step Nesterov) and the parity schedule of $\mathrm{SMM}_{\mathrm{det}}$. The joint spectral radius hugs the upper envelope of the two pure rates.

THEOREM 9.9 *(Linear rate of* $\mathrm{SMM}_{\mathrm{det}}$*). Let* f *be* μ *strongly convex and* L *smooth with minimizer* x^**, put* $\kappa = L/\mu$*, and run* $\mathrm{SMM}_{\mathrm{det}}$ *with step* $\alpha = 1/L$ *and* $\beta \in [0,1)$*. For every even* t*,*

$$\|x_{t+2} - x^*\| \le q\,\|x_t - x^*\|,$$
$$q = q(\beta,\kappa) = \Big(1 - \frac{1}{\kappa}\Big)\max\Big(1 - \frac{1+\beta}{\kappa},\ \beta\Big),$$

and $f(x_{t+2}) - f^* \le q^2\,(f(x_t) - f^*)$*.*

Proof. By Lemma 7.5, $x_{t+2} = G_{1/L}(G_{(1+\beta)/L}(x_t))$. For a step $h \in (0, 2/L)$ and f twice differentiable, $G_h(x) - x^* = x - x^* - h(\nabla f(x) - \nabla f(x^*)) = (I - h\bar{H})(x - x^*)$ with $\bar{H} = \int_0^1 \nabla^2 f(x^* + s(x - x^*))\,ds$, whose eigenvalues lie in $[\mu, L]$. Hence $\|G_h(x) - x^*\| \le \max(|1 - h\mu|, |1 - hL|)\|x - x^*\|$; for f that is only once differentiable the same bound follows by smoothing, or directly from the cocoercivity of ∇f. With $h = (1+\beta)/L$ the factor is $\max(1 - (1+\beta)/\kappa, \beta)$, and with $h = 1/L$ it is $1 - 1/\kappa$. Composing gives the distance bound. For function values, Taylor, Hendrickx and Glineur [98, Theorem 3.3] show that a gradient step of length $h \in (0, 2/L)$ satisfies $f(G_h(x)) - f^* \le \max((1 - h\mu)^2, (1 - hL)^2)(f(x) - f^*)$; composing the two steps gives the second bound. □

How good is this? The contraction of each gradient step is tight for the class, so one might worry only about the composition. We computed the exact worst case of two steps of $\mathrm{SMM}_{\mathrm{det}}$ over all μ strongly convex, L smooth functions in every dimension by *performance estimation* [20, 97]: the worst case is the value of a small semidefinite program built from the interpolation conditions of the class (script `ch5b_pep.py`). For $\beta \in \{0.3, \log_2 3 - 1, 0.9\}$ and $\kappa \in \{2, 3, 5, 10, 20, 50, 100\}$ the worst case of $\|x_{t+2} - x^*\|^2 / \|x_t - x^*\|^2$ agreed with q^2 to six digits in all 21 cases. So the distance bound of Theorem 9.9 is the exact worst case, not merely a bound, and explicit functions attain it for every κ. When $\kappa \geq (1+\beta)/(1-\beta)$, which is $\kappa \geq 3.82$ at $\beta = \log_2 3 - 1$, both factors are attained at the eigenvalue μ and the worst case is the quadratic $\frac{\mu}{2}\|x\|^2$. For smaller κ the two factors are attained at different curvatures, and the worst function is not quadratic: the one dimensional function equal to $\frac{L}{2}x^2$ for $x \geq 0$ and to $\frac{\mu}{2}x^2$ for $x < 0$ attains q, since from $x_t = 1$ the long step lands at $-\beta$ and the short step at $-\beta(1 - 1/\kappa)$. The function value bound q^2 is exact only for $\kappa \geq (1+\beta)/(1-\beta)$, where the same quadratic attains it. Below that threshold performance estimation for function values gives a smaller worst case, for example 0.049 against $q^2 = 0.086$ at $\beta = \log_2 3 - 1$ and $\kappa = 2$.

Corollary 9.10 *(Speed relative to gradient descent). For $\kappa \geq (1+\beta)/(1-\beta)$ the per step rate of $\mathrm{SMM}_{\mathrm{det}}$ is $q^{1/2} = ((1 - 1/\kappa)(1 - (1+\beta)/\kappa))^{1/2}$, so that $-\kappa \log q^{1/2} \to (2+\beta)/2$ as $\kappa \to \infty$, against 1 for gradient descent with the same step. This is the effective step multiplier of Corollary 7.9, now valid on the whole class.*

Figure 9.4 plots the normalized rate $-\kappa \log q/2$ against κ. It lies below gradient descent exactly when $\kappa < 1/(1-\beta)$, which is $\kappa < 2.41$ at $\beta = \log_2 3 - 1$: there the long step $(1+\beta)/L$ overshoots the large curvatures and the factor β dominates. On the range of the figure it is largest at $\kappa = (1+\beta)/(1-\beta)$, where the two factors of q balance (1.60 at $\beta = \log_2 3 - 1$), and beyond that point it decreases to $(2+\beta)/2$, which is 1.29 at $\beta = \log_2 3 - 1$, as $\kappa \to \infty$. Open problem S1 asked for exactly this: a linear rate on the whole class with the constant predicted by the quadratic analysis. The answer is that the quadratic constant is exact for $\kappa \geq (1+\beta)/(1-\beta)$, and in particular as $\kappa \to \infty$.

9.6 Continuous time limits

The continuous time view of Chapter 5 describes accelerated methods by second order differential equations, obtained by letting the step size tend to zero while the momentum tends to one at a matched rate: for heavy ball with step s and $\beta = 1 - c\sqrt{s}$, the limit is the damped oscillator $\ddot{x} + c\dot{x} + \nabla f(x) = 0$ on the time scale $t = k\sqrt{s}$, and Nesterov's method with $\beta_k = (k-1)/(k+2)$ gives the equation of Su, Boyd and Candès [95]. With β *fixed* and $\alpha \to 0$ the picture is different, and it is the relevant one for a fixed default such as $\beta = \log_2 3 - 1$.

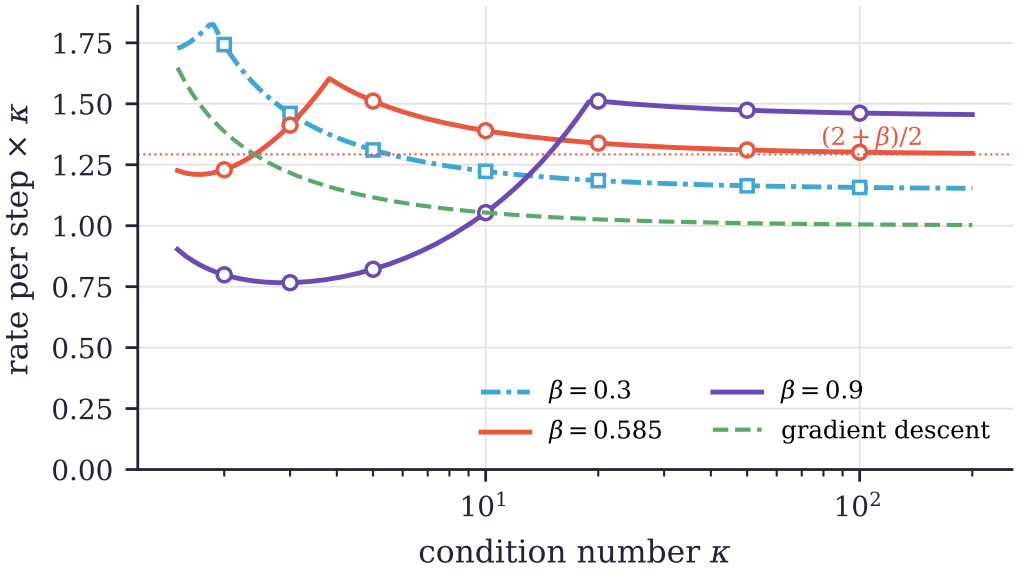


FIGURE 9.4. Worst case per step rate of $\mathrm{SMM}_{\mathrm{det}}$ for the distance to the minimizer on the class of μ strongly convex, L smooth functions, multiplied by κ, from Theorem 9.9 (lines) and from performance estimation (open markers), for three values of β; gradient descent with step $1/L$ is dashed. At large κ the normalized rate of $\mathrm{SMM}_{\mathrm{det}}$ tends to $(2+\beta)/2$.

PROPOSITION 9.11 *(Modified equation for period two steps). Let f be three times differentiable and consider the composition of gradient steps of lengths h_1 and h_2, in either order. To second order in $h = h_1 + h_2$ it agrees with the time h flow of the modified equation*

$$\begin{aligned} \dot{x} &= -\nabla f(x) - \frac{h_1^2 + h_2^2}{2(h_1+h_2)} \nabla^2 f(x) \nabla f(x) \\ &= -\nabla\Big(f + \frac{\bar{h}}{4}\|\nabla f\|^2\Big), \qquad \bar{h} = \frac{h_1^2+h_2^2}{h_1+h_2}. \end{aligned} \tag{9.5}$$

Proof. Write $g = \nabla f$ and $H = \nabla^2 f$. Expanding, $G_{h_2}(G_{h_1}(x)) = x - h_1 g - h_2 g(x - h_1 g) + O(h^3) = x - hg + h_1 h_2 Hg + O(h^3)$, and the term $h_1 h_2 Hg$ is symmetric in the two steps. The time h flow of $\dot{x} = -g - cHg$ is $x - hg - hcHg + \frac{h^2}{2} Hg + O(h^3)$, because the second derivative of the flow is Hg to leading order. Matching the coefficients of Hg gives $-hc + h^2/2 = h_1 h_2$, so $c = (h^2/2 - h_1 h_2)/h = (h_1^2 + h_2^2)/(2h)$. Finally $Hg = \nabla(\frac{1}{2}\|g\|^2)$. □

For a single gradient step of length h the same computation gives the modified function $f + \frac{h}{4}\|\nabla f\|^2$, the implicit gradient regularization of Barrett and Dherin [8]; backward error analysis of this kind is classical in geometric numerical integration [29]. For $\mathrm{SMM}_{\mathrm{det}}$, $h_1 = (1+\beta)\alpha$ and $h_2 = \alpha$, so the iteration

follows the gradient flow of

$$f + \frac{\alpha V(\beta)}{4}\|\nabla f\|^2, \qquad V(\beta) = \frac{(1+\beta)^2+1}{2+\beta},$$

on the time scale $(2+\beta)\alpha/2$ per iteration; $V(\beta)$ is the slow mode noise amplification of Corollary 8.5. Two conclusions follow.

First, Switching MM has no inertial limit. Whatever the value of $\beta < 1$, its continuous time limit is a first order gradient flow, sped up by the factor $(2+\beta)/2$ and implicitly regularized. This is the continuous counterpart of Remark 7.11: the method carries one gradient of memory, not a velocity, and no rescaling turns it into a damped oscillator.

Second, the implicit regularization strength has a familiar ratio. Gradient descent run at the same speed, with step $(2+\beta)\alpha/2$, has regularization coefficient $(2+\beta)\alpha/8$, while $\mathrm{SMM}_{\mathrm{det}}$ has $V(\beta)\alpha/4$. Their ratio is

$$\frac{2V(\beta)}{2+\beta} = \frac{2\big((1+\beta)^2+1\big)}{(2+\beta)^2} = 1 + \frac{\beta^2}{(2+\beta)^2},$$

exactly the noise floor penalty of Proposition 8.9 and of the gain weight theorem, Theorem 8.12. This is not a coincidence. Both quantities are the ratio of the mean square step to the square of the mean step, $\frac{1}{2}(h_1^2+h_2^2)/(\frac{1}{2}(h_1+h_2))^2$, which Cauchy and Schwarz bounds below by 1. Unequal steps cost the same factor whether one measures the stationary noise or the implicit bias.

Notes

The stability of switched systems under arbitrary and constrained switching is surveyed by Liberzon [64]; the joint spectral radius is due to Rota and Strang [88], and Jungers [41] covers both its computation and the theorem of Berger and Wang. The difference inequality method for the inverse Collatz tree is due to Krasikov [50], with improvements by Applegate and Lagarias [3] and the sharpest results by Krasikov and Lagarias [51]; the survey volume [55] gives the history. The doubling chain system, Theorem 9.3 and its certified exponents are our simplified version and are far from the state of the art. The view of gradient descent as an MM algorithm through the descent lemma is standard [40, 60]; the name momentum MM, the monotonicity and joint spectral radius propositions, and the rate theorem are ours, with the function value step bound taken from Taylor, Hendrickx and Glineur [98]. For finite sets of rank one matrices, Ahmadi and Parrilo [1] reduce the joint spectral radius in the same way to products around cycles, a maximum cycle mean problem; Proposition 9.6, with one matrix of rank one, is a variant of their device. Performance estimation was introduced by Drori and Teboulle [20] and made exact for smooth strongly convex functions by Taylor, Hendrickx and Glineur [97]. Modified equations and backward error analysis are treated in [29], and their use for implicit regularization in [8].

Exercises

Exercise 9.12. Show that under arbitrary switching the system (9.1) can realize every sequence of modes, that while $u_n \geq 0$ every increment lies in $\{-\log 2\} \cup (\log \frac{3}{2}, \log 2]$, and that a run of mode 1 increments is strictly decreasing. Show also that for the actual Collatz signal no two consecutive steps of the standard map T_0 are odd. What does this say about the frequencies achievable under T_0?

Exercise 9.13. Compute by hand the operator Φ_g of Theorem 9.3 for $k = 1$, with classes $1, 2 \bmod 3$ and lifts modulo 9, and find the largest g for which it has a positive supersolution. Compare with the values in the table.

Exercise 9.14. Verify the mean field equation $3^{g-1} = 2^g(2^g - 1)$ and show that it has a unique root in $(0, 1)$. Explain why the doubling chain construction cannot give an exponent above this root for typical roots.

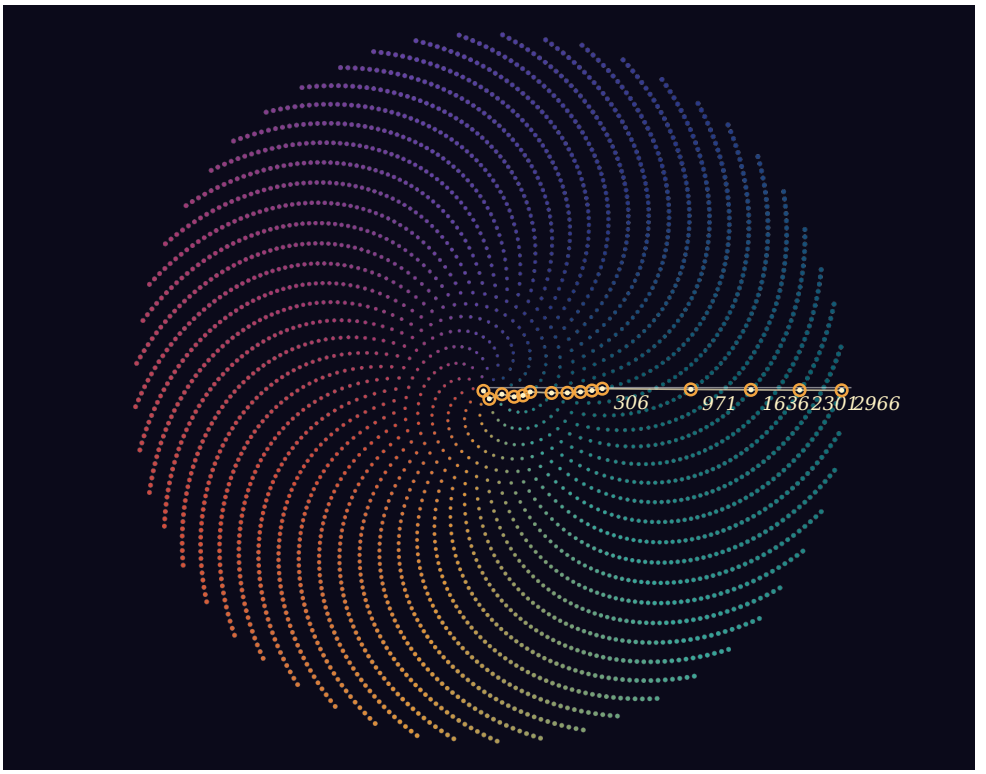
Exercise 9.15. Show that Proposition 9.5 fails for every step heavy ball: construct an L smooth convex quadratic in one dimension and a starting point for which $f(x_{t+2}) > f(x_t)$ for some even t with $\beta = 0.585$ and step $1/L$.

Exercise 9.16. Compute τ_0, τ_1, τ_2 of Proposition 9.6 in closed form and show that $|\tau_1|^{1/2}$ is the per step rate of the periodic schedule. For which a does the supremum in the proposition occur at $j = 0$?

Exercise 9.17. Using Theorem 9.9, find the value of β that minimizes $q(\beta, \kappa)$ for a given κ, and show that it tends to 1 as $\kappa \to \infty$. Compare the resulting rate with gradient descent and with Nesterov's optimal $1 - 1/\sqrt{\kappa}$.

Part III

Cycles and Divergence



The sunflower of $r \log_2 3$ modulo 1 for $r \leq 3000$: angle is the fractional part, radius $\sqrt{r}$; the gold rings are the rungs of the one sided ladder that control every cycle bound.

Chapter 10

Cycles and Continued Fractions

The Collatz conjecture asserts two things at once: no orbit of a positive integer escapes to infinity, and no orbit settles into a cycle other than $1 \mapsto 4 \mapsto 2 \mapsto 1$. Chapter 1 called these the divergence half and the cycle half. The two halves are very different in character. The divergence half is a statement about the long run behaviour of a single orbit, and all the probabilistic machinery of the earlier chapters bears on it only through ensembles. The cycle half is much more rigid. A cycle is a finite object, it satisfies an exact multiplicative identity, and that identity turns the question into one about how well the number $\log_2 3$ can be approximated by rationals.

This chapter develops that reduction completely. We begin with the cycles that do exist, on the negative integers and on the rationals with odd denominator, because they show that the cycle identity is not vacuous. We then prove the identity and its consequence, an upper bound on the least element of a cycle in terms of the one sided distance from $r \log_2 3$ up to the next integer, where r is the number of odd elements. To exploit it we need continued fractions, and we give a primer that is complete in itself, including the classical theorem identifying the best one sided approximations with the intermediate fractions of one parity [31, Theorem 4.5(ii)]. The heart of the chapter is a short *mediant lemma*, a self contained proof of a classical fact, which shows that the first value of r not excluded by a given verification range must be a best one sided approximation. This makes the search for the sharp cycle bound short and rigorous: with the verification range 2^{68}, and equally with the current range 2^{71}, every nontrivial cycle has at least 72 057 431 991 odd elements. We compare this with the cruder bounds one obtains without the lemma, place it in the literature (where sharper numerical bounds exist and where we claim no priority), and then prove that no Diophantine argument of this kind can finish the job, a point that Halbeisen and Hungerbühler [30, §5] made earlier in another form (Remark 10.17): for every verification range infinitely many r survive. What is missing is arithmetic information about the parity word, and a window lemma makes precise how much would suffice. We also give an exact integer certificate for the sharp bound.

Two further sections connect the chapter to recent work. The first treats a sibling problem of Erdős, settled in 2026 by an automated prover, in which good rational approximations to $\log_2 3$ play the opposite role. The second describes a small library of machine checked statements in the Lean proof assistant and the path

from it to a fully verified cycle bound.

Throughout, $\theta = \log_2 3 = 1.584963\ldots$, log is the natural logarithm, and we use the three maps of Definition 2.1: the standard map T_0, the accelerated map T and the Syracuse map S.

10.1 The cycles that exist

10.1.1 The trivial cycle

The orbit of 1 is periodic under all three maps: $1 \mapsto 4 \mapsto 2 \mapsto 1$ under T_0, $1 \mapsto 2 \mapsto 1$ under T, and $S(1) = 1$. It has one odd element and two halvings, so in the notation used below $r = 1$ and $s = 2$, with period $r + s = 3$ under T_0, period $s = 2$ under T and period $r = 1$ under S. The conjecture is that every positive integer eventually enters this cycle; the cycle half asks only that there be no other.

10.1.2 Cycles on the negative integers

The maps T_0, T and S make sense on all of $\mathbb{Z}$, and on the negative integers there are several cycles. Under $n \mapsto -n$ the standard map on the negative integers becomes the map $n \mapsto n/2$ for even n and $n \mapsto 3n - 1$ for odd n on the positive integers, the so called $3x - 1$ problem, so these cycles are cycles of that problem in disguise. We ran T_0 from every starting value $n = -1, -2, \ldots, -10^6$ and recorded the cycle that each orbit enters (script `ch08_negative.py`). Exactly three cycles occur, and every one of the 10^6 starting values falls into one of them.

Table 10.1. The cycles of T_0 entered by orbits of $n = -1, \ldots, -10^6$. Here r is the number of odd elements and s the number of halvings, so the period under T_0 is $r + s$. The last column counts the starting values that reach the cycle.

odd elements	r	s	$r+s$	orbits
-1	1	1	2	327 679
$-5, -7$	2	3	5	323 351
$-17, -25, -37, -55, -41, -61, -91$	7	11	18	348 970

The second cycle is $-5 \mapsto -14 \mapsto -7 \mapsto -20 \mapsto -10 \mapsto -5$ and the third, of period 18 under T_0, starts $-17 \mapsto -50 \mapsto -25 \mapsto -74 \mapsto \cdots$. In each case the product of the factors $3 + 1/x$ over the odd elements x is an exact power of two: 2 for the first cycle, $(3 - \frac{1}{5})(3 - \frac{1}{7}) = 8$ for the second, and 2^{11} for the third (we checked the last in exact rational arithmetic). This is the cycle identity of Lemma 10.1 below. Because the elements are negative the factors are less than 3, so $2^s < 3^r$, that is, $s < r\theta$: indeed $3 < 2\theta = 3.17$ and $11 < 7\theta = 11.09$. For positive cycles the inequality goes the other way, and that change of sign is the whole difficulty. The negative cycles exist because 2^s may undershoot 3^r by a little;

a positive cycle would need 2^s to overshoot 3^r by very little, and the rest of this chapter quantifies how little.

10.1.3 Rational cycles

Solving the affine identity (2.4) for a point that returns to itself, as in the proof of Corollary 2.14, gives, for any parity word w of length k with m ones, the number

$$n(w) = \frac{c(w)}{2^k - 3^m},$$

where $c(w)$ is the sum (2.6). The identity (2.4) holds in the ring of rationals with odd denominator, in which parity is the parity of the numerator, and reading it backwards shows that $n(w)$ is a periodic point of T with parity word w, repeated. For example $w = (1,0,0)$ has $c = 1$ and $n = 1/(8-3) = 1/5$, and indeed $\frac{1}{5} \mapsto \frac{4}{5} \mapsto \frac{2}{5} \mapsto \frac{1}{5}$. Every word thus gives a rational cycle, and the integer cycles are exactly the words for which $2^k - 3^m$ divides $c(w)$. Lagarias [54] studied the set of these rational cycles systematically, organizing them by denominator; for a denominator d coprime to 6 they correspond to the integer cycles, with elements prime to d, of the map that sends x to $(3x+d)/2$ for odd x and to $x/2$ for even x, and he showed, among other things, that cycles exist for many denominators. We shall not need this theory. Its lesson for us is that there is no shortage of periodic parity words; what is rare is the divisibility that makes $n(w)$ an integer, and for positive integers the inequality $3^m < 2^k$ of Corollary 2.14 combined with the size of the elements is the only handle we have.

10.2 The cycle identity

It is most convenient to describe a cycle by its odd elements. Suppose S has a cycle of positive odd integers $x_1, \dots, x_r$, with $x_{i+1} = S(x_i)$ (indices modulo r), and let $k_i = v_2(3x_i + 1) \geq 1$ be the halving exponents, so that

$$x_{i+1} = \frac{3x_i + 1}{2^{k_i}}. \tag{10.1}$$

Put $s = \sum_i k_i$, the total number of halvings in one turn of the cycle.

Lemma 10.1 *(Cycle identity). For a cycle of S as above,*

$$\prod_{i=1}^{r} \Big(3 + \frac{1}{x_i}\Big) = 2^s, \tag{10.2}$$

and therefore, writing $M = \min_i x_i$ for the least element,

$$\Lambda := s\log 2 - r\log 3 = \sum_{i=1}^{r} \log\Big(1 + \frac{1}{3x_i}\Big) \in \Big(0,\ \frac{r}{3M}\Big). \tag{10.3}$$

Equivalently,

$$0 < s - r\theta < \frac{r}{3M \log 2'}, \tag{10.4}$$

and in particular $s/r > \theta$.

Proof. Multiply the r relations (10.1) around the cycle. The product of the left sides is $\prod_i x_i$ and the product of the right sides is $\prod_i (3x_i + 1)/2^s$, so $2^s = \prod_i (3x_i + 1)/x_i$, which is (10.2). Taking logarithms and writing $\log(3 + 1/x_i) = \log 3 + \log(1 + 1/(3x_i))$ gives the middle expression in (10.3). Each summand is positive, so $\Lambda > 0$, and since $\log(1 + u) < u$ for $u > 0$ and $1/(3x_i) \le 1/(3M)$, we get $\Lambda < r/(3M)$. Dividing by $\log 2$ gives (10.4). □

The same identity can be read off from the affine form of Proposition 2.11. If n is an element of a cycle of T of length k with m odd steps, then (2.7) with $T^{(k)}(n) = n$ says that $2^k/3^m$ is the product of $1 + 1/(3y)$ over the odd elements y of the cycle. That is (10.2) with $m = r$ and $k = s$.

Remark 10.2 *(A pitfall: bookkeeping under the accelerated map).* The three maps count the same cycle differently, and it is easy to mix the counts. Under the Syracuse map S the cycle has period r. Under the accelerated map T each odd step already contains one halving, so the cycle has s steps in all, of which r are odd and $s - r$ are even. Under the standard map T_0 every halving is a separate step and so is every tripling, so the period is $r + s$. For the trivial cycle these are 1, 2 and 3. The linear form can be written either as $s \log 2 - r \log 3$ or as $s' \log 2 - r \log(3/2)$ with $s' = s - r$ the number of even steps of T, and the two are equal. The danger lies in the derived quantities. If one works with s' and then reports s', or s, as "the period" of the cycle, the answer is wrong for T_0, whose period is $r + s = 2r + s'$. And the feasibility condition must be stated for the right pair: it is $s/r > \theta$, equivalently $s'/r > \theta - 1 = \log_2(3/2)$, together with the constraint $s' \ge 0$ that comes from $k_i \ge 1$. Lower bounds on cycle length in the literature are quoted in all three conventions, and comparisons between them are meaningless unless the convention is fixed. In this chapter r is the number of odd elements and s the number of halvings, so that a cycle has r steps of S, s steps of T (of which r are odd) and $r + s$ steps of T_0.

For fixed r the strongest conclusion comes from the smallest admissible Λ. Since s must be an integer with $s > r\theta$, we have $s \ge \lceil r\theta \rceil$ and so

$$\Lambda \ge \Lambda_r := \delta(r) \log 2, \qquad \delta(r) := \lceil r\theta \rceil - r\theta \in (0, 1). \tag{10.5}$$

(The value $\delta(r)$ is never 0 because θ is irrational: $2^s = 3^r$ is impossible for $r \ge 1$.) Combining with (10.3) gives the basic inequality of the chapter:

$$M < \frac{r}{3\Lambda_r} = \frac{r}{3\,\delta(r) \log 2} =: F(r). \tag{10.6}$$

The function F depends only on θ and r. A cycle with r odd elements must have its least element below $F(r)$. On the other hand, if every positive integer up to some bound reaches 1, then every element of a nontrivial cycle exceeds that bound. These two facts squeeze the cycle.

Definition 10.3 *(Verification range and exclusion).* We say that X is a *verified range* if every element of every nontrivial cycle of positive integers is at least X; this holds, for example, if every positive integer $n < X$ is known to reach 1. Given X, put $\varepsilon = \varepsilon(X) = 1/(3X\log 2)$ and say that $r \geq 1$ is *excluded* if $\delta(r) \geq r\varepsilon$, equivalently $F(r) \leq X$.

Corollary 10.4. *If X is a verified range, then a nontrivial cycle with r odd elements satisfies $\delta(r) < r\varepsilon$, so r is not excluded.*

Proof. By (10.4), $\delta(r) \leq s - r\theta < r/(3M\log 2) \leq r/(3X\log 2) = r\varepsilon$. □

By Bařina [6], every positive integer below 2^{68} reaches 1, so $X = 2^{68} = 2.95148 \times 10^{20}$ is a verified range. Bařina [7] has since extended the verification to 2^{71}; we keep $X = 2^{68}$ in the examples, and Section 10.6 shows that the larger range gives the same sharp bound. The question is now purely arithmetic: which r satisfy $\delta(r) < r\varepsilon$? The inequality says that $r\theta$ lies just *below* an integer, closer than $r\varepsilon$. To find all such r we need the theory of rational approximation, and specifically its one sided form.

10.3 A primer on continued fractions

This section collects what we need about continued fractions, with proofs. The material is classical; good references are the books of Khinchin [43], Hardy and Wright [32] and Rockett and Szüsz [87], the last of which treats one sided approximation in detail.

10.3.1 Convergents and their recurrences

Let $\theta > 0$ be irrational. Define $\theta_0 = \theta$ and, recursively, $a_k = \lfloor \theta_k \rfloor$ and $\theta_{k+1} = 1/(\theta_k - a_k) > 1$. Then $a_0 \geq 0$, $a_k \geq 1$ for $k \geq 1$, and for every k

$$\theta = a_0 + \cfrac{1}{a_1 + \cfrac{1}{\ddots + \cfrac{1}{a_k + \cfrac{1}{\theta_{k+1}}}}} =: [a_0; a_1, \ldots, a_k, \theta_{k+1}].$$

The numbers a_k are the *partial quotients* and θ_k the *complete quotients*. Define integers p_k, q_k by

$$\begin{aligned} p_k &= a_k p_{k-1} + p_{k-2}, \qquad q_k = a_k q_{k-1} + q_{k-2}, \\ p_{-2} &= 0, \quad p_{-1} = 1, \quad q_{-2} = 1, \quad q_{-1} = 0. \end{aligned} \tag{10.7}$$

Thus $p_0 = a_0$, $q_0 = 1$, $q_1 = a_1$, and $1 = q_0 \le q_1 < q_2 < q_3 < \cdots$. The fractions p_k/q_k are the *convergents*.

Lemma 10.5 *(Basic identities). For every $k \ge 0$:*

(i) $\begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_k & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} p_k & p_{k-1} \\ q_k & q_{k-1} \end{pmatrix}$;

(ii) $p_k q_{k-1} - p_{k-1} q_k = (-1)^{k-1}$, *and this also holds for $k = -1$;*

(iii) $\theta = \dfrac{\theta_{k+1} p_k + p_{k-1}}{\theta_{k+1} q_k + q_{k-1}}$;

(iv) $q_k \theta - p_k = \dfrac{(-1)^k}{\theta_{k+1} q_k + q_{k-1}}$. *In particular $p_k/q_k < \theta$ for even k and $p_k/q_k > \theta$ for odd k, and the error $\varepsilon_k := |q_k \theta - p_k|$ satisfies*

$$\frac{1}{q_{k+1} + q_k} < \varepsilon_k < \frac{1}{q_{k+1}};$$

(v) $\varepsilon_{k+1} = \varepsilon_{k-1} - a_{k+1} \varepsilon_k$, *where $\varepsilon_{-1} = |q_{-1}\theta - p_{-1}| = 1$.*

Proof. (i) is induction on k using (10.7), since multiplying the matrix $\left(\begin{smallmatrix} p_{k-1} & p_{k-2} \\ q_{k-1} & q_{k-2} \end{smallmatrix}\right)$ on the right by $\left(\begin{smallmatrix} a_k & 1 \\ 1 & 0 \end{smallmatrix}\right)$ gives $\left(\begin{smallmatrix} p_k & p_{k-1} \\ q_k & q_{k-1} \end{smallmatrix}\right)$. (ii) Take determinants in (i): each factor has determinant -1; for $k = -1$, $p_{-1}q_{-2} - p_{-2}q_{-1} = 1$. (iii) The matrix $\left(\begin{smallmatrix} a & 1 \\ 1 & 0 \end{smallmatrix}\right)$ acts on t as the Möbius map $t \mapsto a + 1/t$, and composition of Möbius maps corresponds to matrix multiplication, so by (i) the map $t \mapsto [a_0; a_1, \ldots, a_k, t]$ is $t \mapsto (t p_k + p_{k-1})/(t q_k + q_{k-1})$; put $t = \theta_{k+1}$. (iv) From (iii), $q_k\theta - p_k = (q_k p_{k-1} - p_k q_{k-1})/(\theta_{k+1} q_k + q_{k-1})$, and the numerator is $(-1)^k$ by (ii). Since $a_{k+1} < \theta_{k+1} < a_{k+1} + 1$, the denominator lies strictly between $a_{k+1}q_k + q_{k-1} = q_{k+1}$ and $q_{k+1} + q_k$. (v) Multiply the recurrence (10.7) at index $k+1$ by θ and subtract: $q_{k+1}\theta - p_{k+1} = a_{k+1}(q_k\theta - p_k) + (q_{k-1}\theta - p_{k-1})$. By (iv), and for $k = 0$ by $q_{-1}\theta - p_{-1} = -1$, the outer two terms have the same sign and the middle one the opposite sign, which gives the stated relation between absolute values. □

10.3.2 Best approximation

Write $\|t\|$ for the distance from t to the nearest integer. The next theorem, due in essence to Lagrange, is the best approximation property: no denominator smaller than q_{k+1} approximates θ better than q_k.

THEOREM 10.6 *(Best approximation). Let $k \geq 0$, $1 \leq r < q_{k+1}$ and $s \in \mathbb{Z}$. Then $|r\theta - s| \geq \varepsilon_k$. In particular $\|r\theta\| \geq \varepsilon_k \geq \|q_k\theta\|$.*

Proof. By Lemma 10.5(ii) the matrix $\left(\begin{smallmatrix} p_k & p_{k+1} \\ q_k & q_{k+1} \end{smallmatrix}\right)$ has determinant ± 1, so there are integers x, y with $s = xp_k + yp_{k+1}$ and $r = xq_k + yq_{k+1}$. Then $r\theta - s = x(q_k\theta - p_k) + y(q_{k+1}\theta - p_{k+1})$. If $y = 0$ then $r = xq_k$ with $x \geq 1$, and $|r\theta - s| = x\varepsilon_k \geq \varepsilon_k$. If $x = 0$ then $r = yq_{k+1}$, which is impossible for $1 \leq r < q_{k+1}$. If x and y are both positive then $r \geq q_k + q_{k+1} > q_{k+1}$, and if both are negative then $r < 0$; both are impossible. So x and y are nonzero with opposite signs. Since $q_k\theta - p_k$ and $q_{k+1}\theta - p_{k+1}$ also have opposite signs, by Lemma 10.5(iv), the two terms $x(q_k\theta - p_k)$ and $y(q_{k+1}\theta - p_{k+1})$ have the same sign, and $|r\theta - s| \geq |x|\varepsilon_k \geq \varepsilon_k$. □

10.3.3 Intermediate fractions

Between two convergents on the same side of θ one may interpolate. For $k \geq 0$ and $0 \leq j \leq a_{k+1}$ define

$$u_{k,j} = q_{k-1} + jq_k, \qquad P_{k,j} = p_{k-1} + jp_k. \tag{10.8}$$

For $j = 0$ this is the convergent p_{k-1}/q_{k-1}, for $j = a_{k+1}$ it is p_{k+1}/q_{k+1}, and the fractions $P_{k,j}/u_{k,j}$ with $0 < j < a_{k+1}$ are the *intermediate fractions*, or *semiconvergents*. We call the denominators $u_{k,j}$ the *rungs* of the *semiconvergent ladder*: the ladder from q_{k-1} to q_{k+1} climbs in steps of q_k,

$$q_{k-1},\ q_{k-1} + q_k,\ q_{k-1} + 2q_k,\ \ldots,\ q_{k-1} + a_{k+1}q_k = q_{k+1}.$$

By Lemma 10.5(iv) and (v),

$$P_{k,j} - u_{k,j}\theta = (-1)^k(\varepsilon_{k-1} - j\varepsilon_k), \qquad \varepsilon_{k-1} - j\varepsilon_k \geq \varepsilon_{k+1} > 0, \tag{10.9}$$

so every rung of the ladder from q_{k-1} lies on the same side of θ as p_{k-1}/q_{k-1}, and the error decreases linearly in j. For even k the ladder approximates θ from above, for odd k from below.

10.3.4 One sided best approximation

For the cycle problem the relevant quantity is the one sided distance $\delta(r) = \lceil r\theta \rceil - r\theta$ of (10.5), the distance from $r\theta$ up to the next integer. It is small when s/r, with $s = \lceil r\theta \rceil$, approximates θ well *from above*.

DEFINITION 10.7 *(One sided records).* We call $r \geq 1$ a *δ record*, or a best one sided approximation from above, if $\delta(r) < \delta(r')$ for every $1 \leq r' < r$.

THEOREM 10.8 *(One sided best approximations). The δ records are exactly the rungs $u_{k,j} = q_{k-1} + jq_k$ with $k \geq 0$ even and $1 \leq j \leq a_{k+1}$. For such a rung, $\lceil u_{k,j}\theta \rceil = P_{k,j}$ and $\delta(u_{k,j}) = \varepsilon_{k-1} - j\varepsilon_k$.*

This classical fact is the one sided analogue of Theorem 10.6; Hančl and Turek [31, Theorem 4.5(ii)] give a statement with proof, and Rockett and Szüsz [87] treat the theory of one sided approximation. The same proof with the roles of the parities exchanged shows that the best approximations from below are $r = 1$ and the rungs with $k \geq 1$ odd. We give the short proof, since the whole sharp bound rests on it.

Proof. Fix an even $k \geq 0$ and $0 \leq j < a_{k+1}$, and write $u = u_{k,j}$, $P = P_{k,j}$, $e = P - u\theta$ and $d = \varepsilon_k = q_k\theta - p_k > 0$ (positive because k is even). By (10.9), $e = \varepsilon_{k-1} - jd > 0$. We claim that

$$1 \leq r < u + q_k,\ r \neq u \quad \Longrightarrow \quad \delta(r) > e. \tag{10.10}$$

The matrix $\left(\begin{smallmatrix} P & p_k \\ u & q_k \end{smallmatrix}\right)$ has determinant $p_{k-1}q_k - p_kq_{k-1} = (-1)^k = 1$, so for $s = \lceil r\theta \rceil$ there are integers x, y with $s = xP + yp_k$ and $r = xu + yq_k$, and then

$$\delta(r) = s - r\theta = xe - yd \in (0, 1).$$

If $x \leq 0$ then positivity of $xe - yd$ forces $y < 0$, and then $r = xu + yq_k < 0$, which is impossible. So $x \geq 1$. If also $y \geq 1$ then $r \geq u + q_k$, excluded by hypothesis. Hence $y \leq 0$ and $\delta(r) = xe + |y|d \geq e$, with equality only when $x = 1$ and $y = 0$, that is $r = u$. This proves (10.10).

Now suppose $u \geq 1$ (this excludes only $k = 0$, $j = 0$, where $u = q_{-1} = 0$). Then $e < 1$: for $k = 0$, $e = 1 - j\varepsilon_0 < 1$ since $j \geq 1$, and for $k \geq 2$, $e \leq \varepsilon_{k-1} < 1/q_k \leq 1$. As $e \in (0, 1)$ and $P - u\theta = e$, we have $\lceil u\theta \rceil = P$ and $\delta(u) = e$. By (10.10), every $r < u$ has $\delta(r) > \delta(u)$, so u is a δ record, and every r with $u < r < u + q_k$ has $\delta(r) > \delta(u)$ with $u < r$, so r is not a δ record. The rung $u + q_k = u_{k,j+1}$ is covered by the case $j + 1$ if $j + 1 < a_{k+1}$, and if $j + 1 = a_{k+1}$ it equals $q_{k+1} = u_{k+2,0}$, the first rung of the next ladder, which is covered with $k + 2$ in place of k and $j = 0$. The half open intervals $[u_{k,j}, u_{k,j} + q_k)$, over even k and $0 \leq j < a_{k+1}$, therefore tile $[0, \infty)$, their left endpoints other than 0 are δ records, and no other integer is. The left endpoints other than 0 are precisely the $u_{k,j}$ with k even and $1 \leq j \leq a_{k+1}$ (the endpoint $u_{k+2,0} = u_{k,a_{k+1}}$ being counted once). □

10.3.5 The continued fraction of $\log_2 3$

We computed the partial quotients of $\theta = \log_2 3$ at 300 significant digits and confirmed the first 60 of them at 150 digits (script `ch08_compute.py`):

$$\log_2 3 = [1; 1, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, 1, 1, 55, 1, 4, 3, 1, 1, 15, 1, \dots].$$

The convergents are listed in Table 10.2. The two large partial quotients $a_9 = 23$ and $a_{14} = 55$ produce unusually good approximations, at $q_8 = 665$ and $q_{13} = 190\,537$ respectively; note that the good approximation at q_8 is from below and that at q_{13} from above, a distinction that will matter.

TABLE 10.2. Convergents p_k/q_k of $\theta = \log_2 3$. The side column records whether p_k/q_k lies below or above θ; the last column is $\varepsilon_k = |q_k\theta - p_k|$.

k	a_k	p_k	q_k	side	ε_k
0	1	1	1	below	0.585
1	1	2	1	above	0.415
2	1	3	2	below	0.170
3	2	8	5	above	7.52×10^{-2}
4	2	19	12	below	1.96×10^{-2}
5	3	65	41	above	1.65×10^{-2}
6	1	84	53	below	3.01×10^{-3}
7	5	485	306	above	1.47×10^{-3}
8	2	1054	665	below	6.30×10^{-5}
9	23	24727	15601	above	2.62×10^{-5}
10	2	50508	31867	below	1.05×10^{-5}
11	2	125743	79335	above	5.29×10^{-6}
12	1	176251	111202	below	5.19×10^{-6}
13	1	301994	190537	above	9.31×10^{-8}
14	55	16785921	10590737	below	7.55×10^{-8}
15	1	17087915	10781274	above	1.76×10^{-8}
16	4	85137581	53715833	below	5.01×10^{-9}
17	3	272500658	171928773	above	2.58×10^{-9}
18	1	357638239	225644606	below	2.43×10^{-9}
19	1	630138897	397573379	above	1.53×10^{-10}
20	15	9809721694	6189245291	below	1.38×10^{-10}
21	1	10439860591	6586818670	above	1.46×10^{-11}
22	9	103768467013	65470613321	below	6.65×10^{-12}
23	2	217976794617	137528045312	above	1.30×10^{-12}

By Theorem 10.8 the δ records of $\log_2 3$ begin

$$1;\ 3,5;\ 17,29,41;\ 94,147,200,253,306;\ 971,1636,\dots,15601;\ \dots,$$

the groups being the ladders with $k = 0,2,4,6,8$: from $q_{-1} = 0$ in steps of $q_0 = 1$ (one rung, since $a_1 = 1$), from $q_1 = 1$ in steps of $q_2 = 2$, from $q_3 = 5$ in steps of $q_4 = 12$, from $q_5 = 41$ in steps of $q_6 = 53$, and from $q_7 = 306$ in 23 steps of $q_8 = 665$ up to $q_9 = 15\,601$. We generated all rungs below 2×10^{13} from the continued fraction at 80 digits and found 53 of them; independently, a direct scan of $\delta(r)$ for every $r \le 2 \times 10^5$ produced exactly the same list of records below 2×10^5 (script `ladder.py`), and an exact integer scan to 3×10^6 (script `ch08_records.py`) agreed as well.

10.4 The elementary bound

10.4.1 The block bound

The first and simplest use of continued fractions ignores the side from which $r\theta$ is approximated.

Lemma 10.9 *(Block bound). For every $r \geq 1$, $\Lambda_r \geq \|r\theta\| \log 2$. Consequently, if $q_k \leq r < q_{k+1}$, then*

$$F(r) = \frac{r}{3\Lambda_r} < \frac{q_{k+1}}{3\,\|q_k\theta\| \log 2} =: B_k. \tag{10.11}$$

Proof. Write $\{t\}$ for the fractional part. Then $\|r\theta\| = \min(\{r\theta\}, 1 - \{r\theta\}) \leq 1 - \{r\theta\} = \delta(r)$, which is the first claim. For $r < q_{k+1}$, Theorem 10.6 gives $\|r\theta\| \geq \|q_k\theta\|$, hence $\Lambda_r \geq \|q_k\theta\| \log 2$, and $r < q_{k+1}$ bounds the numerator. □

The block bound is in substance Crandall's criterion [18]; see [53, Theorem I] and [92, Lemma 1].

Table 10.3. The block bound B_k of (10.11). A nontrivial cycle with $q_k \leq r < q_{k+1}$ odd elements would have least element below B_k. For $k \leq 7$ the values are 1.16, 2.32, 14.2, 76.8, 1009, 1541, 4.88×10^4 and 2.17×10^5. The first row exceeding $2^{68} = 2.95 \times 10^{20}$ is $k = 21$.

k	q_k	q_{k+1}	B_k
8	665	15 601	1.19×10^8
9	15 601	31 867	5.84×10^8
10	31 867	79 335	3.64×10^9
11	79 335	111 202	1.01×10^{10}
12	111 202	190 537	1.76×10^{10}
13	190 537	10 590 737	5.47×10^{13}
14	10 590 737	10 781 274	6.87×10^{13}
15	10 781 274	53 715 833	1.47×10^{15}
16	53 715 833	171 928 773	1.65×10^{16}
17	171 928 773	225 644 606	4.20×10^{16}
18	225 644 606	397 573 379	7.87×10^{16}
19	397 573 379	6 189 245 291	1.95×10^{19}
20	6 189 245 291	6 586 818 670	2.29×10^{19}
21	6 586 818 670	65 470 613 321	2.16×10^{21}
22	65 470 613 321	137 528 045 312	9.94×10^{21}

Table 10.3 was computed with the script `ch08_bounds.py`, at 150 digits. Since every B_k with $k \leq 20$ is below 2^{68}, every $r < q_{21} = 6\,586\,818\,670$ is excluded. This is already a respectable bound, obtained from nothing more than Theorem 10.6. It gives away a good deal, however, in two ways. It covers a whole block $[q_k, q_{k+1})$

by its worst case, combining the largest numerator q_{k+1} with the smallest distance $\|q_k\theta\|$; and it uses the two sided distance $\|r\theta\|$ where the problem only cares about the one sided $\delta(r)$.

10.4.2 Why convergents alone are not enough

Remark 10.10 *(A tempting but false argument).* One would like to evaluate $F(r)$ exactly, rather than bound it by blocks, and only at a few values of r. A natural argument runs: "Λ_r is smallest when $r\theta$ is closest to an integer, and by the best approximation property this happens at the convergent denominators; so F can set a new record only at a convergent denominator, and it suffices to evaluate F at the q_k." The conclusion is false. The flaw is that $\delta(r)$ is a one sided quantity, while Theorem 10.6 concerns the two sided distance. The two differ exactly when $r\theta$ lies just *above* an integer, which is what happens at every convergent from below. Since the convergents alternate sides, half of them are useless for the cycle problem. The most striking case is $p_8/q_8 = 1054/665$, which lies below θ: here $\|665\theta\| = 6.3 \times 10^{-5}$, an excellent two sided approximation, but $\delta(665) = 0.99994$, about as large as a one sided distance can be, and $F(665) = 320$, far below the running maximum. Conversely, the records of F include $r = 3, 17, 29, 94, 147, 200, 253, \ldots$, none of which is a convergent denominator. A direct search over $r \le 4 \times 10^5$ finds 37 records of F, among them the intermediate fractions 3, 17, 29, 94, 147, 200, 253, 971, 1636, 2301, 2966 and 3631. Evaluating F only at convergents therefore misses values of r where it is larger, and any bound so obtained is unjustified. Halbeisen and Hungerbühler [30, §4.4] make the same observation for Eliahou's criterion: the convergents from the wrong side are harmless, and the lengths at which the bound is weakest lie between convergents.

The cure is visible in the list of records. The values 94, 147, 200, 253 advance by $53 = q_6$ from $41 = q_5$ and stop at $306 = q_7$; the values $971, 1636, 2301, \ldots$ advance by $665 = q_8$ from 306 and stop at $15\,601 = q_9$. These are the rungs of the semiconvergent ladders with even k, the best approximations from above of Theorem 10.8. Figure 10.1 shows the one sided distance for $r \le 700$: its record minima step down along the ladder from 41 to 306, while the convergents from below, 12, 53 and 665, have one sided distance close to 1 even though their two sided distance is small. Figure 10.2 shows $F(r)$ on logarithmic axes: the cloud of all $r \le 10^5$, the rungs of the one sided ladder, and the convergents from above and below.

10.5 The mediant lemma

We now show that the rungs are not merely where the records happen to lie in a finite search, but where they must lie. The argument is a single observation about the additivity of δ.

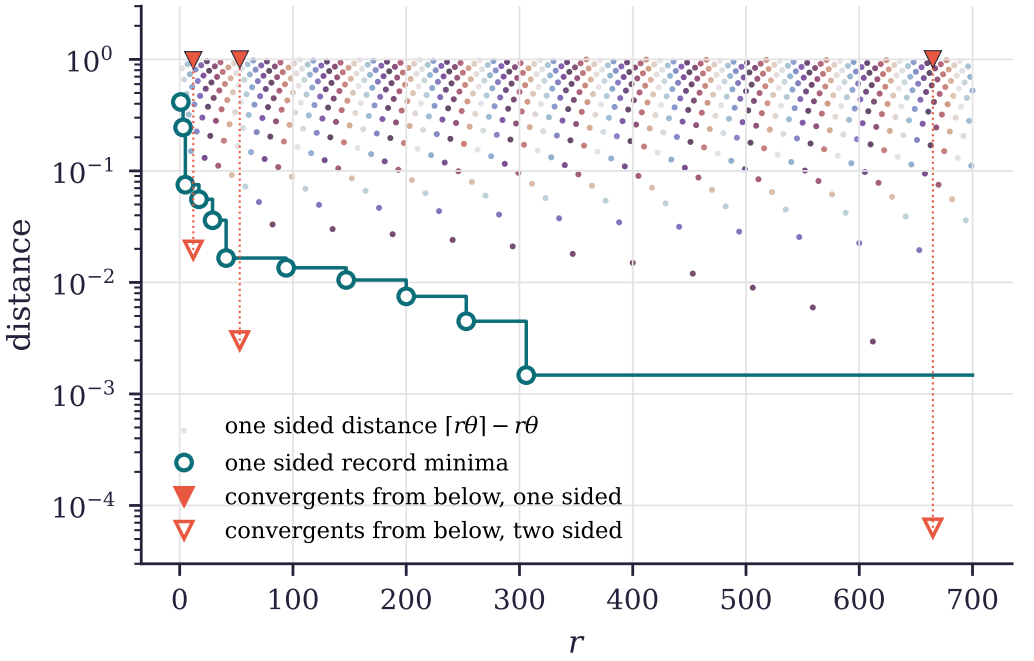


FIGURE 10.1. The one sided distance $\delta(r) = \lceil r\theta \rceil - r\theta$ for $r \leq 700$. Circles mark the record minima, which are the rungs $1, 3, 5, 17, 29, 41, 94, 147, 200, 253, 306$. Triangles mark the convergents from below, $q_4 = 12$, $q_6 = 53$ and $q_8 = 665$: filled at their one sided distance, which is close to 1, and open at their two sided distance $\|q\theta\|$, which is small.

LEMMA 10.11 *(Mediant lemma). Let $1 \leq r' < r$ with $\delta(r') < \delta(r)$, and put $d = r - r'$. Then*

$$\delta(r) = \delta(r') + \delta(d). \tag{10.12}$$

Consequently $\delta(r)/r$ is the mediant of $\delta(r')/r'$ and $\delta(d)/d$, and

$$\frac{\delta(r)}{r} \geq \min\Big(\frac{\delta(r')}{r'}, \frac{\delta(d)}{d}\Big). \tag{10.13}$$

Proof. For every $x \geq 1$, $\delta(x) = \lceil x\theta \rceil - x\theta$ is congruent to $-x\theta$ modulo 1 and lies in $(0,1)$; since $x\theta \notin \mathbb{Z}$, it is the unique number in $[0,1)$ with that property. The number $\delta(r) - \delta(r')$ is congruent to $-(r-r')\theta = -d\theta$ modulo 1, and by hypothesis it lies in $(0,1)$, because both terms lie in $(0,1)$ and the first is larger. By uniqueness it equals $\delta(d)$. For the inequality, $\delta(r)/r = (\delta(r') + \delta(d))/(r' + d)$ is a weighted average of $\delta(r')/r'$ and $\delta(d)/d$, with weights r'/r and d/r, and is therefore at least the smaller of the two. □

The content of (10.12) is that there is no wrap around: the congruence $\delta(r) \equiv \delta(r') + \delta(d)$ modulo 1 always holds, and the hypothesis $\delta(r') < \delta(r)$ upgrades it to an equality. We now draw two consequences.

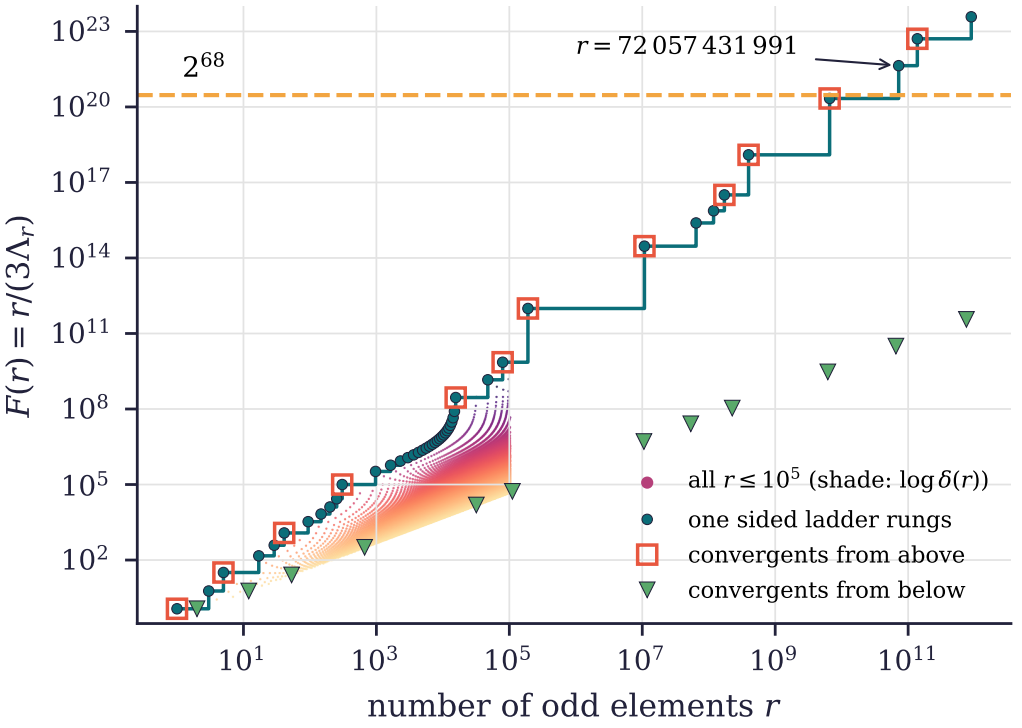


FIGURE 10.2. The bound $F(r) = r/(3\Lambda_r)$ on the least element of a cycle with r odd elements. Shaded by $\log \delta(r)$: all $r \le 10^5$. Staircase: the rungs of the one sided ladder up to 10^{12}, joined as a running maximum. Squares and triangles: convergent denominators from above and below. The dashed line is the verified range 2^{68}; the first rung above it is $r = 72\,057\,431\,991$.

COROLLARY 10.12. *Let $\varepsilon > 0$ and suppose some $r \ge 1$ satisfies $\delta(r) < r\varepsilon$. Then the smallest such r is a δ record.*

Proof. Let r_* be the smallest r with $\delta(r) < r\varepsilon$, and suppose it is not a δ record. Then there is $r' < r_*$ with $\delta(r') \le \delta(r_*)$. Equality is impossible, since it would make $(r_* - r')\theta$ an integer; so $\delta(r') < \delta(r_*)$. Put $d = r_* - r'$, so $1 \le d < r_*$. By minimality of r_*, both $\delta(r') \ge r'\varepsilon$ and $\delta(d) \ge d\varepsilon$. By (10.13), $\delta(r_*)/r_* \ge \varepsilon$, a contradiction. □

COROLLARY 10.13. *If r is a strict record of $r \mapsto r/\delta(r)$, meaning $r/\delta(r) > r'/\delta(r')$ for every $1 \le r' < r$, then r is a δ record.*

Proof. If not, there is $r' < r$ with $\delta(r') < \delta(r)$ (equality being impossible as before). With $d = r - r' < r$, (10.13) gives $r/\delta(r) \le \max(r'/\delta(r'), d/\delta(d))$, contradicting strictness. □

Since $F(r) = r/(3\delta(r)\log 2)$ is a constant multiple of $r/\delta(r)$, Corollary 10.13 combined with Theorem 10.8 locates the records of F exactly.

Proposition 10.14 *(Where the records are). Every r at which $r \mapsto F(r) = r/(3\Lambda_r)$ attains a strict record is a rung of the semiconvergent ladder approximating θ from above, that is, $r = q_{k-1} + jq_k$ with $k \geq 0$ even and $1 \leq j \leq a_{k+1}$. Moreover, for every verified range X, the smallest r that is not excluded in the sense of Definition 10.3 is such a rung.*

Proof. The first statement is Corollary 10.13 combined with Theorem 10.8. The second is Corollary 10.12 with $\varepsilon = 1/(3X\log 2)$, followed by Theorem 10.8. (Such an r exists: by Lemma 10.5(iv), at the convergents from above $\delta(q_k) = p_k - q_k\theta < 1/q_{k+1}$, which is eventually below $q_k\varepsilon$.) □

The converse holds as well, so the strict records of F are exactly the rungs, as the classical theorem [31, Theorem 4.5(ii)] also says. Along the rungs r increases while δ decreases, by Theorem 10.8, so F increases; and for any r' the first point at which F attains its maximum over $[1, r']$ is a strict record, hence a rung $u' \leq r'$, so every rung $u > r'$ has $F(u) > F(u') \geq F(r')$. For $\log_2 3$ every rung up to 72 057 431 991 is accordingly a record of F (the values in Table 10.4 increase), and the exact scan to 3×10^6 found the records of F and the δ records to coincide there. The cycle bound below does not need the converse.

Remark 10.15 *(Ties, and a warning about arithmetic).* Multiples of a rung produce exact ties. If $t\,\delta(q) < 1$ for an integer $t \geq 2$ then $\delta(tq) = t\,\delta(q)$, since $tq\theta = t\lceil q\theta\rceil - t\delta(q)$, so $F(tq) = F(q)$ and the multiple ties with its generator rather than beating it; by Corollary 10.13 it cannot be a strict record. Halbeisen and Hungerbühler [30, §4.4] observe these ties for Eliahou's criterion, at multiples of a convergent denominator. In floating point arithmetic such ties break at random. With Λ_r evaluated in double precision as $s\log 2 - r\log 3$, $s = \lfloor r\theta\rfloor + 1$, a scan of $r \leq 10^6$ reports $918 = 3 \cdot 306$, $46\,803 = 3 \cdot 15\,601$ and $952\,685 = 5 \cdot 190\,537$ as new records (script `ch08_ties.py`); evaluated instead as $(\lceil r\theta\rceil - r\theta)\log 2$, the same scan happens to report no spurious records, the ties breaking the other way or not at all. At 150 digits the differences $\delta(tq) - t\delta(q)$ for 46 803, $571\,611 = 3 \cdot 190\,537$ and 952 685 come out as -2.5×10^{-147}, 0 and 0, but the spurious records do not disappear: a tie is an exact equality between irrational numbers, so rounding decides it at every precision, and the same scan at 150 digits reports $15 = 3 \cdot 5$ and 571 611 as new records in the first form, and 46 803 and 571 611 in the second. The cure is exact arithmetic: $F(r) > F(r')$ if and only if $r\lceil r'\theta\rceil > r'\lceil r\theta\rceil$, a comparison of integers once the ceilings are known, and a scan of $r \leq 10^6$ in this form finds exactly the rungs as its strict records. This is one more reason to prefer an argument that locates the records over a search that merely finds them.

10.6 The cycle length bound

Proposition 10.14 turns the search for the sharp bound into a short computation over the rungs. For $\log_2 3$ there are only 44 rungs up to 72 057 431 991, and Table 10.4 lists the last eleven of them with the value of F at each.

TABLE 10.4. The upper ladder rungs from q_9 to the first rung exceeding 2^{68}, with $s = \lceil r\theta \rceil$ and $F(r) = r/(3\Lambda_r)$, computed at 150 digits. The decomposition column writes each rung as a convergent denominator or as $q_{k-1} + jq_k$.

r	s	$F(r)$	rung
15 601	24 727	2.86×10^{8}	q_9
47 468	75 235	1.45×10^{9}	$q_9 + q_{10}$
79 335	125 743	7.22×10^{9}	q_{11}
190 537	301 994	9.85×10^{11}	$q_{13} = q_{11} + q_{12}$
10 781 274	17 087 915	2.94×10^{14}	$q_{15} = q_{13} + q_{14}$
64 497 107	102 225 496	2.46×10^{15}	$q_{15} + q_{16}$
118 212 940	187 363 077	7.49×10^{15}	$q_{15} + 2q_{16}$
171 928 773	272 500 658	3.20×10^{16}	q_{17}
397 573 379	630 138 897	1.25×10^{18}	$q_{19} = q_{17} + q_{18}$
6 586 818 670	10 439 860 591	2.17×10^{20}	$q_{21} = q_{19} + q_{20}$
72 057 431 991	114 208 327 604	4.36×10^{21}	$q_{21} + q_{22}$

THEOREM 10.16 *(Cycle length bound). Let X be a verified range in the sense of Definition 10.3.*

(i) Block form. *If $B_j \le X$ for $j = 0, 1, \dots, K$, then every nontrivial cycle has $r \ge q_{K+1}$ odd elements.*

(ii) Sharp form. *Every nontrivial cycle has $r \ge r_*(X)$ odd elements, where $r_*(X)$ is the smallest rung u of the upper ladder with $F(u) > X$.*

With $X = 2^{68}$, verified by Bařina [6], part (i) gives

$$r \ge q_{21} = 6\,586\,818\,670,$$

and part (ii) gives

$$r \ge 72\,057\,431\,991, \qquad s \ge 114\,208\,327\,604.$$

Thus a nontrivial cycle has at least 7.2×10^{10} steps of S, at least 1.14×10^{11} steps of T, and at least 1.86×10^{11} steps of T_0.

Proof. By Corollary 10.4 the number r of odd elements of a nontrivial cycle is not excluded, so $F(r) > X$.

(i) If $1 \le r < q_{K+1}$, then $q_j \le r < q_{j+1}$ for some $0 \le j \le K$ (recall $q_0 = 1$), and Lemma 10.9 gives $F(r) < B_j \le X$, so r is excluded. With $X = 2^{68}$, Table 10.3 and the values for $k \le 7$ in its caption show $B_j \le B_{20} = 2.29 \times 10^{19} < 2^{68}$ for all $j \le 20$, so $K = 20$ is admissible and $r \ge q_{21}$.

(ii) By Proposition 10.14, the smallest nonexcluded r is an upper ladder rung u, and by definition of exclusion it is the smallest rung with $F(u) > X$. So every $r < r_*(X)$ is excluded. With $X = 2^{68}$, the 44 rungs up to 72 057 431 991 were evaluated at 150 digits (script `ch08_bounds.py`); all before the last have $F(u) \le$

$2.1689 \times 10^{20} < 2^{68} = 2.9515 \times 10^{20}$, the maximum being attained at $u = q_{21}$, while $F(72\,057\,431\,991) = 4.3585 \times 10^{21} > 2^{68}$. Hence $r_*(2^{68}) = 72\,057\,431\,991$. For the number of halvings, $s > r\theta \geq r_*\theta$ and s is an integer, so $s \geq \lceil r_*\theta \rceil = 114\,208\,327\,604$. The step counts follow from Remark 10.2: r steps of S, s of T and $r + s \geq 186\,265\,759\,595$ of T_0. □

The first part uses only the property of best approximation, and the second part uses the one sided theorem and the mediant lemma, and the numerical evaluation involves margins that are enormous compared with the precision used: the closest call is the ratio $2^{68}/F(q_{21}) = 1.36$, and the values of δ involved are of order 10^{-11}, computed with errors below 10^{-130}. The factor of about eleven between the two parts measures what the block bound gives away by covering a whole interval with a single worst case.

With the verification range 2^{71} of Bařina [7], part (i) improves to $r \geq q_{22} = 65\,470\,613\,321$, since $B_{21} = 2.16 \times 10^{21} < 2^{71} = 2.36 \times 10^{21}$, while part (ii) is unchanged, since $F(72\,057\,431\,991) = 4.36 \times 10^{21} > 2^{71}$.

10.6.1 Exact versus crude

There is an intermediate argument, which uses the δ records but not the mediant lemma, and it can be much weaker. Let $q'_1 < q'_2 < \cdots$ be the δ records. If a cycle has r odd elements and q'_j is the largest record not exceeding r, then $\delta(r) \geq \delta(q'_j)$ by the definition of a record, and $\delta(r) < r\varepsilon$ by Corollary 10.4, so $r > \delta(q'_j)/\varepsilon$ and $r \geq q'_j$. Minimizing over the unknown j,

$$r \geq \min_j \max\Big(q'_j,\ \frac{\delta(q'_j)}{\varepsilon}\Big). \tag{10.14}$$

The same bound appears in our unpublished note on AlphaProof Nexus [78]. It is valid, but loose. The reason is that it uses the single value $\delta(q'_j)$ as a lower bound for δ on the whole block $[q'_j, q'_{j+1})$, so it cannot exclude any r in that block beyond $\delta(q'_j)/\varepsilon$, even though $\delta(r)$ there is in fact much larger. The mediant lemma shows that nothing inside a block needs to be checked at all: only the rungs matter, and at each rung one compares the exact δ with $r\varepsilon$. Table 10.5 compares the two for several verification ranges (script `ch08_crude.py`, at 80 digits).

At 2^{40} the crude bound is fifty times weaker: it is determined by the record $q_{13} = 190\,537$, with $\delta(q_{13})/\varepsilon = 212\,780.2$, whereas the next rung that is actually feasible is $q_{15} = 10\,781\,274$. At 2^{68} the crude bound is determined by q_{21}, with $\delta(q_{21})/\varepsilon = 8\,963\,457\,696.8$, and it is eight times weaker than the exact value. At 2^{60} the two coincide: the record $q_{19} = 397\,573\,379$ is itself feasible, and the preceding record q_{17} has $\delta(q_{17})/\varepsilon \approx 6.2 \times 10^9$, which already exceeds q_{19}, so the minimum in (10.14) is attained at q_{19} with value q_{19}.

The table also shows how slowly the exact bound responds to verification. Moving from 2^{68} to 2^{71}, the step taken by Bařina [7], gives the exact

TABLE 10.5. Lower bounds on the number r of odd elements of a nontrivial cycle, for several verification ranges X: the exact first nonexcluded r of Theorem 10.16(ii), with the corresponding $s = \lceil r\theta \rceil$, and the crude bound (10.14). The ranges up to 2^{71} are verified [6, 7]; the row 2^{72} shows what the method would give.

X	exact $r_*(X)$	s	crude bound
2^{40}	10 781 274	17 087 915	212 781
2^{60}	397 573 379	630 138 897	397 573 379
2^{68}	72 057 431 991	114 208 327 604	8 963 457 697
2^{71}	72 057 431 991	114 208 327 604	71 707 661 575
2^{72}	137 528 045 312	217 976 794 617	78 073 328 038

bound no improvement whatever, because the rung $r = 72\,057\,431\,991$ has $F(r) = 4.3585 \times 10^{21} = 2^{71.884}$: any verified range below about $2^{71.9}$ leaves that rung feasible. For the sharp form of Theorem 10.16, a verification past that point, for instance to 2^{72}, would move the bound to the next rung, $r \geq 137\,528\,045\,312 = q_{23}$, and further progress would require passing $F(q_{23}) = 5.10 \times 10^{22}$. The bound $r \geq 137\,528\,045\,312$ itself already holds unconditionally: it is the current record, which follows from Hercher's sharper accounting of the cycle together with the verification to 2^{71} (Remark 10.17). The return on computation is stepwise, in a way dictated entirely by the continued fraction; over many steps the bound grows roughly like the square root of the verified range, from 1.08×10^7 at 2^{40} to 7.2×10^{10} at 2^{68}.

10.6.2 Verification thresholds

Since $r_*(X)$ is a step function of the verification range, it is natural to ask exactly where the steps are. The first rung u not excluded at X changes precisely when $\log_2 X$ crosses $\log_2 F(u)$ for the current rung u, so the thresholds are the values $\log_2 F(u)$ along the sequence of successive r_*. Table 10.6 lists them for $60 \leq \log_2 X \leq 90$ and Figure 10.3 draws the staircase (script `ch08_profile.py`, at 700 digits).

Three features stand out. The range 2^{68} of Theorem 10.16 sits just above a threshold, at 67.556, which is why it yields the rung 72 057 431 991 rather than 6 586 818 670. The current range 2^{71} [7] lies in the same step, and so does the range $1536 \cdot 2^{60} = 2^{70.585}$ at which Hercher [38, Corollary 29] obtains $r \geq 1.375 \times 10^{11}$, now the record; so the difference between our bound and that record comes entirely from his sharper accounting of the cycle, not from a larger range. And the steps between 2^{78} and 2^{83} are close together, because the partial quotient $a_{25} = 7$ produces a ladder of seven nearby rungs $q_{23} + jq_{24} = 137\,528\,045\,312 + j \cdot 753\,110\,839\,881$, $1 \leq j \leq 7$, of which the table shows the first three and the last; the intermediate rungs occupy $[80.287,\ 82.796)$.

TABLE 10.6. The sharp bound $r_*(X)$ as a function of the verification range. Each row gives the rung r_*, the corresponding $s = \lceil r_*\theta \rceil$ and the interval of $\log_2 X$ on which it is the bound.

r_*	s	range of $\log_2 X$
397 573 379	630 138 897	[54.830, 60.119)
6 586 818 670	10 439 860 591	[60.119, 67.556)
72 057 431 991	114 208 327 604	[67.556, 71.884)
137 528 045 312	217 976 794 617	[71.884, 75.433)
890 638 885 193	1 411 629 234 715	[75.433, 78.333)
1 643 749 725 074	2 605 281 674 813	[78.333, 79.456)
2 396 860 564 955	3 798 934 114 911	[79.456, 80.287)
5 409 303 924 479	8 573 543 875 303	[82.796, 84.500)
11 571 718 688 839	18 340 740 190 704	[84.500, 87.951)
64 021 008 208 555	101 470 897 268 921	[87.951, 90.590)

10.6.3 The literature

REMARK 10.17 *(Attribution).* The method of this section is not new, and we claim no priority for any numerical bound. It goes back to Crandall [18], whose criterion is in substance the block bound of Lemma 10.9, and to Eliahou [21]. Steiner [94] showed in 1977 that there is no nontrivial cycle consisting of a single ascending run followed by a single descending run (a 1 cycle), using the continued fraction of $\log_2 3$ together with Baker's theory of linear forms in logarithms. Eliahou [21] combined the continued fraction of θ with verification to 2^{40} and showed in 1993 that a nontrivial cycle has at least 17 087 915 elements under T; our method reproduces this figure exactly, as the value of s at $r_*(2^{40}) = 10\,781\,274$ in Table 10.5, which we regard as a check on the implementation. The figure $q_{21} = 6\,586\,818\,670$ of Theorem 10.16(i) is also in the literature: Lagarias's overview [56, §6.1, item (W2)] reports it from Eliahou's criterion with the smaller verification range $20 \cdot 2^{58} \approx 5.76 \times 10^{18}$ of item (W1). The bound $r \geq 72\,057\,431\,991$ of Theorem 10.16(ii) coincides with the record $r > 7.2 \times 10^{10}$ that Hercher [38] credits to Hercher and Puchert [37, 84]; with the verification range $87 \cdot 2^{60}$, Hercher [37] had shown that a nontrivial cycle has at least $p_{21} + q_{21} = 17\,026\,679\,261$ elements under T_0. Simons and de Weger [92] extended Steiner's approach to m cycles, cycles with m ascending runs, ruling them out for $m \leq 68$ by combining continued fractions and lattice reduction with an explicit lower bound for $s \log 2 - r \log 3$ derived from Rhin's Padé approximation measure [85], and Hercher [38, Theorem 23] carried this to $m \leq 91$. Hercher [38, Corollary 29] also showed, by a sharper accounting of the cycle structure than the crude inequality $\log(1 + 1/(3x_i)) \leq 1/(3M)$ used in Lemma 10.1, that verification up to $3 \cdot 2^{69} = 1536 \cdot 2^{60}$ would give $r \geq 1.375 \times 10^{11}$; with the crude inequality one needs $3781 \cdot 2^{60}$, the value 4.36×10^{21} of F at

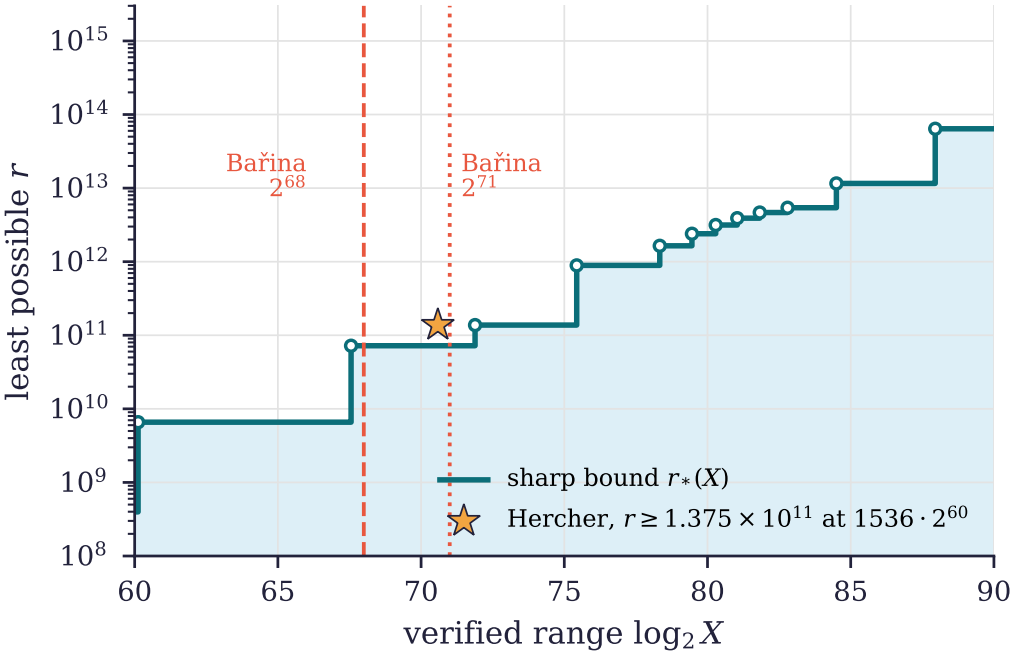


FIGURE 10.3. The sharp bound $r_*(X)$ against the verification range $\log_2 X$. Every step is a rung of the one sided ladder of $\log_2 3$. The dashed line marks the range 2^{68} of Bařina [6] and the dotted line his current range 2^{71} [7], which lies in the same step. The star is the bound $r \geq 1.375 \times 10^{11}$ that Hercher's sharper accounting of the cycle gives at $1536 \cdot 2^{60}$; conditional when it was published, it is the current record since the verification to 2^{71}.

$r = 72\,057\,431\,991$ [38, Remark 28]. Bařina [7] has since verified the conjecture up to $2^{71} > 3 \cdot 2^{69}$, so this bound now holds unconditionally and is the current record: every nontrivial cycle has $r \geq 137\,528\,045\,312 = q_{23}$ odd elements, and at least $p_{23} + q_{23} = 355\,504\,839\,929$ elements under T_0, as Bařina notes [7, §6]. The record structure itself is classical: the first nonexcluded r is the least denominator of a fraction s/r in the interval $(\theta, \theta + \varepsilon)$, and the least denominator fraction in an interval is read off the continued fraction expansion, a lemma Hercher states and uses (his Lemma 22) and which underlies Eliahou's computation; in the form of Theorem 10.8 and Proposition 10.14 it is the description of the best one sided approximations stated and proved in [31, Theorem 4.5(ii)]. The mediant lemma (Lemma 10.11) reduces the question to the δ records in two lines, and Theorem 10.8 identifies those with the rungs; together they keep the certificate of Section 10.8 short. Halbeisen and Hungerbühler [30] go further in another direction: they replace the crude inequality by the optimal criterion, which is attained by the balanced parity word, and they observe both the harmlessness of the convergents from the wrong side (Remark 10.10) and the exact ties at

multiples (Remark 10.15). What we add is that certificate and the analysis of Section 10.7, which shows exactly what Diophantine information cannot do (a point that Halbeisen and Hungerbühler [30, §5] make in another form: rational cycles satisfy the same inequalities, and their least elements grow with the length) and what arithmetic information would suffice.

Remark 10.18 *(Comparison with Baker's method).* Baker's theory gives lower bounds for linear forms in logarithms of algebraic numbers, and for two logarithms the explicit estimates of Laurent, Mignotte and Nesterenko [62, Corollary 2] give, for the form $\Lambda = s\log 2 - r\log 3$, a bound of the shape $\log\Lambda \geq -C(\log r)^2$ with an explicit absolute constant C; precisely, $\log\Lambda \geq -24.34\,\log 3\,\big(\max\{\log b' + 0.14,\,21\}\big)^2$ with $b' = r + s/\log 3 \approx 2.44\,r$. For $r \leq 4.6\times 10^8$ the maximum is 21 and the bound is $-24.34\cdot\log 3\cdot 441 \approx -11\,792$, so it gives only $\Lambda \geq e^{-11800}$ there, and $\Lambda \geq e^{-11900}$ for $r \leq 5\times 10^8$: a bound on M that is astronomically larger than anything verification can reach, whereas the actual values of Λ_r near $r = 10^5$, from Table 10.2, are of order 10^{-6}. The reason is structural. Baker's theorem bounds Λ uniformly in r, with no knowledge of which r are dangerous; here we enumerate the dangerous r, so the exact value of Λ_r is available. Transcendence methods become indispensable only when a bound valid for all r simultaneously is wanted, as for the m cycle results of Steiner, Simons and de Weger, and Hercher, where r is unbounded.

10.7 What Diophantine approximation cannot do

One might believe that the cycle half of the conjecture is, at bottom, a question about how well $\log_2 3$ can be approximated by rationals, and that a sufficiently strong effective irrationality measure for $\log_2 3$ would settle it. This section shows that the belief is false, quantifies what such a measure would give, and identifies the missing ingredient.

Remark 10.19 *(What an irrationality measure gives).* Suppose ν is an effective irrationality measure for $\log_2 3$ with constant c, meaning that $|\log_2 3 - s/r| \geq c\,r^{-\nu}$ for all integers s and $r \geq 1$. Then $\delta(r) = s - r\theta = r(s/r - \theta) \geq c\,r^{1-\nu}$ with $s = \lceil r\theta\rceil$, and (10.6) becomes

$$M < F(r) \leq \frac{r^{\nu}}{3c\log 2}. \tag{10.15}$$

Since $\nu \geq 2$ for every irrational number (Lemma 10.5(iv)), the right side grows at least like r^2. Combined with $M > X$, the bound excludes precisely the r with $r^{\nu} < 3cX\log 2$, a finite range, and nothing beyond it. In particular no irrationality measure, however good, bounds M uniformly in r.

The finite range is not a weakness of irrationality measures in particular. It is forced by the structure of the problem.

Proposition 10.20 *(Diophantine insufficiency). For every $X \geq 1$ there are infinitely many $r \geq 1$ with*

$$0 < \delta(r) < \frac{r}{3X \log 2},$$

that is, infinitely many r that are not excluded at verification range X.

Proof. By Theorem 10.8 the one sided records $u_1 < u_2 < \cdots$ form an infinite sequence along which $\delta(u_i)$ decreases strictly. Moreover $\delta(u_i) \to 0$: the convergents p_k/q_k with k odd lie above θ, so $\lceil q_k\theta \rceil = p_k$ and $\delta(q_k) = p_k - q_k\theta < 1/q_{k+1}$ by Lemma 10.5(iv), and these q_k are among the records. Put $\varepsilon = 1/(3X \log 2)$. Once $\delta(u_i) < \varepsilon$, which holds for all large i, we have $\delta(u_i) < \varepsilon \leq u_i\varepsilon$, so u_i is not excluded. □

In words: the inequality (10.6), a verified range and any information about the number $\log_2 3$ alone cannot together exclude all nontrivial cycles.

How large is the gap in practice? The best conceivable constant for $\nu = 2$ consistent with the rungs computed so far is $c_2 = \min_u u\,\delta(u) = 0.01773$, attained at $u = 190\,537$ (the rung before the partial quotient 55), and the corresponding exclusion is $r < (3c_2X \log 2)^{1/2}$. Table 10.7 compares this with the exact $r_*(X)$. Even this idealized measure, far better than any effective measure known (the explicit bound of Rhin [85], in the form used by Simons and de Weger [92, Lemma 12], corresponds to $\nu \approx 14.3$), is beaten by the ladder computation by factors from about two to more than fifty, because the ladder uses the exact value of δ at each dangerous r while the measure uses a worst case. With exponents $\nu \geq 3$ the best conceivable constant is $c_\nu = \log_2(4/3) = 0.4150$, attained at $r = 1$, and at $X = 2^{68}$ it excludes only $r < 6.3 \times 10^6$ for $\nu = 3$ and $r < 1.3 \times 10^5$ for $\nu = 4$.

Table 10.7. Exclusion by an idealized measure with $\nu = 2$ and $c_2 = 0.01773$ versus the exact ladder bound.

X	measure excludes $r <$	exact $r_*(X)$	ratio
2^{40}	2.01×10^5	1.08×10^7	53.5
2^{60}	2.06×10^8	3.98×10^8	1.93
2^{68}	3.30×10^9	7.21×10^{10}	21.8
2^{72}	1.32×10^{10}	1.38×10^{11}	10.4
2^{80}	2.11×10^{11}	2.40×10^{12}	11.4

Figure 10.4 shows the empirical irrationality profile of $\log_2 3$: for each convergent the exponent $\nu_k = -\log|\theta - p_k/q_k| / \log q_k$, together with the next partial quotient. The exponents settle near 2 (their mean over $50 \leq k \leq 198$ is 2.010), with excursions exactly where a large partial quotient follows; the largest partial quotients among the first 199 are 55 (twice), 49, 37 and 33. This is what one sees for a typical real number, and it says nothing provable about $\log_2 3$. In particular the profile gives no evidence either way on whether the partial

quotients of $\log_2 3$ are bounded; and by Proposition 10.20, even bounded partial quotients would not finish the cycle half.

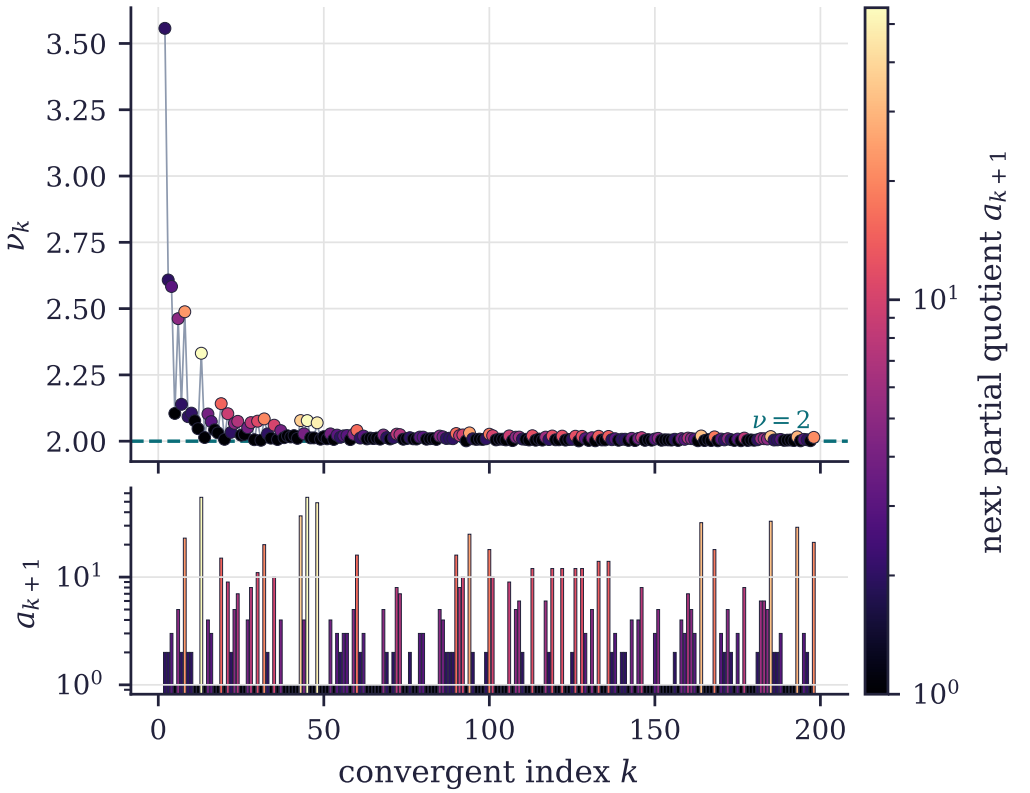


FIGURE 10.4. Empirical irrationality profile of $\log_2 3$. Top: the exponent ν_k of the kth convergent, coloured by the next partial quotient. Bottom: the partial quotients a_{k+1} on a logarithmic scale. Large partial quotients produce the visible excursions above $\nu = 2$.

10.7.1 The window lemma

What, then, is missing? The inequality (10.6) is an *upper* bound on the least element of a cycle. The affine identity of Chapter 2 gives a matching *lower* bound, and the two together confine the least element to a window whose width is the real measure of our ignorance. In the rest of this section we write $m = r$ for the number of odd steps and $k = s$ for the number of steps of T, as in Chapter 2, so that k is no longer a continued fraction index.

LEMMA 10.21 *(Window lemma). Let $x_1, \dots, x_r$ be the odd elements of a cycle of the accelerated map T with $r = m$ odd steps and $k = s$ steps in all, and put $\Lambda = k \log 2 -$*

$m \log 3 > 0$. *Then every element x of the cycle (odd or even) satisfies*

$$\frac{3^m - 2^m}{2^k - 3^m} \;\le\; x, \qquad \text{and} \qquad M = \min_i x_i \;<\; \frac{m}{3\Lambda} \;\le\; F(m), \tag{10.16}$$

where $(3^m - 2^m)/(2^k - 3^m) = (1 - (2/3)^m)/(e^\Lambda - 1)$, *and* $m/(3\Lambda) = F(m)$ *exactly when* $k = \lceil m\theta \rceil$. *In particular the least element lies in the window* $\big[(1 - (2/3)^m)/(e^\Lambda - 1),\ m/(3\Lambda)\big)$, *whose endpoints have ratio* $\frac{m}{3} \cdot \frac{e^\Lambda - 1}{\Lambda}(1 - (2/3)^m)^{-1}$, *which is* $\frac{m}{3}(1 + o(1))$ *as* $\Lambda \to 0$ *and* $m \to \infty$.

Proof. Let x be any element of the cycle. Applying (2.4) to the k steps that return x to itself gives $2^k x = 3^m x + c_k(x)$, so $x = c_k(x)/(2^k - 3^m)$, the denominator being positive because $\Lambda > 0$. By the recursion (2.5), an odd step at time i contributes 2^i, multiplied by 3 once for each later odd step, so $c_k(x) = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{i_j}$, where $0 \le i_0 < i_1 < \cdots < i_{m-1} \le k - 1$ are the times of the odd steps. Since $i_j \ge j$, every term is at least the corresponding term with $i_j = j$, hence $c_k(x) \ge \sum_{j=0}^{m-1} 3^{m-1-j} 2^j = 3^m - 2^m$. This gives the lower bound, and dividing numerator and denominator by 3^m gives the second form. For the upper bound, the proof of Lemma 10.1 with $r = m$ gives $\Lambda < \sum_i 1/(3x_i) \le m/(3M)$, since $\log(1 + u) < u$; and $\Lambda \ge \Lambda_m = \delta(m) \log 2$ by (10.5), with equality exactly when $k = \lceil m\theta \rceil$, so $m/(3\Lambda) \le F(m)$. The ratio of the endpoints is the quotient of the two expressions, and $(e^\Lambda - 1)/\Lambda \to 1$, $(2/3)^m \to 0$. □

For the trivial cycle, $m = 1$ and $k = 2$, the window is $[1,\ 1.159)$ and the cycle sits at its left end. For the hypothetical cycles at the ladder rungs the window is enormous: at the rung $r_* = 72\,057\,431\,991$, with k the corresponding s, the lower end is $1.8 \times 10^{11} \approx 2^{37.4}$ and the upper end $4.4 \times 10^{21} = 2^{71.9}$, a factor $r/3$ apart as the lemma predicts (Figure 10.5). The verified range 2^{68} removes the lower part of the window, and what is left is exactly the gap that the sharp bound of Theorem 10.16 measures.

Corollary 10.22 *(What arithmetic would suffice). Suppose that for some function f every nontrivial cycle with m odd steps and k steps in all satisfied the sharpened upper bound $M < m/(3\Lambda f(m))$, where $\Lambda = k \log 2 - m \log 3$ is the value at that cycle.*

(i) If $f(m) \ge \frac{m}{3} \cdot \frac{e^\Lambda - 1}{\Lambda}\big(1 - (2/3)^m\big)^{-1}$ for every pair (m, k) with $m\theta < k < 2m$, then there are no nontrivial cycles.

(ii) If $f(m) \ge \frac{m}{3}(1 + \eta)$ for some $\eta > 0$ and all $m \ge m_0$, where $(1 + \eta)(1 - (2/3)^{m_0}) > 1$, let $\Lambda_0 > 0$ solve $(e^{\Lambda_0} - 1)/\Lambda_0 = (1 + \eta)(1 - (2/3)^{m_0})$. Then every nontrivial cycle with $m \ge m_0$ odd steps has least element $M < C = 1/((1 + \eta)\Lambda_0)$, which is about $1/(2\eta)$ when η is small and m_0 large; so a verified range $X \ge \max\big(C, \max_{m < m_0} F(m)\big)$ excludes all nontrivial cycles, with no further use of Diophantine approximation.

Proof. A nontrivial cycle contains no element 1, so every $x_i \geq 3$ and (10.2) gives $2^k = \prod_i (3 + 1/x_i) < 4^m$; with $\Lambda > 0$ this is $m\theta < k < 2m$. (i) The hypothesis puts the upper end $m/(3\Lambda f(m))$ of the sharpened window at or below the lower end $(1-(2/3)^m)/(e^\Lambda - 1)$ of (10.16), which is impossible. (ii) The function $(e^\Lambda - 1)/\Lambda$ increases from 1 to ∞ on $(0, \infty)$. For a cycle with $m \geq m_0$ and $\Lambda < \Lambda_0$ it is therefore below $(1+\eta)(1-(2/3)^{m_0}) \leq \frac{3f(m)}{m}(1-(2/3)^m)$, and the window is empty as in (i). For a cycle with $m \geq m_0$ and $\Lambda \geq \Lambda_0$ the hypothesis gives $M < m/(3\Lambda_0 f(m)) \leq 1/((1+\eta)\Lambda_0) = C$. For $m < m_0$, Lemma 10.21 gives $M < F(m)$. So a nontrivial cycle would have $M < X$, whereas every element of a nontrivial cycle is at least X. □

The corollary converts the vague hope that "the arithmetic of the parity word should help" into a precise target: an improvement of the upper bound by a factor slightly more than $m/3$. A factor $\frac{m}{3}(1+\eta)$ for all large m would confine the least element of every nontrivial cycle with many odd steps below about $1/(2\eta)$, and a finite verification would then exclude all nontrivial cycles. Raising the lower end by such a factor, or splitting the factor between the two ends, empties the window only when Λ is small; the cycles with Λ bounded below would still need an upper bound on M that does not grow with m. By contrast, Proposition 10.20 shows that no improvement of the Diophantine side alone can work.

10.7.2 An exploration of parity words

To see where actual cycles of rational numbers sit inside the window, we enumerated all parity words with $m \leq 20$ ones and $\lceil m\theta \rceil \leq k \leq 2m$ letters (sampling 10^6 words when there are more than 3×10^6), computed $x = c_k(w)/(2^k - 3^m)$ for each, and recorded how close $c_k(w)$ comes to a multiple of $D = 2^k - 3^m$ (script `ch08_window.py`). Every word defines a cycle of T on the rationals with odd denominator, the rational cycles of Lagarias [54], and an integer cycle corresponds to $D \mid c_k(w)$. Three observations emerged.

First, the only integer hits for $m \leq 20$ are the words of the trivial cycle $\{1, 2\}$ repeated, at $k = 2m$; this is of course consistent with Theorem 10.16. The closest approaches of $c_k(w)/D$ to an integer are of the size predicted by a model in which the residues $c_k(w)$ mod D of the words examined are independent and uniform on the nonzero classes: over the 92 shapes (m, k) with at least 50 words, the closest approach divided by its expected value under the model has mean 1.1 and median 0.66, where the model gives 1 and about $\log 2 = 0.69$, and the median drifts only from 0.87 for $m \leq 12$ to 0.63 for $m \geq 13$ while the expected value itself falls from 8×10^{-2} to 2×10^{-7} (Figure 10.6, right). We found no sign of a structural obstruction at this scale, in either direction.

Second, the least element of a rational cycle can sit nearly anywhere in the window: over all rotation classes with $m \leq 11$ the ratio of the least element to the lower end of the window attains 1 (the lower bound of Lemma 10.21 is sharp,

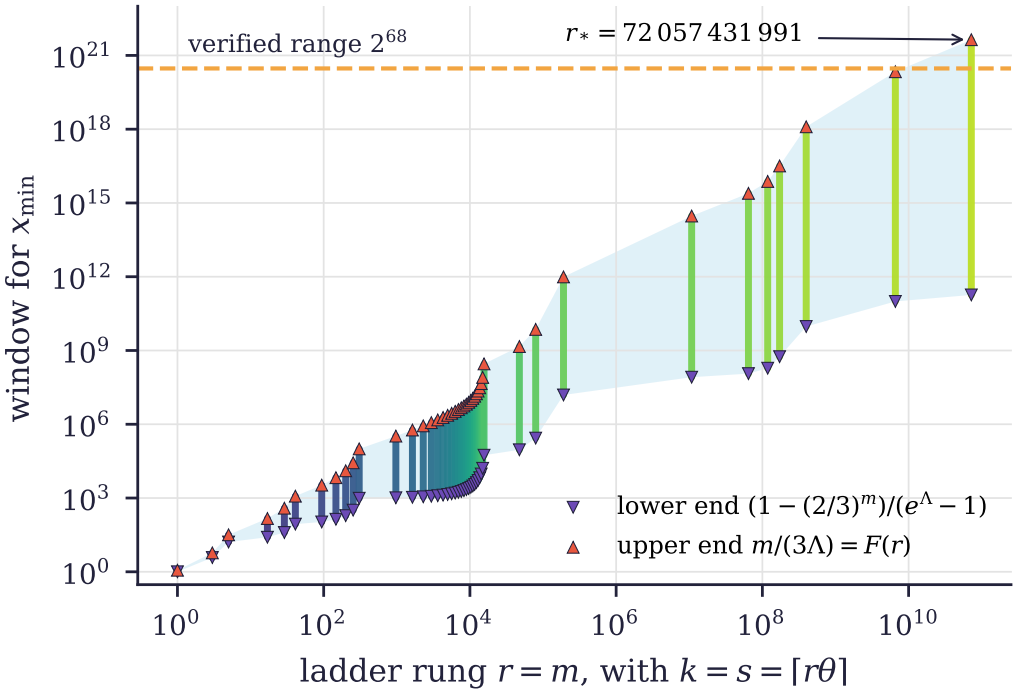


FIGURE 10.5. The window of Lemma 10.21 for the hypothetical cycles at every rung of the one sided ladder up to $r_*(2^{68})$. Downward triangles are the lower ends $(1-(2/3)^m)/(e^\Lambda-1)$, upward triangles the upper ends $F(m) = m/(3\Lambda)$; the vertical segments are the windows. Only the rung $r_* = 72\,057\,431\,991$ reaches above the verified range 2^{68}.

attained by the word with all odd steps first), and the least element never exceeds 0.87 of the upper end $m/(3\Lambda)$ of its window: the largest ratio, $3\log(4/3) = 0.863$, is that of the trivial cycle repeated, and for $k < 2m$ the ratio stays below 0.79, while the harmonic mean of the odd elements, which is what the proof of Lemma 10.21 actually bounds, reaches 0.995 of the upper end (Figure 10.6, left).

Third, for the balanced words, the Christoffel words with m ones spread as evenly as possible among k letters, which are the natural candidates for a cycle with k/m close to θ, the least and greatest odd elements sit at nearly fixed fractions of the upper end when $k = \lceil m\theta \rceil$: for the 1154 primitive balanced words ($\gcd(m,k) = 1$) with $100 \le m \le 2000$, $x_{\min}/F(m)$ lies in $[0.719, 0.724]$ and the greatest odd element in $[1.435, 1.444]\,F(m)$, close to $1/(2\log 2) = 0.7213$ and $1/\log 2 = 1.4427$. (When $\gcd(m,k) > 1$ the balanced word is a shorter one repeated and gives the same rational cycle; thus $m = 400$ and 800 repeat $m = 200$.) A heuristic explains the constants. For a balanced word the odd elements of the cycle are nearly a geometric progression in $\log_2$, filling an interval $[x_{\min}, 2x_{\min}]$ with $\log_2 x$ nearly uniformly distributed; then the mean

of $1/x_i$ is about $x_{\min}^{-1}\int_0^1 2^{-u}\,du = 1/(2x_{\min}\log 2)$, and $\Lambda \approx \sum_i 1/(3x_i)$ gives $F(m) = m/(3\Lambda) \approx 2\log 2\, x_{\min}$. We record this as an observation, not a theorem: it says that for balanced words the least element sits a fixed fraction below the top of the window, so balanced words are as far as possible from the factor $m/3$ that Corollary 10.22 asks for. Any arithmetic proof that there are no cycles must therefore treat balanced words by divisibility, not by size.

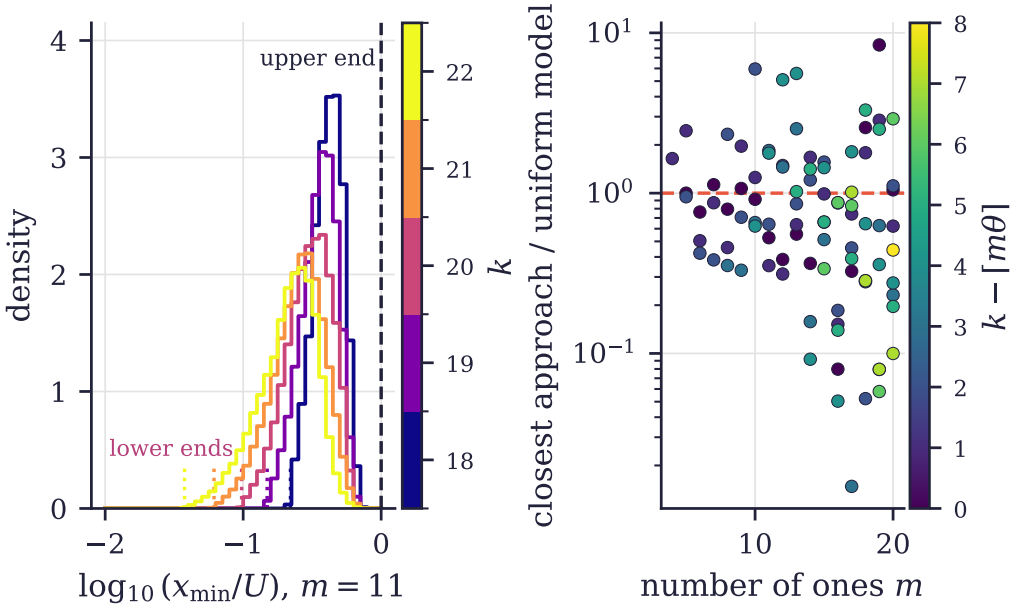


FIGURE 10.6. Left: distribution of $\log_{10}(x_{\min}/U)$, where $U = m/(3\Lambda)$ is the upper end of the window for the given k, over rotation classes of rational cycles with $m = 11$, for each k from $\lceil m\theta \rceil = 18$ to $2m = 22$ (colour); the lower ends of the windows are marked (dotted). Right: the closest approach of $c_k(w)/D$ to an integer, relative to its expected value when the residues of the words examined are independent and uniform on the nonzero classes modulo D, for the shapes with at least 50 words, against the number of ones m, coloured by $k - \lceil m\theta \rceil$.

10.8 An exact certificate

The bound of Theorem 10.16(ii) was computed in floating point, at 150 digits. For a result of this kind it is worth having a certificate that uses no floating point at all: a list of integer inequalities that anyone can check, and that a proof assistant could check. The mediant lemma and Theorem 10.8 make such a certificate short, and we have built one (script `ch08_certificate.py`, which runs in eleven seconds).

Step 1: an enclosure of $\log_2 3$. Take the convergents

$$\frac{a}{b} = \frac{1\,193\,652\,440\,098}{753\,110\,839\,881}, \qquad \frac{a'}{b'} = \frac{217\,976\,794\,617}{137\,528\,045\,312},$$

which lie on opposite sides of θ. The inequalities $2^a < 3^b$ and $3^{b'} < 2^{a'}$ certify $a/b < \theta < a'/b'$, an interval of width exactly $1/(bb') = 9.65 \times 10^{-24}$. The powers are far too large to form, so each comparison is made by enclosing 3^b between $L \cdot 2^e$ and $U \cdot 2^e$ with 320 bit integer mantissas, computed by repeated squaring with downward and upward truncation; every truncation is itself an integer inequality, and 39 squarings suffice.

Step 2: an enclosure of $\log 2$. The partial sums of $\sum_{n \geq 1} 1/(n2^n) = \log 2$ are lower bounds, and adding the tail bound $1/((N+1)2^N)$ gives upper bounds; we use an enclosure of width 9×10^{-30}. The exclusions in fact survive the crude replacement of $\log 2$ by $69/100$, and the final feasibility check survives $7/10$, which follow from the sum to $n = 7$: $1163/1680 > 69/100$ and $1163/1680 + 1/1024 < 7/10$.

Step 3: the partial quotients. The common prefix of the continued fraction expansions of the two rational endpoints, less its last term, is certified to be the beginning of the expansion of θ:

$$[1; 1, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, 1, 1, 55, 1, 4, 3, 1, 1, 15, 1, 9].$$

Step 4: the rungs. From these partial quotients the ladder rungs are generated exactly, 44 of them up to 72 057 431 991. For each rung $u < r_*$ the exclusion $\delta(u) \geq u\varepsilon$ is checked as an inequality between rationals built from the enclosures, never using more than 248 bit integers; the smallest certified margin is $\delta(u)/(u\varepsilon) > 1.355$ (true value 1.361), at $u = 6\,586\,818\,670$. At r_* itself the opposite inequality is certified from the upper end of the enclosure, $\delta(r_*) < 7.97 \times 10^{-12}$, so that $\delta(r_*)/(r_*\varepsilon) < 0.068$ (true value 0.0677), and the bound is attained by the method.

Step 5: the records. The enclosures of δ are certified to decrease strictly along the rungs, as Theorem 10.8 predicts, a consistency check on the partial quotients; by that theorem the rungs are the records, and with the mediant lemma this certifies that no other r needs checking. As an independent check, an exact integer scan of all $r \leq 2 \times 10^6$ finds exactly the 37 records predicted by the ladder.

The certificate is the bridge to a machine checked proof (Section 10.10): every step is an inequality between integers of at most a few hundred bits, plus two exponentiations handled by a verified squaring routine.

10.9 A sibling problem: Erdős 125

Good rational approximations to θ, the raw material of the previous sections, appear in a different role in a problem from the Erdős collection that was settled in 2026.

10.9.1 The problem

Let A be the set of nonnegative integers whose base 3 digits all lie in $\{0, 1\}$, and B the set of nonnegative integers whose base 4 digits all lie in $\{0, 1\}$. Then $|A \cap$

TABLE 10.8. Selected rungs of the certificate at $X = 2^{68}$: the rung u, its numerator $s = \lceil u\theta \rceil$, a certified bound for $\delta(u)$ (a lower bound for $u < r_*$, an upper bound at $u = r_*$), and the margin $\delta(u)/(u\varepsilon)$, which must exceed 1 for exclusion.

u	s	$\delta(u)$	margin
15 601	24 727	$> 2.62 \times 10^{-5}$	$> 1.0 \times 10^{12}$
190 537	301 994	$> 9.30 \times 10^{-8}$	$> 2.99 \times 10^{8}$
10 781 274	17 087 915	$> 1.76 \times 10^{-8}$	$> 1.0 \times 10^{6}$
171 928 773	272 500 658	$> 2.58 \times 10^{-9}$	> 9214
397 573 379	630 138 897	$> 1.52 \times 10^{-10}$	> 235.7
6 586 818 670	10 439 860 591	$> 1.45 \times 10^{-11}$	> 1.355
72 057 431 991	114 208 327 604	$< 7.97 \times 10^{-12}$	< 0.068

$[0, 3^n - 1]| = 2^n$ and $|B \cap [0, 4^n - 1]| = 2^n$, so $|A \cap [0, N]|$ is of order $N^{\log 2/\log 3} = N^{0.631}$ and $|B \cap [0, N]|$ of order $N^{1/2}$. The number of pairs (a, b) with $a + b \le N$ is therefore of order $N^{1.131}$, which exceeds N by a power, and positive density of the sumset $A + B$ is the naive guess. Burr, Erdős, Graham and Li [15] asked whether $A + B$ has positive density; in the form recorded as problem 125 of the Erdős problem collection, the question is whether it has positive lower density. Melfi [68] proved that $|(A + B) \cap [1, x]| \gg x^{0.965}$, and Hasler and Melfi [36] improved the exponent to 0.97777 and showed that the lower density is at most $1015/1458 = 0.696$.

In 2026 the question was settled negatively by AlphaProof Nexus, an agent that searches for proofs in the Lean proof assistant, developed at Google DeepMind by Tsoukalas, Kovsharov, Shirobokov, Surina and collaborators [101]: the lower density of $A + B$ is zero. The same work reports that the agent solved 9 of 353 open Erdős problems (formalized in Lean) and 44 of 492 conjectures from the OEIS. The proof proceeds by thinning along windows in which a power of 3 is close to a power of 4, that is, along good rational approximations to the ratio $\log 4/\log 3 = 2/\theta$; an early very good one is $3^{53} \approx 4^{42}$, with ratio 1.0021. DeepMind's work does not address the Collatz problem; the link we draw is through this Diophantine mechanism.

10.9.2 An exact computation

We computed the counting function $|(A + B) \cap [0, N]|$ exactly for every $N \le 4^{13} = 67\,108\,864$, by marking each sum $a + b \le N$ in a bit array, as reported in our note [78]. For this chapter we reran the computation independently (script `ch08_erdos125.py`; the run to 4^{13} takes about sixteen seconds) and reproduced every entry of Table 10.9 to the four digits shown, as well as the location and value of the minimum.

Three things are visible in Figure 10.7. The density stays well above the Hasler and Melfi bound throughout the range, so finite computation gives no hint that the

TABLE 10.9. Density $|(A+B)\cap[0,N]|/(N+1)$ at powers of 4 and of 3. The exact minimum over $10^4 \le N \le 4^{13}$ is 0.763392, at $N = 3^{15}-1 = 14\,348\,906$.

N	density	N	density
4^7	0.8593	3^{10}	0.7785
4^8	0.7913	3^{11}	0.8372
4^9	0.8372	3^{12}	0.8583
4^{10}	0.8688	3^{13}	0.8742
4^{11}	0.8068	3^{14}	0.8147
4^{12}	0.7836	3^{15}	0.7634
4^{13}	0.8382	3^{16}	0.8312

lower density is zero: the decay that the proof exploits operates at the scale of the good approximations to $2/\theta$, and the first very good one, $3^{53} \approx 4^{42}$, is far beyond any computation. The minima sit just below powers of 3, where a new block of A is about to begin and the previous blocks have been exhausted. And the deepest dips occur when a power of 3 lies a little below a power of 4: $3^{10}/4^8 = 0.90$ and $3^{15}/4^{12} = 0.855$ are such cases, and they give the two lowest values in the table. This is the thinning mechanism of the proof seen at small scale: when 3^m and 4^k are close, the two digit structures align, and the gaps of A and of B overlap rather than fill each other.

10.9.3 The analogy

Both problems turn on good rational approximations to a ratio of logarithms, but they use them in opposite ways. For Erdős 125 the coincidences are $3^m \approx 4^k = 2^{2k}$, that is, approximations $2k/m$ to θ from either side; they produce windows in which $A+B$ thins, and the proof uses infinitely many such windows to drive the lower density to zero. Only their existence matters, and it follows from soft arguments, such as the pigeonhole principle or Lemma 10.5; the example $3^{53} \approx 4^{42}$ is the convergent $84/53$, which lies below θ and is useless for cycles (Remark 10.10). For Collatz cycles the good approximations $3^r \approx 2^s$ with s/r just above θ are exactly what a cycle needs; their records bound the length of a cycle (Theorem 10.16), but by Proposition 10.20 they are good enough infinitely often, whatever the verification range, so no information about the approximations of θ alone, not even bounded partial quotients, can exclude every cycle. In one problem the existence of good approximations is the engine of the proof; in the other it is an obstruction that Diophantine approximation cannot remove, and what is missing is arithmetic information about the parity word (Corollary 10.22). That asymmetry is a fair summary of why the cycle half is hard.

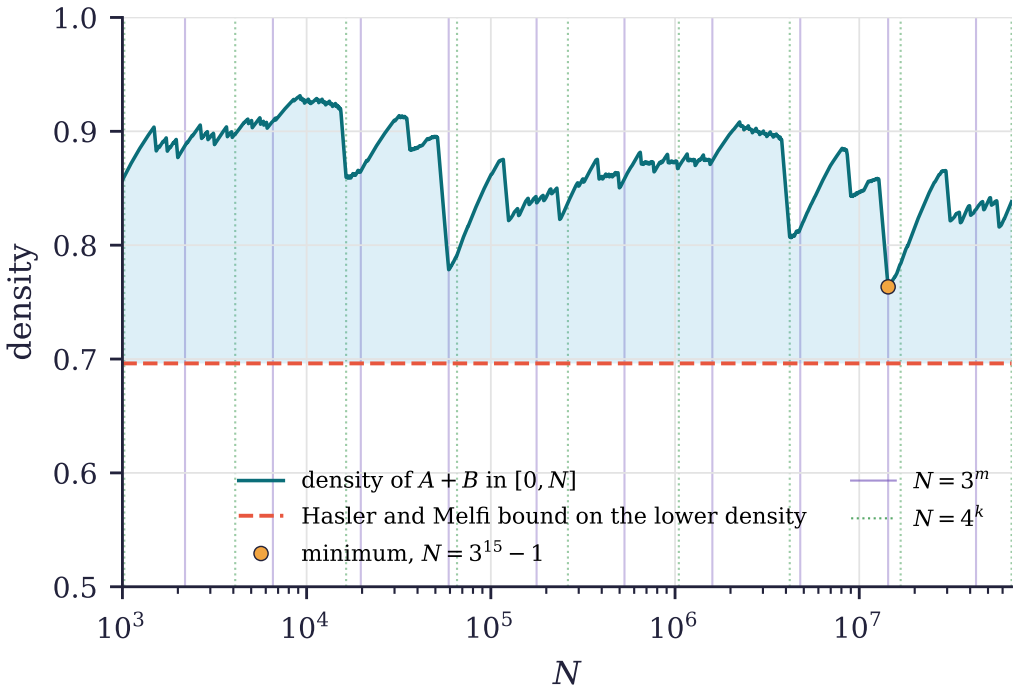


FIGURE 10.7. Density of $A + B$ in $[0, N]$ for $10^3 \le N \le 4^{13}$, with vertical lines at the powers of 3 (solid) and of 4 (dotted) and the Hasler and Melfi upper bound 1015/1458 on the lower density.

10.10 MACHINE CHECKED STATEMENTS

A bound such as Theorem 10.16 is assembled from a short chain of elementary steps and a finite computation, which makes it an attractive target for formal verification. In our note [78] we wrote two small files in Lean 4 [19], using only the core library (Lean 4.15, without Mathlib). The restriction is severe: there are no real numbers, no rationals and no limits, so every statement must be phrased in natural number arithmetic, and every proof uses only induction, case analysis, linear arithmetic and the kernel. It is also useful, since it forces exactly the logarithm free formulations a verified cycle bound will need. Both files compile; the first produces no warnings, the second one warning for each open statement. They are in the folder `lean` of the book's repository (Appendix B).

10.10.1 What is proved

The definitions are the accelerated map, its iterates and the number of odd steps.

```
def T (x : Nat) : Nat :=
  if x % 2 = 0 then x / 2 else (3 * x + 1) / 2
def Tn : Nat → Nat → Nat
  | 0, x => x
```

```
  | k + 1, x => T (Tn k x)
def odd : Nat → Nat → Nat
  | 0, _ => 0
  | k + 1, x => odd k x + (if Tn k x % 2 = 1 then 1 else 0)
```

The four theorems are the following.

```
theorem B1 (k x : Nat) : ∃ c : Nat,
    2 ^ k * Tn k x = 3 ^ (odd k x) * x + c ∧
    (odd k x > 0 → c > 0)
theorem B2 {k x : Nat} (hx : x > 0) (hk : k > 0)
    (hcyc : Tn k x = x) (ho : odd k x > 0) :
    3 ^ (odd k x) < 2 ^ k
theorem B3 : ∀ x, 1 ≤ x → x ≤ 10000 →
    ∃ j, j ≤ 400 ∧ Tn j x = 1
theorem B4 : Tn 1 1 = 2 ∧ Tn 2 1 = 1 ∧
    ∀ k, Tn k 1 = 1 ∨ Tn k 1 = 2
```

B1 is the affine identity of Proposition 2.11, proved by induction on k through a stronger lemma that names the remainder by the explicit recursion $c_{k+1} = 3c_k + 2^k$ after an odd step and $c_{k+1} = c_k$ after an even one; an inductive hypothesis asserting only the existence of some remainder is awkward to propagate. B2 is the cycle inequality $3^m < 2^k$ of Corollary 2.14, the lower half of (10.4). B3 checks that every starting value from 1 to 10^4 reaches 1 within 400 steps of T (the true maximum is 165 steps, at $x = 6171$, by our own count). It starts at $x = 1$ because 0 is a fixed point of T, so the statement over all $x \leq 10^4$ would be false. It is proved by a Boolean evaluator together with a kernel checked soundness lemma, and the evaluation is run natively rather than in the kernel, so the proof depends on the axiom `Lean.ofReduceBool`, which amounts to trusting the compiled evaluator. B4 describes the trivial cycle. The axiom report printed by Lean is: B1 uses `propext` and `Quot.sound`; B2 also uses `Classical.choice`; B3 uses `propext`, `Quot.sound` and `Lean.ofReduceBool`; B4 uses no axioms. The proofs were written by an AI coding agent working against the compiler, and we then recompiled both files independently and read every statement against its informal meaning, which is where the two details just mentioned were caught.

10.10.2 What is stated

The open statements are the conjecture, its descent form, and the cycle problem.

```
theorem O1_collatz : ∀ x, x ≥ 1 → ∃ k, Tn k x = 1
theorem O3_descent : ∀ x, x ≥ 2 → ∃ k, k > 0 ∧ Tn k x < x
theorem O4_no_cycles : ¬ ∃ x k, x > 2 ∧ k > 0 ∧ Tn k x = x
```

The names O1, O3 and O4 are those of the Lean files and are unrelated to the numbering of the problems in Chapter 15. O3 implies O1 by strong induction, and O1 implies O3 because an orbit that reaches 1 passes below its starting value. O4 excludes every cycle of positive integers other than $\{1, 2\}$; the hypothesis $x > 2$ rules out the elements of the trivial cycle, and $x = 0$ is excluded with them.

10.10.3 From statements to a verified cycle bound

What separates these files from a verified form of Theorem 10.16 is the upper half of (10.4) and the continued fraction computation. Both are within reach of the core library once they are stated without logarithms. For the first, if every element of a cycle of T with k steps and m odd steps is at least X, then by (10.2)

$$2^k = \prod_{i=1}^{m}\Big(3+\frac{1}{x_i}\Big) \le \Big(3+\frac{1}{X}\Big)^m, \quad \text{that is} \quad 2^k(3X)^m \le 3^m(3X+1)^m,$$

an inequality between natural numbers which, combined with B2, is the whole content of Lemma 10.1 and Corollary 10.4.

For the second, the exclusion of all $m < r_*(X)$ can be replaced by the short certificate of Section 10.8. The proof of Theorem 10.16(ii) uses three ingredients: the identities of Lemma 10.5(ii) for the listed pairs (p_i, q_i) of convergents, which are integer arithmetic; the signs of $q_i\theta - p_i$, which are comparisons between 3^{q_i} and 2^{p_i}; and, at each rung u, the inequality $\delta(u) \ge u\varepsilon$. Checking a comparison such as 3^q against 2^p for q near 10^{11} by direct exponentiation is out of the question, but interval arithmetic on rational approximations of $\log_2 3$, proved once, reduces each check to arithmetic on integers of modest size, as in Steps 1 to 4 of Section 10.8; since the quantities $\delta(u)$ involved are of order 10^{-11} and the margins are at least a factor 1.36, a rational enclosure of $\log_2 3$ to a few dozen digits suffices. The mediant lemma and Theorem 10.8 keep the certificate short: instead of checking $\delta(m) \ge m\varepsilon$ for each of the some 7×10^{10} values $m < r_*$, Corollary 10.12 and the lattice argument of Theorem 10.8 reduce the task to the 44 rungs up to $r_*(2^{68})$, and there are only 53 rungs below 2×10^{13} in all. This is a well posed task for an automated prover of the AlphaProof Nexus type, and it would make the cycle bound the first part of the programme of this book to be fully machine checked. The remaining trusted input would be the verification range itself, which rests on a large computation of a different kind.

10.11 A refined open problem

The cycle length bounds above use the cycle structure only through the crude inequality $\log(1 + 1/(3x_i)) \le 1/(3M)$, and the window lemma adds only the lower bound $c_k \ge 3^m - 2^m$. For any cycle with m odd steps, k steps in all and smallest element $x_{\min}$, Lemma 10.21 gives

$$x_{\min} < \frac{m}{3\Lambda} \le \frac{m}{3\log 2\;\delta(m)}, \qquad \Lambda = k\log 2 - m\log 3.$$

The remaining information in a cycle lies in its parity vector. A cycle of T is described by a word of length k with m ones, and by Corollary 2.14 each element is $x = c_k(x)/(2^k - 3^m)$, where the affine constant $c_k(x)$ of (2.6) is a sum of terms $3^j 2^i$ indexed by the positions of the ones in the word. The thinning argument of

the Erdős 125 proof controls how two digit expansions interact over a window in which $3^m \approx 4^k$, and it is natural to ask whether a similar block decomposition of the parity word can be brought to bear here.

Problem *(Arithmetic sharpening of the cycle identity).* Find a function f, derived from the arithmetic of the parity vector rather than from the size of the elements, such that every nontrivial cycle with m odd steps and k steps in all satisfies

$$x_{\min} < \frac{m}{3\Lambda\, f(m)}, \qquad \Lambda = k\log 2 - m\log 3.$$

By Corollary 10.22, any f with $f(m) \geq \frac{m}{3}(1+\eta)$ for large m would confine the least element of every nontrivial cycle with many odd steps below about $1/(2\eta)$, and so reduce the cycle half to a finite verification; one may also try to raise the lower end $(1-(2/3)^m)/(e^{\Lambda}-1)$ of the window by a factor slightly more than $m/3$, which would empty the window whenever Λ is small. More concretely: does a block decomposition of the parity word, in the style of the thinning argument for Erdős 125, force $c_k(x)$ to be large compared with $3^m - 2^m$, or constrain it modulo $2^k - 3^m$, in a way that the size of x alone does not?

The target is demanding. For balanced words the least element already sits at about 0.72 of the upper end of the window, so no size argument can gain the factor $m/3$ there, and the exploration of parity words found residues $c_k(w)$ mod $(2^k - 3^m)$ behaving like uniform random variables. A proof along these lines must therefore exploit divisibility, and would amount to showing that $2^k - 3^m$ never divides $c_k(w)$ except for the trivial word. Any partial result with f bounded below by a constant $b > 1$ would lower every threshold of Table 10.6 by at least the factor b, which is worth at least as much as a verification range larger by the factor b, at every range at once; a result with $f(m) \to \infty$ would eventually gain more than any fixed increase of the verification range. We collect the problem, with the others raised in the book, in Chapter 15.

Notes

The cycle identity and the bound $M < r/(3\Lambda_r)$ go back at least to the 1970s; the use of continued fractions of $\log_2 3$ for cycle bounds is due to Crandall [18], whose criterion is in substance the block bound of Lemma 10.9, to Steiner [94] and, with larger verification ranges, to Eliahou [21]. The surveys of Lagarias [53] and the collection [55] describe the history, including the negative cycles and the $3x-1$ problem, and Lagarias [54] treats the rational cycles. Simons and de Weger [92] and Hercher [38] are the current state of the art for m cycles and for the length bound, Hercher's record [38, Corollary 29] being unconditional since the verification to 2^{71}. The verification ranges 2^{68} and 2^{71} are due to Bařina [6, 7]; see also the account of earlier verification efforts in [55]. For continued fractions, Khinchin [43] remains the most economical introduction, Hardy and Wright [32]

give the best approximation theorems, and Rockett and Szüsz [87] treat one sided approximation and the role of the intermediate fractions. Linear forms in two logarithms are treated by Laurent, Mignotte and Nesterenko [62], and the general theory by Baker [5]. The record structure of Proposition 10.14 is classical, equivalent to finding the least denominator fraction in an interval just above $\log_2 3$ (Lemma 22 of Hercher [38]), and is the description of the best one sided approximations that Hančl and Turek [31, Theorem 4.5(ii)] state and prove; the mediant lemma and Theorem 10.8 give a short self contained proof of the part we need. Halbeisen and Hungerbühler [30] give the optimal criterion, attained by the balanced parity word, and observe for Eliahou's criterion that the dangerous lengths lie between convergents, that convergents from the wrong side are harmless and that multiples tie exactly [30, §4.4]. The comparison of exact and crude bounds, the integer certificate, the threshold table, the insufficiency proposition and the window corollary are ours, although the structural point of the insufficiency proposition is made in another form in [30, §5]; the lower bound $c_k \geq 3^m - 2^m$ underlying the window is elementary and is implicit in the classical analyses of cycles by Böhm and Sontacchi [12] and by Steiner. The Erdős 125 problem is from Burr, Erdős, Graham and Li [15], and its resolution is in Tsoukalas and collaborators [101]; the exact density computation and the Lean files are from our unpublished note [78]. For Lean 4 see de Moura and Ullrich [19].

Exercises

Exercise 10.23. Verify the cycle identity (10.2) for the three negative cycles of Table 10.1, and show that for a cycle of negative integers the inequality is reversed: $2^s < 3^r$. Show that the conjugation $n \mapsto -n$ carries T_0 on the negative integers to the map $n \mapsto n/2$ (n even), $n \mapsto 3n - 1$ (n odd) on the positive integers, and list the corresponding cycles of this map.

Exercise 10.24. Using (2.4), find the rational periodic points of T whose parity words are $(1,0)$, $(1,1,0)$, $(1,0,0)$ and $(1,1,0,0)$, and check which of them are integers. Show that for every word w with $2^k \neq 3^m$ the number $c(w)/(2^k - 3^m)$ is a rational number with odd denominator whose orbit under T is periodic with parity word w.

Exercise 10.25. Prove that the best approximations from below, the r with $\{r\theta\} < \{r'\theta\}$ for every $1 \leq r' < r$, are exactly $r = 1$ and the rungs $q_{k-1} + jq_k$ with $k \geq 1$ odd and $1 \leq j \leq a_{k+1}$, by adapting the proof of Theorem 10.8. List them for $\log_2 3$ below 700.

Exercise 10.26. Show that if q is a rung and $t \geq 2$ is an integer with $t\,\delta(q) < 1$, then $\delta(tq) = t\,\delta(q)$ and $F(tq) = F(q)$. Deduce directly, without the mediant

lemma, that tq is never a strict record of F. Which multiples of $q_9 = 15\,601$ satisfy the hypothesis?

Exercise 10.27. Show that for even $i \geq 0$ one has $\delta(q_i) = 1 - \varepsilon_i$, so that $F(q_i) = q_i/(3(1-\varepsilon_i)\log 2)$, only slightly more than $q_i/(3\log 2)$, and explain the triangles of Figure 10.2. Then, for a cycle of T with k steps and m odd steps all of whose elements are at least X, prove the logarithm free inequality $2^k(3X)^m \leq 3^m(3X+1)^m$, and deduce from it and $\log(1+u) < u$ for $u > 0$ that $\delta(m) < m/(3X\log 2)$.

Exercise 10.28. For the Erdős 125 sets, prove that $|A \cap [0, 3^n - 1]| = |B \cap [0, 4^n - 1]| = 2^n$ and that the number of pairs $(a, b) \in A \times B$ with $a + b \leq N$ is of order $N^{\log 2/\log 3 + 1/2}$. Using `ch08_erdos125.py` or your own code, compute the density of $A + B$ in $[0, N]$ for $N = 3^m - 1$ and $N = 4^k$ up to 4^{10}, and relate the depth of each dip to the ratio $3^m/4^k$ for the nearest powers.

Exercise 10.29. Show that the lower bound of Lemma 10.21 is attained exactly by the rotation of a parity word whose m ones come first, and compute the window for $m = 2$, $k = 4$. Is there an integer in it? Which rational cycle sits at its lower end?

Exercise 10.30. Let $\nu = 2$ and suppose $|\log_2 3 - s/r| \geq c/r^2$ for all r. Show that the verification range X then excludes exactly the $r < (3cX\log 2)^{1/2}$ that the inequality (10.15) can handle, and that Proposition 10.20 still produces infinitely many nonexcluded r. Conclude that bounded partial quotients for $\log_2 3$ would not by themselves exclude all cycles.

Chapter 11

The Divergence Half

An orbit of the Collatz map that never reaches 1 either enters a cycle or grows without bound. The previous chapter dealt with cycles: a cycle is an exact multiplicative coincidence, continued fractions of $\log_2 3$ control how long it would have to be, and Diophantine information alone cannot exclude it (Proposition 10.20). This chapter deals with the other possibility. A divergent orbit is not a coincidence at all. It is an orbit along which the halvings are, on average, too few, and the question of whether one exists is a question about averages along a single deterministic sequence.

The stochastic model of Chapter 3 answers that question for an idealized random orbit, and the answer is no: a random walk with drift $\log(3/4)$ per stride does not escape. The difficulty, which has been met in every chapter so far, is that the model describes ensembles of starting values and the conjecture concerns each starting value separately. In this chapter we make the gap as precise as we can. We prove a Cesàro identity that turns the divergence half into a statement about the running mean of the halving exponents (Proposition 11.3); we show that the obvious empirical test of that statement carries no information at all; we formulate the statement that would be needed (Conjecture 11.6) and relate its threshold $\log_2 3$ to the momentum coefficient of Chapter 7; we study the residues of Syracuse orbits modulo powers of 3, where the model chain, in which the halving exponents are independent draws, can be computed exactly (Lemma 11.8); and we describe Tao's theorem, which is the deepest result in this direction, and the reason it stops short of every integer. The chapter ends with three routes to the divergence half and the points at which they fail.

Throughout, x_0 is an odd positive integer, $y_j = S^{(j)}(x_0)$ are the successive odd iterates under the Syracuse map, and

$$k_j = v_2(3y_{j-1} + 1), \qquad \bar{k}_N = \frac{1}{N}\sum_{j=1}^{N} k_j,$$

so that $y_j = (3y_{j-1} + 1)/2^{k_j}$.

11.1 Divergent orbits and what is known

Call the orbit of x_0 *divergent* if $\sup_j y_j = \infty$. No divergent orbit is known, and very little is known about what one would look like. We collect the elementary facts.

Lemma 11.1 *(Divergent orbits tend to infinity). If the orbit of x_0 is divergent, then $y_j \to \infty$, the values y_j are pairwise distinct, and no y_j is an integer known to reach 1; in particular $y_j > 2^{71}$ for all j.*

Proof. If $y_i = y_j$ for some $i < j$, the orbit is periodic from i on and hence bounded, so the values are distinct. A sequence of distinct positive integers takes each value at most once, so for every B only finitely many y_j are at most B, and $y_j \to \infty$. If some y_j reached 1 the orbit would be bounded. Every integer below 2^{71} reaches 1 by the computation of Bařina [7], so no y_j lies below 2^{71}, and $y_j \neq 2^{71}$ since y_j is odd. □

Beyond this, the known results about divergence are of two kinds. Density theorems say that the set of starting values with badly behaved orbits is small: Terras [99, 100] and, independently, Everett [22] (Theorem 2.26) show that almost every integer, in natural density, falls below its starting value; Korec [49] shows that almost every orbit falls below n^γ for each $\gamma > \log 3/\log 4$; and Tao [96, Theorem 1.3] shows that, for every function f tending to infinity, almost every orbit, in logarithmic density, falls below $f(n)$. None of these statements implies that the set of starting values with divergent orbits has density zero, let alone that it is empty, because an orbit that falls below its starting value may rise again, and its later values are not uniformly distributed in any residue class. Counting theorems, of which the strongest is due to Krasikov and Lagarias [51], give lower bounds for the number of integers that provably reach 1; we return to them in Section 11.7.

The second kind of result is structural: a divergent orbit, if one exists, must be atypical in a precise statistical sense. The rest of this section and the next make that precise.

Remark 11.2 *(The 2 adic picture).* By Theorem 2.18, the accelerated map on the 2 adic integers is conjugate to the shift on parity sequences, and Haar measure corresponds to fair coin flips (Proposition 2.19). By the strong law of large numbers, Haar almost every 2 adic integer has a T orbit in which the frequency of odd steps is $\frac{1}{2}$. We shall see in Corollary 11.4 that a divergent orbit must have a frequency of odd steps of at least $1/\log_2 3 \approx 0.631$ in the limit. The set of 2 adic integers with this property has Haar measure zero. This is no contradiction, since the positive integers are themselves a set of Haar measure zero, but it is a measure of how atypical a divergent orbit must be. Counting parity words gives a quantitative version: among the 2^n words of length n, the number with at least qn ones, $q = 1/\log_2 3$, is $2^{nH(q)+o(n)}$, where $H(q) = -q\log_2 q - (1-q)\log_2(1-q) \approx 0.9500$. The fraction of residue classes modulo 2^n whose first n steps of T contain at least $n/\log_2 3$ odd steps therefore decays like $2^{-0.05n}$, the rate $e^{-D} \approx 0.9659$ per step of Theorem 2.26 and Section 3.4.

11.2 The Cesàro identity

The following identity is the telescoped form of the stride formula (2.3). It is elementary, and its value lies entirely in what it does and does not say.

Proposition 11.3 *(Cesàro identity). For every odd x_0 and every $N \geq 1$,*

$$\log y_N - \log x_0 = \sum_{j=1}^{N} \log\Big(3 + \frac{1}{y_{j-1}}\Big) - N\bar{k}_N \log 2. \tag{11.1}$$

Equivalently,

$$\bar{k}_N = \log_2 3 + \frac{1}{N}\Big(\log_2 x_0 - \log_2 y_N + \sum_{j=1}^{N} \log_2\Big(1 + \frac{1}{3y_{j-1}}\Big)\Big). \tag{11.2}$$

Consequently, if the orbit of $x_0 > 1$ reaches 1 at stride N, then

$$\bar{k}_N \;>\; \log_2 3 + \frac{\log_2 x_0}{N} \;>\; \log_2 3. \tag{11.3}$$

Proof. Taking logarithms in $2^{k_j} y_j = 3y_{j-1} + 1$ gives $\log y_j - \log y_{j-1} = \log(3 + 1/y_{j-1}) - k_j \log 2$, and summing over $j = 1, \ldots, N$ gives (11.1). Writing $\log(3 + 1/y) = \log 3 + \log(1 + 1/(3y))$, dividing by $N \log 2$ and rearranging gives (11.2). If $y_N = 1$ then $\log_2 y_N = 0$ and every correction term $\log_2(1 + 1/(3y_{j-1}))$ is positive, which gives (11.3). □

The identity converts the conjecture into a statement about the running mean of the halving exponents. Its consequence for divergent orbits is the following.

Corollary 11.4 *(The running mean along a divergent orbit). If the orbit of x_0 is divergent, then*

$$\limsup_{N \to \infty} \bar{k}_N \;\leq\; \log_2 3.$$

Equivalently, the fraction of odd steps among the first n steps of the accelerated map satisfies $\liminf_n(\textit{fraction}) \geq 1/\log_2 3 \approx 0.6309$, *and among the first n steps of the standard map* $\liminf_n(\textit{fraction}) \geq 1/(1 + \log_2 3) \approx 0.3869$.

Proof. By Lemma 11.1, $y_j \to \infty$, so the correction terms $\log_2(1 + 1/(3y_{j-1}))$ tend to zero and so does their Cesàro mean; and $\log_2 y_N > 0$ for all large N. Inserting both facts in (11.2) gives $\bar{k}_N \leq \log_2 3 + o(1)$. The first N strides contain N odd steps and $N\bar{k}_N$ steps of T, or $N(1 + \bar{k}_N)$ steps of T_0. If step n of T falls in stride $N + 1$, then at least N of the first n steps are odd and $n \leq (N+1)\bar{k}_{N+1} \leq (N+1)(\log_2 3 + o(1))$, so the fraction of odd steps is at least $N/((N+1)(\log_2 3 + o(1)))$, which tends to $1/\log_2 3$. The same count with $1 + \bar{k}_{N+1}$ in place of $\bar{k}_{N+1}$ gives the statement for T_0. □

The typical values are $\bar{k}_N \to 2$, an odd fraction of $\frac{1}{2}$ for T and $\frac{1}{3}$ for T_0 (Proposition 3.7). A divergent orbit must sustain more than 63 percent odd steps under T, against a typical 50.

11.2.1 A pitfall: the identity is not a test

Since a divergent orbit must have $\limsup \bar{k}_N \le \log_2 3$, it is natural to compute $\bar{k}_N$ along many orbits and observe that it stays above $\log_2 3$. The computation is easy and the result is striking.

Remark 11.5 *(A tempting but uninformative computation).* For all 2500 odd starting values $3 \le x_0 \le 5001$ we computed $\bar{k}_N$ at the stride N at which the orbit reaches 1. The minimum is 1.7073, the mean is 2.2473, and every value lies above the threshold $\log_2 3 = 1.5850$; the identity (11.1) holds along every orbit with a maximum rounding residual of about 3×10^{-14}. One might present this as evidence for the divergence half. It is not evidence of any kind. The orbits in the scan were selected by the property that they reach 1, and by (11.3) every such orbit satisfies $\bar{k}_N > \log_2 3$ identically. No outcome of the computation could have been different. The minimum 1.7073 is $70/41$, attained at $x_0 = 27$ (Example 2.5), and it measures the spread of $\bar{k}$ over terminating orbits; it says nothing about orbits that do not terminate.

The same lesson appears in a sharper form for long orbits. For three odd starting values chosen at random near 2^{1000}, 2^{2000} and 2^{4000}, the orbits reached 1 after $N = 2773$, 4566 and 9507 strides, with final means $\bar{k}_N = 1.94555$, 2.02300 and 2.00568. By (11.2) with $y_N = 1$, these are

$$\bar{k}_N = \log_2 3 + \frac{\log_2 x_0}{N} + \frac{1}{N}\sum_j \log_2\Big(1 + \frac{1}{3y_{j-1}}\Big),$$

and the last term is below 10^{-4} in all three cases; the values of $\log_2 3 + (\log_2 x_0)/N$ are 1.94547, 2.02293 and 2.00565. So the final mean of the halving exponents carries no information beyond the number of strides, which is itself a reformulation of the total stopping time. What does carry information is the behaviour of $\bar{k}_N$ along the way, at strides N when the orbit is still far from 1, and that is what Figure 11.1 shows.

11.3 The Cesàro liminf conjecture

The divergence half of the conjecture can now be stated in the form the identity suggests.

Conjecture 11.6 *(Cesàro liminf). For every odd x_0,*

$$\liminf_{N\to\infty} \frac{1}{N}\sum_{j=1}^{N} k_j \;>\; \log_2 3.$$

Along an orbit that reaches 1 the exponents are eventually all equal to 2, since $S(1) = 1$ with $k = 2$, and along an orbit that enters a cycle with r odd elements

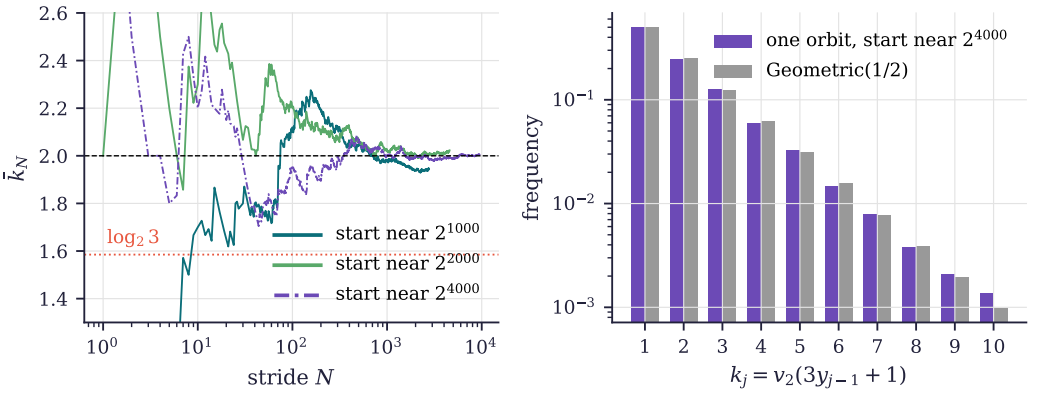


FIGURE 11.1. Left: the running mean $\bar{k}_N$ along three orbits started at random odd integers near 2^{1000}, 2^{2000} and 2^{4000}, with the Geometric mean 2 (dashed) and the threshold $\log_2 3$ (dotted). Right: the frequencies of the halving exponents along the orbit from near 2^{4000} against the Geometric$(\frac{1}{2})$ law, on a logarithmic scale.

and s halvings in one turn their mean tends to $s/r > \log_2 3$ (Lemma 10.1); so Conjecture 11.6 holds for every bounded orbit. By Corollary 11.4, every divergent orbit has $\liminf \bar{k}_N \le \limsup \bar{k}_N \le \log_2 3$, so Conjecture 11.6 holds if and only if there is no divergent orbit. It is therefore exactly equivalent to the divergence half of the Collatz conjecture, and its interest lies not in being weaker but in pointing at a mechanism. Under the Geometric law of Proposition 2.15 the exponent at a uniformly random odd residue has mean 2, and if the exponents along an orbit obeyed the strong law of large numbers we would have $\bar{k}_N \to 2 > \log_2 3$, with a margin of $2 - \log_2 3 \approx 0.415$. The margin belongs to the model: along a divergent orbit there is none, by Corollary 11.4. A proof of Conjecture 11.6 would need much less than a Geometric law for the exponents along unbounded orbits, or full equidistribution of their residues (Section 11.4): one needs only that along no orbit $\limsup_N \bar{k}_N \le \log_2 3$, that is, that the mean of the exponents cannot fall short of 2 by $2 - \log_2 3 \approx 0.415$ or more in the limit.

The threshold has a second reading. Write

$$\log_2 3 = 1 + \beta, \qquad \beta = \frac{\log(3/2)}{\log 2} \approx 0.585,$$

where β is the Switching MM momentum coefficient of Chapter 7. Every stride contains one forced halving, since $k_j \ge 1$, and the conjecture asks that the *extra* halvings $k_j - 1$ exceed β on average. In the optimization reading of the book, the tripling and the forced halving together form the overshoot $\log(3/2)$, and β is that overshoot measured in units of one halving; the extra halvings are the corrective descent. The Geometric law gives $\mathbb{E}[k - 1] = 1$, so the correction exceeds the overshoot by a factor $1/\beta \approx 1.71$. The divergence half asks that this inequality, which holds in expectation, hold along every orbit in the long run.

11.4 Equidistribution of the halving exponent along orbits

The right panel of Figure 11.1 shows the empirical law of k_j along the single orbit from near 2^{4000}. Over its 9507 strides the frequencies of $k = 1, 2, 3, 4, 5$ are 0.5026, 0.2462, 0.1271, 0.0596, 0.0330, against 0.5, 0.25, 0.125, 0.0625, 0.03125; the sample mean is 2.0057 and the sample variance 2.0913, against 2 and 2. The running means in the left panel settle near 2 and stay well clear of $\log_2 3$ once the orbit is past its first few dozen strides.

Part of this agreement is a theorem and part of it is not. Extending Proposition 3.2 to strides: if x_0 is uniformly distributed over the odd residues modulo 2^K, the parity vector of length K is uniform over the words that begin with 1, so the halving exponents $k_1, k_2, \ldots$ are independent Geometric$(\frac{1}{2})$ for as long as their sum stays below K. A starting value near 2^{4000} thus controls roughly the first 2000 strides, about a fifth of its orbit. After that the exponents are determined by bits that the orbit has manufactured itself, through the multiplications by 3 and the additions of 1, and no theorem says they remain Geometric. The figure says that they do, and so do the other long orbits computed in this book, such as the giant orbit of the gallery (Plate 5); this is the phenomenon that has never been proved. Note also that the theorem concerns an ensemble of starting values, whereas the figure shows one orbit. Even over the first 2000 strides, Proposition 3.2 says only that most residue classes produce Geometric statistics, not that the particular orbit drawn does. The same is true of the quantitative form due to Tao, which shows that the first halving exponents of a starting value drawn at random from a long range are close in total variation to independent Geometric$(\frac{1}{2})$ draws [96, Proposition 1.9].

What one would like is a statement of the form: for every odd x_0 whose orbit is divergent, and every m, the residues $y_j \bmod 2^m$ are asymptotically equidistributed over the odd classes in the Cesàro sense. Since $y_{j-1} \bmod 2^m$ determines $\min(k_j, m)$, Proposition 2.15 would then give $\liminf_N \bar{k}_N \geq 2 - 2^{1-m}$ for every m, so $\liminf_N \bar{k}_N \geq 2$ along such an orbit, against $\limsup_N \bar{k}_N \leq \log_2 3$ from Corollary 11.4; the statement is therefore equivalent to the divergence half, as Conjecture 11.6 is. It cannot be asked of every orbit: an orbit that reaches 1 stays there, so its residues are eventually constant and are not equidistributed for any $m \geq 2$. For orbits that reach 1 the natural statement concerns random starting values, as problem O2 of Chapter 15 does for the residues modulo 3^a.

11.5 The 3 adic side: the model chain modulo 3^a

The 2 adic residues of the iterates determine the halving exponents; the 3 adic residues are where the multiplication by 3 leaves its mark, and it is on the 3 adic side that Tao found the leverage for his theorem. Here the picture is clean enough to compute exactly.

Since $3y + 1 \equiv 1 \pmod 3$ and 2 is invertible modulo 3, every Syracuse iterate

$S(y)$ is prime to 3. Fix $a \geq 1$ and $M = 3^a$, and let U_a be the set of $2 \cdot 3^{a-1}$ units modulo M. For $r \in U_a$ define the matrix

$$P_a(s,r) = \sum_{k \geq 1} 2^{-k} \mathbb{I}\big[(3r+1)2^{-k} \equiv s \pmod{M}\big], \qquad r, s \in U_a, \tag{11.4}$$

in which $\mathbb{I}$ is the indicator function, 2^{-k} inside it denotes the inverse of 2^k modulo M, and the weight 2^{-k} in front of it is the real number. Column r of P_a is the law of $S(y)$ mod M when $y \equiv r \pmod{M}$ and the halving exponent is an independent Geometric($\frac{1}{2}$) variable.

Lemma 11.7 *(The one step matrix is exact). For each $r \in U_a$ and $s \in U_a$,*

$$\lim_{X \to \infty} \frac{\#\{y \leq X : \ y \text{ odd}, \ y \equiv r, \ S(y) \equiv s \pmod{M}\}}{\#\{y \leq X : \ y \text{ odd}, \ y \equiv r \pmod{M}\}} = P_a(s,r).$$

More generally, for each fixed J, as $X \to \infty$ the joint law of $(y \bmod M, k_1, \ldots, k_J)$ for y uniform over the odd integers up to X converges to the product of the uniform law and J independent Geometric($\frac{1}{2}$) laws.

Proof. Fix exponents $k_1, \ldots, k_J$ with sum s_J. By Theorem 2.8 (in its Syracuse form, of which Proposition 2.15 is the one stride case), the odd y with these exponents form a single residue class modulo 2^{s_J+1}. By the Chinese remainder theorem its intersection with a class modulo 3^a is a single class modulo $2^{s_J+1}3^a$, whose density among the odd integers is $2^{-s_J} \cdot 3^{-a}$. This is the product law. Given $k_1 = k$ and $y \equiv r$, the residue of $S(y) = (3y+1)2^{-k}$ modulo M is $(3r+1)2^{-k}$, which gives the first statement. □

So the one step transition frequencies are not a model: they are exact limits. A model enters only when the matrix is iterated, since iterating assumes that the exponent at each stride is independent of the current residue and of the past along the orbit. We call the Markov chain on U_a with transition matrix P_a, in which the halving exponents are independent Geometric($\frac{1}{2}$) draws at every stride, the *model chain*. By the second half of Lemma 11.7 its assumption is again exact for an ensemble of starting values over any fixed number of strides, and the residues y_j mod M for a uniformly random large y_0 then form, in the limit, a Markov chain with transition matrix P_a. For a single orbit nothing of the kind is known: along an orbit the exponents are determined by the iterates, and the Syracuse map does not induce a Markov chain on residues modulo M.

Lemma 11.8 *(Exact mixing in a strides). For $0 \leq j \leq a$, column r of P_a^j depends on r only through $r \bmod 3^{a-j}$. In particular $P_a^a = \pi_a e^\top$, where e is the vector of ones: all columns of P_a^a equal the stationary law π_a of P_a, so the model chain is exactly stationary after a strides from every starting residue. Consequently the eigenvalue 1 of P_a is simple and every other eigenvalue is zero: the characteristic polynomial of P_a is $\lambda^{|U_a|-1}(\lambda - 1)$.*

Proof. Let $K_1, K_2, \ldots$ be independent Geometric($\frac{1}{2}$) and define the random map $\phi_i(r) = (3r+1)2^{-K_i} \bmod M$. By induction,

$$\phi_j \circ \cdots \circ \phi_1(r) \equiv 3^j 2^{-(K_1+\cdots+K_j)} r + c_j \pmod{M},$$

where c_j depends on $K_1, \ldots, K_j$ only. The first term depends on r only through $r \bmod 3^{a-j}$, and vanishes modulo M for $j = a$. Column r of P_a^j is the law of this composition, which proves the first two claims. The matrix $P_a^a = \pi_a e^\top$ has rank one and trace $e^\top \pi_a = 1$, so its eigenvalues, counted with multiplicity, are 1 once and 0 otherwise. They are the ath powers of the eigenvalues of P_a, and 1 is an eigenvalue of P_a because $e^\top P_a = e^\top$; hence 1 is simple and every other eigenvalue of P_a is zero. □

The identity in the proof is Tao's [96, eq. (1.21)]: the residue of the mth Syracuse iterate modulo 3^a depends only on the halving exponents once $m \geq a$. His recursion [96, Lemma 1.12] is one step of the model chain, and he notes that the law of his Syracuse random variable is the unique stationary measure of the 3 adic chain $x \mapsto (3x+1)/2^k$ with $\mathbb{P}(k) = 2^{-k}$ [96, Remark 1.13]. What Lemma 11.8 adds is a restatement in the form of a finite Markov chain, $P_a^a = \pi_a e^\top$, from which the spectrum follows. The order matters: the converse inference fails, since a zero spectrum off the eigenvalue 1 only makes $P_a - \pi_a e^\top$ nilpotent, which by itself would allow the chain as many as $|U_a| - 1$ strides to become stationary. Nor does the proof use the law of the halving exponents beyond their independence of the current residue; that independence is the model's assumption, and it fails along an actual orbit, where the exponents are determined by the iterates.

The stationary law π_a is the law of the *Syracuse random variable*

$$\sum_{m=1}^{a} 3^{m-1} 2^{-(K_1+\cdots+K_m)} \bmod 3^a$$

of Tao [96, eq. (1.22)], written in the index reversed form of [96, Remark 1.13]; the reversal of the order of the K_i relative to the proof does not change the law. For $a = 1$ it is $(\frac{1}{3}, \frac{2}{3})$ on the residues $1, 2$, since $S(y) \equiv 2^{-k} \equiv 2^k \pmod 3$ and k is odd with probability $\frac{2}{3}$. For $a = 2$, on the residues $1, 2, 4, 5, 7, 8$ modulo 9, it is

$$\pi_2 = \tfrac{1}{63}\,(8,\ 16,\ 11,\ 4,\ 2,\ 22).$$

These laws are far from uniform, and the distance from uniform does not shrink with a: the total variation distance from π_a to the uniform law on U_a is 0.278, 0.346, 0.371, 0.389, 0.399, 0.407 for $a = 2, \ldots, 7$. For $a = 3$ the most likely unit, 26, is 29 times as likely as the least likely, 25; only 5 of the 18 units have probability within a factor $\frac{3}{2}$ of $\frac{1}{18}$, and the median probability is 0.038, against the mean $\frac{1}{18} \approx 0.056$. Tao states that his Syracuse random variables are not uniformly distributed ([96, Remark 1.15] and the paragraph before his Proposition 1.14); what he proves is approximate uniformity at fine 3 adic scales (Section 11.6).

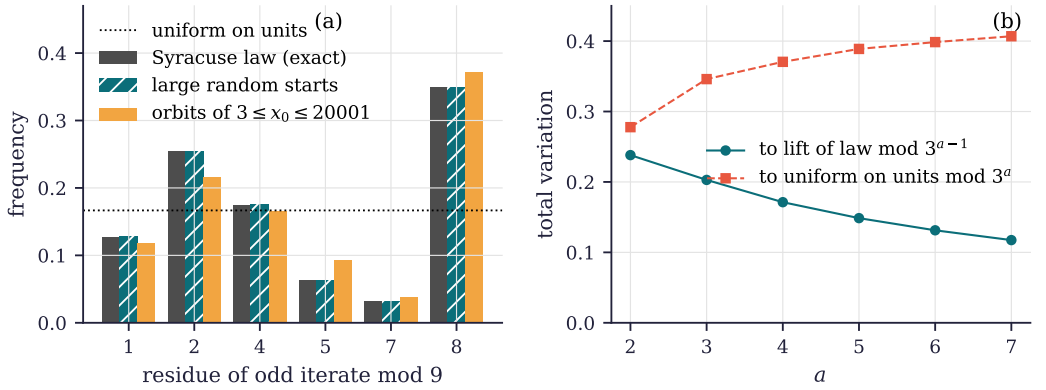


FIGURE 11.2. Left: frequencies of the residues modulo 9 of odd Syracuse iterates, against the exact Syracuse law π_2 and the uniform law on units (dotted). In each group of three bars, the left bar is π_2, the middle bar 3000 random starting values near 2^{511}, strides 3 to 102, and the right bar all odd iterates of the orbits of $3 \le x_0 \le 20001$, pooled over orbits that share their tails. Right: total variation distance from π_a to the lift of π_{a-1} (circles) and to the uniform law on units modulo 3^a (squares).

Figure 11.2 compares π_2 with residue frequencies along actual orbits. For 3000 random starting values near 2^{511}, using strides 3 to 102, the empirical law is within total variation 0.0009 of π_2; this is a regime in which Lemma 11.7 applies, and the agreement is a theorem up to sampling error. For the orbits of the small odd starting values $3 \le x_0 \le 20001$, run to 1, the distance is 0.058; but these orbits share their tails, so their pooled counts are not samples from the model law: their roughly 317 000 odd iterates take only about 12 000 distinct values, and 94 percent of the orbits end with the same final stride, from 5 through 16 to 1.

REMARK 11.9 *(A pitfall: the wrong null hypothesis).* It is natural to test the 3 adic equidistribution of Collatz orbits by comparing residue frequencies modulo 9 with the uniform law on the six units, or worse on all nine residues, with a chi squared test. Such a test rejects decisively for every long orbit in this chapter, and the rejection means nothing. Residues divisible by 3 never occur after the first stride, and the correct null hypothesis on the units is the Syracuse law π_2, which puts weight $22/63 \approx 0.35$ on the residue 8 and $2/63 \approx 0.03$ on the residue 7. Equidistribution in the sense that matters for Tao's theorem is equidistribution with respect to the Syracuse law, not with respect to Haar measure on the units.

11.5.1 A pitfall: odd representatives modulo 27

Since every Syracuse iterate is odd, one might build the chain modulo 27 on the nine odd representatives

$$1,\ 5,\ 7,\ 11,\ 13,\ 17,\ 19,\ 23,\ 25$$

TABLE 11.1. Transition matrices on residues modulo 3^a. Left block: the matrix restricted to odd representatives, second eigenvalue modulus from the empirical matrix (odd $y \le 10^6$) and from the limiting matrix. Right block: the model chain on all units; rank of P_a^j for $j = a-1, a$, and the largest total variation distance between a column of the empirical matrix and the corresponding column of P_a.

	Odd representatives			All units		
3^a	States	Empirical	Limit	States	Ranks	Column TV
9	3	0.0000	0.0000	6	2, 1	$< 10^{-4}$
27	9	0.5198	0.5202	18	2, 1	0.0002
81	27	0.7593	0.7596	54	2, 1	0.0006
243	81	0.7462	0.7461	162	2, 1	0.0014

of the units in $[1, 27)$, recording a transition $r \to s$ whenever an odd $y \equiv r$ has $S(y) \equiv s \pmod{27}$ with both r and s in this set. The resulting matrix appears to have a respectable spectral gap, and it is worth seeing why the gap means nothing.

REMARK 11.10 *(A spurious spectral gap).* Let $\hat{P}$ be the empirical transition matrix of the nine state chain just described, estimated from all odd $y \le 10^6$ not divisible by 3, with each column normalized to sum to one. Its eigenvalue of second largest modulus has modulus 0.5198, and the limiting matrix, obtained by restricting P_3 of (11.4) to the nine states and renormalizing its columns, has a complex conjugate pair of eigenvalues of modulus 0.5202; taken at face value, this is a spectral gap of about 0.48. It is not the mixing rate of the model chain, nor of anything else attached to the dynamics. Since 27 is odd, every residue class modulo 27 contains integers of both parities, so whether the representative of y in $[0, 27)$ is odd says nothing about y, and the set of odd y with odd representatives is not invariant under S: $55 \equiv 1 \pmod{27}$, while $S(55) = 83 \equiv 2 \pmod{27}$. The restricted matrix conditions, at every stride, on a set that the dynamics leaves, and its eigenvalues are artefacts of the conditioning. The model chain is P_3 on all eighteen units, which by Lemma 11.8 is exactly stationary after three strides, so that its eigenvalues other than 1 are all zero.

The empirical and limiting values agree, as they must, since by Lemma 11.7 the empirical matrix converges to the restricted limiting matrix; the agreement shows only that the restricted matrix has been computed correctly. Table 11.1 compares the restricted matrices with the model chains for the moduli 9, 27, 81 and 243.

The table was computed as follows. For each modulus we formed the counts of pairs $(y \bmod 3^a, S(y) \bmod 3^a)$ over odd $y \le 10^6$ not divisible by 3, normalized each column to sum to one, and computed eigenvalues with a standard dense solver; the limiting matrices are given by (11.4) with the sum over k folded modulo the order $2 \cdot 3^{a-1}$ of 2 modulo 3^a; ranks were computed with tolerance 10^{-9}. The

restricted matrices have second eigenvalues that do not decrease with the modulus and bear no relation to the mixing of the model chain, which is exactly stationary after a strides, as Lemma 11.8 predicts, while the empirical one step matrices are close to the limiting ones, as Lemma 11.7 predicts. The column distances grow with the modulus only because the number of samples per column falls, from 55555 at modulus 9 to 2057 at modulus 243; the empirical second eigenvalues of the full matrices on all units, which are pure sampling noise around zero, grow for the same reason.

Remark 11.11 *(Total variation).* For the model chain, $d_{\mathrm{TV}}(P_a^j \delta_r, \pi_a) = 0$ for every $j \geq a$ and every starting residue r, by Lemma 11.8. This is not a statement about an orbit. It says that if the stride exponents along an orbit behaved like independent Geometric variables, independent of the current residue, residues modulo 3^a would forget their initial value after a strides; whether the exponents do so behave is the question. Along a single orbit the residues $y_j \bmod 3^a$ are a deterministic sequence, and no property of a Markov chain says whether a particular deterministic sequence equidistributes.

11.6 Tao's theorem

The strongest result on the divergence side is the theorem of Tao [96, Theorem 1.3], stated in Theorem 1.3: for any f with $f(n) \to \infty$, the orbit of n attains a value below $f(n)$ for all n in a set of logarithmic density one. We describe the structure of the proof, because it shows both how far the ideas of this chapter can be pushed and where they stop.

Reduction to the Syracuse map. Tao works with the Syracuse map on odd integers and proves the corresponding statement for $\mathrm{Syr}_{\min}(N) = \min_j S^{(j)}(N)$; the statement for the standard map follows. The sense of "almost all" is logarithmic density, in which a set A is weighted by $\sum_{n \in A, n \leq x} 1/n$. Tao adopts it for technical reasons [96]: the proof moves an orbit from one scale to a lower one by a multiplicative factor that, after m steps, is known only up to a further factor $\exp(O(m^{1/2}))$ [96, eq. (1.17)], and the measure dn/n is approximately invariant under multiplicative rescaling, whereas counting measure is not. He regards natural density as plausibly attainable by his methods with a stronger fine scale mixing estimate [96, Remarks 1.4 and 1.16].

Syracuse random variables. The first m Syracuse steps from N can be written in affine form, as in Proposition 2.11,

$$S^{(m)}(N) = 3^m 2^{-(k_1 + \cdots + k_m)} N + \sum_{i=1}^{m} 3^{m-i} 2^{-(k_i + \cdots + k_m)}.$$

Reducing modulo 3^m kills the first term, so $S^{(m)}(N) \bmod 3^m$ is a function of the exponents only [96, eq. (1.21)]. When N is random, the exponents are close to

independent Geometric($\frac{1}{2}$) variables, by Lemma 11.7 for a fixed number of strides and quantitatively, in total variation, by [96, Proposition 1.9], and $S^{(m)}(N) \bmod 3^m$ is close in law to the Syracuse random variable [96, eq. (1.22)], whose law is π_m. This is exactly the mixing of Lemma 11.8, seen from the side of the iterate.

Fine scale mixing. The heart of the argument is an estimate saying that the law π_n, though far from uniform at every scale, becomes uniform at fine scales relative to coarse ones. Precisely, Tao proves [96, Proposition 1.14] that for $1 \le m \le n$ the sum $\sum_{Y \bmod 3^n} \left|\pi_n(Y) - 3^{m-n}\pi_m(Y \bmod 3^m)\right|$ is bounded by $O_A(m^{-A})$ for every $A > 0$: given the residue modulo 3^m, the finer digits are nearly uniform. The proof goes through the Fourier coefficients of π_n at frequencies prime to 3, which Tao shows decay faster than any power of n, using a careful analysis of the random walk formed by the partial sums of the exponents. The right panel of Figure 11.2 shows the case $m = n - 1$ for small n: the total variation distance between π_a and the uniform lift of π_{a-1} is 0.238, 0.203, 0.171, 0.149, 0.131, 0.117 for $a = 2, \dots, 7$, decreasing steadily, while the distance from π_a to the uniform law on units increases.

Iteration across scales. With fine scale mixing in hand, Tao shows that if N is drawn logarithmically from a range of the form $[y, y^\alpha]$, for a suitable fixed $\alpha > 1$, then the first time the orbit enters a lower range lands at a value whose law, after reduction modulo a power of 3, is close to a Syracuse law, nearly independently of where in the original range N started. That near independence is what allows the step to be repeated: the output of one descent is distributed well enough to serve as the input to the next. Iterating from scale N down to any scale tending to infinity gives the theorem.

Why it does not reach every integer. Every step of this argument is an averaging over starting values. The mixing estimates say that the residues of $S^{(m)}(N)$ are well distributed for most N, with an exceptional set of small logarithmic density at each scale, and the exceptional sets accumulate. The theorem says nothing about any particular N, and cannot, because a single orbit is a single point of the probability space. There are also two specific obstacles. The conclusion is that orbits become small, below $f(n)$, not that they reach 1; to close that gap one would want f bounded, and with f bounded the exceptional set is not controlled; Tao expects that sharpening to be almost as hard as the conjecture itself [96, Remark 1.4]. And the density is logarithmic, not natural; as noted above, this restriction may well be technical [96, Remark 1.16]. But even with natural density the conclusion would concern almost every integer rather than every one, and neither this limitation nor the first is one that a sharper estimate would remove. They reflect the same gap between ensemble and orbit that the Cesàro identity isolates: a density statement, however strong, is a statement about a measure, and the conjecture is a statement about every point.

11.7 The Krasikov and Lagarias counting bound

A different line of attack counts integers from below. Let $\pi_1(x)$ be the number of $n \le x$ whose orbit reaches 1 (not to be confused with the stationary law π_1 of Section 11.5). Krasikov and Lagarias [51] proved that $\pi_1(x) \ge x^{0.84}$ for all sufficiently large x. Their method works on the tree of preimages of 1: every m has the preimage $2m$, and also $(m-1)/3$ when $m \equiv 4 \pmod 6$, and the growth of this tree can be bounded below by a system of difference inequalities indexed by residue classes modulo a power of 3, which is then solved numerically. The method gives an exponent strictly less than one, so the bound does not give the integers that reach 1 a positive density. Chapter 9 develops a simplified version of the method in full, with certified exponents up to 0.477.

The counting bound is complementary to the density theorems. It concerns integers that genuinely reach 1, not merely integers whose orbits become small, but it reaches only a thin set of them. In the stochastic model the tree of preimages is a branching random walk [59] whose growth is consistent with the conjecture, which asserts $\pi_1(x) = \lfloor x \rfloor$; the gap between the exponent 0.84 and the exponent 1 is the gap between what can be proved about the tree and what is believed. Neither kind of theorem excludes a divergent orbit, which would be a single infinite path in the forward direction that never enters the tree.

11.8 Single orbits in practice

Open problem O2 asks whether, for a random starting value, the residues y_j mod 3^a along its orbit, followed until it reaches 1 or, for $x_0 \le X$, for at most a fixed multiple of $\log X$ strides (Chapter 15 gives the rule), equidistribute according to the stationary law π_a of the model chain. It cannot be posed for every orbit: the orbit of $(4^m - 1)/3$ reaches 1 after one stride. We cannot prove anything of the kind, but we can look, using the correct null hypothesis of Remark 11.9. We followed 24 orbits from random odd starting values near 2^{1000}, 2^{3000} and 2^{10000}, which take between about 2000 and 25000 Syracuse strides to reach 1, and recorded the running empirical distribution of y_j mod 3^5 over the 162 classes prime to 3 (script `ch9_orbits.py`, with π_5 computed from the exact transition matrix). As a yardstick we drew independent samples of the same sizes from π_5.

The result is in Figure 11.3. The total variation distance between the running distribution and π_5 decays along every orbit at exactly the rate of independent sampling, like $j^{-1/2}$: about 0.09 after 2400 strides, 0.05 after 7000 and 0.03 after 24000, against 0.091, 0.053 and 0.029 for independent draws. The orbits scatter about this benchmark somewhat more than independent draws would, and about as much as the model chain does: its successive residues are dependent, its running distance has the same mean as for independent draws but a standard deviation a quarter to a third larger, and the final distances of the 24 orbits are spread evenly over its law, from its 2nd to its 99th percentile (script

`ch9_orbits_check.py`). The residues modulo 9 show no drift or persistent bias over the length of an orbit from 2^{10000}. Along single orbits, then, the running distribution of the residues approaches the stationary law at the rate of independent sampling, which is what O2 asks for; the mean curve cannot tell independent residues from the dependent residues of the model chain. This is evidence, not proof, and it concerns orbits that do reach 1; the hypothetical divergent orbit, which by Corollary 11.4 would need $\limsup_N \bar{k}_N \le \log_2 3$, would have to depart persistently from the model in its halving exponents, which drive these residues.

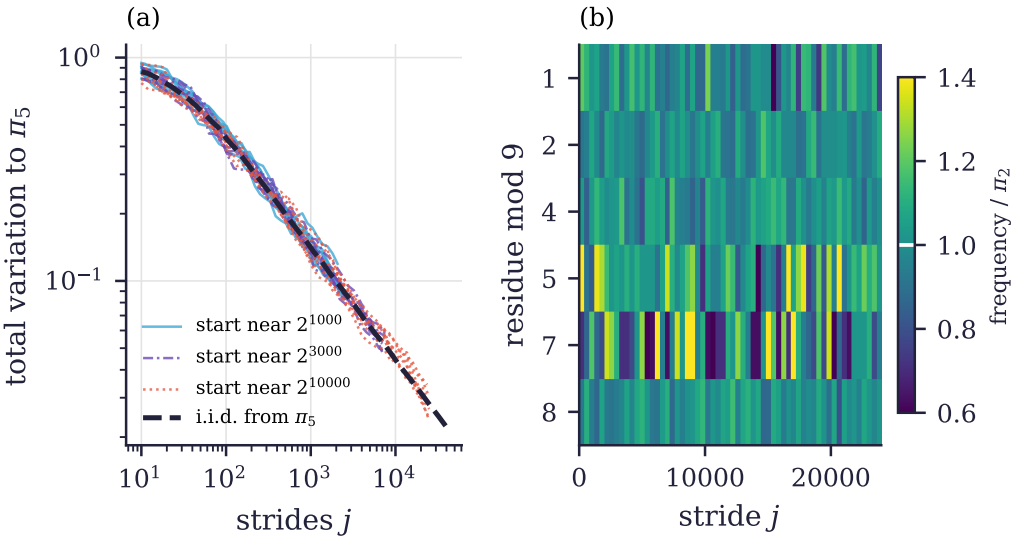


FIGURE 11.3. Equidistribution along single orbits. (a) Total variation distance between the empirical distribution of y_j mod 3^5 along an orbit and the stationary law π_5, for 24 orbits started near 2^{1000}, 2^{3000} and 2^{10000}, against the mean distance for independent samples from π_5 (dashed). (b) Residue frequencies modulo 9 along one orbit from near 2^{10000}, in windows of 400 strides, relative to π_2.

11.9 THREE ROUTES AND WHERE THEY FAIL

We end by comparing three routes to the divergence half, each natural, each developed somewhere in this book, and each failing.

Equidistribution. If the iterates y_j were equidistributed modulo 2^m, for every m, along every divergent orbit, then $\liminf_N \bar{k}_N \ge 2$ along it by Proposition 2.15, which Corollary 11.4 forbids; there would be no divergent orbit, and Conjecture 11.6 would follow with room to spare. No such statement can be asked of every orbit, since an orbit that reaches 1 is eventually constant. What is known concerns random starting values. Their first halving exponents are

close in total variation to independent Geometric($\frac{1}{2}$) draws [96, Proposition 1.9], and the deep input of Tao's theorem is a fine scale mixing estimate modulo powers of 3 [96, Proposition 1.14]. The theorem establishes a form of equidistribution, but on the 3 adic side, in logarithmic density over starting values, and with an exceptional set. Lemma 11.8 gives the finite analogue, exact mixing of the model chain. Every one of these statements is about an ensemble or about a Markov chain built from one; the passage from an ensemble to a single orbit is the step that is missing.

A Lyapunov certificate. One would like a function W and a constant $\rho < 1$ with $W(S(x)) \le \rho W(x)$ for every odd $x \ge 3$, which would give geometric descent with no probabilistic content. Theorem 4.8 rules this out for every W increasing in x, since $S(x) > x$ on the whole class $x \equiv 3 \pmod 4$, and Corollary 4.15 rules out such W at every fixed number of strides, since ascents can be chained. A certificate would have to depend on the arithmetic of x. Certificates of that kind exist if the conjecture holds, for then $W(x) = 2^{N(x)}$, where $N(x)$ is the number of strides from x to 1, satisfies $W(S(x)) = \frac{1}{2}W(x)$; but such a certificate, like the total stopping time it is built from, presupposes the conjecture. The functions that do decrease, $\log x$ and x^p for $0 < p < 1$, decrease in expectation over a residue class, which is again an ensemble.

Uniform integrability. Optional stopping for the martingale formed by the iterates of the model does not get started: stopped when it first drops to 1 or below, the martingale is not uniformly integrable (Proposition 3.17), and for the deterministic map the corresponding invariant function, the indicator of reaching 1, exists trivially, while showing that it equals 1 everywhere is the conjecture itself (Remark 3.18). Above all, every martingale here lives on a probability space of random parities. For a deterministic orbit there is no filtration with respect to which $\log y_N - N \log \frac{3}{4}$ is a martingale.

Two of the three failures have the same form: equidistribution and uniform integrability each replace a statement about a specific trajectory by a statement about a measure, and in each case the replacement is the step that cannot be justified. A pointwise certificate involves no measure and fails differently: no increasing W works at any fixed number of strides, and the certificates that would work presuppose the answer. The heuristic is not weak. Its predictions match the data closely in the tests of this book and of the literature, the total stopping time slope to within about two percent and the variance of the stopping times to about ten percent (Chapter 3). What is missing is a mechanism that transfers statistics from an ensemble to an individual orbit, in the presence of a deterministic map that, as far as anyone knows, has no special structure that would supply one. The Cesàro identity makes the target precise: an orbit that reaches 1 automatically has $\bar{k}_N > \log_2 3$, and an orbit that diverges automatically has $\limsup \bar{k}_N \le \log_2 3$. The divergence half of the Collatz conjecture is the assertion that the second never happens, and the margin of 0.415 between the Geometric mean 2 of the model and

the threshold $\log_2 3$ is, so far, a margin that no argument has been able to transfer to a single orbit.

Notes

The observation that a divergent orbit must contain a fraction of odd steps of at least $1/\log_2 3$ is elementary and has been made many times; it is the divergence counterpart of the cycle inequality $3^m < 2^k$ of Chapter 10. The Cesàro identity is the stride formula (2.3) summed along an orbit, and the pitfall of Remark 11.5 is an immediate consequence of it. The statement that almost every orbit falls below its starting value is due to Terras [99, 100] and Everett [22]; the exponent $\log 3/\log 4$ is due to Korec [49]. The survey of Lagarias [53] and the volume [55] contain much more on the 2 adic picture of Remark 11.2.

Tao's theorem appeared in [96]; the paper is long but very clearly written, and its introduction contains an excellent account of the difficulties described in Section 11.6. The Syracuse random variables and the fine scale mixing estimate are defined and proved there, and so are the identities behind the model chain of Section 11.5: the loss of memory [96, eq. (1.21)], the one step recursion [96, Lemma 1.12] and the stationary law [96, Remark 1.13]. Lemma 11.8 restates them in the form of a finite Markov chain. The counting bound $\pi_1(x) \geq x^{0.84}$ is from Krasikov and Lagarias [51]. The branching random walk model for the tree of preimages is in Lagarias and Weiss [59].

All numerical results in this chapter were computed by the authors with the scripts `ch9_spectral.py`, `ch9_table.py`, `ch9_figures.py`, `ch9_numbers.py`, `ch9_orbits.py` and `ch9_orbits_check.py` of the code appendix, and the scan of Remark 11.5 with `ch9_cesaro_scan.py`.

Exercises

Exercise 11.12. State and prove the analogue of Proposition 11.3 for the standard map T_0, in terms of the number of odd steps among the first n steps. Deduce the second frequency statement of Corollary 11.4 directly.

Exercise 11.13. For the $5x+1$ map with Syracuse form $y \mapsto (5y+1)/2^{v_2(5y+1)}$, derive the Cesàro identity and show that an orbit that reaches 1 must have $\bar{k}_N > \log_2 5$. Under the Geometric law the mean exponent is still 2. What does this suggest about typical orbits of the $5x+1$ map?

Exercise 11.14. Compute the matrix P_2 of (11.4) on the six units modulo 9, check that P_2^2 has identical columns, and verify that they equal $\frac{1}{63}(8, 16, 11, 4, 2, 22)$.

Exercise 11.15. Let R be the operation that deletes from a column stochastic matrix on the eighteen units modulo 27 the rows and columns of the nine units with even representatives in $[0, 27)$, and renormalizes the remaining columns.

Show that the limiting nine state matrix of Remark 11.10 is $R(P_3)$, that $R(P_3^3)$ has rank one, and that $R(P_3)^3$ does not. Conclude that the eigenvalues of the restricted matrix need not reflect the exact mixing of Lemma 11.8.

Exercise 11.16. Let $q \in (\frac{1}{2}, 1)$. Show that the number of binary words of length n with at least qn ones is $2^{nH(q)+o(n)}$. Using Theorem 2.8, deduce that the proportion of residue classes modulo 2^n whose first n accelerated steps contain at least $n/\log_2 3$ odd steps tends to zero exponentially, and compute the rate.

Exercise 11.17. Let $K_1, K_2, \ldots$ be independent Geometric$(\frac{1}{2})$. Compute the law of $\sum_{m=1}^{3} 3^{m-1} 2^{-(K_1+\cdots+K_m)}$ mod 27 numerically, and compare it with the stationary law of P_3. Then compute the fine scale discrepancy $\sum_Y |\pi_3(Y) - \frac{1}{3}\pi_2(Y \bmod 9)|$.

Chapter 12

The Path to Collatz

Over the preceding chapters we have accumulated a good deal of machinery: parity vectors and the 2 adic conjugacy, martingales and their failure of uniform integrability, Lyapunov functions and the impossibility theorems for monotone ones, an optimizer, a cycle bound and a Cesàro conjecture. This chapter steps back and asks what all of it says about a proof. We set out what a proof of the conjecture would have to contain, where each piece of the book sits relative to that, and which statements would close each half.

12.1 The conjecture splits in two

Fix an odd $n \in \mathbb{N}$ and follow its forward orbit under the Syracuse map. Exactly one of three things happens. The orbit reaches 1; or it is unbounded; or it is eventually periodic on a cycle not containing 1. The conjecture asserts that the second and third possibilities never occur, and these are genuinely separate problems.

The *divergence half* asks whether $\log y_N$ can drift upward forever. By the Cesàro identity (Proposition 11.3),

$$\log y_N - \log y_0 = \sum_{j=1}^{N} \log\Big(3 + \frac{1}{y_{j-1}}\Big) - N\bar{k}_N \log 2,$$

so along an unbounded orbit the running mean $\bar{k}_N$ of the halving exponents satisfies $\limsup_N \bar{k}_N \le \log_2 3 \approx 1.585$ (Corollary 11.4). Under the Geometric$(\frac{1}{2})$ law of Proposition 2.15 the mean exponent is 2, which leaves a heuristic margin of 0.415; along an unbounded orbit there is no margin at all, so it describes the model only. The difficulty is that the law describes the exponent when the argument is drawn uniformly from residue classes, whereas what is needed is a statement about the exponent when the argument is the previous iterate of the same map. This is a law of large numbers for a deterministic sequence.

The *cycle half* asks whether the orbit can close up. A cycle with r odd elements $x_1, \dots, x_r$ and s halvings imposes the exact multiplicative constraint $\prod_i (3 + 1/x_i) = 2^s$ of Lemma 10.1: a product of r numbers each slightly above 3 is exactly a power of 2. Taking logarithms, $s \log 2 - r \log 3$ must be positive and smaller than $r/(3M)$, where M is the least element. No probability is involved;

the tools are Diophantine, although Diophantine information alone bounds the length of a cycle without excluding one (Section 12.3).

The two halves differ in kind, not only in technique. The first is a law of large numbers we cannot prove. The second is a sequence of finite verifications that we can carry further and further but never complete by computation alone.

A remark on notation joins the two halves to Part II. The threshold $\log_2 3$ that governs divergence equals $1+\beta$, where $\beta = \log(3/2)/\log 2$ is the momentum coefficient of Switching MM (Chapter 7). In the language of the accelerated map, in which one halving is absorbed into every odd step, the divergence half asks whether the extra halvings per odd step average more than β, and the cycle half asks whether s/r can come within $1/(3M\log 2)$ of $1+\beta$ from above. By construction the same number sets the inertia of the optimizer; it is also the drift threshold of the orbit and the Diophantine target of the cycle problem.

12.2 Closing the divergence half

Conjecture 11.6 is the precise statement required. It is equivalent to the divergence half, and both directions are immediate (Section 11.3): every bounded orbit satisfies it, while by the Cesàro identity every unbounded orbit has $\limsup_N \bar{k}_N \le \log_2 3$ (Corollary 11.4), so the conjecture holds exactly when no orbit is unbounded. Its value lies in the mechanism it names. Three routes suggest themselves, and we show where each fails.

Equidistribution. Suppose that along an unbounded orbit the iterates were equidistributed over the odd residues modulo 2^m, for every fixed m. Then $\liminf_N \bar{k}_N \ge 2$ along it, which Corollary 11.4 forbids; so if this held along every unbounded orbit there would be no unbounded orbit, and the conjecture would follow. The hypothesis cannot be asked of every orbit, since an orbit that reaches 1 is eventually constant. What is known concerns random starting values rather than a fixed orbit (Chapter 11), which is also how problem O2 of Chapter 15 is posed, and the passage from one to the other is the whole difficulty. For a starting value drawn uniformly modulo 2^m the parity vector of its first m steps is uniformly distributed over the 2^m words (Theorem 2.8; Terras and Everett [22, 99], see [53, Theorem B]); for a random odd starting value its first halving exponents are close in total variation to independent Geometric$(\frac{1}{2})$ draws [96, Proposition 1.9], and the deep input in Tao's theorem is a fine scale mixing estimate modulo powers of 3 [96, Proposition 1.14]. The finite analogue in Chapter 11 makes the point sharply. The model chain on residues modulo 3^a, in which the halving exponents are independent draws, mixes *exactly* in a strides (Lemma 11.8), to a law that is far from uniform; but this, like the rest, is a statement about an ensemble of starting values, averaged by the residue classes, and none of it constrains the residues visited by one trajectory.

A Lyapunov certificate. One would like a function W bounded away from zero

and a constant $\rho < 1$ with $W(S(x)) \leq \rho\, W(x)$ for every odd $x \geq 3$, which would give geometric descent with no probabilistic content. Such a certificate exists if and only if the conjecture holds (Remarks 4.3 and 4.10), so the question is whether one can be written down without knowing the answer, and the natural candidates increase with x. Theorem 4.8 shows that no increasing W works: whenever $x \equiv 3 \pmod 4$ the stride ascends, and half of all odd integers have that form. Composing strides does not help. Corollary 4.15 shows that for every fixed horizon J there are infinitely many x (those congruent to -1 modulo 2^{J+1}) whose first J strides all ascend, so no increasing certificate over a fixed number of strides exists either, and a nonincreasing one fails at $x = 5$, since $S^{(J)}(5) = 1$ for every $J \geq 1$. A certificate must therefore be nonmonotone, or let the horizon depend on x, as the stopping time does.

Uniform integrability. If the martingale $\{x_n\}$ of the accelerated model, stopped when it first drops to 1 or below, were uniformly integrable, optional stopping would give $\mathbb{E}[x_\tau] = x_0$, which is false since $x_\tau \leq 1 < x_0$. Proposition 3.17 turns this into a proof that uniform integrability fails. In the model the conditional probability of termination is a uniformly integrable martingale, but it is the constant 1 (Proposition 3.16), so there is nothing to certify. For the deterministic map its counterpart is the indicator of reaching 1, which exists trivially; showing that it equals 1 everywhere is the conjecture itself (Remark 3.18).

The first and third routes replace a statement about a specific trajectory with a statement about an ensemble, and each time the replacement is the step that cannot be justified; the second avoids the ensemble, but descent at a fixed horizon fails pointwise for every monotone W. We regard this as the real content of the divergence half. It is not that the heuristic is weak: the model predicts the total stopping time slope to within about two percent and, as Chapter 3 shows, its dispersion to within about ten percent. It is that no mechanism is known for transferring ensemble statistics to individual orbits in this setting.

12.3 Closing the cycle half

Here the mechanism is better understood, but it can only bound the length of a cycle, never exclude one. Lemma 10.1 converts the existence of a cycle with r odd elements and least element M into the inequality $M \leq r/(3\Lambda_r)$, where $\Lambda_r = (\lceil r \log_2 3 \rceil - r \log_2 3) \log 2$ is $\log 2$ times the one sided distance from $r \log_2 3$ up to the next integer. Two facts then combine. Exhaustive computation gives a lower bound on M, since every integer below the verified range reaches 1 (2^{68} in Theorem 10.16, now 2^{71} by Bařina [7]). Continued fraction theory gives an upper bound on $r/(3\Lambda_r)$, since Λ_r can be small only when r is a good one sided approximation denominator of $\log_2 3$.

The mediant lemma of Chapter 10 is what makes the second fact exact. The one sided distance $\delta(r) = \lceil r \log_2 3 \rceil - r \log_2 3$ is additive modulo 1, so if a smaller r'

has $\delta(r') < \delta(r)$ then $\delta(r)$ splits as $\delta(r') + \delta(r - r')$ and $\delta(r)/r$ is a mediant of two smaller ratios. It follows that the first r not excluded by the verified range, and every record of $r/\delta(r)$, is a record of δ, and Theorem 10.8 identifies these records with the rungs of the semiconvergent ladder, which is the classical description of the best one sided approximations (Hančl and Turek [31, Theorem 4.5(ii)]). There are only 53 rungs below 2×10^{13}, the bound is evaluated at each, and Theorem 10.16 follows without any unverified proposition: every nontrivial cycle has $r \geq 72\,057\,431\,991$ odd elements.

The gap is that this excludes finitely many r at a time. A large enough improvement in the verification range moves the exclusion forward to a later rung, and the excluded range grows roughly like the square root of the verified range: 2^{40} gives $r \geq 1.08 \times 10^7$ and 2^{68} gives $r \geq 7.2 \times 10^{10}$. The table in Chapter 10 shows the steps: verification to 2^{71} [7] gains nothing over 2^{68} for this inequality, because the next obstruction sits at $2^{71.88}$; verification to 2^{72} would move the bound to 137 528 045 312, the current record, which Hercher's sharper accounting of the cycle already obtains from $3 \cdot 2^{69} < 2^{71}$ [38, Corollary 29]; and then nothing further happens until beyond 2^{75}. Can a bound on Λ_r valid for all r at once, an effective irrationality measure for $\log_2 3$, close the gap? It cannot. Remark 10.19 shows that a measure ν with constant c yields only $M \leq r^\nu/(3c \log 2)$, which grows at least like r^2, and Proposition 10.20 shows more: for every verification range there are infinitely many r that the Diophantine inequality cannot exclude, whatever is known about $\log_2 3$. Even an idealized measure with $\nu = 2$ and the best constant consistent with the computed partial quotients is beaten by the exact ladder computation by a factor of twenty at 2^{68}.

So the cycle half is *not* a question about one real number. What is missing is arithmetic information about the parity word. The window lemma of Chapter 10 makes this precise. The affine identity of Chapter 2 writes every element of a cycle as $c_k(x)/(2^k - 3^m)$, with $c_k(x)$ a sum of terms $3^j 2^i$ indexed by the parity word, and the trivial bound $c_k \geq 3^m - 2^m$ gives a lower bound for the least element which, together with the upper bound $M < r/(3\Lambda)$, confines M to a window whose ends are a factor of about $r/3$ apart. An arithmetic improvement of the upper end by a factor slightly more than $r/3$, together with a verification up to a fixed bound, would exclude every nontrivial cycle (Corollary 10.22); raising the lower end alone empties the window only for cycles with small Λ. The exploration of parity words in Chapter 10 suggests why this is hard: for balanced words the least element sits, in our computations, at a nearly fixed fraction, about 0.72, of the top of the window, and the residues of c_k modulo $2^k - 3^m$ look uniformly random. A proof must exploit divisibility. The Erdős 125 analogy suggests where to look: there the good approximations $3^m \approx 4^k$ thin the sumset and drive its lower density to zero; for cycles, approximations of the same constant, 2^k slightly above 3^m, are what a cycle needs, and the question is whether the digit structure of the word prevents $2^k - 3^m$ from ever dividing c_k.

12.4 Where the optimization framework fits

The framework does not prove either half. What it supplies is the vocabulary in which the halves separate cleanly, and it collects in one place the constants a proposed mechanism must reproduce: the drift $\delta^* = \log(\sqrt{3}/2)$, the step variance $\sigma^2 \approx 0.302$, the total stopping time slope $1/|\delta^*| \approx 6.95$, the stride constants $\log(5/3)$ and $|\log(3/4)|$ of Definition 5.3, the parity autocorrelations $(-\frac{1}{2})^h$ and the long run variance factor $\frac{1}{3}$ of the model for the standard map T_0 (Proposition 3.12 and Corollary 3.14). These matter for a practical reason. When one proposes a mechanism for the conjecture, the first test is whether it reproduces the known constants. A mechanism that predicts a drift other than $\log(\sqrt{3}/2)$ per step, or a total stopping time slope other than about 6.95, for the accelerated map, can be discarded without further work.

It also records negative results. Theorem 4.8 and Corollary 4.15 rule out monotone certificates over a bounded horizon, Proposition 4.19 rules out the natural majorization scheme, and Proposition 3.17 rules out the naive martingale argument. The pitfall remarks scattered through the book rule out a further collection of arguments that look like progress. The Collatz iteration is not a descent method, and no monotone reparametrization makes it one.

And the framework gave something back. The arithmetic of the map, read as an optimizer, produced Switching MM, a method whose structure (cyclic steps of sizes α and $(1+\beta)\alpha$ on quadratics, Theorem 7.6) and whose noise behaviour (amplification bounded by $5/3$ for every $\beta < 1$, Theorem 8.4 and Propositions 8.6 and 8.7) can be analysed exactly. The constant β does not prove anything about Collatz orbits; it is simply a well motivated, tuning free default, and Chapter 8 is careful about how much that is worth.

12.5 Verification as a partner

A last theme runs through Part III. The cycle bound is assembled from three steps: the affine identity, the cycle inequality $3^m < 2^k$ with its upper companion, and a finite computation with continued fractions. The first step and the lower half $3^m < 2^k$ of the second are now machine checked in Lean (Chapter 10); the upper half is an inequality between natural numbers within reach of the same tools; and the mediant lemma and Theorem 10.8 reduce the third to a short certificate: a list of ladder rungs u and numerators $\lceil u \log_2 3 \rceil$ together with rational bounds on $\log_2 3$ that verify each inequality by interval arithmetic. Formalizing the upper half, the mediant lemma, Theorem 10.8 and a verified squaring routine, and replaying that certificate in Lean, would make the cycle bound machine checked, conditional only on the verification below 2^{68} (project P1 of Chapter 15). The divergence half offers no such target, and the contrast is itself informative: every finite part of the cycle half is within reach of verification, while the divergence half, which requires a law of large numbers for one orbit, is not.

The next chapter takes up the computational side of this accounting: which parts of the programme computation can finish, which it provably cannot, and what the optimization reading says about where to look.

Notes

The split into a divergence half and a cycle half is classical; see the surveys of Lagarias [53, 56]. The observation that the probabilistic attacks on the divergence half replace an orbit by an ensemble is implicit in the parity heuristic of Crandall [18] and the stochastic models of Lagarias and Weiss [59] and explicit in the discussion surrounding Tao's theorem [96]. The reduction of the cycle half to Diophantine approximation of $\log_2 3$ goes back to the work of Crandall [18], Steiner [94] and Eliahou [21]. That it cannot suffice is a point Halbeisen and Hungerbühler [30, §5] made in another form, since rational cycles satisfy the same inequalities; the proof of Proposition 10.20, and the window corollary, are ours.

Exercises

Exercise 12.1. Show that the three possibilities for a Syracuse orbit (reaching 1, being unbounded, being eventually periodic on a cycle not containing 1) are mutually exclusive and exhaustive.

Exercise 12.2. Show that along a cycle with r odd elements and s halvings the averages $\bar{k}_N$ converge to s/r, and that $0 < s/r - \log_2 3 < 1/(3M \log 2)$. Conclude that the Cesàro condition is satisfied by every cycle, so that no argument based on Cesàro averages alone can exclude cycles.

Exercise 12.3. Suppose an effective irrationality measure $\nu = 3$ with constant $c = 10^{-3}$ were known for $\log_2 3$. Combine Remark 10.19 with the verified range $X = 2^{68}$ to show that it excludes exactly the cycle lengths $r < (3cX \log 2)^{1/3}$, about 8.5×10^5, far fewer than Theorem 10.16. Explain why no choice of (ν, c) can exclude all r.

Exercise 12.4. Show that if $\liminf_N y_N < \infty$ along a Syracuse orbit then the orbit is eventually periodic. Deduce that an orbit is unbounded if and only if $y_N \to \infty$, so that the divergence half concerns orbits that escape to infinity, not merely orbits that are unbounded along a subsequence.

Exercise 12.5. List the constants of Section 12.4 and, for each, state one proposed mechanism for the conjecture that it would falsify if the mechanism predicted a different value.

Chapter 13

A Computational Path

Chapter 12 asked what a proof of the conjecture would have to contain. This chapter asks a narrower and more practical question: what can computation contribute, and does the optimization reading of Part II tell us where to point it? The answer has three parts. For the cycle half there is a concrete computational programme, each step of which proves something definite, and we lay it out with costs. For the divergence half computation can rule out whole classes of certificates, and we prove a structural obstruction that explains why no finite amount of residue arithmetic can finish the job. And the MM view does give guidance, though mostly by telling us what a certificate must look like and what it cannot be. The next chapter takes up the suggestion that simulation, in the form of nested, split or vertical likelihood sampling of an energy on the integers, might settle the matter in polynomial time.

13.1 What a computation can prove

Three kinds of computation appear in this book, and they prove different kinds of statement.

Verification. Checking that every $n \le X$ reaches 1 is a finite statement. By itself it says nothing about larger integers, but combined with the cycle inequality it excludes every cycle shorter than the sharp bound $r_*(X)$ of Chapter 10, and it gives a lower bound X for the least element of any divergent orbit.

Certificates. A finite object that a short program, or a proof assistant, can check and that implies an infinite statement. The integer certificate of Section 10.8 is one: 44 rungs and inequalities between integers of a few hundred bits imply, together with the verification of the conjecture below 2^{68}, that every nontrivial cycle has at least 72 057 431 991 odd elements. The certified exponents of Theorem 9.3 are another, implying a counting bound at every scale.

Evidence. Statistics of orbits, such as the residue statistics of orbits from random starting values in Section 11.8 or the census of Section 3.6, which prove nothing about the conjecture but constrain any proposed mechanism and point to the statements worth proving.

A computational path to the conjecture would have to end in the second kind. The question is whether a finite certificate of either half can exist.

13.2 The cycle half: a programme

For the cycle half the answer is mixed. By Proposition 10.20, no certificate built from verification and Diophantine approximation alone can exclude all cycles. But every finite range of cycle lengths can be certified, and the window lemma identifies the arithmetic statement that would close the rest. The programme has five steps, summarized in Table 13.1.

Table 13.1. A computational programme for the cycle half. Each step proves a definite statement; only the last would finish the half.

Step	What it proves	Status
Verification past $2^{71.88}$	every cycle has $r \geq 137\,528\,045\,312$, by the cycle inequality alone	bound already known [38]; verification needs about twice the range 2^{71}
Lean proof of the bound (project P1)	$r \geq 72\,057\,431\,991$, machine checked given the verification below 2^{68}	certificate designed, 44 rungs
Modular sieve on $c_k(w)$	exclusion of whole shapes (m, k)	excludes shapes only for $m \leq 8$ in our runs
Window sharpening	a factor $f(m)$ on every row of Table 10.6	open
Divisibility theorem	no word with $2^k - 3^m \mid c_k(w)$ other than those of the trivial cycle	the cycle half

The first two steps are engineering. Bařina's verification to 2^{71} [7] gains nothing for the cycle inequality, since Table 10.6 puts the next threshold at $2^{71.884}$, although with Hercher's sharper accounting of the cycle it already gives $r \geq 137\,528\,045\,312$ [38, Corollary 29], the current record. Verification to 2^{72}, about twice that range, would give the same bound from the cycle inequality alone; the next threshold after $2^{71.884}$ is at $2^{75.433}$. The Lean proof of the bound is a project of modest size, described as P1 in Chapter 15.

The third step tests whether arithmetic can do more than size. A cycle with m odd steps among k requires $d = 2^k - 3^m$ to divide $c_k(w)$ for its parity word w, so for every prime p dividing d it requires $c_k(w) \equiv 0 \pmod{p}$. The residues of $c_k(w)$ modulo p over all words with m ones can be computed by dynamic programming over positions, with the recursion $c_{t+1} = 3c_t + 2^t$ at an odd step and $c_{t+1} = c_t$ at an even one, in time proportional to pkm. If 0 is unreachable, no cycle of that shape exists. We ran this sieve for every m from 2 to 40, for the four smallest $k > m \log_2 3$, and every prime factor $p < 3000$ of d (script `ch10b_divsieve.py`). It excluded 16 shapes, all with $m \leq 8$; eight of them, such as $(m, k) = (2, 5)$, have $k > 2m$ and are excluded already by $2^k < 4^m$ (proof of Corollary 10.22), and the other

eight, such as $(m,k) = (8,13)$ through $p = 233$, are genuine exclusions. For larger m every residue was reachable for every prime tested. This is what the residue uniformity of Chapter 10 predicts: once the number of words greatly exceeds p, the residues of c_k fill $\mathbb{Z}/p\mathbb{Z}$. The sieve is a valid exclusion tool, and with large prime factors of d it could exclude more, but the state space grows with p, and it offers no route to all m.

The last two steps are mathematics. By Corollary 10.22 an arithmetic improvement of the upper end of the window by a factor slightly more than $m/3$, together with a finite verification, would exclude all cycles, and the exploration of Chapter 10 suggests that for balanced words no size argument can provide it. What remains is a divisibility statement about the numbers $c_k(w)$, a sum of terms $3^j 2^i$ indexed by a binary word. Computation can test conjectures of that kind on all words up to a given length, and it can find the right statement to prove; it cannot prove it.

13.3 The divergence half: what residues cannot do

For the divergence half the natural computational strategy is a residue certificate: partition the positive integers into residue classes modulo 2^k, and for each class exhibit a number of steps after which every member has decreased. Terras' theorem makes this possible for most classes: a class whose parity vector satisfies $3^{m_j} < 2^j$ for some $j \le k$ has an affine form $T^{(j)}(n) = (3^{m_j} n + c_j)/2^j$ that is below n for all large n in the class. The classes that are not certified at level k are those with coefficient stopping time above k, and Proposition 2.25 counts them exactly. Figure 13.1 shows the count up to $k = 400$ (script `ch10b_sieve.py`): 64 classes at $k = 10$, 6.4×10^9 at $k = 40$, 3.0×10^{26} at $k = 100$ and 3.0×10^{111} at $k = 400$. The proportion of uncovered classes tends to zero, which is the density theorem of Terras and Everett (Theorem 2.26), but their number grows like 2^{hk} with

$$h = H\big(1/\log_2 3\big) = 0.9500,$$

the binary entropy of the threshold frequency of odd steps. A class survives exactly when every prefix of its parity word has a proportion of ones above $1/\log_2 3$ (Proposition 2.25), and the number of such words is $2^{(h+o(1))k}$; the upper bound 2^{hk} is Lagarias's exponential form of the density theorem, $1 - F(k) \le 2^{-(1-h)k}$ [53, Theorem D]. A residue sieve never closes.

Theorem 6.3 gives the same conclusion from the side of optimization: the residue graph has cycles of positive mean weight, so no potential $\log x + \varphi(x \bmod 2^k)$ with bounded φ decreases at every step for all large x. The structural reason is 2 adic. Let $E \subset \mathbb{Z}_2$ be the set of 2 adic integers whose parity sequence under T satisfies $3^{m_j} > 2^j$ for every j, the points with infinite coefficient stopping time.

Proposition 13.1 *(What residue certificates can and cannot do).*

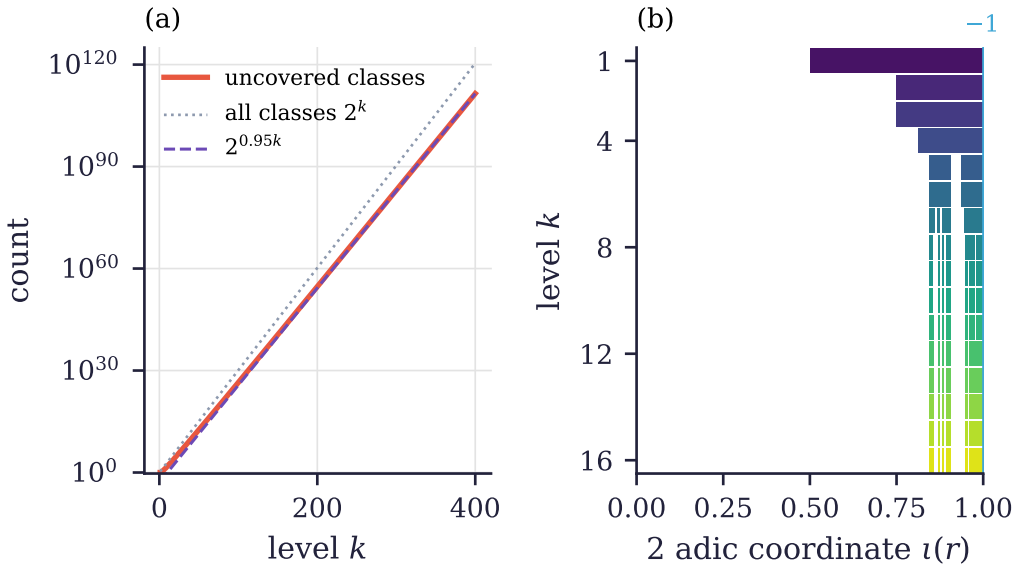


FIGURE 13.1. The residue sieve. (a) The number of classes modulo 2^k not certified to descend within k steps, computed exactly up to $k = 400$, against all 2^k classes and the growth rate $2^{0.95k}$. (b) The uncovered classes at levels 1 to 16 drawn in the 2 adic coordinate $\iota(r) = \sum r_i 2^{-i-1}$; they shrink to a closed set of measure zero that contains -1, the point at the right edge, whose orbit never descends.

(i) *E is a closed subset of $\mathbb{Z}_2$ of Haar measure zero, it is the intersection over k of the unions of uncovered classes at level k, and it contains -1, -5 and -17, one point of each of the three known cycles of T on the negative integers.*

(ii) *Every uncovered class at every level contains infinitely many positive integers. Hence no residue certificate of finite level covers all integers $n \geq 2$.*

(iii) *If a positive integer $n \geq 2$ lies in E, its orbit never falls below n. Hence the descent form of the conjecture implies $E \cap \{2, 3, \ldots\} = \emptyset$, and conversely this together with the coefficient stopping time conjecture of Chapter 2 implies the descent form.*

Proof. (i) The uncovered set at level k, E_k, is a finite union of balls $r + 2^k\mathbb{Z}_2$, hence closed, and $E = \bigcap_k E_k$ because the parity vector of a 2 adic integer of length k depends only on its class modulo 2^k. Its measure is $\lim_k(1 - F(k)) = 0$ by Theorem 2.26. The point -1 is fixed by T and odd, so $m_j = j$ and $3^j > 2^j$ for all j. For a point of a negative cycle with r odd steps in a period of s steps, $3^r > 2^s$ by the reversed cycle identity of Chapter 10, so the ratio $3^{m_j}/2^j$ is multiplied by $3^r/2^s > 1$ over each period; hence it exceeds 1 for every j as soon as it does for $j \leq s$, and a direct check over one period confirms this for -5 (period 3) and -17 (period 11), though not for the other points of those cycles. (ii) A class $r + 2^k\mathbb{Z}$ contains $r + 2^k t$ for every $t \geq 1$. (iii) If $n \in E$ then $T^{(j)}(n) = (3^{m_j}n + c_j)/2^j > n$ for

every j, since $c_j \geq 0$. The converse uses the conjecture of Terras that the coefficient stopping time equals the stopping time for every $n \geq 2$. □

The proposition isolates exactly what computation on residues can deliver: the descent form of the conjecture implies that no integer $n \geq 2$ lies in the measure zero closed set E, and the converse holds if the coefficient stopping time conjecture of Terras holds (Remark 2.27). Positive integers are the 2 adic integers whose digits are eventually all 0, while the known points of E, such as $-1 = \ldots 1111$, have digits eventually all 1 or eventually periodic. The remaining statement is an integrality property, a statement about how E meets the eventually zero sequences, and no finite residue computation can establish it, because every finite level leaves balls of E_k that contain both integers and points of E.

13.4 Automated termination provers

The most ambitious computational attack to date takes the certificate idea to its logical end. Yolcu, Aaronson and Heule [109] encode the Collatz map as a string rewriting system acting on mixed binary and ternary representations, prove that termination of the system is equivalent to the conjecture, and search for termination proofs by matrix interpretations, a standard technique in which each rewriting rule must decrease a weight computed by multiplying small matrices. Their prover found automatic proofs of nontrivial weakenings of the conjecture, but not of the conjecture itself.

In the language of this book a matrix interpretation is a Lyapunov function of a special kind: the weight of a word is a linear function of a product of matrices indexed by its letters, so it is a potential that reads the digits of n through a finite automaton. Proposition 13.1 explains why such potentials face the same obstruction as residue certificates when they read only boundedly many digits, and why the mixed representation matters: a ternary digit records information about n that its binary residues do not, so an automaton reading both can in principle see the integrality that separates the positive integers from E. That is the most promising feature of the approach, and the reason it is a natural place for the large scale computation of a coming decade.

13.5 Does the MM view offer guidance?

It does, in several specific ways.

The potential. The martingale duality of Chapter 3 constrains the potential: in the Bernoulli model for T, x_n itself has no drift, $\log x_n$ has drift $\log(\sqrt{3}/2)$, and x_n^θ decreases in expectation exactly when $0 < \theta < 1$ (Proposition 3.4). Any certificate whose leading term grows at least linearly in n, such as one built on n or on n^θ with $\theta \geq 1$, fights the martingale, which is why linear majorizers fail (Proposition 4.19); the powers n^θ with $0 < \theta < 1$ decrease in expectation but, like

$\log n$, only on average over residue classes. Matrix interpretations measure size multiplicatively, through products, which is the right geometry.

The horizon. Theorem 4.8 and Corollary 4.15 and the comments after them show that no monotone certificate with a fixed horizon exists: for every J there are integers whose first J strides all ascend, which defeats every nondecreasing potential, and $S^{(J)}(5) = 1$ for every $J \geq 1$, which defeats every nonincreasing one. An MM certificate built on a monotone potential such as $\log x$ must therefore majorize over a *variable* horizon, the number of steps to the first descent, which is the stopping time. The greedy scheme "iterate until the potential has decreased, then majorize afresh" is exactly the stopping time decomposition of Chapter 2, and the conjecture in descent form is the statement that this greedy MM is well defined at every $n \geq 2$. The MM language thus turns the conjecture into a termination statement for a specific algorithm, which is precisely the form a termination prover needs.

The switching. For Switching MM the joint spectral radius of the two modes, computed exactly by Proposition 9.6, is below one at every parameter value computed, and at all values if Conjecture 9.7 holds; every switching sequence is then safe (Corollary 9.8). For the Collatz switched system (9.1) the opposite holds: some switching sequences diverge, the all odd one for instance, and by the theorem of Terras and Everett (Theorem 2.8) every parity word of every length is realized by a residue class containing arbitrarily large integers. So there is no constraint on the switching signal at any finite horizon, and the constrained switching methods of control theory, which certify stability by restricting the admissible switching sequences to the language of a finite automaton, have nothing to work with. Whatever constrains the Collatz signal couples the parity sequence to the size and the digits of the state. This is the switched systems form of Proposition 13.1.

The potential, pressed. Chapter 6 pushes the majorization idea to its limit and proves three impossibility results for potentials that decrease at every step: no potential $\log x + \varphi(x \bmod 2^k)$ with bounded φ decreases at every step for all large x, for any k, because the cycles of T on the negative integers are positive cycles of every residue graph (Theorem 6.3); any pointwise potential that decreases at every stride must give integers with j trailing ones a surplus growing like $j \log(3/2)$ (Proposition 6.4); and the block potentials that supply this surplus fail, in the model, at a fraction of odd integers that no weight brings below 0.143 (Proposition 6.6). The same chapter shows that the cycle inequality is the bound of the multiplicative majorizer summed around the cycle, so for the cycle half majorization is a reformulation and not a method.

The memory. Accelerated methods need Lyapunov functions of two consecutive iterates (Section 9.4), and one might hope that a potential with memory, depending on the recent history of the orbit, could certify Collatz descent. A

memory of bounded length, however, adds only boundedly many parities, since x_{n-1} is determined by x_n and the parity of x_{n-1}, and a potential $\log x_n + \varphi$ with φ bounded, whatever part of that history and of the residues of x_n it reads, still increases along the long runs of odd steps from integers 2 adically close to -1, as in Theorem 6.3. Memory helps only if it records unbounded arithmetic information, the digits that decide integrality.

Polyak. A reader of Chapters 5 and 9 may ask whether Polyak's results, which underlie the momentum method, have anything to offer here. Four of them are relevant and none of them transfers. The heavy ball method [80] is what Switching MM specializes, and its analysis is exactly the switched system of Chapter 9; it tells us how the Collatz schedule behaves as an optimizer, not how the integers behave. The gradient dominance, or Polyak and Łojasiewicz, inequality $\|\nabla f\|^2 \geq 2\mu(f - f^*)$ [81] is the condition that makes linear rates possible without convexity, and it is a pointwise condition: it would require $\log x$ to decrease at every step by a fixed fraction of its distance to the minimum, which the ascents of Theorem 4.8 rule out. The Collatz drift is additive and holds only in expectation, which is Foster's condition for Markov chains, not Polyak's for descent methods. Polyak and Ruppert averaging [82, 83, 89], the device by which a noisy iteration achieves the optimal rate by averaging its iterates, corresponds under the logarithm to the Cesàro criterion of Chapter 11: it is the right object to look at and it is the statement that remains unproved. Finally the Polyak step size $\alpha_k = (f(x_k) - f^*)/\|\nabla f(x_k)\|^2$ [81] needs the optimal value, which for Collatz is $f^* = \log 1 = 0$ and is known; but the Collatz step sizes are not ours to choose, the parity sequence chooses them, and a method that adapts its step to the distance from the optimum has nothing to adapt. Polyak's results are therefore useful for the optimizer in Part II and irrelevant to the conjecture in Part III, which is the same division the rest of the book makes.

So the optimization reading does not shorten the path, but it marks it. It says which potentials can work, that the horizon of a monotone certificate must be variable, that parity information of bounded length can never suffice, and that the missing ingredient is an integrality argument, a way of using the fact that n has a terminating binary expansion. These are also features of the most serious computational programme now in existence.

Notes

The 2 adic formulation of the stopping time problem is due to Terras and to Lagarias [53], and Bernstein [10] gives a noniterative form. The rewriting approach and the matrix interpretation prover are from Yolcu, Aaronson and Heule [109]. The residue sieve counts to $k = 400$ and the divisibility sieve experiment are ours. The exponent $h = H(1/\log_2 3)$ gives the decay rate $1 - h$ of Lagarias's exponential form of the density theorem of Terras and Everett,

$1 - F(k) \le 2^{-(1-h)k}$ [53, Theorem D]. Proposition 13.1 in the form stated is ours, although each ingredient of the proposition is classical.

Exercises

Exercise 13.2. Using the recursion of Proposition 2.25, show that the number of uncovered classes modulo 2^k is equal to the number of binary words of length k all of whose prefixes have a proportion of ones above $1/\log_2 3$, and deduce that it grows like $2^{(h+o(1))k}$ with $h = H(1/\log_2 3)$ (the binomial tail gives the upper bound).

Exercise 13.3. Run the divisibility sieve of Section 13.2 for $(m,k) = (8,13)$ by hand or by machine, find the residues of $c_{13}(w)$ modulo 233 that occur, and confirm that 0 is not among them. How many words are there, and how does that compare with 233?

Exercise 13.4. Show that the rational number $1/5$ is periodic under the accelerated map on $\mathbb{Z}_2$ with odd denominators allowed, and decide whether it lies in E.

Exercise 13.5. Explain why a potential $V(n) = \log n + \varphi(n \bmod 2^k)$ with φ bounded cannot certify descent over a fixed horizon J for all $n \ge 2$, by exhibiting integers in the class $-1 \bmod 2^{J+1}$ for which V increases over J steps once n is large.

Chapter 14

Samplers and the Horizon

Chapter 13 asked what computation can prove. This chapter asks what simulation can find. The idea, suggested by the Bayesian computation literature, is to treat the problem as one of rare event simulation. Define an energy on the integers, say the total stopping time, declare the integers with high energy to be the interesting ones, and use a sampler that climbs the energy landscape to find them or to measure how many there are. Nested sampling, the vertical likelihood sampler and split sampling are three such methods, all built to estimate a normalizing constant by moving through a sequence of level sets rather than by sampling the base measure directly, and, given exact draws from each level set, all three reach regions of exponentially small prior mass in a number of steps that is polynomial in the depth. The question this chapter answers is whether such a method can say anything about the conjecture in polynomial time. It can do one thing exactly, which we prove, and beyond a horizon a local sampler leaves the tree of merged orbits it starts in only about as often as rejection sampling finds a new one, which we explain heuristically in the Bernoulli model and observe in practice.

14.1 One identity, three samplers

Let π be a probability measure on a space $\mathcal{X}$ and let $L : \mathcal{X} \to [0, \infty]$ be an energy. Write

$$Z(m) = \pi\{L > m\}, \qquad m \geq 0,$$

for the prior volume of the level set above m. The quantity every method estimates is the expected energy, or more generally $\mathbb{E}_\pi[g(L)]$ for a monotone g, through the layer cake identity

$$\mathbb{E}_\pi[L] = \int_0^\infty Z(m)\, dm. \tag{14.1}$$

The common device is a level variable. Fix a weight, a measure ω on $[0, \infty)$ with cumulative weight $W(l) = \omega([0, l))$ finite for every l, and consider the joint law

$$\pi_\omega(dx, dm) \;\propto\; [\, L(x) > m \,]\, \pi(dx)\, \omega(dm), \tag{14.2}$$

which is a probability law when its total mass $\int Z\, d\omega = \mathbb{E}_\pi[W(L)]$ is finite. Its two marginals are read off at once: integrating out m gives x the law proportional to $W(L(x))\pi(dx)$, and integrating out x gives m the law proportional to $Z(m)\, \omega(dm)$. Polson and Scott [79] call a Markov chain that alternates the two conditionals, m

given x and x given m, a weighted slice sampler, and show that vertical likelihood and nested sampling are choices of the weight; Birge, Chang and Polson [11] show the same for split sampling.

(i) *Vertical likelihood.* Take the cumulative weight $W = 1/Z$, that is a unit atom at $m = 0$ together with the density $(1/Z)'$. The level marginal is then proportional to $-Z'(m)/Z(m)$ away from 0, which is flat in the coordinate $\log(1/Z(m))$: the chain spends equal time in each factor e of prior volume, and the x marginal is π reweighted by $W(L(x)) = 1/Z(L(x))$, which grows like e^{DL} when $Z(m) \approx e^{-Dm}$. With this weight the augmented law has infinite mass when L is continuous, since $Z(L)$ is then uniform on $(0,1)$, and Polson and Scott floor the weight, $W(m) = 1/\max(\eta, Z(m))$, to keep the law proper [79, eq. (19)]. Since Z is unknown one uses a running estimate $\hat{Z}$ in its place, and recovers $Z(m)$ from the chain by importance weighting each sample x with $1/W(L(x))$. The conditional of m given x is explicit: draw T uniform on $(0, 1/\max(\eta, Z(L(x))))$ and set $m = 0$ if $T \leq 1$ and $m = Z^{-1}(1/T)$ otherwise [79, Lemma 3].

(ii) *Nested sampling.* Skilling [93] discretizes the same weight $W = 1/Z$. A population of N live points is drawn from π; at each iteration the point of lowest energy L_i is removed and replaced by a draw from π conditioned on $\{L > L_i\}$, obtained by running a π invariant Markov kernel inside the level set. The prior volume above L_i shrinks by a factor with mean $N/(N+1) \approx e^{-1/N}$, whose logarithm has mean exactly $-1/N$, per iteration, which is the flat level marginal in logarithmic coordinates, and (14.1) is estimated by the quadrature $\sum_i (X_{i-1} - X_i)\, g(L_i)$ with $X_i = e^{-i/N}$.

(iii) *Split sampling.* Birge, Chang and Polson [11] fix the levels adaptively at quantiles instead: at stage t the sampler runs inside $\{L > m_{t-1}\}$, sets m_t to the $(1-\rho)$ quantile of the energies it sees, and records the conditional fraction $p_t \approx \rho$ above m_t. The product $\prod_{s \leq t} p_s$ estimates $Z(m_t)$; this is the product estimator, which they show is split sampling with the discrete weights $\omega_t = 1/Z(m_t)$ (their Theorem 1). If the stage estimates are independent with relative variances v_t, the relative variance of the product is $\prod_t (1+v_t) - 1$; with N independent samples per stage $v_t \approx (1-\rho)/(\rho N)$, and the product is about $t(1-\rho)/(\rho N)$, linear in the number of stages $t = \log(1/Z)/\log(1/\rho)$, when N is large compared with t. The Rao Blackwellized estimate $\hat{Z}(m) = \sum_{x:L(x)>m} \Omega(L(x))^{-1} / \sum_x \Omega(L(x))^{-1}$, with $\Omega = W$ the cumulative weight actually used, recovers the whole curve Z from one run; with $\Omega = 1/Z$ the sampling distribution of the method matches that of nested sampling, which is their Theorem 2.

In every case the work per sample is polynomial in the depth, provided one ingredient: a kernel that moves freely inside each level set $\{L > m\}$. The cost of a

method is the cost of that kernel, and the obstruction we will meet is that for the natural Collatz energy no local kernel moves freely at depth.

14.2 The Collatz prior and two energies

The only natural prior on the integers is the one the 2 adic formulation provides, the Haar measure μ on $\mathbb{Z}_2$ of Chapter 2, under which the residue modulo 2^k is uniform for every k and the parity digits of the orbit are independent fair coins (Theorem 2.8). Two energies suggest themselves. The first is the coefficient stopping time τ of Definition 2.22, the first k at which $3^{m_k} < 2^k$, which is a function of the parity word of length k and so of the residue class modulo 2^k. The second is the total stopping time σ_∞, the first k with $T^{(k)}(n) = 1$, counted in steps of T (in Chapter 1 the same symbol counts steps of T_0), which is the quantity the conjecture is about. The two are separated by a horizon.

Proposition 14.1 *(Two energies).*

(i) For every m the level set $\{\tau > m\} \subset \mathbb{Z}_2$ is a union of residue classes modulo 2^m, hence open and closed, and its Haar measure is $Z_\tau(m) = 1 - F(m) = 2^{-m} \sum_j N_m(j)$, computable exactly in $O(m^2)$ arithmetic operations by Proposition 2.25. The sets decrease in m and their intersection $\{\tau = \infty\}$ has measure zero.

(ii) For every m the set $\{x \in \mathbb{Z}_2 : \sigma_\infty(x) \le m\}$ is finite, and it contains at most one positive integer in each residue class modulo 2^m. Consequently $\{\sigma_\infty > m\}$ has Haar measure one for every m, and so does the intersection $\{\sigma_\infty = \infty\}$ of these sets: the energy σ_∞ is almost surely infinite under μ, and any simulation with this energy must use a uniform prior on a finite range, which we take to be $[2, 2^B)$, with $Z_\sigma(m) = \#\{2 \le n < 2^B : \sigma_\infty(n) > m\}/(2^B - 2)$.

Proof. (i) is the content of Section 2.2 and Proposition 2.25: $\tau(n) > m$ is decided by the first m parities, which are a bijective function of n mod 2^m, and the density theorem of Chapter 2 says $1 - F(m) \to 0$. For (ii), every $y \in \mathbb{Z}_2$ has exactly the two preimages $2y$ and $(2y-1)/3$ under T, so $\{\sigma_\infty \le m\}$, which lies in $\bigcup_{k \le m} T^{-k}(1)$, has at most $2^{m+1} - 1$ points. Now fix a parity word w of length m with j ones. By the Terras formula $T^{(m)}(n) = (3^j n + c_w)/2^m$ for every n in the class of w, and $\sigma_\infty(n) \le m$ forces $T^{(m)}(n) \in \{1, 2\}$, because the orbit alternates between 1 and 2 once it reaches 1; so $n = (2^m - c_w)/3^j$ or $n = (2^{m+1} - c_w)/3^j$. These differ by $2^m/3^j$, so both are integers only when $j = 0$; then w is the zero word, $c_w = 0$, and of 2^m and 2^{m+1} only the first has $\sigma_\infty \le m$. Hence each class contains at most one positive integer with $\sigma_\infty \le m$, the complement of a finite set has full Haar measure, and so does a countable intersection of such complements. □

The horizon is the following. The event $\{\tau > m\}$ reads only the first m parities, which are a bijective function of the residue modulo 2^m, so its level sets have the clopen structure a sampler needs: a residue class is a cylinder, and moving

inside a union of cylinders is easy. An integer below 2^B, on the other hand, carries no residue information beyond its first B parities, which determine it by Terras' bijection, so beyond its bit length its orbit has no unread bits, and on the box the level sets of σ_∞ are sets of particular integers with no 2 adic description at all.

14.3 Perfect split sampling for the coefficient stopping time

On the near side of the horizon simulation is not merely polynomial but exact.

Theorem 14.2 *(Perfect split sampling). Fix a depth $K \geq 1$, let the levels be the integers $0 \leq k \leq K$, let $Z_k = 1 - F(k)$ be the exact volumes of Proposition 14.1(i), write $\tau_K = \min(\tau, K+1)$, and let the discrete weights be*

$$\omega_0 = 1, \qquad \omega_k = \frac{1}{Z_k} - \frac{1}{Z_{k-1}} \quad (1 \leq k \leq K),$$

$$\text{so that} \qquad W(l) = \sum_{k<l} \omega_k = \frac{1}{Z_{l-1}} \quad (1 \leq l \leq K+1).$$

Then the augmented law $\pi_\omega(x,k) \propto [\tau_K(x) > k]\, \mu(dx)\, \omega_k$ has the finite mass $1 + \sum_{k=1}^{K} \lambda_k$, where $\lambda_k = Z_k \omega_k = 1 - Z_k / Z_{k-1}$ is the hazard, and its Gibbs sampler can be run exactly, with no Markov chain inside the level sets:

(i) given x, the level k is drawn from ω restricted to $\{0, \ldots, \tau_K(x) - 1\}$, a finite discrete law;

(ii) given k, a point with law μ restricted to $\{\tau > k\}$ is drawn exactly, as far as τ_K can see it, which is to say its residue modulo 2^K. An alive word of length k is drawn uniformly by backward sampling of the dynamic programme of Proposition 2.25: starting from the terminal count $N_k(j)$ chosen with probability $N_k(j) / \sum_{j'} N_k(j')$, the parities are generated from the last to the first with the probability $N_{i-1}(j) / N_i(j)$ of a zero and $N_{i-1}(j-1) / N_i(j)$ of a one. The word is followed by $K - k$ independent fair parities, and the word of length K is converted to a residue modulo 2^K by the Terras bijection. The cost is $O(K)$ per draw after an $O(K^2)$ table, since step (i) reads $\tau_K(x)$ from the word itself.

The level marginal of the chain is proportional to 1 at $k = 0$ and to λ_k for $1 \leq k \leq K$, and the x marginal is proportional to μ weighted by $1 / Z_{\tau_K(x)-1}$, so that the event $\{\tau > K\}$ has stationary probability $1 / (1 + \sum_{k \leq K} \lambda_k)$. Every $k \leq K$ with $R_k > 0$, in the notation of Proposition 2.25(ii), is visited with the positive frequency $\lambda_k / (1 + \sum_{k' \leq K} \lambda_{k'})$. Without the truncation the augmented law has infinite mass, since $\prod_{k \leq K} (1 - \lambda_k) = Z_K \to 0$ forces $\sum_k \lambda_k = \infty$, and every level is then visited with long run frequency zero.

Proof. The weights are nonnegative because Z is nonincreasing, and W telescopes as stated. The conditional of k given x is ω restricted to $k < \tau_K(x)$ by (14.2); the conditional of x given $k \leq K$ is μ restricted to $\{\tau_K > k\} = \{\tau > k\}$. Under μ the parities are independent fair coins (Theorem 2.8), so under this conditional

ALGORITHM 14.1. Perfect split sampling for the coefficient stopping time

Input: depth K, number of sweeps S

compute $N_i(j)$ for $i \le K$ by Proposition 2.25; set $Z_i = 2^{-i} \sum_j N_i(j)$
set $\omega_0 \leftarrow 1$ and $\omega_i \leftarrow 1/Z_i - 1/Z_{i-1}$ for $1 \le i \le K$
draw x, a uniform residue modulo 2^K, and set $k \leftarrow 0$
for $s = 1, \dots, S$ *do*
    draw k from $\{0, \dots, \min(\tau(x), K+1) - 1\}$ with probability $\propto \omega_k$ ▷ level given point
    draw j with probability $N_k(j) / \sum_{j'} N_k(j')$ ▷ number of odd steps
    for $i = k, k-1, \dots, 1$ *do*
        with probability $N_{i-1}(j)/N_i(j)$ set $w_i \leftarrow 0$, else $w_i \leftarrow 1$ and $j \leftarrow j - 1$
    end for
    $x \leftarrow$ the residue modulo 2^K whose parity word is $w_1 \cdots w_k$ followed by $K - k$ fair parities (Terras) ▷ point given level
    record (x, k)
end for

law the first k parities form a uniform alive word of length k, the later parities are independent fair coins, and τ_K depends only on the first K of them, that is on the residue modulo 2^K. The counts $N_i(j)$ of Proposition 2.25 count the alive words of length i with j ones, and a word is alive if and only if every prefix is alive, so the stated backward probabilities are the exact conditionals of a uniformly chosen alive word, by induction on i. The marginals follow from (14.2): the event $\{\tau > K\}$ has μ measure Z_K and weight $W(K+1) = 1/Z_K$. The level process is itself a Markov chain, since the point drawn at level k is fresh, and it is reversible with respect to $\nu(k) = Z_k \omega_k$, because its transition probabilities satisfy $\nu(k)P(k,l) = \omega_k \omega_l \sum_{t > \max(k,l)} \mu\{\tau_K = t\} W(t)^{-1}$, which is symmetric in k and l (with τ in place of τ_K when there is no truncation). It is irreducible on the levels with $\nu(k) > 0$, which are 0 and the k with $Z_k < Z_{k-1}$, that is with $R_k > 0$, because 0 is reached from every level in one step and every such level from 0; and it is aperiodic, because it can stay at 0. With the truncation ν has finite mass, and the ergodic theorem gives the stated frequencies. Without it ν has infinite mass; a positive recurrent chain would have a stationary probability law, which for an irreducible recurrent chain is the unique invariant measure up to a constant factor, and ν is not proportional to a probability law, so the chain is transient or null recurrent and every level has long run frequency zero. □

The theorem is a complete answer to the simulation question on the near side of the horizon. Algorithm 14.1 states the chain in full. We ran it at depth $K = 400$ for twenty thousand sweeps, and Figure 14.2(b) shows the first three thousand. The level moves up at 46% of the sweeps and down at 46%, with mean 167 (the stationary mean is 157); its maximum 400 is the depth K of the run, not a property

of the walk. A rejection sampler would need $1/Z_{400} \approx 8.6 \times 10^8$ draws from μ to see one residue with $\tau > 400$. The chain returns such a residue in 5.6% of its sweeps, about one in eighteen (the stationary probability is $1/(1+\sum_{k\le 400}\lambda_k) = 0.049$), each sweep costing $O(K)$, and each such residue is an exact uniform draw from the residues modulo 2^{400} with $\tau > 400$; the level itself reaches the top level 400 in about one sweep in four hundred. That is polynomial time sampling of a set of exponentially small measure, and the samples are exact, not approximate. It is also, we stress, information already contained in Proposition 2.25, whose backward sampling at the single level 400 gives independent exact draws at the same cost: the Monte Carlo is a way of exploring the nest, not of measuring it.

Panel (a) of the figure gives the volumes themselves. On the stopping levels $k = \lfloor j \log_2 3 \rfloor + 1$, the k with $R_k > 0$ by Proposition 2.25(ii), the hazard $\lambda_k = 1 - Z_k/Z_{k-1}$ is positive and elsewhere it is zero, so Z_τ is a staircase; over $100 \le k \le 400$ it follows the fit $Z_\tau(k) \approx Ck^{-1.355}e^{-Dk}$ with

$$D = \log 2 - H\big(1/\log_2 3\big) = 0.03469,$$

the Chernoff rate for the proportion of ones in a fair word to exceed $1/\log_2 3$, where H is the entropy in nats; in terms of the residue sieve exponent $h = 0.95$ of Section 13.3, $D = (1-h)\log 2$, since $2^{(h-1)k} = e^{-Dk}$. The polynomial correction is the ballot factor, of order $k^{-3/2}$ asymptotically by the heuristic of Section 2.6; the fitted exponent is smaller at these depths, and in the exact recursion it rises to 1.48 over $1000 \le k \le 6000$ and 1.50 over $6000 \le k \le 20\,000$. On the stopping levels with $100 < k \le 400$ the hazard averages 0.063 (0.055 over $100 < k \le 6000$), against the value $1 - e^{-D\log_2 3} = 0.053$ that a pure exponential with one stopping level every $\log_2 3$ steps would give; the excess is the polynomial factor, which makes the staircase steeper than exponential at finite depth. One consequence is a precise statement about which expectations exist: the partition function $\mathbb{E}_\mu[e^{\beta\tau}] = \sum_k (Z_{k-1} - Z_k)e^{\beta k}$ converges if and only if $\beta \le D$. It diverges for $\beta > D$, because it is at least $e^{\beta k}Z_k$ for every k and $Z_k = e^{-Dk+o(k)}$, the exponent of Theorem 2.26 being sharp (Section 2.6). At $\beta = D$ it converges: write $X_k = m_k \log 3 - k \log 2$, a random walk under μ with steps $\log(3/2)$ and $-\log 2$, so that τ is the first k with $X_k < 0$; the step ξ has $\mathbb{E}e^{\theta\xi}$ minimal at $\theta_* = 0.488$, where it equals e^{-D}, so $e^{\theta_* X_k + Dk}$ is a martingale, and since $-\log 2 \le X_\tau < 0$ optional stopping gives $\mathbb{E}_\mu[e^{D\tau}] \le 2^{\theta_*} = 1.40$ (the exact recursion to depth 20 000 gives about 1.25). So a Boltzmann weighting of the coefficient stopping time has a finite temperature barrier at $\beta = D$, and no annealing scheme can be run past it. The expected coefficient stopping time itself is $\mathbb{E}_\mu[\tau] = \sum_k Z_k = 3.4927$, the layer cake sum of the staircase.

14.4 Beyond the horizon: the total stopping time

On the far side the picture changes, and not in the way one first expects. The naive prediction is that a local move in n, a change of one bit, produces an integer

whose orbit is unrelated to the orbit of n beyond the bit length, so that inside a deep level set every proposal is rejected with probability about $1 - Z_\sigma(m)$ and the sampler is a rejection sampler. That prediction is wrong, and the reason it is wrong is the first structural fact about σ_∞ that simulation reveals: orbits of nearby integers merge.

Lemma 14.3 *(Merging). For every $k \geq 0$, $T^{(3)}(8k+4) = T^{(3)}(8k+5) = 3k+2$. More generally, if $n \equiv n' \pmod{2^i}$ then the first i parities of n and n' agree and*

$$T^{(i)}(n') - T^{(i)}(n) = 3^j \, \frac{n' - n}{2^i},$$

where j is the number of odd steps among the first i (Lemma 2.7). In particular flipping bit i of n moves the orbit at step i by exactly $\pm 3^j$; and if at some step the two orbits occupy a pair $\{8k+4, 8k+5\}$ with $k \geq 1$, they coincide three steps later, so that $\sigma_\infty(n') = \sigma_\infty(n)$.

Proof. $8k+4 \mapsto 4k+2 \mapsto 2k+1 \mapsto 3k+2$ and $8k+5 \mapsto 12k+8 \mapsto 6k+4 \mapsto 3k+2$. For $k \geq 1$ none of these values is 1 or 2, so if the two orbits are at $8k+4$ and $8k+5$ after s steps, neither has reached 1, and both total stopping times equal $s + 3 + \sigma_\infty(3k+2)$; for $k = 0$ the clause fails, since $\sigma_\infty(4) = 2$ and $\sigma_\infty(5) = 4$. The general formula is Lemma 2.7, or the Terras affine form $T^{(i)}(n) = (3^j n + c_w)/2^i$ applied to two integers in the same class of w. □

Merging is the mechanism that makes the Collatz graph a tree with many confluences rather than a bundle of separate paths, and it is why 12 and 13 both reach 5, and 28 and 29 both reach 11, after three steps. For a sampler it has a precise consequence.

Proposition 14.4 *(Coalescence). Let n and n' be positive integers whose orbits meet at a point $x \geq 3$, reached from n after s steps and from n' after s' steps. Then $\sigma_\infty(n') = \sigma_\infty(n) - s + s'$. In particular, if $s = s'$, as for the merging pairs of Lemma 14.3, then n' lies in every level set $\{\sigma_\infty > m\}$ that contains n.*

Proof. An orbit that reaches 1 stays in $\{1, 2\}$ from then on, so neither orbit reaches 1 before it reaches x. Hence $\sigma_\infty(n) = s + \sigma_\infty(x)$ and $\sigma_\infty(n') = s' + \sigma_\infty(x)$. □

The condition $s = s'$ matters. The integers 27 and 31 differ in one bit and their orbits meet, since $27 \mapsto 41 \mapsto 62 \mapsto 31$, but with $s = 3$ and $s' = 0$, so $\sigma_\infty(31) = \sigma_\infty(27) - 3 = 67$, and 31 is not in the level set $\{\sigma_\infty > 69\}$ that contains 27. Fix a threshold R and call two integers of $\{\sigma_\infty > m\}$ equivalent if their orbits pass through a common point of remaining stopping time at least $m - R$; the classes of this relation are the *coalescence classes* of the level set, and we take $R = 100$ below.

Remark 14.5 *(Coalescence or independence, a heuristic).* Let $n < 2^B$ lie in $\{\sigma_\infty > m\}$ and let n' differ from n in one bit. If the orbits of n and n' meet at a point of remaining stopping time at least $m - R$, Proposition 14.4 gives $\sigma_\infty(n')$, and n' joins

the class of n when $s = s'$. If they do not, the first B parities of n' determine it, and the Bernoulli model of Chapter 3 treats the later parities of its orbit as fair coins independent of the orbit of n. The model probability that n' lies in the level set is then about $Z_{\mathrm{mod}}(m - B)$, where Z_{mod} is the tail of the total stopping time in the model, and $Z_{\mathrm{mod}}(m - B) \sim e^{DB} Z_{\mathrm{mod}}(m)$ as $m \to \infty$, because in the model the total stopping time is a first passage time of the same random walk as τ and has the same rate D. So a kernel of single bit flips restricted to $\{\sigma_\infty > m\}$ should leave the coalescence class of its starting point only at proposals of probability of order $e^{DB} Z_\sigma(m)$, with the census value Z_σ standing in for the model tail. This is a heuristic and not a theorem: the independence after step B is an assumption of the model, and nothing here controls when the orbits of two integers meet.

Both halves of the dichotomy are visible in the data. Panel (a) of Figure 14.1 shows, for each level m, the fraction of single bit flips of integers in the level set on $[2, 2^{26})$ that stay in it. Flips of the thirteen high bits behave as the naive prediction suggests: their acceptance falls with $Z_\sigma(m)$ and is zero from $m = 380$ on, because a change high in the binary expansion displaces the orbit by 3^j with j large, and such a displacement rarely lands on a merging pair. Flips of the thirteen low bits accept at a rate that falls only slowly, from 18% at $m = 200$ to about 5% between $m = 360$ and $m = 420$, while Z_σ falls by more than four orders of magnitude: those are the coalescences. At $m = 400$ the overall acceptance is 2.6% against the acceptance rate 3.8×10^{-6} of rejection sampling, a factor of nearly seven thousand, and every one of the accepted moves stays in the coalescence class of its starting point (with the threshold $R = B = 26$ in place of $R = 100$, 170 of the 172 do).

At the deepest levels the confinement is exact in the data. Table 14.1 decomposes each level set into coalescence classes, two integers being in the same class if their orbits meet at a point whose remaining stopping time is at least $m - 100$ (the meeting point may lie above the box, and usually does). The 26 708 integers above 300 fall into 1 281 classes, the largest with 1 659 members; the 257 above 400 fall into 34, with 83 in the largest; the 24 above 440 fall into six. And of the 172 accepted single bit flips among the integers above 400 (the 86 pairs that differ in one bit, each counted in both directions), all 172 stay in the same class. Not one crosses between classes, as the heuristic suggests, since a crossing would be an independent orbit of length above 400 found by a single proposal. At $m = 300$ and $m = 350$ some accepted flips do cross, 1 112 of 27 896 and 22 of 2 848, that is 1.6×10^{-3} and 2.5×10^{-4} of all proposals, against the values 9.8×10^{-4} and 1.3×10^{-4} of $e^{DB} Z_\sigma(m)$ in the heuristic.

This is the right description of the far side of the horizon. A deep level set is not a sparse set of unrelated integers; it is a small number of trees, each the set of all tributaries of one slow orbit, and a local sampler stays in the tree it starts in, leaving it only rarely, at a rate per proposal of the order of the acceptance rate of rejection sampling. Inside the tree it is confined further, to the few members within a few bit flips of its current point: the 86 pairs that differ in one bit join

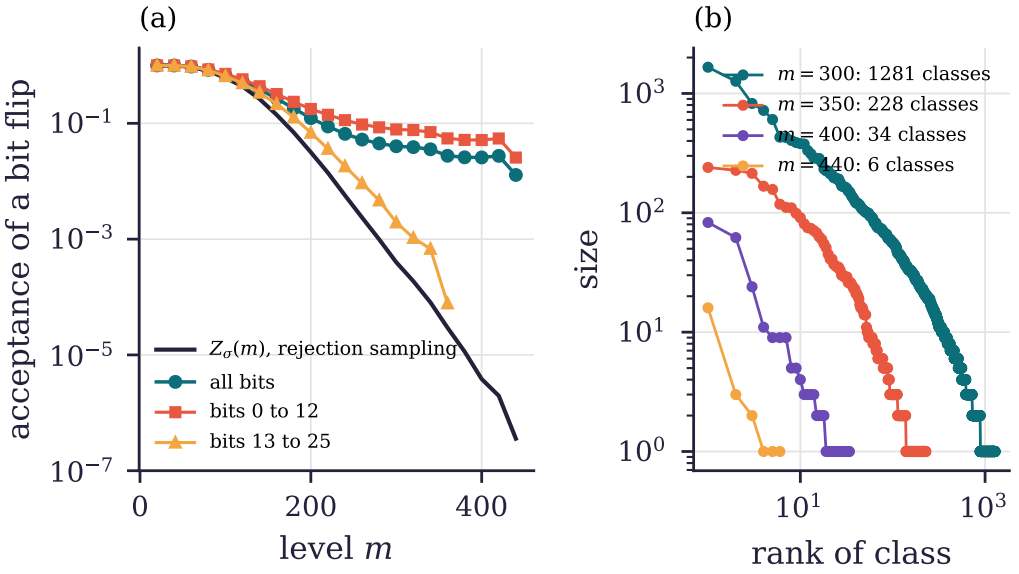


FIGURE 14.1. Coalescence or independence. (a) Acceptance rate of a single bit flip inside the level set $\{\sigma_\infty > m\} \cap [2, 2^{26})$, for all bits, for the low bits and for the high bits, against the acceptance rate $Z_\sigma(m)$ of rejection sampling. Low bit flips coalesce and keep accepting; high bit flips rarely coalesce and stop. (b) Sizes of the coalescence classes of the level sets at $m = 300, 350, 400, 440$, ranked. The 257 integers above 400 form 34 classes, 83 of them in the largest; the 24 above 440 form six.

the 257 integers above 400 into at least 171 separate pieces, and each stalled stage of split sampling reported below visited between one and twenty three distinct integers. The number of trees is what shrinks with the depth, not the acceptance rate: 34 at 400, six at 440. A level set sampler therefore does not degrade into rejection sampling. It does something more specific. It stays in the tree it starts in, and in fact among a handful of its members, and it cannot climb above the maximum of σ_∞ on that tree, because every integer it can reach shares the tail of the same orbit. Its depth is bounded by the deepest tree its initial population happens to contain, and new trees are found only by independent draws from the prior or by the rare crossing proposals, at about the acceptance rate of rejection sampling.

How the trees multiply with the size of the box is shown in Table 14.2, which repeats the decomposition on $[2, 2^B)$ for $B = 22, 24, 26$. At a fixed level and over $22 \le B \le 26$ the number of trees grows with B at a rate between 2^B and $2^{1.4B}$, faster at deeper levels, while the mean tree size grows roughly like $2^{0.4B}$ and, at fixed B, falls steadily with the depth, from 208 at $m = 200$ to 7.6 at $m = 400$ for $B = 26$. These are local exponents: at a fixed level the number of trees is bounded in B, since each tree has its own root, an integer with $\sigma_\infty = m - 100$, and all

TABLE 14.1. Coalescence classes of the level sets $\{\sigma_\infty > m\}$ on $[2, 2^{26})$, two integers being equivalent if their orbits merge no later than the point where the remaining stopping time equals $m - 100$. The last two columns count the accepted single bit flips, that is the ordered pairs of integers in the level set that differ in one bit, and how many of them stay in the same class.

m	integers	classes	largest	accepted flips	same class
300	26 708	1 281	1 659	27 896	26 784
350	3 432	228	240	2 848	2 826
400	257	34	83	172	172
440	24	6	16	8	8

such integers are at most 2^{m-100}, because a step of T at most halves. A tree is the set of predecessors below 2^B of one point x at remaining depth $m - 100$ that lie more than a hundred steps above x, so its size is a count on the inverse tree of Chapter 9 and the number of trees is the number of such points x that are reached from the box by an orbit segment of more than a hundred steps. At the deeper levels the root x usually lies above the box: at $m = 400$ the roots of the two largest trees are 146 053 012 and 1 330 038 578, well beyond 2^{26}, because a slow orbit of a box integer climbs out of the box long before its remaining stopping time falls to 300. The trees are also mostly tributary: of the 83 members of the largest tree at $m = 400$ only 14 lie on the orbit of its deepest member, the rest being integers whose orbits join that orbit later.

The deepest member of a tree is sometimes a delay record of Chapter 1 (for 2 of the 34 trees at $m = 400$), but the correspondence is loose. At $m = 400$ three of the 34 trees contain a record, including the two largest, whose deepest members are 55 285 737 ($\sigma_\infty = 446$) and the record 36 791 535 (466); the overall record 63 728 127, with $\sigma_\infty = 592$, is a tree of one member at $m = 440$, because its orbit leaves the box at once and nothing below 2^{26} joins it in time. The records are the deepest trunks; the large trees are the trunks that stay low enough, for long enough, to collect tributaries. A naive model of tree size, in which each backward path of d steps with j odd steps from x exists with the integrality probability 3^{-j}, predicts fewer than four members for every root at $m \geq 350$ and is wrong by three orders of magnitude or more: the median ratio of observed to predicted size is about 1.6×10^3 at $m = 350$ and 10^4 at $m = 400$, where the largest tree has 83 members against a prediction of 0.012. Conditioned on being reached from the box by a slow orbit, a root is reached by many, and the conditioning is the likely explanation. We do not have a model that predicts either the number of trees or their sizes, and we record the question as project P9 of Chapter 15.

TABLE 14.2. Trees of slow integers on $[2, 2^B)$: the number of integers with $\sigma_\infty > m$, the number of coalescence classes, and their mean and largest size, for three box sizes.

B	m	integers	trees	mean size	largest
22	200	36 482	647	56.4	2 224
22	250	2 558	204	12.5	198
22	300	201	29	6.9	65
24	200	291 361	2 539	114.8	15 153
24	250	28 429	1 047	27.2	1 320
24	300	2 610	184	14.2	381
24	350	259	33	7.8	42
26	200	2 125 551	10 235	207.7	110 856
26	250	253 975	5 682	44.7	8 424
26	300	26 708	1 281	20.8	1 659
26	350	3 432	228	15.1	240
26	400	257	34	7.6	83

14.5 The three samplers against the census

We tested the three samplers on the prior uniform on $[2, 2^{26})$, with energy the total stopping time of T and the exact Z_σ from the census of Chapter 3 (the maximum is $\sigma_\infty = 592$, at $n = 63\,728\,127$). Each sampler had a budget of at most 2×10^5 energy evaluations per run, and the level set samplers used a kernel step that flips one to three random bits. Naive sampling used exactly 2×10^5 independent draws; nested sampling ran 200 live points for 1600 iterations with 120 kernel steps per replacement (192 200 evaluations); split sampling took 1200 samples per stage, ten kernel steps apart, with $\rho = e^{-1}$, and stopped at the first stage that found no sample above its quantile (between 105 300 and 176 510 evaluations); and the vertical likelihood sampler ran 12 000 sweeps of 16 kernel steps (192 000 evaluations) with a floor $\eta = 10^{-7}$ and the running estimate of Z refreshed every 2000 sweeps. Table 14.3 reports the medians of six independent runs.

Up to $m = 200$ all four estimates agree with the truth to within a factor of about two and a half. At $m = 300$, where Z_σ is 4×10^{-4} and the budget contains some eighty naive hits, naive and nested sampling are within 15% of the truth and split sampling is low by a factor of about two, while the vertical likelihood estimate is already low by a factor of ten; the level set methods bring nothing here, because the levels are still cheap. Beyond $m = 300$ the level set samplers stop, for two different reasons. Split sampling stopped in every run at a level between 303 and 338, in a stage whose chain never left one coalescence class: it visited between one and twenty three distinct integers, all with the same total stopping time, so that no sample lay above the quantile. That is the confinement of the tree picture, and the estimate beyond that level is the product of the stage fractions found so far. Nested sampling stopped for want of budget: its 1600 iterations shrink the prior volume

TABLE 14.3. Estimates of $Z_\sigma(m) = \mathbb{P}\{\sigma_\infty > m\}$ on $[2, 2^{26})$, medians of six runs at a budget of at most 2×10^5 energy evaluations each. The exact values are from the census. From $m = 400$ on, nested and split sampling report the prior mass of their last level, the vertical likelihood estimate is zero in four of the six runs at $m = 400$, and so in the median, and in all six at $m = 450$ and 500, and naive sampling sees at most two integers above 400 in a run and none above 450.

m	exact	naive	nested	split	vertical
100	6.13×10^{-1}	6.13×10^{-1}	6.47×10^{-1}	4.57×10^{-1}	5.21×10^{-1}
200	3.17×10^{-2}	3.18×10^{-2}	3.37×10^{-2}	3.25×10^{-2}	1.25×10^{-2}
300	3.98×10^{-4}	3.75×10^{-4}	4.56×10^{-4}	1.94×10^{-4}	3.85×10^{-5}
400	3.83×10^{-6}	5.0×10^{-6}	3.35×10^{-4}	2.54×10^{-5}	0
450	1.04×10^{-7}	0	3.35×10^{-4}	2.54×10^{-5}	0
500	1.49×10^{-8}	0	3.35×10^{-4}	2.54×10^{-5}	0

to $e^{-1600/200} = 3.35 \times 10^{-4}$, which its six runs reached at thresholds between 297 and 322, and its estimate of $Z_\sigma(m)$ for every m above the last threshold is that final volume. At the end its live points occupied between 12 and 36 of the 573 to 1 331 coalescence classes at the final threshold, and in every run some occupied class had a member above 400, the deepest lying between 419 and 466; reaching $m = 400$, where $Z_\sigma = e^{-12.5}$, would have needed about 2500 iterations, more than the budget allows, so the table does not show whether nested sampling would be trapped in its trees. The vertical likelihood chain visited energies up to between 381 and 446 in the six runs, but four runs never went above 400, so its median estimate there is zero. Naive sampling by contrast sees about one integer above 400 per run, since $2 \times 10^5 \times 3.8 \times 10^{-6} \approx 0.8$ (the six runs see 2, 1, 1, 0, 2 and 0), and is the best of the four there, because an independent draw is not confined to a tree. The compression factors that make nested and split sampling polynomial in the depth presuppose that the constrained kernel delivers an independent point of the level set; in the stage where split sampling stopped it delivered another tributary of the same orbit, or the same integer again.

The expected value that (14.1) delivers in this setting is the mean total stopping time on the box, $\mathbb{E}[\sigma_\infty] = 116.9$ at $B = 26$, and the Boltzmann evidence $\mathbb{E}[e^{\beta\sigma_\infty}]$, which nested sampling (a separate run with 400 live points, 6000 iterations and 20 single bit flips per replacement, script `ch10b_nested.py`) estimates at $\beta = 0.02$ as 16.2 against the exact 15.4: good enough, and irrelevant, because both are averages over the integers that do reach 1.

14.6 What simulation can and cannot decide

We can now state the verdict with some precision.

(i) The conjecture is a universal statement: no positive integer lies in every set of the nest $\{\sigma_\infty > m\}$. A sampler explores the nest at finite depth. Under

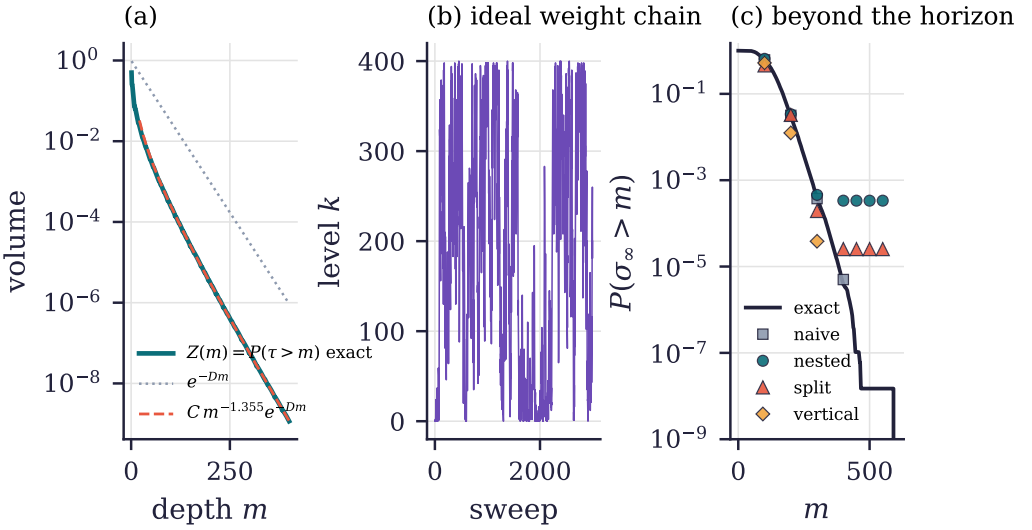


FIGURE 14.2. Simulation on both sides of the horizon. (a) The exact staircase $Z_\tau(m) = \mathbb{P}\{\tau > m\}$ under Haar measure, against e^{-Dm} and the fit $Cm^{-1.355}e^{-Dm}$ over $100 \le m \le 400$. (b) The level of the perfect split sampler of Theorem 14.2 at depth $K = 400$, over the first 3000 of twenty thousand sweeps: a free random walk on the stopping levels up to the depth 400 of the run, where the prior mass is 1.2×10^{-9}. (c) The same methods applied to the total stopping time on $[2, 2^{26})$: exact Z_σ against the medians of naive, nested, split and vertical likelihood estimates. Naive and nested sampling track the truth to depth 300, split sampling within a factor of about two there, and the vertical likelihood sampler only to depth 200, within a factor of 2.5; beyond depth 300 the level set estimates stop changing or vanish.

Haar measure this nest carries no signal at any depth: every $\{\sigma_\infty > m\}$ has measure one by Proposition 14.1(ii), and so does the limit $\{\sigma_\infty = \infty\}$, which contains 0 and every negative integer; the positive integers in the limit form a countable set, hence a Haar null set, and the conjecture says that there are none. The residue nest $\{\tau > m\}$, which a sampler can navigate, decreases to the set $E = \{\tau = \infty\}$, which has Haar measure zero by Proposition 14.1(i), contains -1, -5 and -17, and contains no integer $n \ge 2$ if the conjecture is true (Proposition 13.1(iii)). Either way the target is a null set. No Monte Carlo method samples a null set, and a method that is polynomial in the depth is polynomial in a parameter that must go to infinity.

(ii) What simulation does in polynomial time is sample and measure the exceptional classes at depth m: the residues with $\tau > m$, exactly and with no Markov chain, by Theorem 14.2. These are the classes the residue sieve of Section 13.3 enumerates, and the sampler is the randomized version of the sieve. Any computation that needs typical members of the exceptional classes, for instance to test a candidate potential or to seed a search for a

certificate, has an exact tool.

(iii) Beyond the horizon, which is the bit length of the integers being sampled, a deep level set is a small number of trees of merged orbits (Proposition 14.4, Remark 14.5, Table 14.1). A local sampler stays among a few members of one tree and leaves the tree only rarely, at about the acceptance rate of rejection sampling, so its depth is essentially the depth of the trees its initial population contains, and new trees are found by independent draws or by rare crossings, at about that rate. Estimating the number of slow integers below 2^B at levels where $Z_\sigma(m)$ is far below the reciprocal of the sampling budget therefore costs what it costs the census, and the census is exact.

(iv) The energy whose level sets a sampler can sample exactly at every depth is a function of a finite parity word; the energy the conjecture concerns is not. The gap between the two is the integrality gap of Proposition 13.1 once more: a sampler moves through residue classes, and the conjecture is about integers.

So the answer to the question that opened this chapter is no. Simulation does not decide the conjecture in polynomial time, or in any time, and the obstruction is not one of efficiency but of what the level sets are. What simulation offers, and this is worth having, is three things: the perfect sampler of Theorem 14.2 and the hazard description of the staircase, which make the exceptional set at every finite depth a computable and samplable object; the tree decomposition of the deep level sets, which says that, in the boxes we computed, the slow integers are a small number of long orbits together with their tributaries; and a heuristic account, borne out by the census, of why the one method that moves freely between trees, independent sampling, is the one that costs $1/Z_\sigma(m)$.

Notes

Nested sampling is Skilling's [93]; the vertical likelihood view of it, and the weighted slice sampler that places several level set methods in one scheme, are from Polson and Scott [79]; split sampling, with its Rao Blackwellized volume curve and its correspondences with the product estimator and with nested sampling, is from Birge, Chang and Polson [11]; the slice sampler itself is Neal's [70]. The weight $W = 1/Z$ makes the level histogram flat in $\log(1/Z)$, while the flat energy histogram that the Wang and Landau algorithm [104] aims at comes from weighting by the inverse density of states [79, Section 2]. The perfect sampler of Theorem 14.2 is ours as a sampler, but its backward step is the classical recursive method of uniform generation from counts of Nijenhuis and Wilf [74]; the proposition on the two energies, the coalescence heuristic and the tree decomposition of the deep level sets are ours; the merging of consecutive integers under the map is classical (Garner [26] used it to show that infinitely many pairs of consecutive integers have the same total stopping time) and is the

source of the confluences in every picture of the Collatz graph.

Exercises

Exercise 14.6. Implement the backward sampler of Theorem 14.2 and draw ten residues modulo 2^{200} with $\tau > 200$. Verify directly that each has more than $200/\log_2 3$ ones among its first 200 parities, and estimate from a thousand draws the mean number of ones at depth 200. Compare with the conditional law $j \mapsto N_{200}(j)/\sum_{j'} N_{200}(j')$.

Exercise 14.7. In the Bernoulli model of Chapter 3, show that the tail of the total stopping time has the same exponential rate $D = \log 2 - H(1/\log_2 3)$ as $\mathbb{P}\{\tau > m\}$, and explain why, in the model with starting points uniform on $[2, 2^B)$, the box changes the prefactor but not the rate for $m \gg B$. (For the census itself $Z_\sigma(m) = 0$ for $m \geq 592$, the largest total stopping time in the box for $B = 26$.)

Chapter 15

Open Problems and Computational Projects

Each chapter of this book ends at a precise question. This chapter collects them. We separate three kinds: problems about the Collatz map itself, problems about Switching MM, and computational projects that are well posed, finite and likely to repay effort. Where a problem is stated in an earlier chapter we restate it in its sharpest form and say what is already known. Several problems have been sharpened by results proved in this book, and we flag where a natural formulation turns out to be the wrong one.

15.1 The divergence half

Problem *(O1. Cesàro equidistribution).* Prove Conjecture 11.6: along every Syracuse orbit, $\liminf_N \bar{k}_N > \log_2 3$. Bounded orbits satisfy this automatically (Section 11.3), so the content is in the unbounded ones. Informally, prove that along no orbit can the running mean of the exponents $v_2(3y_j + 1)$ have upper limit at most $\log_2 3 = 1 + \beta$, which by Corollary 11.4 is what an unbounded orbit would need.

This is the divergence half of the conjecture itself, restated (Section 11.3). Proposition 11.3 supplies only the converse implication, which is an identity, and Remark 11.5 explains why numerical checks of $\bar{k}_N > \log_2 3$ on terminating orbits carry no information. A weaker target that would still be new is a statement of the form: along every orbit, the proportion of strides with $k_j = 1$ has upper density strictly less than one. Even this is open, and Corollary 4.15 shows that arbitrarily long runs of $k_j = 1$ do occur.

Problem *(O2. Trajectory mixing).* Fix a and a constant $C > 1/|\log(3/4)|$, which is about 3.48, and let x_0 be drawn uniformly from the odd integers up to X. Show that the empirical distribution of $y_j \bmod 3^a$ over the strides before the orbit reaches 1, stopped after $\lfloor C \log X \rfloor$ strides if the orbit has not reached 1 by then, converges in probability, as $X \to \infty$, to the stationary law π_a of Lemma 11.8. In the model the cap is reached with probability tending to zero; its only role is to make the empirical distribution defined for every starting value.

The problem is posed for random starting values because no version can hold for every orbit: the orbit of $(4^m - 1)/3$ reaches 1 after one stride and stays there. As Lemma 11.8 shows, the model chain on residues modulo 3^a, in which the halving exponents are independent Geometric$(\frac{1}{2})$ draws, mixes *exactly* in a strides,

with every nontrivial eigenvalue zero, so there is no spectral question left for the model. The entire content is the transfer to the map, whose halving exponents are produced by the orbit itself. Tao proves a fine scale form of the transfer for the law of a single iterate of a random starting value [96, Propositions 1.9 and 1.14]; the problem asks for the empirical distribution along the whole orbit.

The evidence is strong. Along orbits of up to about 25000 strides from random starting values near 2^{10000}, the residues modulo 3^5 approach π_5 at the rate of independent sampling, and they scatter about that benchmark about as much as the model chain does (Section 11.8).

Problem *(O3. Certificates with growing horizon).* Theorem 4.8 and Corollary 4.15 rule out certificates $W(S^{(J)}(x)) \le \rho\, W(x)$ with W increasing, for any fixed J, and a decreasing W fails at $x = 5$ for every horizon, since $S^{(J)}(5) = 1$; without monotonicity a certificate bounded away from zero exists exactly when the conjecture holds (Remarks 4.3 and 4.10). Find an increasing certificate whose horizon $J(x)$ grows with x, say $J(x) = O(\log x)$, with $W(S^{(J(x))}(x)) < W(x)$ for every odd $x \ge 3$.

For increasing W the condition reads $S^{(J(x))}(x) < x$, so such a certificate is equivalent to the conjecture together with a bound on stopping times: the problem is the conjecture in a stronger form, not a separate question. Its value is that it separates what is known on average from what is not. The density one version is already known: by the density theorem (Theorem 2.26), for every $C > 0$ the odd x with $S^{(J)}(x) < x$ for some $J \le C \log x$ have density one. The corresponding statement for the total stopping time, that almost every x reaches 1 within $C \log x$ strides, is open for every C; it would show that almost every integer reaches 1, and in the model it holds exactly when $C > 1/|\log(3/4)| \approx 3.48$.

Problem *(O3′. Potentials of unbounded depth).* Chapter 6 shows that every potential $\log x + \varphi(x)$ with φ depending on boundedly many binary digits of x fails, and that the run length potentials $\log x + c\, v_2(x+1)$ fail at a fraction at least 0.143 of odd integers. Determine the infimum, over potentials of depth h (reading the runs of the first h blocks), of the fraction of odd integers at which the potential fails to decrease, whether it tends to zero as $h \to \infty$, and if so how fast. The failure set of such a potential is, up to a set of density zero, a union of residue classes modulo powers of 2, so its density is the total measure of those classes, which by Theorem 2.8 is the probability of failure in the Bernoulli model: the problem is a question about the model, and its answer cannot reveal arithmetic structure beyond it.

Problem *(O4. The third cumulant of total stopping times).* The census of Section 3.6 locates the variance deficit: the deficit of about ten percent is an additive, apparently bounded term of about 2.6 in $\log^2$ units, concentrated in the deterministic final descent of the orbit, which indicates that the model is

asymptotically right about the variance. The skewness is not explained: it is 0.67 at $n \approx 2^{29}$ against 0.96 for the inverse Gaussian law of the model. Prove that the variance deficit is bounded, and explain the third cumulant. As stated, the first part contains the conjecture, since the deficit of a dyadic block $[2^k, 2^{k+1})$ is finite only if every integer in it reaches 1; the question is to prove that the deficit is bounded assuming the conjecture, or unconditionally for the integers of each block whose orbits reach 1 within a fixed multiple of k steps.

Problem *(O5. Uniform integrability)*. Settled by Proposition 3.17: the iterate martingale of the accelerated model, stopped when it first drops to 1 or below, is not uniformly integrable, and Chapter 12 explains why its deterministic counterpart, the indicator of reaching 1, leaves nothing to certify short of the conjecture itself. There is no separate open problem here.

15.2 The cycle half

Problem *(C1. Arithmetic sharpening of the cycle identity)*. Find a function f, derived from the arithmetic of the parity vector rather than the size of the elements, such that every nontrivial cycle with m odd steps, k steps in all and least element $x_{\min}$ satisfies $x_{\min} < m/(3\Lambda f(m))$, where $\Lambda = k\log 2 - m\log 3$ is the value at that cycle; Lemma 10.21 gives this with $f = 1$. By Corollary 10.22, $f(m) \geq \frac{m}{3}(1+\eta)$ for some fixed $\eta > 0$ and all large m would confine the least element of every nontrivial cycle with many odd steps below a constant of about $1/(2\eta)$ for small η, and so reduce the cycle half to a finite verification; any f bounded below by a constant $b > 1$ would lower every threshold of Table 10.6 by at least the factor b.

An effective irrationality measure for $\log_2 3$ cannot do this on its own (Proposition 10.20): a measure with exponent ν and constant c excludes only $r < (3cX\log 2)^{1/\nu}$. On a cycle $x(2^k - 3^m) = c_k(x)$, where $c_k(x)$ is a sum of terms $3^j 2^i$ indexed by the parity word, and the window lemma shows that the least element is confined to an interval with ends a factor of more than $m/3$ apart. In the computations of Chapter 10 the primitive balanced words with $100 \leq m \leq 2000$ sit near the top of the window, at about 0.72 of its upper end (an observation, not a theorem), so arguments that use only the size of the elements, and therefore apply equally to rational cycles, are unlikely to gain the factor there; the problem is really whether $2^k - 3^m$ can divide $c_k(w)$ for a nontrivial word, perhaps attacked by a block decomposition in the style of the thinning argument that settled Erdős 125.

Problem *(C2. Effective measures and the finite range)*. Find explicit ν and c with $|\log_2 3 - s/r| \geq c\, r^{-\nu}$ for all integers s and $r \geq 1$, with ν as close to 2 as possible; the explicit bound of Rhin [85], in the form quoted in Section 10.7, corresponds to $\nu \approx 14.3$. By Remark 10.19 such a pair excludes the cycle lengths $r < (3cX\log 2)^{1/\nu}$ without any enumeration, but this range can never exceed the exact ladder value

$r_*(X)$, and Table 10.7 shows that even the idealized pair $\nu = 2$, $c = 0.01773$ falls short of it by factors from about 2 to more than 50. The interest of the problem is therefore Diophantine, not the cycle bound.

Problem *(C3. Sharper accountings at the next rungs)*. Hercher's record $r \geq 1.375 \times 10^{11}$ [38, Corollary 29], unconditional since Bařina's verification to 2^{71} [7], uses a sharper accounting of the cycle structure than the inequality $\log(1 + 1/(3x_i)) \leq 1/(3M)$. It keeps the form of the cycle inequality: his bound [38, Theorem 27] is $0 < s/r - \log_2 3 < 1/(4X \log 2)$, in place of $1/(3X \log 2)$, so the least length it does not exclude is again a rung, by the mediant lemma of Chapter 10 (Hercher finds it with his lemma on the least denominator of a fraction in an interval [38, Lemma 22]), and every threshold of Table 10.6 falls by the factor $\frac{3}{4}$; his computer assisted Corollary 29 lowers the threshold for the rung 72 057 431 991 further, to $3 \cdot 2^{69}$. Extend that computation to the later rungs: for each rung from 137 528 045 312 on, find the least verification range at which his accounting excludes it.

15.3 Switching MM

Problem *(S1. Rates beyond quadratics)*. Settled for the distance by Theorem 9.9. For μ strongly convex and L smooth f, with $\kappa = L/\mu$, the even iterates of $\mathrm{SMM}_{\mathrm{det}}$ contract the distance to the minimizer by $q = (1 - 1/\kappa) \max(1 - (1 + \beta)/\kappa, \beta)$ per period, which is the exact worst case over the class: explicit one dimensional functions attain it for every κ, and performance estimation confirms it. The function gap contracts by q^2, which is exact for $\kappa \geq (1 + \beta)/(1 - \beta)$, where the quadratic $\frac{\mu}{2}\|x\|^2$ attains it. The remaining question is the function gap for $\kappa < (1 + \beta)/(1 - \beta)$, where performance estimation gives a smaller worst case, for example 0.049 against $q^2 = 0.086$ at $\beta = \log_2 3 - 1$ and $\kappa = 2$: find it in closed form and describe the worst function.

Problem *(S2. Stochastic analysis)*. Extend Theorem 8.4 from quadratics to μ strongly convex, L smooth objectives under a bounded variance oracle: show that the stationary mean square error of $\mathrm{SMM}_{\mathrm{det}}$ is at most $\frac{5}{3}$ times that of gradient descent with the same step, up to terms vanishing with the step size, for every $\beta < 1$.

Partial progress. On quadratics the result holds for any uncorrelated noise, and Proposition 8.15 extends it to multiplicative (curvature) noise: under the least squares scaling the slow mode ratio is unchanged, and under the diffusive scaling a linear periodic method satisfying the hypotheses of Theorem 8.12, with gradient gains c_i (Definition 8.11), has the mean square stability threshold $\epsilon^* = 2 \sum_i c_i / \sum_i c_i^2$, which is 1.472 for $\mathrm{SMM}_{\mathrm{det}}$ against 0.830 for every step momentum at $\beta = \log_2 3 - 1$. What remains open is the nonquadratic case and large multiplicative noise at large step, where the bound 5/3 fails.

Problem *(S3. The best period two ratio).* Settled for quadratics by Proposition 8.14. Every period two ratio r between 1 and the Chebyshev ratio $r_{\mathrm{ch}}(\kappa)$, correctly scaled, lies on the rate and noise frontier $R = 1 - 4\kappa/((2-N)(\kappa+1)^2)$, with noise floor $N = 2 - 1/G(r)$ and $G(r) = (1+r)^2/(4r)$, and larger ratios are dominated by the Chebyshev ratio in both criteria; the Chebyshev ratio is the fast end, gradient descent the quiet end, and the SMM ratio $1+\beta$ sits near the quiet end with a 5.4 percent gain in the rate exponent for a 5.1 percent rise in noise. The open part is the choice of scale without knowledge of μ: the optimal scale is $\alpha L \to (2+\beta)/(1+\beta)$, not 1.

Problem *(S4. Adaptive switching).* $\mathrm{SMM}_{\mathrm{state}}$ suppresses injection when progress is adequate. Find a principled switching rule, using only L and observed function values, that retains the uniform noise bound of Proposition 8.6 while closing the noiseless gap to every step momentum identified in Chapter 8.

Partial progress. On quadratics, at every parameter value computed, no switching rule is unstable or much slower than the worse of gradient descent and every step Nesterov: the joint spectral radius of the descent and injection matrices, computed exactly by Proposition 9.6, is below one, and at $\beta = \log_2 3 - 1$ and $\beta = 0.9$ it exceeds the larger pure rate by at most 0.006. A proof of the transient bound Conjecture 9.7 would make the stability unconditional, at all parameter values. The noise side of the problem, a rule whose stationary variance is provably bounded by $\frac{5}{3}$ of gradient descent, remains open.

15.4 Computational projects

The following projects are finite, well specified, and suited to a collaborator with computing resources. Each comes with a clear success criterion.

P1. A machine checked cycle bound. Formalize the bound of Theorem 10.16 in Lean. The affine identity and the inequality $3^m < 2^k$ are already proved in Lean 4 using only its core library (Section 10.10), and Section 10.8 gives an exact integer certificate: an enclosure of $\log_2 3$ by two convergents certified with 320 bit mantissas, an enclosure of $\log 2$, the 44 ladder rungs up to 72 057 431 991, and an exclusion inequality at each rung using integers of at most 248 bits. What remains is to formalize the upper half of the cycle inequality in the logarithm free form $2^k(3X)^m \le 3^m(3X+1)^m$, the mediant lemma, Theorem 10.8 and a verified squaring routine, and to replay the certificate. Success means a Lean file, with no `sorry`, whose main theorem states that every nontrivial cycle of the accelerated map on the positive integers has at least 72 057 431 991 odd steps, conditional only on a stated hypothesis encoding the verified range: every positive integer below 2^{68} reaches 1.

P2. Thresholds and the window. The verification thresholds at which the sharp bound moves are now computed for $60 \le \log_2 X \le 90$ (Table 10.6); the next step

past the current range 2^{71} [7] is at $2^{71.884}$. The open computational question is the window of Lemma 10.21: for the hypothetical cycles at the rungs, compute the largest lower bound for the least element that can be proved from the structure of $c_k(w)$ for words with the prescribed (m,k) that satisfy a further restriction, for instance compatibility with the first steps of an orbit starting above 2^{71}. Without a restriction nothing beyond the trivial bound $3^m - 2^m$ for $c_k(w)$ is possible, since the word $1^m 0^{k-m}$ attains it.

P3. Orbits from very large starting values. The study of Section 11.8 followed orbits from near 2^{10000} with residues modulo 3^5. Extend it to starting values near 2^{10^5} and beyond, to moduli 3^a with $a \leq 8$, to joint statistics of consecutive residues, and to the dyadic digits of the iterates, and look for any systematic departure from the model chain of Lemma 11.8.

P4. Total stopping time cumulants at scale. Extend the exact census of Section 3.6 beyond 2^{30} (memory permitting) and the sampled version to 10^{15}, test whether the additive variance deficit saturates, and fit the third cumulant.

P5. Switching MM at scale. Run $\mathrm{SMM}_{\mathrm{det}}$ against every step Nesterov and heavy ball at $\beta \in \{0.585, 0.9\}$, and against Adam, on minibatch training of standard models where the gradient noise is genuine, with the base step size of gradient descent and the three momentum methods fixed at the common value $\alpha = 1/L$, at which all four are stable and every eigendirection of the Hessian has $a = \alpha\lambda \leq 1$. Report final loss and its variance over seeds, and compare the measured noise floors with the prediction of Theorem 8.4 evaluated on the spectrum of the Hessian at the solution, whose ratio to gradient descent lies between 1 and $(2 + 2\beta + \beta^2)/(2 + \beta)$ on every eigendirection (Proposition 8.7). Larger common steps are possible only below $2(1 + \beta)/((1 + 2\beta)L)$, the stability limit of every step Nesterov, where the denominator of (8.9) vanishes and which lies below the limit $(2 + \beta)/((1 + \beta)L)$ of $\mathrm{SMM}_{\mathrm{det}}$ (Corollary 7.8); above $a = 1$ Proposition 8.7 does not apply; (8.13) still holds there, since the proof of Theorem 8.4 uses only $|\rho| < 1$, and at $\beta = 0.585$ it gives a ratio above $(2 + 2\beta + \beta^2)/(2 + \beta)$ once $a > 1.41$, reaching 1.55 at the Nesterov limit.

P6. Periodic schedules in general. Settled in the slow mode by the gain weight theorem (Theorem 8.12): no stable linear periodic first order method beats gradient descent at matched speed, and the penalty is the Cauchy and Schwarz ratio $p \sum c_i^2 / (\sum c_i)^2$ of the gradient gains. What remains computational is the regime away from the slow mode: for a given spectrum and noise level, find the period p schedule minimizing the summed stationary variance at a prescribed worst case rate, and compare it with the period two frontier of Proposition 8.14.

P7. Erdős 125 *beyond* 4^{13}. Extend the exact density computation of Chapter 10 to 4^{15} or beyond with a bit array of a few gigabytes, and test whether the successive minima just below powers of 3 track the quality of the approximations $3^m \approx 4^k$ quantitatively.

P8. Perfect samplers beyond the coefficient stopping time. The sampler of Theorem 14.2 is exact because the level sets of the coefficient stopping time τ are unions of residue classes. Find other energies with this property whose level sets admit a dynamic programme of size polynomial in the depth, for instance the number of descents in the first k steps or the coefficient stopping time of $T^{(j)}$ for fixed j, give the dynamic programme for each, and use the samplers to measure the frequency with which the block potentials of Proposition 6.6 fail at depth k, as a function of k.

P9. Trees of slow integers. Table 14.1 shows that the 257 integers below 2^{26} with total stopping time above 400, counted in steps of T as in Chapter 14, are 34 trees of merged orbits, and the 24 above 440 are six. Table 14.2 counts the trees of $\{\sigma_\infty > m\} \cap [2, 2^B)$ for $B = 22$, 24 and 26: over this range their number grows with B roughly like 2^{cB} with c between 1 and 1.4, and their mean size like $2^{0.4B}$. These exponents are local: at fixed m each tree has its own root, an integer with $\sigma_\infty = m - 100$, and there are only finitely many such integers, all at most 2^{m-100} since a step of T at most halves, so the number of trees is bounded in B. Explain the exponents observed for $22 \le B \le 26$, or find how the number and the size of the trees grow when m and B increase together, for instance with m/B fixed. The number of trees is the number of points at remaining depth $m - 100$ reached from the box by an orbit segment of more than a hundred steps, a statistic of the merging mechanism of Lemma 14.3 that has not, to our knowledge, been studied; the mean size is a count on the inverse tree of Chapter 9, and whether the certified exponents of that chapter bear on it, when, at the deeper levels, the roots usually lie above the box, is part of the problem. Relate the roots of the trees to the delay records of Chapter 1.

Notes

Problems O1, O3, C2 and the cycle formulation are restatements of questions going back to Terras [99], Steiner [94] and Eliahou [21], sharpened by the results of this book. Problem C1 and the Erdős 125 analogy come from our note on the AlphaProof Nexus system [78, 101]. Problems S1 to S4 and P5, P6 grew out of the analysis of Chapters 7 and 8, and P8 and P9 out of Chapter 14.

A Gallery of Orbits

The pictures in this gallery are computed, not drawn. Every point, cell and curve comes from exact integer arithmetic on the maps of Chapter 1 and Chapter 2 (in one plate, on the sumset of Chapter 10), and every number quoted in a caption was produced by the script that made the picture (the files `art_*.py` in the `scripts` directory). We have allowed ourselves colour and a dark ground, but nothing decorative has been added: every colour encodes a computed quantity. Throughout, T_0 is the standard map ($3n+1$ for odd n, $n/2$ for even n), $\sigma_\infty(n)$ is the number of steps of T_0 that take n to 1, its total stopping time (as in Chapter 1; Chapters 3, 14 and 15 count steps of T instead), and $M(n)$ is the largest value on the orbit.

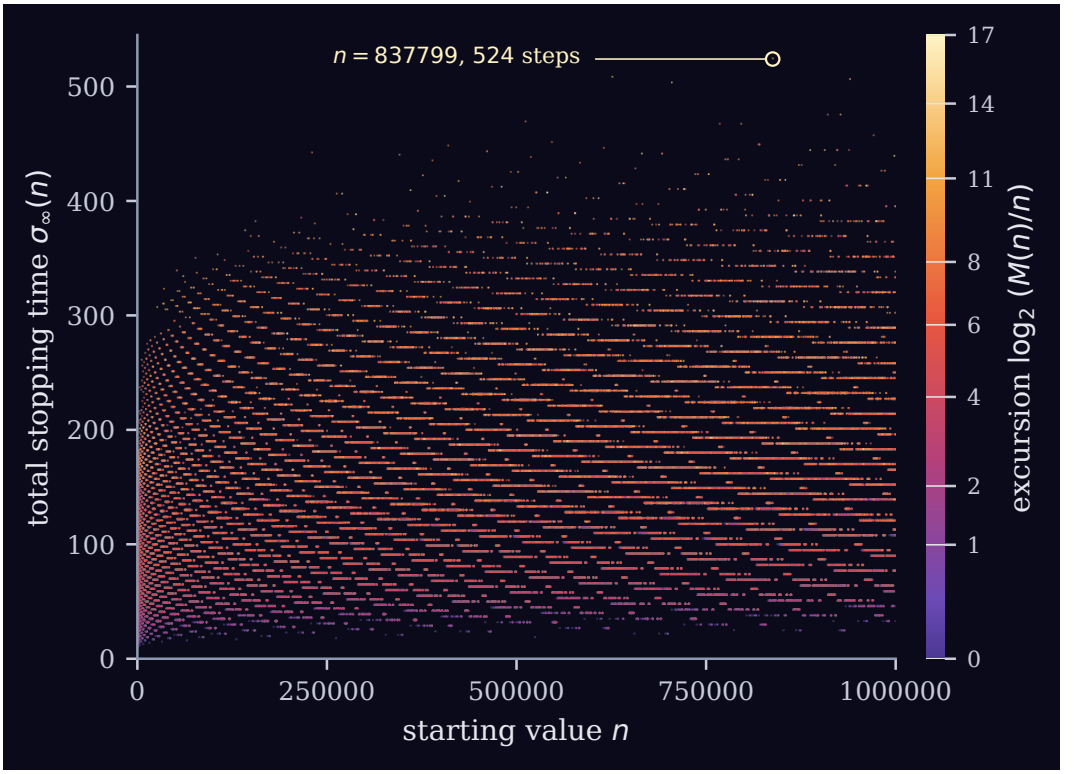


PLATE 1. The total stopping time landscape. Each of the integers $2 \le n \le 10^6$ is one point at height $\sigma_\infty(n)$, coloured by the excursion $\log_2(M(n)/n)$; the longest orbit belongs to $n = 837799$, with 524 steps, and the largest excursion to $n = 159487$, which climbs to $M(n) = 17\,202\,377\,752$, a factor of about $2^{16.7}$. The horizontal streaks are runs of consecutive integers whose orbits merge after a few steps, and the slow upward drift of the cloud is the logarithmic growth of total stopping times discussed in Chapter 1 and Chapter 3.

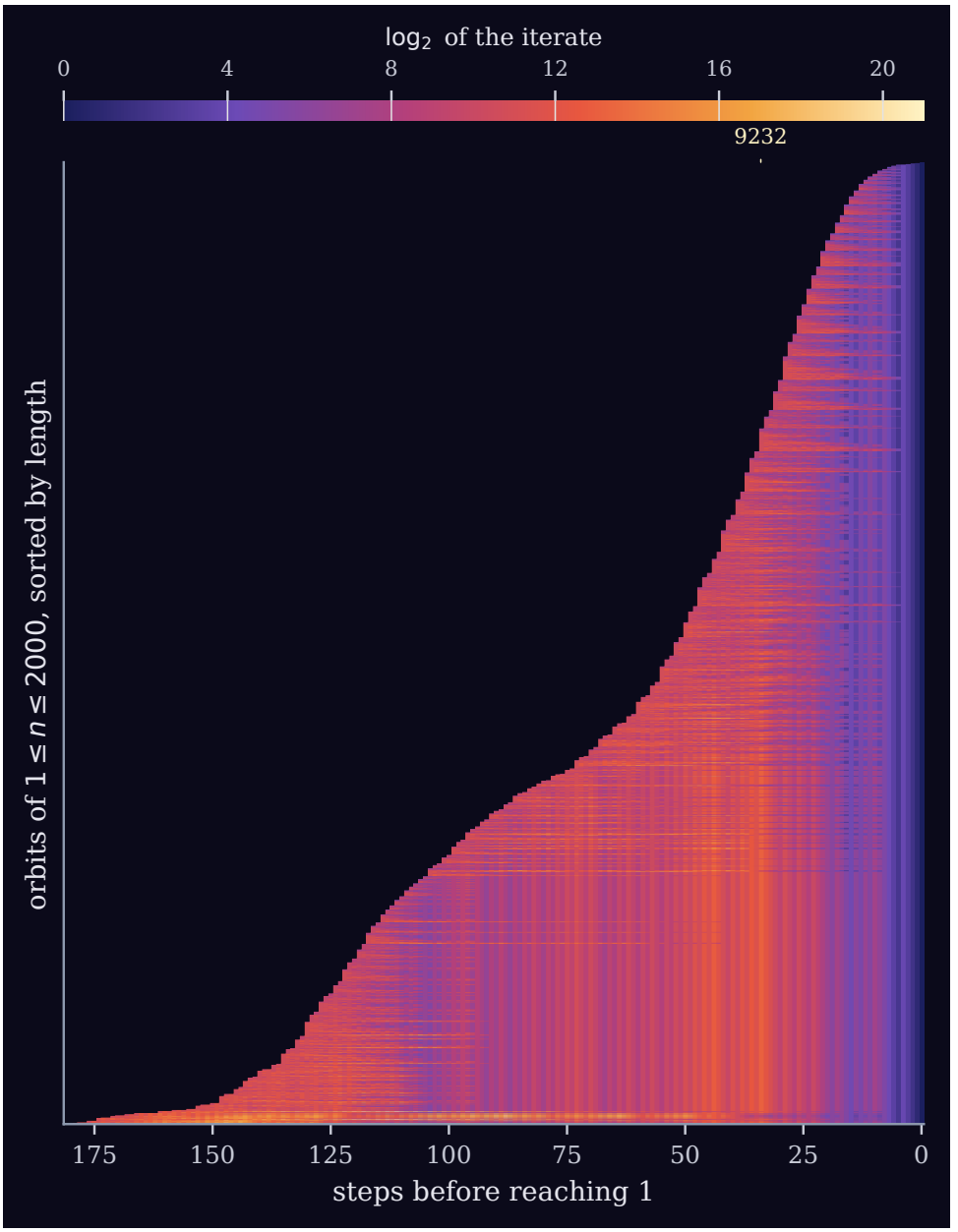


PLATE 2. The orbits of $1 \le n \le 2000$, one per row, written from right to left so that every orbit ends at 1 in the last column; colour is $\log_2$ of the iterate, and the rows are sorted by orbit length, so the left edge is the sorted list of total stopping times, rising to 181 steps for $n = 1161$. The bright vertical band 34 steps from the end is the value 9232, through which 772 of the 2000 orbits pass: shared tails line up as columns, which is the inverse tree of the next plate seen edge on.

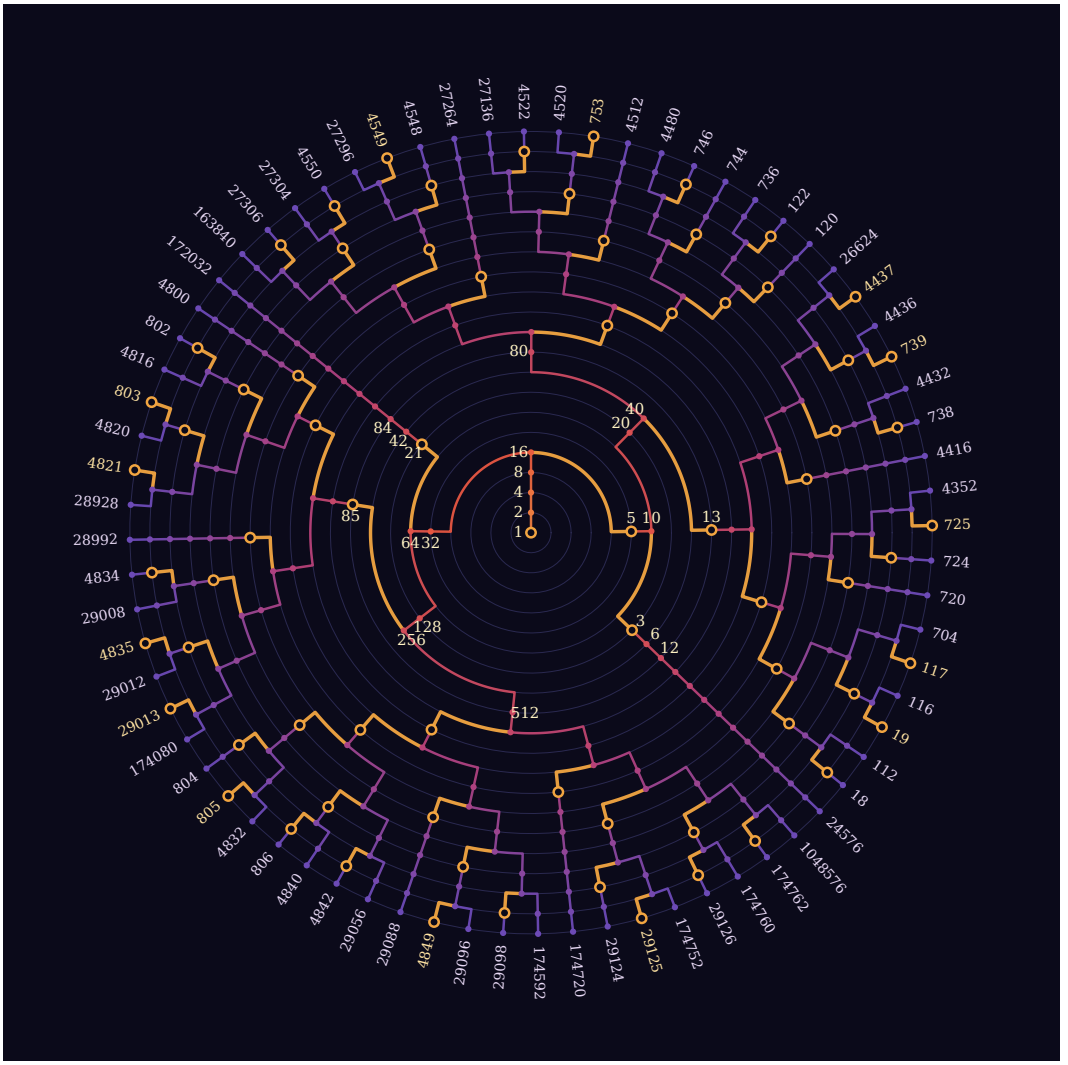


PLATE 3. The inverse tree of 1 to depth 20. The children of m are $2m$ and, when $m \equiv 4 \pmod 6$ and $m > 4$, the odd number $(m-1)/3$; radius is depth, so the ring at radius d holds exactly the n with $\sigma_\infty(n) = d$. The tree has 342 nodes; its 72 leaves, the numbers with total stopping time 20, run from 18 to $2^{20} = 1048576$, and the open circles mark the odd children, the only places where the tree branches; the branching random walk model of this tree, due to Lagarias and Weiss [59], is mentioned in Chapter 3.

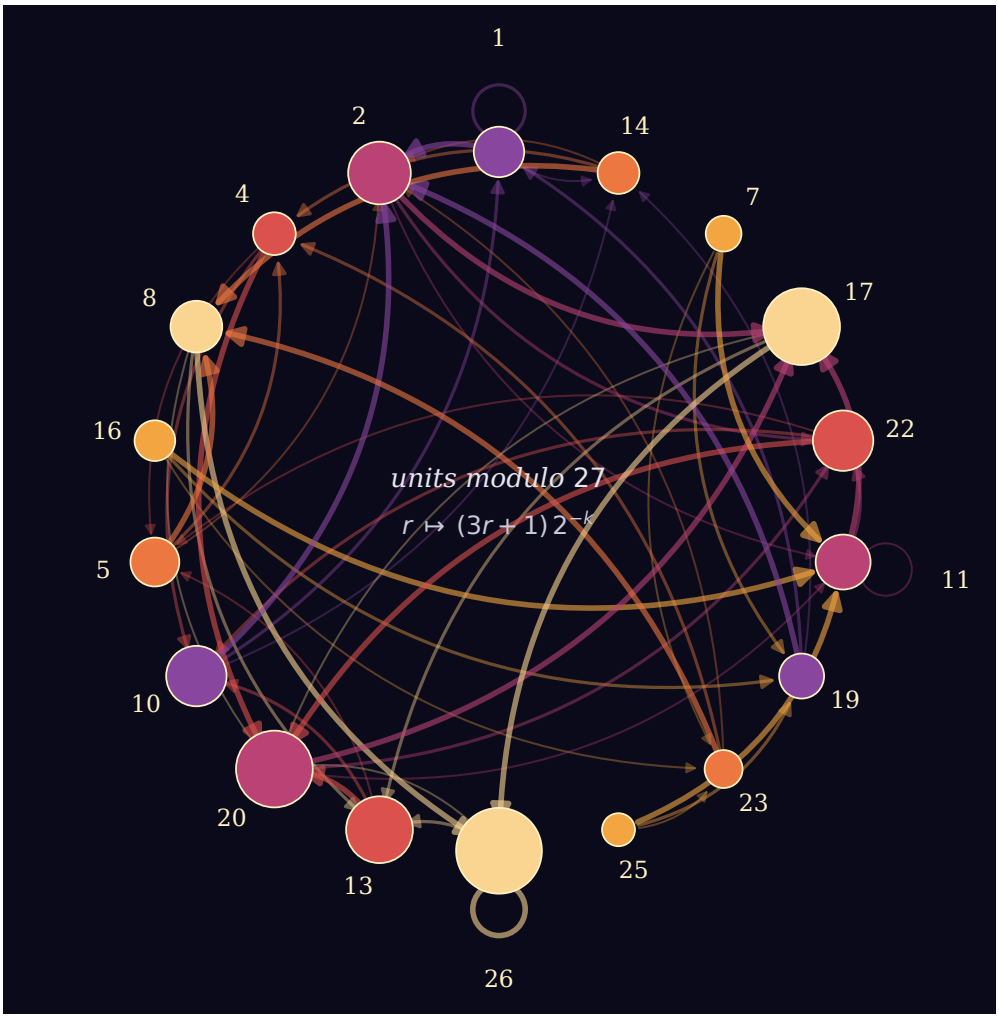


PLATE 4. The model chain on the 18 units modulo 27, in which the halving exponents are independent Geometric($\frac{1}{2}$) draws: r moves to $(3r+1)2^{-k}$ with probability 2^{-k}, which for one stride from a uniformly random odd $y \equiv r$ holds exactly in the limit (Chapter 11); with the units placed in the order $2^0, 2^1, \ldots, 2^{17}$, division by 2 is one step clockwise, so the out edges of r fan clockwise from $3r+1$. Edges with $k \leq 3$ are shown, and the three units r, $r+9$, $r+18$ share a shade because their common residue modulo 9 alone determines the out edges; node area is the exact stationary law, largest at 26 ($46570/262143 \approx 0.18$, with a loop of probability about $\frac{1}{2}$) and smallest at 25 ($1598/262143 \approx 0.006$), and it projects onto the law $\frac{1}{63}(8, 16, 11, 4, 2, 22)$ modulo 9.

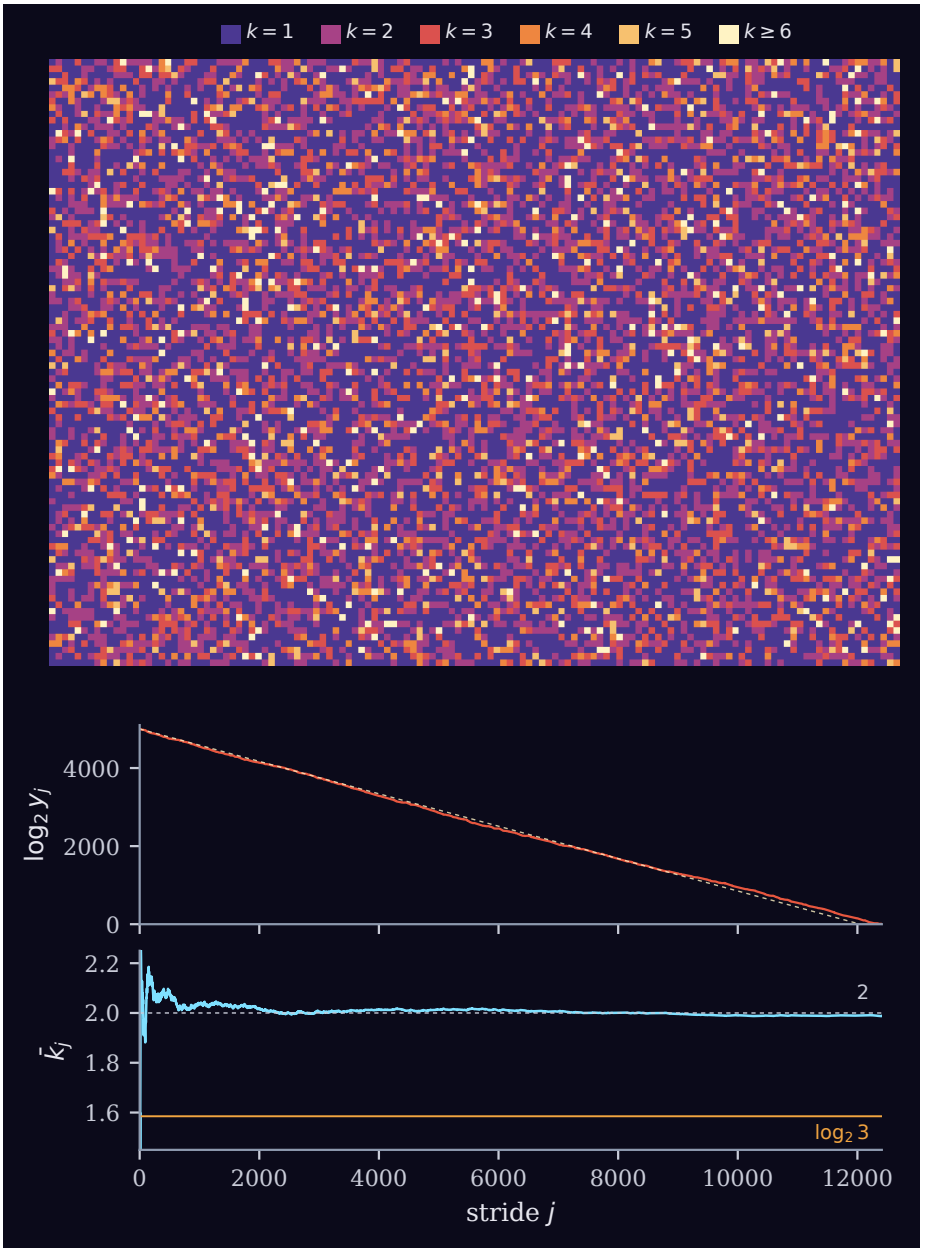


PLATE 5. A giant orbit. Starting from $x_0 = 3^{3155} \approx 2^{5000.56}$, the Syracuse map takes 12408 strides (37075 steps of T_0) to reach 1. The mosaic shows the exponents $k_j = v_2(3y_{j-1}+1)$ in reading order, 132 to a row, with frequencies 0.504, 0.249 and 0.128 for $k = 1, 2, 3$, close to the Geometric law of Chapter 2; the descent of $\log_2 y_j$ follows the drift line of slope $\log_2 3 - 2$ per stride (dashed), and the running mean $\bar{k}_j$ settles near 2, ending at 1.98799; it lies at or below the threshold $\log_2 3$ of Chapter 11 only for $j \le 10$.

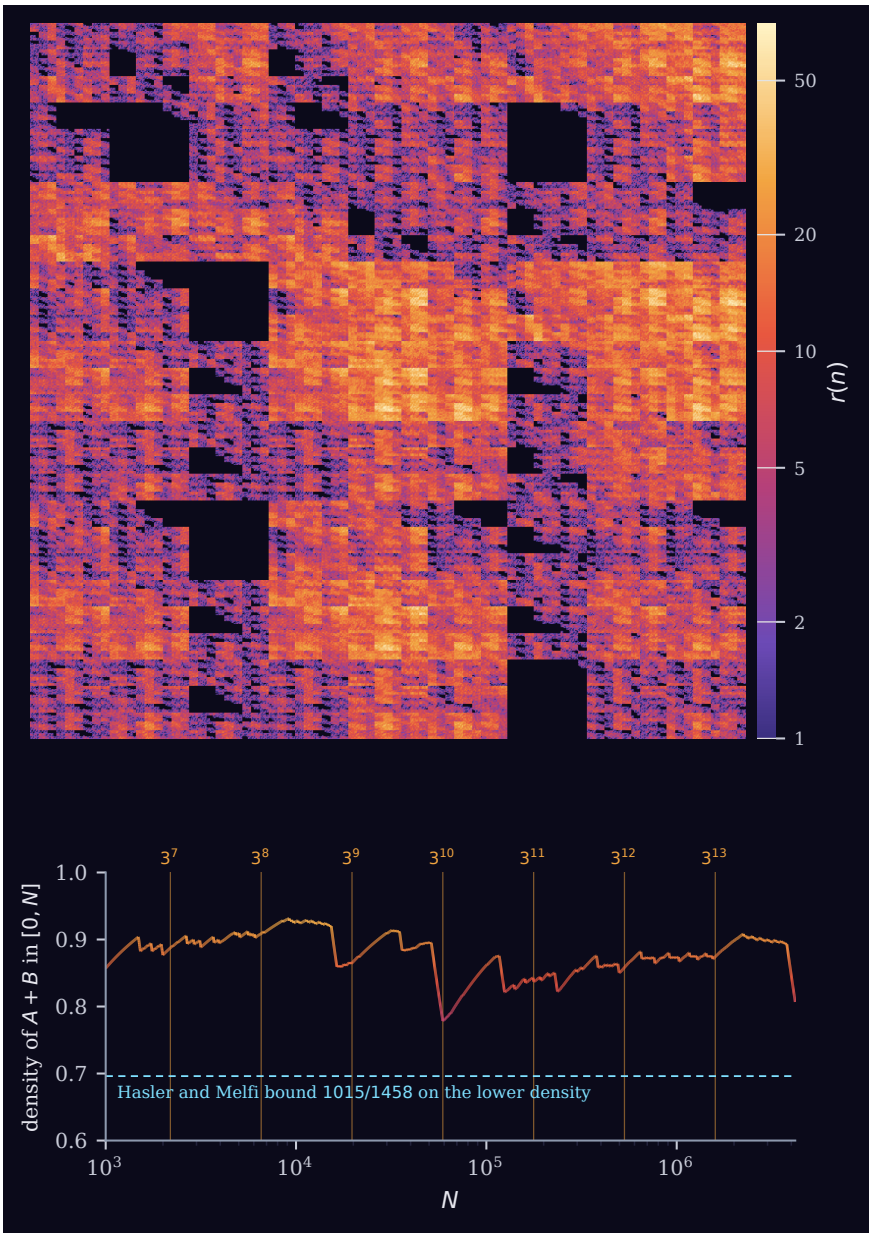


PLATE 6. The Erdős 125 sumset of Chapter 10, with A the numbers whose base 3 digits are 0 or 1 and B the same in base 4. Top: the number $r(n)$ of representations $n = a + b$ for $n < 3^{12}$, placed on a 729×729 square with column and row read from the even and odd position base 3 digits of n; dark holes are gaps of $A + B$, which covers 85.83 percent of this range, and $r(n)$ reaches 70. Bottom: the density of $A + B$ in $[0, N]$ for $10^3 \le N \le 4^{11}$, whose lowest value, 0.7785, occurs at $N = 3^{10} - 1$, well above $1015/1458 \approx 0.696$, the upper bound of Hasler and Melfi [36] on the lower density.

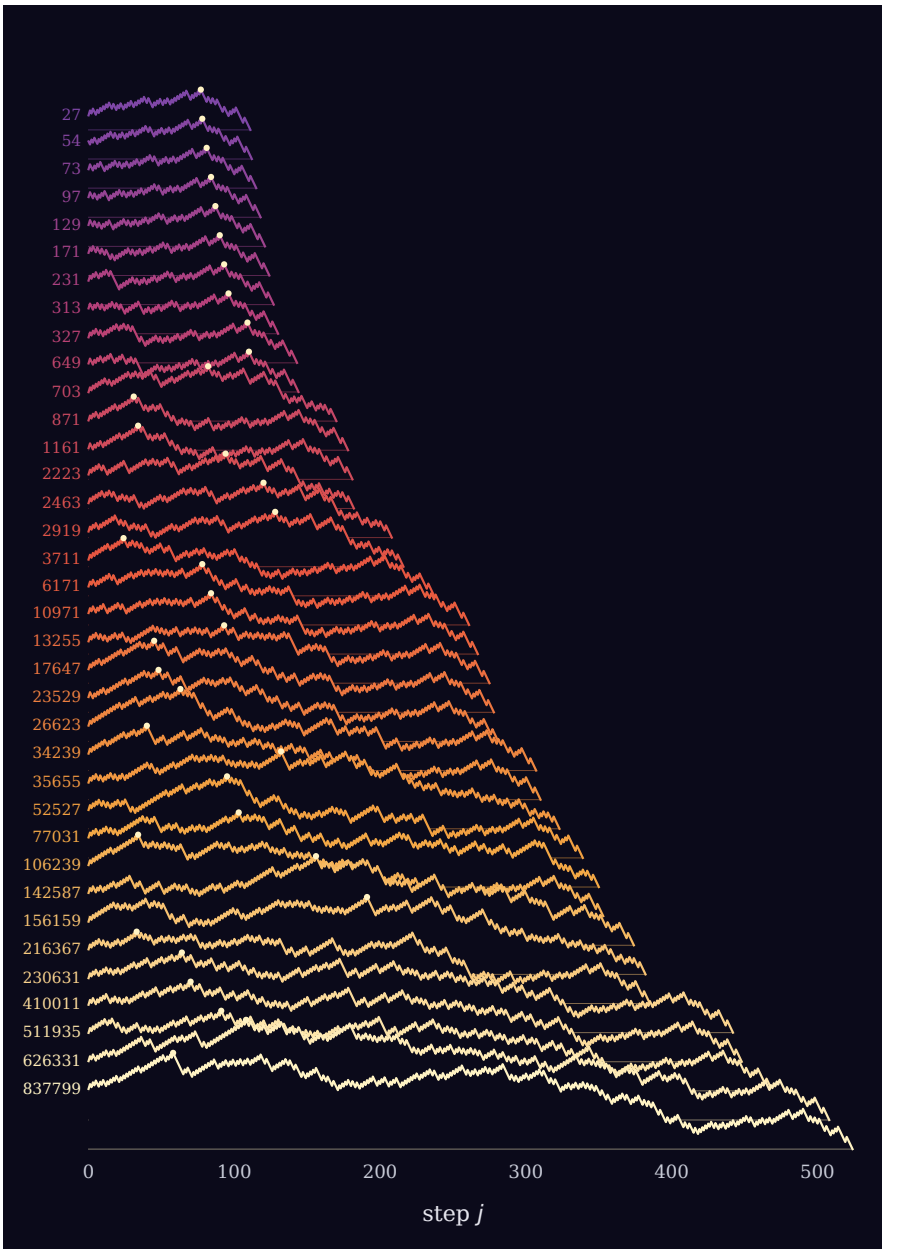


PLATE 7. Ridgelines of the total stopping time records. Up to 10^6 there are 44 integers whose total stopping time exceeds that of every smaller integer; for the 36 of them from 27 (111 steps) to 837799 (524 steps) each ridge traces $\log_2 T_0^{(j)}(n)$ until the orbit reaches 1, with a pale dot at its maximum. The highest summit belongs to 626331, which climbs to 7 222 283 188; the saw teeth are the alternation of rises by about $\log_2 3$ and falls by 1 whose net balance is the descent problem of Chapter 4.

Appendix A

Numerical Constants

Table A.1 collects the main constants used in the book, to six significant figures unless stated otherwise. All were recomputed for this book with the scripts of Appendix B.

The continued fraction of $\log_2 3$

To the precision of our computations (300 digits, with the first 60 partial quotients confirmed at 150),

$$\log_2 3 = [1; 1, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, 1, 1,$$
$$55, 1, 4, 3, 1, 1, 15, 1, 9, 2, 5, \dots].$$

The large partial quotients 23 and 55 make the convergents just before them, $p_8/q_8 = \frac{1054}{665}$ and $p_{13}/q_{13} = \frac{301994}{190537}$, exceptionally good two sided approximations. Only the second lies above $\log_2 3$ and counts for the one sided cycle bound of Chapter 10, which jumps from $r = 190\,537$ to the next rung of the ladder, $r = 10\,781\,274$; the first lies below, with $\lceil 665 \log_2 3 \rceil - 665 \log_2 3 = 0.99994$, and is useless there (Remark 10.10). With the verified range $X = 2^n$, the exact lower bounds on the number of odd elements r and halvings s of a nontrivial cycle are given in Table A.2. The established range is $X = 2^{71}$, by Bařina [7], and there the bound is the same as at 2^{68}. The last row is nevertheless the current record, $r \geq 137\,528\,045\,312$ [7, §6]: Hercher's sharper accounting of the cycle [38, Corollary 29] reaches it already from $X = 3 \cdot 2^{69} < 2^{71}$.

TABLE A.1. Numerical constants.

Constant	Value	Meaning
$\log 2$	0.693147	one halving
$\log(3/2)$	0.405465	asymptotic odd step of T
$\log(3/4)$	-0.287682	drift per stride of S, model
$\frac{1}{3}\log(3/4)$	-0.0958940	drift per step of T_0, model
$\delta^* = \log(\sqrt{3}/2)$	-0.143841	drift per step of T
σ^2	0.301737	step variance of T
$2(\log 2)^2$	0.960906	stride variance of S, model
$\mathbb{E}[\Delta^2]$	1.04367	second moment of a stride, model
$\frac{2}{27}(\log 6)^2$	0.237808	long run step variance of T_0, model
$1/\lvert\delta^*\rvert$	6.95212	total stopping time slope, T, model
	6.91960	the same, observed for $n \le 10^4$
$\sigma^2/\lvert\delta^*\rvert^3$	101.387	variance slope, T, model
$3/\lvert\log(3/4)\rvert$	10.4282	total stopping time slope, T_0, model
$1/\lvert\log(3/4)\rvert$	3.47606	total stopping time slope, S, model
$C = \sigma^2/(\delta^*)^2$	14.5836	central limit constant
D	0.0346882	Chernoff rate, density theorem
e^{-D}	0.965907	Chernoff factor per step
$1/\log_2 3$	0.630930	critical frequency of odd steps, T
$H(1/\log_2 3)$ in bits	0.949956	residue sieve exponent h
$\log_2 3$	1.58496	Cesàro threshold
$\beta = \log_2 3 - 1$	0.584963	Switching MM momentum
$L = \log(5/3)$	0.510826	largest stride ascent
$\log(10/3)$	1.20397	largest step ascent of T_0
$\kappa = L/\lvert\log(3/4)\rvert$	1.77566	stride ratio (an analogy)
$\kappa_* = \log 2/\lvert\log(3/4)\rvert$	2.40942	per step ratio (an analogy)
$\log(10/3)/\lvert\log(3/4)\rvert$	4.18508	step ratio of T_0 (an analogy)
$\rho(h)$	$(-\frac{1}{2})^h$	parity autocorrelation, model
$\mathcal{A}$	$1/3$	long run variance factor
$(2+2\beta+\beta^2)/(2+\beta)$	1.35867	SMM noise amplification
$1/(1-\beta)$	2.40942	every step amplification
$(2+\beta)/2$	1.29248	SMM effective step multiplier
$(2+\beta)/(1+\beta)$	1.63093	SMM stability limit for $\alpha\lambda$
θ^*	1	Wald root, accelerated model

TABLE A.2. Exact cycle bounds from the ladder of $\log_2 3$, as functions of the verified range X. Only $X \leq 2^{71}$ is established; the last row shows what a larger range would give by this method.

X	$r \geq$	$s \geq$
2^{40}	10 781 274	17 087 915
2^{60}	397 573 379	630 138 897
2^{68}	72 057 431 991	114 208 327 604
2^{71}	72 057 431 991	114 208 327 604
2^{72}	137 528 045 312	217 976 794 617

Appendix B

Code

Every number computed in this book was produced by a short Python script; numbers quoted from the literature are cited where they appear. The scripts use the standard scientific stack (mpmath, NumPy, SciPy, SymPy and Matplotlib), with Numba for the census of Section 3.6 and for the gallery, CVXPY with the Clarabel solver for the performance estimation of Chapter 9, and scikit-learn only to load the data set of Section 8.9; they run in seconds to minutes on a laptop. They are in the folder `book/scripts` of the book's repository, `https://github.com/VadimSokolov/collatz`, and are released under the MIT License. The two Lean files of Section 10.10 are in its folder `lean`. This appendix lists two of them in full and indexes the rest.

B.1 The exact cycle bound

The script below computes the partial quotients of $\log_2 3$ at eighty digits, builds the candidate rungs $q_{k-1} + jq_k$, keeps those that are records of the one sided distance $\delta(r) = \lceil r \log_2 3 \rceil - r \log_2 3$, and returns the first rung not excluded by a verified range X. By Theorem 10.8 and the mediant lemma (Lemma 10.11) this is the exact bound. Its output reproduces Eliahou's bound $s \geq 17\,087\,915$ [21] at $X = 2^{40}$ (with $r \geq 10\,781\,274$) and gives $r \geq 72\,057\,431\,991$ at $X = 2^{68}$, unchanged at the current range $X = 2^{71}$ [7].

```
# Exact lower bound on the number r of odd elements
# of a nontrivial Syracuse cycle, given that every
# n <= X reaches 1.  By the mediant lemma only rungs
# of the one sided ladder of theta = log_2 3 need
# to be checked.
from mpmath import mp, mpf, log, ceil, floor
mp.dps = 80
theta = log(3) / log(2)

def partial_quotients(x, n=40):
    a = []
    for _ in range(n):
        ai = int(floor(x)); a.append(ai)
        x = 1 / (x - ai)
    return a

a = partial_quotients(theta)
```

```
q = [0, 1]                    # q_{-1} = 0, q_0 = 1
for ai in a[1:]:
    q.append(ai * q[-1] + q[-2])

# candidate rungs q_{k-1} + j q_k,
# then keep the records of delta
cands = sorted({q[k-1] + j * q[k]
                for k in range(1, len(q) - 1)
                for j in range(a[k] + 1)} - {0})
delta = lambda r: ceil(r * theta) - r * theta
rungs, best = [], mpf(2)
for r in cands:
    d = delta(r)
    if d < best:
        best = d; rungs.append((r, d))

def first_not_excluded(X):
    eps = 1 / (3 * X * log(2))
    return next(r for r, d in rungs if d < r * eps)

for e in (40, 60, 68, 71, 72):
    r = first_not_excluded(mpf(2) ** e)
    s = int(ceil(r * theta))
    print(f"X = 2^{e}:  r >= {r},  s >= {s}")
```

B.2 The noise amplification of Switching MM

The second script derives the slow mode amplification factor of Theorem 8.4 symbolically. On one eigendirection with $a = \alpha\lambda$, two steps of $\mathrm{SMM}_{\mathrm{det}}$ starting from an even index contract the error by $\rho = (1-a)(1-(1+\beta)a)$ and add noise of variance $\alpha^2\sigma^2[(1-a)^2(1+\beta)^2+1]$.

```
# Exact stationary variance of Switching MM on one
# eigendirection of a quadratic, a = alpha * lambda,
# unit noise (alpha^2 sigma^2 = 1).
import sympy as sp
a, b = sp.symbols('a b', positive=True)
# two step contraction and two step noise variance
rho = (1 - a) * (1 - (1 + b) * a)
q = (1 - a)**2 * (1 + b)**2 + 1
v_even = q / (1 - rho**2)
# odd phase: one descent step after an even time
v_odd = (1 - a)**2 * v_even + 1
v_avg = sp.simplify((v_even + v_odd) / 2)
# gradient descent gives 1/2
slow = sp.simplify(sp.limit(a * v_avg, a, 0))
print("slow mode a*Var:", slow)
V = sp.simplify(2 * slow)
print("amplification vs GD:", V)
print("at beta = log2(3) - 1:",
```

```
      float(V.subs(b, sp.log(3, 2) - 1)))
print("limit beta -> 1:", V.subs(b, 1))
```

Its output is the factor $((1+\beta)^2+1)/(2+\beta)$, equal to 1.3587 at $\beta = \log_2 3 - 1$ and to $5/3$ in the limit $\beta \to 1$.

B.3 Index of scripts

The scripts are grouped by the chapter that uses them, in the order of the book, and a script that reads the output of another is listed after it. In particular `ch3_o4_decomp.py` writes the table of total stopping times of T below 2^{26} that `ch3_o4_end.py` and most scripts of Chapter 14 read.

Script	Purpose
Chapter 1	
`ch01_ch02_figures.py`	orbit of 27, total stopping times, residue tree of Chapter 2
`ch01_ch02_numbers.py`	worked orbits and delay records
`ch1_slopes.py`	least squares slopes of the total stopping time against $\log n$ for T_0 and T, over $n \le 10^4$ and $n \le 10^5$ (Section 1.3)
Chapter 2	
`ch02_stopping.py`	Terras bijection, Chernoff bound
`ch02_exact.py`	exact stopping counts by dynamic programming
`ch02_extra.py`	orbit of 27 by strides, $c_{70}(27)$, means over odd $n < 2^{20}$, values of τ
`ch02_checks.py`	Geometric counts below 2^{20}, parity words of T_0
`ch02_figures.py`	Syracuse increment, sunburst, stopping heat map, 2 adic shift
`ch2_stopdecay.py`	decay of $1 - F(k)$, binomial tail and Chernoff bound (Figure 2.6)
Chapter 3	
`ch3_stats.py`	Lyapunov parameters, total stopping time CLT, parity autocorrelations, excursion tail
`ch3_stopping.py`	total stopping time slope and standardized histogram
`ch3_figures.py`	model paths, autocorrelations, Ville bound
`ch3_o4_model.py`	exact law of the model passage time
`ch3_model_means.py`	exact mean and variance of the model passage time
`ch3_o4_decomp.py`	exact census to 2^{30}, variance decomposition
`ch3_o4_census.py`	exact census of total stopping times to 2^{30}, skewness
`ch3_o4_end.py`	where the variance deficit lives

Script	Purpose
Chapter 4	
`ch4_ch5_numbers.py`	two stride scans and other numbers of Chapters 4 and 5, crossings of the optimal coefficient
`ch4_exact.py`	multi stride ascent densities, passage time of the stride model
`ch4_figures.py`	strides, chained ascents
Chapter 5	
`ch5_numbers.py`	stride constants and their sensitivity, orbit of 27
`ch5_parrondo.py`	capital dependent Parrondo games
`ch5_nag_check.py`	optimal constant momentum on quadratics, numerical check
`ch5_figure.py`	optimal Nesterov coefficient against κ (Figure 5.1)
Chapter 6	
`ch5c_mm.py`	residue potentials, positive cycles of G_5, block maps, run length potentials
`ch5c_figures.py`	residue graph, block descent and run length potential (Figures 6.1 and 6.2)
Chapter 7	
`smm_rate.py`	rate algebra of Switching MM
`ch6_checks.py`	two step identity, defect, period two Chebyshev schedule
`ch6_contraction.py`	per step contraction factors and their crossings (Figure 7.2)
`ch6_trajectory.py`	two dimensional trajectories (Figure 7.1)
`ch6_ablation.py`	frequency ablation run to convergence
Chapter 8	
`smm_variance_book.py`	the second listing above
`smm_var.py`	symbolic and numerical stationary variances of SMM, NAG and HB
`smm_sim.py`	simulated slow mode variance against the leading term
`ch7_variance.py`	exact stationary variances, amplification figure, predicted noise ratios
`ch7_gainweight.py`	gain weight theorem, periodic Lyapunov checks
`ch7_gainweight_fig.py`	gain weight figure (Figure 8.2)
`ch7_pareto.py`	period two rate and noise frontier
`ch7_multiplicative.py`	curvature noise, stability thresholds
`ch7_experiments.py`	sweep, ablation, noise, logistic and minibatch runs
`ch7_experiments_fig.py`	figures of the sweep, the ablation, the logistic runs and the minibatch runs

Script	Purpose
`ch7_noisefig.py`	noise experiment against theory (Figure 8.7)
`ch7_reconcile.py`	observed and predicted noise ratios
`ch7_extra.py`	closed forms, bounds, logistic predictions
Chapter 9	
`ch5b_tree.py`	certified exponents for the inverse tree
`ch5b_tree_exact.py`	the same certificates checked in exact integer arithmetic
`ch5b_counts.py`	exact counting functions on the tree
`ch5b_chains.py`	large scale growth of the doubling chain construction
`ch5b_jsr.py`	exact joint spectral radius
`ch5b_pep.py`	performance estimation for two steps
`ch5b_transient.py`	transient grid behind the transient bound conjecture
`ch5b_figures.py`	figures of the chapter
Chapter 10	
`cycle_bound_book.py`	the first listing above
`ch08_negative.py`	cycles on the negative integers
`ch08_compute.py`	continued fraction at 300 digits, confirmed at 150, and the convergents
`ladder.py`	ladder rungs, exact and crude bounds
`ch08_records.py`	records of $r/(3\Lambda_r)$ in exact arithmetic, against the ladder
`ch08_bounds.py`	block bounds B_k and the sharp bound at 150 digits
`ch08_ties.py`	ties in double and 150 digit arithmetic, exact record scan
`ch08_crude.py`	exact versus crude bound
`ch08_figures.py`	figures of the records and the one sided distances
`ch08_profile.py`	thresholds, insufficiency, irrationality profile
`ch08_window.py`	window lemma and parity word exploration
`ch08_certificate.py`	exact integer certificate at 2^{68}
`ch08_erdos125.py`	density of the Erdős 125 sumset
`ch08_erdos125_fig.py`	figure of the Erdős 125 density
`ch08_misc.py`	small checks: the longest orbit of T up to 10^4, the rational cycle of $\frac{1}{5}$
Chapter 11	
`ch9_cesaro_scan.py`	terminal $\bar{k}_N$ for odd $3 \le x_0 \le 5001$ (Remark 11.5)
`ch9_figures.py`	running $\bar{k}_N$ and the Syracuse law modulo 9 (Figures 11.1 and 11.2)
`ch9_numbers.py`	the Cesàro identity along three long orbits, odd step thresholds
`ch9_spectral.py`	model chains modulo 3^a, empirical transitions

Script	Purpose
`ch9_table.py`	restricted and full transition matrices modulo 3^a (Table 11.1)
`ch9_orbits.py`	residues along single orbits from 2^{10000}
`ch9_orbits_check.py`	single orbits against the model chain, chi squared checks
Chapter 13	
`ch10b_sieve.py`	residue sieve counts and entropy exponent
`ch10b_divsieve.py`	divisibility sieve for cycles
`ch10b_figures.py`	residue sieve counts and the uncovered classes in the 2 adic coordinate (Figure 13.1)
Chapter 14	
`ch10b_nested.py`	exact partition function, nested sampling runs, evidence at $\beta = 0.02$
`ch10b_nested2.py`	decay of the level set volumes, nested search for the slowest integers
`ch10c_perfect.py`	perfect split sampling for the coefficient stopping time
`ch10c_ladder.py`	exact staircase, hazards, fits, partition function at $\beta = D$
`ch10c_montecarlo.py`	nested, split and vertical likelihood samplers
`ch10c_acceptance.py`	bit flip acceptance inside level sets
`ch10c_clusters.py`	coalescence classes of the level sets
`ch10c_trees.py`	trees by box size and level
`ch10c_treerecords.py`	trunks, tributaries and delay records
`ch10c_treemodel.py`	naive model of tree sizes
`ch10c_figures.py`	staircase, perfect sampler and sampler comparison (Figure 14.2)
`ch10c_acceptance_fig.py`	acceptance and class sizes (Figure 14.1)
Utilities and the gallery	
`bookstyle.py`	shared style of the figures
`make_index_tags.py`	places index entries in the text
`art_*.py`	cover, part openers and gallery plates

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